Characters of $GL_n(\mathbb{F}_q)$ and the Boson-Fermion correspondence

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Chapter 1

Notions of commutative algebra

1.1 Modules and rings

1.1.1 Basic definitions

Definition 1.1.1. [1] Let R be a ring and M an abelian group written additively. Then M is said to be an $R-$module if there is a map $R \times M \to M : (r, m) \mapsto rm$ satisfying:

(1) $r(sm) = (rs)m$

(2) $(r + s)m = rm + sm$

(3) $r(m + n) = rm + rn$

(4) $1m = m$

for any $r, s \in R$ and $m, n \in M$.

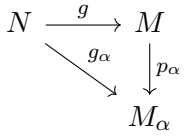
This is the notion of left $R-$module. Right $R-$modules are defined in a similar way. If R is a commutative ring, these two notions are equivalent. In this section, modules are all "left" (see remarks after definition (1.1.16)).

Definition 1.1.2. [1] Given two $R-$modules M, N, a *morphism* $f : M \to N$ from M to N, is a map satisfying $f(m + n) = f(m) + f(n)$ and $rf(m) = f(rm)$ for all $m, n \in M$ and $r \in R$. The set of morphisms from M to N is written as $\mathrm{Hom}_R(M, N)$.

Notation 1.1.3. [7] With these two definitions we have the notion of *category of* $R-$modules, which will be denoted by $R - \overline{\mathbf{mod}}$.

Definition 1.1.4. [7] Given a family $\{M_\alpha\}_{\alpha\in A}$ of $R-$modules, the product $M = \prod_{\alpha\in A} M_\alpha$ is defined to be the set of maps $f : A \to \bigcup_{\alpha\in A} M_\alpha$ with the property $f(\alpha) \in M_\alpha,\ \forall\alpha \in A$. So, $\prod_{\alpha\in A} M_\alpha$ inherits a structure of a $R-$module by defining rf to be the map $\alpha \mapsto rf(\alpha)$. The product is equipped with canonical projections $p_\alpha : M \to M_\alpha : f \mapsto f(\alpha)$.

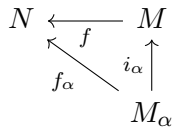
The product $M = \prod_{\alpha\in A} M_\alpha$ is characterized the following universal property: Suppose that we are given a $R-$module N with a collection of homomorphisms $g_\alpha : N \to M_\alpha$, then there exist a unique homomorphism $g : N \to M$ such that the diagram



commutes for all $\alpha \in A$ [7].

Definition 1.1.5. [7] Given a family $\{M_\alpha\}_{\alpha\in A}$ of $R-$modules, the direct sum $M = \bigoplus_{\alpha\in A} M_\alpha$ is defined to be the submodule of $\prod_{\alpha\in A} M_\alpha$ consisting of maps $f : A \to \bigcup_{\alpha\in A} M_\alpha$ with finite support, i.e. $f(\alpha) = 0$ for all but a finite number of $\alpha \in A$. The direct sum M is equipped with canonical injections $i_\alpha : M_\alpha \to M$, where $i_\alpha(m_\alpha) \in M$ satisfies $i_\alpha(m_\alpha)(\beta) = 0,\ \beta \neq \alpha$ while $i_\alpha(m_\alpha)(\alpha) = m_\alpha$.

The direct sum $M = \bigoplus_{\alpha\in A} M_\alpha$ satisfies the following universal property: Given a $R-$module N with a family of maps $f_\alpha : M_\alpha \to N$, then there exists an unique homomorphism $f : M \to N$ such that the diagram



commutes for all $\alpha \in A$ [7].

Definition 1.1.6. [1] Let S be a subset of a $R-$module M. Then S is said to be $R-$*free* if whenever $m_1, \dots m_k \in S$ and

$$r_1 m_1 + \cdots + r_k m_k = 0$$

for some $r_1, \dots, r_k \in R$. Then $r_i = 0 \quad \forall i = 1, \dots, k$.

Definition 1.1.7. [1] A subset S of an $R-$module M is said to be a set of generators of M, if for every $m \in M$, we have $r_1, \dots, r_k \in R$ and $m_1, \dots, m_k \in S$ such that $m = \sum_{i=1}^{k} r_i m_i$.

Notation 1.1.8. Let M be an $R-$module with a finite set of generators $\{m_1, \dots, m_k\}$. In this case we will write $M = \sum_{i=1}^{k} Rm_i$.

Definition 1.1.9. [1] An $R-$module M is said to be *cyclic* if it is generated by a unique element $m \in M$. In this case we can just write $M = Rm$.

Definition 1.1.10. [1] An $R-$module M is said to be a *free* $R-$module if it has a $R-$free set of generators S. We will say that S is an $R-$basis of M and that M is *freely generated* on S.

Corollary 1.1.11. [1] Let $M = \sum_{i=1}^{k} Rm_i$ be a free $R-$module. Then

$$M \cong \bigoplus_{i=1}^{k} Rm_i \cong \bigoplus_{i=1}^{k} R = R^k$$

Proof: Define the surjective homomorphism $R^k \to M : (r_1, \dots, r_k) \mapsto r_1 m_1 + \cdots + r_k m_k$. Since $\{m_1, \dots, m_k\}$ is free, this map is injective and we have an isomorphism.

Definition 1.1.12. Let $(A, \leq)$ be a partially ordered set, $\{M_\alpha\}_{\alpha \in A}$ a family of $R-$modules and $\{f_{\alpha\beta}\}_{\alpha \leq \beta}$ a family of homomorphisms $f_{\alpha\beta} : M_\alpha \to M_\beta$ satisfying $f_{\alpha\alpha} = \mathbb{1}_{M_\alpha}$ $\forall \alpha \in A$ and $f_{\alpha\gamma} = f_{\beta\gamma} \circ f_{\alpha\beta}$ for $\alpha \leq \beta \leq \gamma$. Then we say that the pair $(\{M_\alpha\}_{\alpha \in A}, \{f_{\alpha\beta}\})$ is a *directed system*. The direct limit of a directed system $\{(M_\alpha\}_{\alpha \in A}, \{f_{\alpha\beta}\})$ is defined to be the module

$$\varinjlim M_\alpha = \bigcup_{\alpha \in A} M_\alpha \Big/ \sim \tag{1.1}$$

where $m_\alpha \in M_\alpha \sim m_\beta \in M_\beta$ if there exists $\gamma \geq \alpha, \beta$ such that $f_{\alpha\gamma}(m_\alpha) = f_{\beta\gamma}(m_\beta)$.

Let $\alpha, \beta \in A$ and take $\gamma \geq \alpha, \beta$. Then, we define the sum

$$[m_\alpha] + [m_\beta] = [f_{\alpha\gamma}(m_\alpha) + f_{\beta\gamma}(m_\beta)] \tag{1.2}$$

Moreover, the $R-$action on $\varinjlim M_\alpha$ is defined as $r[m_\alpha] = [rm_\alpha]$. All these operations are clearly well defined.

1.1.2 Tensor product of modules

The main purpose of this subsection is to construct the tensor product modules.

Definition 1.1.13. [1] Let M (resp. N) be right (resp. left) $R-$modules and P an abelian group. A map $f : M \times N \to P$ is said to be *balanced* if it satisfies

(1) $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$

(2) $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$

(3) $f(mr, n) = f(m, rn)$

for all $r \in R$ and $m_i \in M,\ n_i \in N$.

This concept can be seen as an extension of the concept of bilinear maps from linear algebra.

Theorem 1.1.14. [1] Let M (resp. N) be right (resp. left) $R-$modules. Then there exist an abelian group $M \otimes_R N$, called the *tensor product* between M and N, with the following properties.

(1) There exist a balanced map $\pi : M \times N \to M \otimes_R N$ such that every element of $M \otimes_R N$ is of the form $\sum_{i=1}^{k} \pi(m_i, n_i)$ where $m_i \in M$ and $n_i \in N$.

(2) Every balanced map $f : M \times N \to P$ factors through $M \otimes_R N$, i.e. there exist $\tilde{f} : M \otimes_R N \to P$ such that $\tilde{f} \circ \pi = f$.

Proof: Let $A = \bigoplus_{M \times N} \mathbb{Z}$, i.e. A is the free $\mathbb{Z}-$module generated by $M \times N$. Writing $\overline{(m,n)} \in A$ in place of $i_{(m,n)}(m,n)$, let $G \subset A$ be the submodule generated by elements of the form:

$$\begin{cases} \overline{(m_1 + m_2, n)} - \overline{(m_1, n)} - \overline{(m_2, n)} \\ \overline{(m, n_1 + n_2)} - \overline{(m, n_1)} - \overline{(m, n_2)} \\ \overline{(mr, n)} - \overline{(m, rn)} \end{cases} \tag{1.3}$$

So we define the tensor product to be the $\mathbb{Z}-$module $M \otimes_R N := A/G$. By definition, it is clear that $\pi : M \times N \to M \otimes_R N : (m,n) \mapsto \overline{(m,n)} + G$ is balanced, and that $M \otimes_R N$ is generated by sums of the form $\sum_{i=1}^{k} \pi(m_i, n_i)$ where $m_i \in M$. Now, let $f : M \times N \to P$ be a balanced map. Since A is the free $\mathbb{Z}-$module generated by $M \times N$, we have a map $\bar{f} : A \to P : \overline{(m,n)} \mapsto f(m,n)$. But since f is balanced $G \subset \mathrm{Ker}\bar{f}$, so by the isomorphism theorem we get a $\mathbb{Z}-$module homomorphism $\tilde{f} : A/G = M \otimes_R N \to P : \pi(m,n) \mapsto f(m,n)$.

Notation 1.1.15. From now on, in the conditions of the previous theorem, we will write $m \otimes n$ instead of $\pi(m,n)$.

It's important to note the properties (1) and (2) in theorem (1.1.14) characterize the tensor product uniquely. In fact, suppose that $M \otimes'_R N$ is another abelian group that satisfies both properties (1) and (2). Let $\pi' : M \times N \to M \otimes'_R N$ be the balanced map such that $M \otimes'_R N$ is generated by elements of the form $\sum_{i=1}^{k} \pi'(m_i, n_i)$ for $m_i \in M$ and $n_i \in N$. So we have the induced homomorphisms $\overline{\pi} : M \otimes_R N \to M \otimes'_R N$ and $\overline{\pi}' : M \otimes'_R N \to M \otimes_R N$ such that

$$\overline{\pi}'(\pi(m,n)) = \pi'(m,n) \quad \text{and} \quad \overline{\pi}(\pi'(m,n)) = \pi(m,n) \tag{1.4}$$

By the property (2), both maps $\overline{\pi}$ and $\overline{\pi}'$ are surjective and therefore, inverses of each other by the last equation. Particularly, this proves that $\overline{\pi} : M \otimes_R N \to M \otimes'_R N$ is an isomorphism of groups.

Definition 1.1.16. [1] Let S and R be rings. A $(S,R)-$bimodule M is a module that is both a left $S-$module and a right $R-$module that satisfies the additional property

$$(sm)r = s(mr) \quad \forall s \in S, r \in R \text{ (“associativity of actions”)} \tag{1.5}$$

An important observation is that if R is a commutative ring, then there is no difference between left $R-$modules and right $R-$modules. In fact, any left (or right) $R-$module is a $(R,R)-$bimodule. For example, if M is a left $R-$module, we can just define $mr := rm$ turning M into a $(R,R)-$bimodule. The associativity property is a direct consequence of the commutativity of R. Therefore, when working with modules over commutative rings we will not specify left or right. For example, any vector space V over a field $\mathbb{F}$ is a $(\mathbb{F},\mathbb{F})-$bimodule and if W is another vector space over the same field, we conclude that $V \otimes_{\mathbb{F}} W$ is also a vector space over $\mathbb{F}$.

Now, let M be an $(S,R)-$bimodule and N a right $R-$module. We claim that $M \otimes_R N$ has a structure of a left $S-$module. To see this, just define for $s \in S$ the map $f_s : M \times N \to M \otimes_R N : (m,n) \mapsto sm \otimes n$. This map is clearly balanced, so this define an action $s(m \otimes n) := \tilde{f}_s(m \otimes n) = (sm) \otimes n$. In summary, we have:

Theorem 1.1.17. [1] Let M be a $(S,R)-$bimodule and N a right $R-$module. The tensor product of modules $M \otimes_R N$ is a left $S-$module with action

$$s(m \otimes n) = (sm) \otimes n. \tag{1.6}$$

Similarly, if N is a left $S-$module, then $N \otimes_S M$ is a right $R-$module with action $(n \otimes m)r = n \otimes (mr)$. The proof is exactly the same as for the module $M \otimes_R N$.

Now we come to important application of the definition of tensor product. If R is a ring contained in another ring S, then we can define a functor $\mathrm{Ind}_R^S : R - \overline{\mathbf{mod}} \to S - \overline{\mathbf{mod}}$ called *induction*. The functor is defined as follows, for each left $R-$module M, we let $\mathrm{Ind}_R^S M := S \otimes_R M$. Since S is clearly a $(S,R)-$bimodule, the abelian group $\mathrm{Ind}_R^S M$ is also a left $S-$module. Also, for $f \in \mathrm{Hom}_R(M,N)$ we have

$$\mathrm{Ind}_R^S f : \mathrm{Ind}_R^S M \to \mathrm{Ind}_R^S N : s \otimes m \mapsto s \otimes f(m). \tag{1.7}$$

An important property of induction is that it is transitive.

Theorem 1.1.18. Let $R \subset S \subset T$ be rings and M a left $R-$module. Then we have an isomorphism of left $T-$modules

$$\mathrm{Ind}_S^T \mathrm{Ind}_R^S M \cong \mathrm{Ind}_R^T M \tag{1.8}$$

To show this, we will need some lemmas.

Lemma 1.1.19. Let $N \cong N'$ be two isomorphic left $R-$modules and M a $(S,R)-$bimodule. Then we have an isomorphism of left $S-$modules $M \otimes_R N \cong M \otimes_R N'$.

Proof: Let $f : N \to N'$ be an isomorphism of left $R-$modules and g its inverse. Then $M \times N \to M \otimes_R N' : (m,n) \mapsto m \otimes f(n)$ is balanced, so we have an induced homomorphism of groups $\varphi : M \otimes_R N \to M \otimes_R N' : m \otimes n \mapsto m \otimes f(n)$. It's clear that φ is also a homomorphism of modules. By the same reasoning, we can construct a homomorphism $\psi : m \otimes n' \mapsto m \otimes g(n')$ from $M \otimes_R N'$ to $M \otimes_R N$. But φ and ψ are clearly inverses of each other, so we get that φ is an isomorphism and we have proved the theorem.

Lemma 1.1.20. Let M be a left $R-$module. Since R is a $(R,R)-$bimodule, we can form the tensor product $R \otimes_R M$. We claim that $R \otimes_R M \cong M$ as left $R-$modules.

Proof: Define the map $R \times M \to M : (r,m) \mapsto rm$. This map is clearly balanced so we have a homomorphism of abelian groups $\varphi : R \otimes_R M \to M : r \otimes m \mapsto rm$. Conversely, the map $\psi : S \to R \otimes_R M : m \mapsto 1 \otimes m$ is clearly a group homomorphism. Also we have that $\psi \circ \varphi = 1_{R \otimes_R M}$ because

$$\psi \circ \varphi(r \otimes m) = \psi(rm) = 1 \otimes rm = r \otimes m. \tag{1.9}$$

Moreover

$$\varphi \circ \psi(m) = \varphi(1 \otimes m) = 1m = m \tag{1.10}$$

so $\psi \circ \varphi = 1_{R \otimes_R M}$ and $\varphi \circ \psi = 1_M$. This proves that $R \otimes_R M$ and M are isomorphic as abelian groups. But both φ and ψ are left $R-$modules homomorphisms. So we get that we have in fact, φ is an isomorphism of left $R-$modules.

Lemma 1.1.21. Let M be a right $R-$module, N a $(R,S)-$bimodule and L a left $S-$module. Then we have an isomorphism of abelian groups $M \otimes_R (N \otimes_S L) \cong (M \otimes_R N) \otimes_S L$.

Proof: Fix $m \in M$ and define the map $f_m : N \times L \to (M \otimes_R N) \otimes_S L : (n,l) \mapsto (m \otimes n) \otimes l$. This map is clearly balanced, so we have a well defined homomorphism of groups

$$\tilde{f_m} : N \otimes_S L \to (M \otimes_R N) \otimes_S L : n \otimes l \mapsto (m \otimes n) \otimes l. \tag{1.11}$$

This shows that the map

$$M \times N \otimes_S L \to (M \otimes_R N) \otimes_S L : (m, n \otimes l) \mapsto \tilde{f}_m(n \otimes l) \tag{1.12}$$

is well defined. This map is clearly balanced, so we have an induced homomorphism of groups

$$\varphi : M \otimes_R (N \otimes_S L) \to (M \otimes_R N) \otimes_S L : m \otimes (n \otimes l) \mapsto \tilde{f_m}(n \otimes l) = (m \otimes n) \otimes l \tag{1.13}$$

which is also a left $R-$module homomorphism. Similarly, we can show that

$$\psi : (M \otimes_R N) \otimes_S L \to M \otimes_R (N \otimes_S L) : (m \otimes n) \otimes l \mapsto m \otimes (n \otimes l) \tag{1.14}$$

is a well defined left $R-$module homomorphism. Since ψ and φ are inverses of each other, we have proved the theorem.

Now we can proceed to prove theorem (1.1.18). By definition, we just need to check that $T \otimes_S (S \otimes_R M) \cong T \otimes_R M$ as left $T-$modules. But

$$\begin{aligned} T \otimes_S (S \otimes_R M) &\cong (T \otimes_S S) \otimes_R M \quad \text{(by lemma (1.1.21))} \\ &\cong T \otimes_R M \quad \text{(by lemma (1.1.19) and lemma (1.1.21))} \end{aligned} \tag{1.15}$$

Before we finish this subsection, we will need one more useful property of tensor products.

Lemma 1.1.22. Let M_j (resp. N_j) be right (resp. left) $R-$modules $j = 1, 2$. If $T_j \in \mathrm{Hom}(M_j, N_j)$, then the map

$$T_1 \otimes T_2 : M_1 \otimes_R M_2 \to N_1 \otimes_R N_2 : m_1 \otimes m_2 \mapsto T_1(m_1) \otimes T_2(m_2) \tag{1.16}$$

is a well defined homomorphism of groups.

Proof: We just need to check that the map $(m_1, m_2) \mapsto T_1(m_1) \otimes T_2(m_2)$ is balanced, which is clear.

Theorem 1.1.23. Let $M = M_1 \oplus M_2$ be a right $R-$module and N a left $R-$module. Then, we have the following group isomorphism $M \otimes_R N \cong M_1 \otimes_R N \oplus M_2 \otimes_R N$.

Proof: The direct sum $M = M_1 \oplus M_2$ is characterized by the existence of module homomorphisms $i_j : M_j \to M$ and $p_j : M \to M_j$ such that

$$\begin{cases} p_j \circ i_j = \mathbb{1}_{M_j} \quad j = 1, 2 \\ p_k \circ i_j = 0 \quad \text{for} \quad k \neq j \\ \sum_{j=1}^{2} i_j \circ p_j = \mathbb{1}_M \end{cases} \tag{1.17}$$

Now, we define $\tilde{p_j} = i_j \otimes \mathbb{1}_N$ and $\tilde{p_j} = p_j \otimes \mathbb{1}_N$. So we get

$$\begin{aligned} &\tilde{p_j} \circ \tilde{i_j} = (p_j \circ i_j) \otimes \mathbb{1}_N = \mathbb{1}_{M_j} \otimes \mathbb{1}_N = 1_{M_j \otimes_R N} \quad j = 1, 2 \\ &\tilde{p_k} \circ \tilde{i_j} = (p_k \circ i_j) \otimes \mathbb{1}_N = 0 \otimes \mathbb{1}_n = 0 \quad k \neq j \\ &\sum_{j=1}^{2} \tilde{i_j} \circ \tilde{p_j} = \left(\sum_{j=1}^{2} i_j \circ p_j \right) \otimes \mathbb{1}_N = \mathbb{1}_M \otimes \mathbb{1}_N = \mathbb{1}_{M \otimes_R N} \end{aligned} \tag{1.18}$$

Therefore $M \otimes_R N \cong M_1 \otimes_R N \oplus M_2 \otimes_R N$.

Corollary 1.1.24. Let V and W be two vector spaces of finite dimension over a field $\mathbb{F}$. Then $\dim_{\mathbb{F}}(V \otimes_{\mathbb{F}} W) = \dim_{\mathbb{F}} V \dim_{\mathbb{F}} V$. Moreover, if $\{v_i\}$ and $\{w_j\}$ are basis for the vector spaces V and W respectively, then $\{v_i \otimes w_j\}$ is a basis for $V \otimes_{\mathbb{F}} W$.

Proof: Let $r = \dim_{\mathbb{F}} V$, then we get $V \cong \mathbb{F}^r$ and applying the theorem (1.1.23) multiple times, we get the vector space isomorphism isomorphism

$$V \otimes_{\mathbb{F}} W \cong \bigoplus_{j=1}^{r} \mathbb{F} \otimes_{\mathbb{F}} W \tag{1.19}$$

But, by the lemma (1.1.20) we have $\mathbb{F} \otimes_{\mathbb{F}} W \cong W$. So we have

$$V \otimes_{\mathbb{F}} W \cong \bigoplus_{j=1}^{r} W \tag{1.20}$$

and $\dim_{\mathbb{F}}(V \otimes_{\mathbb{F}} W) = \sum_{j=1}^{r} \dim_{\mathbb{F}} W = r \dim_{\mathbb{F}} W = \dim_{\mathbb{F}} V \dim_{\mathbb{F}} W$. Moreover, any $v \otimes w \in V \otimes_{\mathbb{F}} W$ is a sum of the form $\sum_{i,j} \alpha_i \beta_j v_i \otimes w_j$ and $\text{Span}\{v_i \otimes w_j\} = V \otimes_{\mathbb{F}} W$. But the cardinality of the set $\{v_i \otimes w_j\}$ is exactly the product of dimensions of V and W. Therefore, this set must be a basis of $V \otimes_{\mathbb{F}} W$.

Notation 1.1.25. To ease notation, from now on, when working with tensor products between vector spaces over a field, say $\mathbb{F}$, we will avoid the subscript $\mathbb{F}$ in the product $\otimes$.

Theorem 1.1.26. [1] Let A and A' be algebras over a field $\mathbb{F}$. We claim that $A \otimes A'$ is an algebra over $\mathbb{F}$ with product defined componentwise:

$$\left(\sum_i a_i \otimes a_i'\right)\left(\sum_j b_j \otimes b_j'\right) = \sum_{i,j} a_i b_j \otimes a_i' b_j'. \tag{1.21}$$

Proof: We will show that if $\sum_{i=1}^{n} a_i \otimes a_i' = 0$, then $\sum_{i,j} a_i b_j \otimes a_i' b_j' = 0$. This shows that the product is well defined and associativity follows from associativity of A and A'. Let $\{v_k\}$ (resp. $\{v_l'\}$) be a basis of A (resp. A'), so $a_i = \sum_k \alpha_{ik} v_k$ and $a_i' = \sum_l \alpha_{il}' v_l'$. Therefore

$$\begin{aligned} &\sum_{i=1}^{n} a_i \otimes a_i' = 0 \Rightarrow \\ &\quad \sum_{i=1}^{n} \sum_{k,l} \alpha_{ik} \alpha_{il}' v_k \otimes v_l' = 0 \Rightarrow \\ &\quad \sum_{i=1}^{n} \alpha_{ik} \alpha_{il}' = 0 \ \forall k, l \ \text{ (because of corollary (1.1.24))} \end{aligned} \tag{1.22}$$

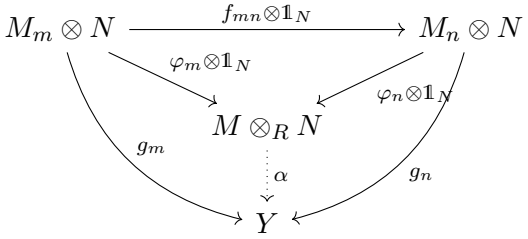
and

$$\begin{aligned}\sum_{i,j} a_i b_j \otimes a'_i b'_j &= \sum_{i,k,l} \alpha_{ik}\alpha'_{il} v_k b_j \otimes v'_l b'_j \\ &= \sum_{k,l} \left(\sum_i \alpha_{ik}\alpha'_{il} \right) v_k b_j \otimes v'_l b'_j = 0.\end{aligned} \tag{1.23}$$

Another important property of tensor products is that its commutes with direct limits.

Theorem 1.1.27. Let (M_n, f_{mn}) be a directed system of right $R-$modules and write $M = \varinjlim M_n$ for the associated limit. Then, if N is a left $R-$module, we have an isomorphism $M \otimes_R N \cong \varinjlim M_n \otimes_R N$.

Proof: Since $(M_n \otimes_R N, f_{mn} \otimes \mathbb{1}_N)$ is a directed system, we can form the direct limit $\varinjlim M_n \otimes_R N$. Let $\varphi_n : M_n \to M : m \mapsto [m]$ be the canonical mappings associated to the direct limit M. We will show that $M \otimes_R N \cong \varinjlim M_n \otimes_R N$ by showing that $(M \otimes_R N, \varphi_n \otimes \mathbb{1}_N)$ is an universal "repelling" target. Said this, we need to show that if (Y, g_n) is a target, then exists a unique α such that the following diagram commutes:



Define the map

$$M \times N \to Y : (m, n) \mapsto g_i(m_i \otimes n) \tag{1.24}$$

where m_i is a representative for m, i.e $m = [m_i] = \varphi_i(m_i)$. This map is well defined. In fact, let $m_i \sim m_j \Rightarrow f_{li}(m_i) = f_{lj}(m_j)$ for some $l \geq i, j$. By the assumption of the pair (Y, g_n) being a target, we have $g_m = g_n \circ (f_{nm} \otimes \mathbb{1}_N)$ whenever $n \geq m$. Therefore $g_i(m_i \otimes n) = g_l(f_{li}(m_i) \otimes n)$ and $g_j(m_j \otimes n) = g_l(f_{lj}(m_j) \otimes n)$, but since $f_{il}(m_i) = f_{jl}(m_j)$ we get that $g_i(m_i \otimes n) = g_j(m_j \otimes n)$. This map is clearly balanced, so we get a well defined group homomorphism

$$\alpha : M \otimes N \to Y : m \otimes n \mapsto g_i(m_i \otimes n) \tag{1.25}$$

where m_i is a representative for $m \in M$. For uniqueness, starting with the previous commutative diagram, the equality of compositions $\alpha \circ (\varphi_i \otimes \mathbb{1}_N) = g_i$ implies the formula previously defined for α. It's clear that if we have in addition that (M_n, f_{mn}) is a directed system of $(S, R)-$modules, then α is a homomorphism of $S-$modules.

Theorem 1.1.28. Let $M = \bigoplus_{\alpha \in A} M_\alpha$ be a direct sum of right $R-$modules. Then, if N is a left $R-$module, we have an isomorphism $M \otimes_R N \cong \bigoplus_{\alpha \in A} M_\alpha \otimes_R N$.

Proof: Let $\mathcal{A}$ be the set consisting of finite subsets of A. For $S \subset A$, write $M_S = \bigoplus_{\alpha \in S} M_\alpha$. Then, we have a canonical monomorphism (in fact an inclusion) $f_{ST} : M_S \to M_T$ whenever $S \subset T$. Let M be the limit of the directed system (M_S, f_{ST}). Now, let Y be a right $R-$module, homomorphisms $g_\alpha : M_\alpha \to Y$ $(\alpha \in A)$ and $i_\alpha := f_{\{\alpha\}A}$. Then we claim that exists a unique homomorphism $g : M \to Y$ such that $g \circ i_\alpha = g_\alpha$. Let $x \in M$, so $x \in M_S$ for some $S \in \mathcal{A}$. In this case we put $g(x) = g_S(x)$ where g_S is the map induced by the universal property of the direct sum $\bigoplus_{\alpha \in S} M_\alpha$. It's clear that g is well defined and that $g \circ i_\alpha = g_\alpha \ \forall \alpha \in A$. Therefore M satisfies the universal property for the direct sum of the modules $\{M_\alpha\}$, so $M \cong \bigoplus_{\alpha \in A} M_\alpha$. By the theorem (1.1.23) we get the result.

1.1.3 Structure theorem for finitely generated modules over a principal ideal domain

Definition 1.1.29. [11] A ring R is called an integral domain, if it is commutative and has no divisors of zero. This means that for $a, b \in R$, then if $ab \in R \Rightarrow a \in R$ or $b \in R$.

Definition 1.1.30. [11] A commutative ring R such that every nonzero element is a unit is called a *field*.

An important property of integral domains that we will need on the next chapter is the following.

Theorem 1.1.31. [11] A finite integral domain is a field.

Proof: Since R is commutative, because it's a integral domain, we just need to show that every nonzero element in R is a unit. Let $R = \{a_1, \ldots, a_n\}$ and $a \in R - \{0\}$. Then, since R is an integral domain, $aa_i \in R$ are all distinct for different i. Therefore $aa_i = 1$ for some i. This shows that every nonzero element $a \in R$ is a unit.

Definition 1.1.32. [4] A ring R is called principal ideal ring if every ideal $I \subset R$ is of the form $I = (a)$ for some $a \in R$. Ideals of this form will be called *principal ideals*.

A useful property of principal ideal rings, is that every ascending chain of ideals $I_1 \subset I_2 \subset \ldots$ must stabilize. This means that exists $i \geq 1$ such that $I_j = I_i$ whenever $j \geq i$. In

fact, $I = \bigcup_i I_i$ is an ideal. Now, since R is principal, we have $I = (a)$. So we see that $a \in I_i$ for some i, which implies that $(a) \subset I_i \Rightarrow (a) = I_i$ and $I_j = I_i = (a)$ for all $j \geq i$. Rings with the property that every ascending chain of ideals must terminate are called *noetherian* rings. From the previous discussion we get the following and important result.

Theorem 1.1.33. [4] Every principal ideal ring is noetherian.

We also have the notion of noetherian modules, that will be useful later.

Definition 1.1.34. [1] A module M is said to be *noetherian* if every increasing chain of submodules $M_1 \subset M_2 \subset \dots$ terminates at some point.

Example 1.1.35. Every finite dimensional vector space is noetherian.

In this section we will be interested in rings that are both principal ideal rings and integral domains.

Definition 1.1.36. [1] A ring R is said to be a principal ideal domain, if it's an integral domain and a principal ideal ring.

Definition 1.1.37. Let R be a ring, a proper ideal $I \subset R$ is said to be *maximal* if whenever $I \subset J$, J an ideal, we have necessarily $J = I$ or $J = R$. Also, a proper ideal $P \subset R$ is said to be *prime* if whenever $ab \in P \Rightarrow a \in P$ or $b \in P$.

An important property of principal ideal domains is the following.

Theorem 1.1.38. If R is a principal ideal domain, then $P \subset R$ is a prime ideal if and only if it is a maximal ideal.

Proof: See ([11] p.17).

Theorem 1.1.39. [1] Let R be a principal ideal ring and M a finitely generated module over R with basis $\{m_1, \dots, m_k\}$. Then, any submodule $N \subset M$ has a basis of cardinality $s \leq k$.

Proof: The result is immediate for $k = 1$. Indeed $M \cong R$ as $R-$modules, and a submodule of R is the same thing as an ideal. So the principal ideal condition implies the

result directly. Now suppose $k > 1$ and the result is true for free $R-$modules generated by $k-1$ elements. Since $M = \bigoplus_{i=1}^{k} Rm_i$, we have the projection homomorphism $p_1 : M \to R$. Since $N \subset M$ is a submodule, $p_1(N) \subset R$ is an ideal. But R is a principal ideal ring, so $p_1(N) = (\sigma)$ and there exists $n_1 \in N$ such that $p_1(n_1) = \sigma$. Now define $M' = \bigoplus_{i=2}^{k} Rm_i$. Then $p_1(M') = 0 \Rightarrow Rn_1 \cap M' = (0)$. Also, if $n \in N$ and $p_1(n) = \alpha\sigma$, then $p_1(n - \alpha n_1) = 0$, so $n - \alpha n_1 \in N \cap M'$. Therefore $N = Rn_1 \oplus N \cap M'$. But, by the induction hypothesis, $N \cap M'$ has a basis of cardinality at most $k-1$ elements and the result follows.

Theorem 1.1.40. [1] Let M be a free $R-$module finitely generated by a basis $\{m_1, \ldots, m_k\}$. The number k is well defined and is called the *rank* of M.

Proof: We have $M \cong \bigoplus_{i=1}^{k} R$. Let $I \subset R$ be a maximal ideal, then $IM \subset M$ is a submodule isomorphic to $\bigoplus_{i=0}^{k} I$. Therefore $M/IM \cong \bigoplus_{i=1}^{k} R/I \cong L^k$ where $L = R/I$ is a field. So $k = \dim_L(M/IM)$ is a well defined number.

Notation 1.1.41. Given a ring R, we will write $M_{mn}(R)$ for the $m \times n$ matrices over R and $M_n(R)$ for $n \times n$ matrices over R.

Definition 1.1.42. Let $A \in \mathrm{M}_n(R)$. Then A is said to be *unimodular* if exists $B \in \mathrm{M}_n(R)$ such that $AB = BA = I$.

Important examples of unimodular matrices, are the *elementary* matrices. These matrices may be used to do elementary operations with the row and columns of another matrix. For example, let $A = (A_{ij}) \in \mathrm{M}_n(R)$ and $\sigma \in S_n$, define $\sigma(A) = (A_{\sigma(i)j})$. Particularly, $(\sigma(I)A)_{ij} = \sum_{k=1}^{n} \delta_{\sigma(i)k} A_{kj} = A_{\sigma(i)j}$. So $\sigma(I)A = \sigma(A)$. This means that multiplication from the left on A by $\sigma(I)$ has the effect of permuting the rows of A by σ. Now, define $Y_{ij}(\alpha) = I + \alpha E_{ij}$ for $i \neq j$. Then, if $A \in \mathrm{M}_n(R)$ is a matrix whose rows are $\{R_1, \ldots, R_n\}$, then multiplication from the left on A by $Y_{ij}(\alpha)$ has the effect of substituting the row R_i by $R_i + \alpha R_j$. Clearly $\sigma(I)\sigma^{-1}(I) = \sigma^{-1}(I)\sigma(I) = I$ and $Y_{ij}(\alpha)Y_{ij}(-\alpha) = Y_{ij}(-\alpha)Y_{ij}(\alpha)Y = I$. So these matrices are unimodular.

Observation: We can use elementary matrices E to do elementary operations on columns of a matrix A. We just need to transpose E and multiply on the right of A. See example 1.1.46.

Definition 1.1.43. [1] For $A, B \in \mathrm{M}_{mn}(R)$, we say that A and B are *equivalent* if there exists $X_1 \in \mathrm{M}_m(R)$ and $X_2 \in \mathrm{M}_n(R)$ such that $B = X_1 A X_2$. In this case we will write $A \sim B$.

Theorem 1.1.44. (Invariant factor theorem for matrices)[1] Let R be a principal ideal domain and A a rectangular matrix with entries in R. Then $A \sim \mathrm{diag}(\delta_1, \delta_2, \ldots, \delta_k, 0, \ldots, 0)$ where $\delta_i \in R - \{0\}$ and $\delta_i | \delta_{i+1}$ $\forall i = 1, \ldots, k-1$. The elements $\delta_1, \ldots, \delta_k$ are called *invariant factors* and are uniquely determined up to unit factors.

Rather than reproduce the proof of this theorem, we offer examples that illustrate the general procedure for determining equivalence.

Definition 1.1.45. [4] A *unique factorization domain* (UFD) is a ring R such that every nonzero element can be written uniquely, up to unit factors, as a product of primes.

Example 1.1.46. Consider

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & -1 \end{pmatrix} \in M_3(\mathbb{Z}) \tag{1.26}$$

This matrix is equivalent to $\mathrm{diag}\{1, 2, 2\}$.

Multiply A on the right by $(13)(I), Y_{21}(-1)$ and on the left $Y_{31}(1)$ to get

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & -2 \\ 0 & 0 & 2 \end{pmatrix} \tag{1.27}$$

Now, multiply this matrix on the right by $Y_{31}(-2)^T$ and $Y_{32}(1)^T$ to obtain $\mathrm{diag}\{1, 2, 2\}$.

The next example illustrates the case where in some step, there is one element that doesn't divide some other element in the matrix. In this case the strategy will be the following: let $A = (a_{ij})$, where for some (i, j) there is (m, n) such that a_{ij} does not divide a_{mn}. Using elementary operations on rows and columns we can assume, without loss of generality, that $(i, j) = (1, 1)$ and $(m, n) = (1, 2)$ or $(2, 1)$. Suppose the former holds. Since R is a principal ideal domain, we can compute the $\gcd(a_{11}, a_{12}) = \beta$. In this case, there are $\gamma_1, \gamma_2 \in R$ such that $\gamma_1 a_{11} + \gamma_2 a_{12} = \beta$. Also, since $\beta | a_{1i}, i = 1, 2$, there exist $\epsilon_i, i = 1, 2$

such that $\epsilon_1\gamma_1 - \epsilon_2\gamma_2 = 1$. In this case, define

$$X = \begin{pmatrix} \gamma_1 & \epsilon_2 & \\ \gamma_2 & \epsilon_1 & \\ & & I \end{pmatrix} \tag{1.28}$$

This matrix is unimodular with inverse given by

$$X^{-1} = \begin{pmatrix} \epsilon_1 & -\epsilon_2 & \\ -\gamma_2 & \epsilon_1 & \\ & & I \end{pmatrix}, \tag{1.29}$$

So $AX \sim A$ and $(AX)_{11} = \beta$. We are working in R a PID, it's known that every PID is a UFD, so each element has a factorization into a finite product of prime elements. Since a_{11} does not divide a_{12}, β has strictly fewer prime factors than a_{11}. Applying this reduction multiple times, we get at some step an element γ at $(1,1)-$position that divide all the others, so we can apply elementary operations with rows and columns as in the example (1.1.46) to get

$$A \sim \begin{pmatrix} \gamma & \\ & B \end{pmatrix} \tag{1.30}$$

Now we repeat this procedure with the smaller matrix B.

Example 1.1.47. Consider

$$A = \begin{pmatrix} t & -1 & 0 \\ 0 & t & -1 \\ 0 & 0 & t \end{pmatrix} \in \mathrm{M}_3(k[[t]]) \tag{1.31}$$

This matrix is equivalent to $\mathrm{diag}(1, 1, t^3)$.

Since $t \not| -1$, we write $\gamma_1 t + \gamma_2(-1) = 1$ where $\gamma_1 = 1$ and $\gamma_2 = t - 1$. So we also have $\gamma_1\epsilon_1 - \gamma_2\epsilon_2 = 1$ for $\epsilon_1 = t$ and $\epsilon_2 = 1$. Therefore, we take

$$X = \begin{pmatrix} 1 & 1 & 0 \\ t-1 & t & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1.32}$$

So

$$A \sim AX = \begin{pmatrix} 1 & 0 & 0 \\ t^2 - t & t^2 & -1 \\ 0 & 0 & t \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & t^2 & -1 \\ 0 & 0 & t \end{pmatrix} \tag{1.33}$$

Now, repeating exactly the same procedure with the smaller matrix $\begin{pmatrix} t^2 & -1 \\ 0 & t \end{pmatrix}$ we get

$$\begin{pmatrix} 1 & 2t^2 \\ t^3 - t & -t^3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & -t^3 \end{pmatrix} \tag{1.34}$$

Finally we conclude that $A \sim \text{diag}(1, 1, -t^3)$. Maybe note that it's equivalent to $\text{diag}(1, 1, t^3)$.

Now let M be a finitely generated free module over a principal ideal domain R and $N \subset M$ a submodule. Then by theorem (1.1.39) N has a basis. If $\{m_1, \ldots, m_k\}$ is a basis of M and $\{n_1, \ldots, n_s\}$ is a basis of N, then we can write

$$(n_1, \ldots, n_s) = (m_1, \ldots, m_k)A \tag{1.35}$$

where $A \in \mathrm{M}_{ks}(R)$. So multiplying A from the left by a unimodular matrix is equivalent to a change of basis of M and multiplication on the right is equivalent to a change of basis of N. In this case, theorem (1.1.44) says that we can choose a basis of $\{m_1, \ldots, m_k\}$ of M such that $\{\delta_1 m_1, \ldots, \delta_s m_s\}$ is a basis for N, where $\delta_i | \delta_{i+1}$ for $i = 1, \ldots, s-1$. For the uniqueness of $\{\delta_i\}$, see [1].

Example 1.1.48. Let $K = \{1, X, Y, XY\}$ be the Klein 4−group, the unique non cyclic group of order 4. It's an abelian group, so it's also a $\mathbb{Z}-$module. In this case

$$K \cong \frac{\mathbb{Z}}{(2)} \oplus \frac{\mathbb{Z}}{(2)} \tag{1.36}$$

To see this, let F be the free $\mathbb{Z}-$module generated by K, so we have a surjective homomorphism

$$f : F \to K : \alpha + \beta X + \gamma Y + \theta XY \mapsto X^\beta Y^\gamma (XY)^\theta \tag{1.37}$$

We are using summation notation on F to differ from the multiplication notation on K and avoid confusion on operations defined on each module. But by theorem (1.1.39), $\text{Ker} f$ is a free submodule of F and in fact, has a basis $\{1, 2X, 2Y, X + Y - XY\}$. We write

$$(1, 2X, 2Y, X + Y - XY) = (1, X, Y, XY) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{1.38}$$

By example (1.1.46) we have

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & -1 \end{pmatrix} \sim \text{diag}(1, 1, 2, 2) \tag{1.39}$$

Therefore, there is a basis of $\{m_1, m_2, m_3, m_4\}$ of M such that $\{m_1, m_2, 2m_3, 2m_4\}$ is a basis of N. And we conclude that

$$K \cong \frac{F}{\text{Ker} f} \cong \frac{\mathbb{Z}m_1}{\mathbb{Z}m_1} \oplus \frac{\mathbb{Z}m_2}{\mathbb{Z}m_2} \oplus \frac{\mathbb{Z}m_3}{\mathbb{Z}2m_3} \oplus \frac{\mathbb{Z}m_4}{\mathbb{Z}2m_4} \cong \frac{\mathbb{Z}}{2\mathbb{Z}} \oplus \frac{\mathbb{Z}}{2\mathbb{Z}} \tag{1.40}$$

Of course $\{m_1, m_2, m_3, m_4\}$ can be found explicitly. By example (1.1.46) we have

$$(m_1, m_2, m_3, m_4) = (1, X, Y, XY)Y_{31}(-1)Y_{21}(1) = (1, X + Y - XY, Y, XY) \tag{1.41}$$

Example 1.1.49. Let $\mathbb{F}$ be a field and

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \in \mathrm{M}_3(\mathbb{F}) \tag{1.42}$$

Since A is nilpotent, the vector space $M = \mathbb{F}^3$ becomes an $\mathbb{F}[[t]]-$module with action $t \cdot v := Av$ for $v \in M$. In this case

$$M \cong \frac{\mathbb{F}[[t]]}{(t^3)} \tag{1.43}$$

To see this, let F be the free $\mathbb{F}[[t]]-$module generated by the canonical basis $\{e_1, e_2, e_3\}$ of F. To avoid confusion, in M we will use the letter $f_i, i = 1, 2, 3$ for this basis. So, we have a surjective homomorphism of modules

$$f : F \to M : f_i \mapsto e_i \tag{1.44}$$

In this case $\ker(f)$ have basis $\{tf_1, tf_2 - f_1, tf_3 - f_2\}$ and

$$(tf_1, tf_2 - f_1, tf_3 - f_2) = (f_1, f_2, f_3) \begin{pmatrix} t & -1 & 0 \\ 0 & t & -1 \\ 0 & 0 & t \end{pmatrix} \tag{1.45}$$

By example (1.1.47), we have

$$\begin{pmatrix} t & -1 & 0 \\ 0 & t & -1 \\ 0 & 0 & t \end{pmatrix} \sim \mathrm{diag}(1, 1, t^3) \tag{1.46}$$

Therefore, there is a basis $\{m_1, m_2, m_3\}$ of F such that $\{m_1, m_2, t^3 m_3\}$ is a basis of $\mathrm{Ker}(f)$ and we conclude that

$$M \cong \frac{F}{\mathrm{Ker}(f)} \cong \frac{\mathbb{F}[[t]]}{(t^3)} \tag{1.47}$$

These examples illustrate the structure theorem for finitely generated modules over a principal integral domain.

Theorem 1.1.50. [1] Let M be a finitely generated module over a principal ideal domain R. Then, there exist elements $\delta_i \in R - \{0\}$, uniquely defined up to unit factors, for $i = 1, \ldots, k$, such that $\delta_i \mid \delta_{i+1}$, and

$$M \cong \frac{R}{R\delta_1} \oplus \cdots \oplus \frac{R}{R\delta_k} \oplus R \oplus \cdots \oplus R. \tag{1.48}$$

The summand $R \oplus \cdots \oplus R$ will be called the *free* part of M.

The general strategy for the proof closely follows the procedure demonstrated in the examples (1.1.48) and (1.1.49). In the case that each δ_i factors as a products of primes, we can refine this result. To do this, we use the next lemma.

Lemma 1.1.51. [1] Let R be a principal ideal domain and $\delta = p_1^{k_1} \cdots p_n^{k_n}$ where $p_1, \ldots, p_n$ are non-associated primes. Then, we have a module isomorphism

$$R/(\delta) \cong R/(p_1^{k_1}) \oplus \cdots \oplus R/(p_n^{k_n}) \tag{1.49}$$

Proof: Let p, q be non-associated primes. Then $(p) \neq (q) \Rightarrow (p) + (q) = R$ because both (p) and (q) are maximal ideals (by theorem (1.1.38)). Therefore, there exist $r, s \in R$ such that $rp + sq = 1$. Now, define the map

$$\varphi : R/(p) \oplus R/(q) \to R/(pq) : (x + (p), y + (q)) \mapsto py + qx + (pq) \tag{1.50}$$

This map is clearly a well defined homomorphism of $R-$modules. Since $R/(pq)$ is cyclic and $\varphi(s + (p), r + (q)) = 1 + (pq)$, we see that φ is surjective. To see that it is injective let $\varphi(x + (p), y + (q)) = 0 \iff py + qx = pqu$ for some $u \in R$. Now, multiplying $rp + sq = 1$ by y, we get $yrp + ysq = y$ which implies that $y + rqx = pqu + ysq \Rightarrow y \in (q)$. Similarly, multiplying $rp + sq = 1$ by x we see that $x \in (p)$. This shows that φ is injective and the rest of the proof follows by induction.

Example 1.1.52. For example, it is a consequence of the *euclidean division algorithm* that the ring of integers $\mathbb{Z}$ is a unique factorization domain. The same goes for the ring of polynomials in one variable $\mathbb{F}[X]$ over a field $\mathbb{F}$.

It's possible to show, using the Zorn's lemma, that any principal ideal domain R is a unique factorization domain (the converse doesn't holds, for example $\mathbb{Z}[X]$ is a unique factorization that is not a principal ideal) [4]. But in this work, we will only work with the rings $\mathbb{Z}$ and $\mathbb{F}[X]$. These rings are both unique factorization domains (by the euclidean algorithm) and principal ideal domains. So, by this reason, we will state these two hypothesis together whenever we need them.

Corollary 1.1.53. [1] If R is a principal ideal domain. Then, any finitely generated $R-$module M is of the form

$$M \cong \bigoplus_{p \text{ prime}} \bigoplus_{i \geq 1} R/(p^{\mu_i(p)}) \tag{1.51}$$

where $\mu : p \in \{\text{primes elements of } M\} \mapsto \mu(p) = (\mu_1(p), \mu_2(p), \dots) \in \mathbb{N}^\infty$ is uniquely defined by M. Also, μ have the property that $\mu_j(p) = 0$ if $j \geq j_p$ for some $j_p \geq 1$ and $\mu(p) = 0$ for all but a finite subset of primes elements p of R.

This corollary will be useful in the classification of conjugacy classes of $GL_n(\mathbb{F}_q)$ in the next chapter (see theorem (2.2.4)).

1.2 Localization

1.2.1 Localization of rings and its properties

Definition 1.2.1. Let R be a ring, $S \subset R$ is said to be a multiplicative system if $1 \in S$ and whenever $x, y \in S \Rightarrow xy \in S$. So this is equivalent to say that S is a multiplicative monoid.

Definition 1.2.2. Let R be a ring and $S \subset \mathrm{Z}(R)$, the center of R, a multiplicative system. Define an equivalence relation on $R \times S$ by

$$(r_1, s_1) \sim (r_2, s_2) \iff \exists\, s \in S \text{ such that } r_1 s_2 s = r_2 s_1 s \tag{1.52}$$

We will write r/s for the equivalence class containing (r, s) under $\sim$.

Theorem 1.2.3. [13] Let R be a commutative ring and $S \subset R$ be a multiplicative system. Define

$$S^{-1}R := (R \times S)/\sim$$

where $\sim$ is as in the previous definition. Then $S^{-1}R$ is a ring with sum and multiplication defined by

$$\frac{x}{s} + \frac{y}{t} = \frac{xt + ys}{st} \text{ and } \frac{x}{s}\frac{y}{t} = \frac{xy}{st} \tag{1.53}$$

Proof: See ([13]).

Lemma 1.2.4. The following map

$$f : R \to S^{-1}R : x \mapsto x/1 \tag{1.54}$$

is a ring homomorphism and it's called *localization map*. In the case that R is a principal ideal domain, f is also injective.

Notation 1.2.5. For $A \subset R$ an arbitrary subset, we will write

$$S^{-1}A = \{x/s \in S^{-1}R \mid x \in A \text{ and } s \in S\} \tag{1.55}$$

Lemma 1.2.6. Let $f : R \to S^{-1}R$ be as above. For any ideal $I \subset S^{-1}R$, then $f^{-1}(I) \subset R$ is an ideal. Moreover $I = S^{-1}f^{-1}(I)$.

Proof: To ease notation, write $\overline{I} := f^{-1}(I)$. Let $x/s \in I$, then since I is an ideal, we have $\frac{s}{1}\frac{x}{s} = \frac{x}{1} \in I \Rightarrow x \in \overline{I}$. So $x/s \in S^{-1}\overline{I}$ and $I \subset S^{-1}\overline{I}$. Conversely, if $x/s \in S^{-1}\overline{I} \Rightarrow x \in \overline{I}$ and $s \in S$. Moreover $x/1 \in I \Rightarrow \frac{1}{s}\frac{x}{1} = \frac{x}{s} \in I$ and we get $S^{-1}\overline{I} \subset I$.

Remark: Lemma (1.2.6) says that any ideal $I \subset S^{-1}R$ is of the form $S^{-1}\overline{I}$ for some ideal $\overline{I} \subset R$.

Since $0 \in S$ implies $S^{-1}R = (0)$, this case isn't interesting. Therefore we will be considering only multiplicative systems $S \subset R - \{0\}$.

Lemma 1.2.7. Let R be an integral domain and $S \subset R - \{0\}$ a multiplicative system. Then $S^{-1}R$ is an integral domain.

Proof: Suppose that $\frac{r_1}{s_1}\frac{r_2}{s_2} = 0$. Then, there exists $s \in S$ such that $r_1 r_2 s = 0 \Rightarrow r_1 r_2 = 0$ because $s \neq 0$. So $r_1 = 0$ or $r_2 = 0 \Rightarrow r_1/s_1 = 0$ or $r_2/s_2 = 0$.

Definition 1.2.8. A ring R is said to be *local* if it has a unique maximal ideal.

Theorem 1.2.9. Let R be a PID and $P \subset R$ a prime ideal. Then $S := R - P$ is a multiplicative system. We claim that $R_P := S^{-1}R$ is a local ring.

Proof: The candidate for the maximal ideal is

$$I := S^{-1}P = \{x/s \in R_P \mid x \in P \text{ and } s \in S\} \tag{1.56}$$

First, let's show that I is a maximal ideal. By lemma (1.1.38), we have that P is a maximal ideal in R. Now, let $I \subset J$, J being an ideal. Then, if $\bar{J} := \{x \in R \mid x/1 \in J\} \Rightarrow \bar{J} \subset R$ is an ideal. In particular $P \subset \bar{J} \Rightarrow \bar{J} = P$. Then by lemma (1.2.6) we have $J = I$ or $J = R_P$. Now, let J be any proper ideal in R_P. Consider two cases: If $\bar{J} \cap S \neq \varnothing$, then there exist $x \in \bar{J} \cap S \Rightarrow x/1 \in \bar{J}$. But $1/x \in R_P$ and since J is an ideal, we have $\frac{1}{x}\frac{x}{1} = \frac{1}{1} \in J$ and $J = R_P$. So we exclude this case. Now, if $\overline{J} \cap S = \varnothing \Rightarrow \bar{J} \subset P$ and $J \subset I$. Therefore I contains all the proper ideals of R_P.

The definition of localization can be extended to modules.

Definition 1.2.10. Let R be a commutative ring, $S \subset R$ a commutative system and M an $R-$module. We define an equivalence relation on $M \times S$ by $(m, s) \sim (n, t)$ if $\exists\, u \in S$ such that $mtu = nsu$. We will write m/s for the equivalence class containing (m, s) in $M \times R$.

Remark: Since R is considered to be a commutative ring, M will be regarded as a $(R, R)-$bimodule. So we can write the R action from either direction.

Theorem 1.2.11. [13] Define

$$S^{-1}M := (S \times R)/\sim \tag{1.57}$$

then $S^{-1}M$ is an $S^{-1}R-$module with sum defined by

$$\frac{m}{s} + \frac{n}{t} = \frac{mt + ns}{st} \tag{1.58}$$

and $S^{-1}R-$action by

$$\frac{x}{t}\frac{m}{s} = \frac{xm}{ts} \tag{1.59}$$

Proof: The proof is completely analogous to that of theorem (1.2.3).

1.2.2 Functorial properties of localization

Both the localization of rings and modules have functorial properties that we will be useful in the next section.

Notation 1.2.12. Let M be an $R-$module. Each $x \in R$ defines a map $x_M : M \to M : m \mapsto xm$. So we have a homomorphism $(\cdot)_M : R \to \mathrm{End}(M) : x \mapsto x_M$. Also, for any subset $S \subset R$, we write S_M for the set of endomorphisms $s_M \in \mathrm{End}(M)$ such that $s \in S$.

Theorem 1.2.13. Let R be a commutative ring and $S \subset R$ a multiplicative system. Then the category of $S^{-1}R-$modules is equivalent to the category of $R-$modules M with the property that $S_M \subset \mathrm{Aut}(M)$.

Proof: Define the functor $F : R - \overline{\mathbf{mod}} \to S^{-1}R - \overline{\mathbf{mod}}$ that associates to each $R-$module M the same set, but with $S^{-1}R-$action defined by $(x/s) \cdot m := x_M \circ s_M^{-1}(m)$ for $m \in M, x \in R$ and $s \in S$. Now define another functor $G : S^{-1}R - \overline{\mathbf{mod}} \to R - \overline{\mathbf{mod}}$ that associates to each $S^{-1}R-$module M, the same set, but with action $x \cdot m := (x/1)m$ for $m \in M$ and $x \in R$.

Theorem 1.2.14. Let R be a commutative ring and $S \subset R$ a multiplicative system. Then, there is a functor

$$G : R - \overline{\mathbf{mod}} \to S^{-1}R - \overline{\mathbf{mod}} : M \mapsto S^{-1}M \tag{1.60}$$

that associates to each $f \in \mathrm{Hom}_R(M, N)$ a $G(f) \in \mathrm{Hom}_{S^{-1}R}(S^{-1}M, S^{-1}N)$, defined by $G(f)(x/s) := f(x)/s$. In fact, is an exact functor.

Proof: The fact that it is a functor is clear, now lets show that this in fact, defines an exact functor. Let

$$L \xrightarrow{u} M \xrightarrow{v} N$$

be an exact sequence. Since F is a functor, we also have a sequence

$$S^{-1}L \xrightarrow{G(u)} S^{-1}M \xrightarrow{G(v)} S^{-1}N$$

In fact, it is clear that we have $\mathrm{Im}(G(u)) \subset \mathrm{Ker}(G(v))$ because $G(v) \circ G(u) = G(vu) = G(0) = 0$. We claim that it is an exact sequence. Now, let $x/s \in \mathrm{Ker}(G(v))$. Then $v(x)/s = 0 \Rightarrow (v(x), 0) \sim (0, 1)$. So, there is a $t \in S$ such that $v(x)t = 0 \Rightarrow v(xt) = 0 \Rightarrow xt \in \mathrm{Ker}(v) = \mathrm{Im}(u)$. So there exist $y \in L$ such that $u(y) = tx$ and $G(u)(y/t) = x/s$.

Corollary 1.2.15. Let $N \subset M$ be a submodule, then $S^{-1}M/S^{-1}N \cong S^{-1}(M/N)$. Particularly, if $I \subset R$ is an ideal, we have $S^{-1}(R/I) \cong S^{-1}R/S^{-1}I$.

We saw two functors related to the localization map. If $S_M \subset \mathrm{Aut}(M)$, the first functor takes an $R-$module M to the same set, but viewed as an $S^{-1}R-$module. The second one takes an $R-$module M to the $S^{-1}R-$module $S^{-1}M$ in general. If the first case applies, there is no difference between these two functors. This is the content of the next theorem.

Theorem 1.2.16. If $S_M \subset \mathrm{Aut}(M)$, then $S^{-1}M \cong M$ as an $S^{-1}R-$modules (or equivalently, as $R-$modules by theorem (1.2.13)).

Proof: We claim that the localization map $f : M \to S^{-1}M : m \mapsto m/1$ is an isomorphism of $S^{-1}R-$modules. To show this, lets make an observation: For $m \in M$ and $s \in S$ we have $(s_M^{-1}(m))/1 = m/s$. In fact $(s_M^{-1}(m), 1) \sim (m, s)$. Therefore $(r/s)f(m) = (r/s)(m/1) = rm/s = (r_M \circ s_M^{-1}(m))/1 = f((r/s) \cdot (m))$. Moreover, $f(s_M^{-1}(m)) = m/s$ which implies that f is surjective. To see injectivity, let $f(m) = 0 \Rightarrow (m, 1) \sim (0, 1) \iff$ there exists $s \in S$ such that $0 = sm = s_M(m)$. But $S_M \subset \mathrm{Aut}(M)$, which implies that $m = 0$.

1.3 Modules over discrete valuation rings

1.3.1 Partitions of positive integers

Definition 1.3.1. [12] A partition is any (finite or infinite) nonincreasing sequence $\lambda = (\lambda_1, \lambda_2, \dots)$ of positive integers such that $\lambda_i = 0$ for $i \gg 1$. We choose not to distinguish between sequences that differ only by a string of zeros at the end. The numbers λ_i are called the *parts* of λ.

We emphasize the fact we don't make any distinction between partitions like

$$(5, 3, 1), (5, 3, 1, 0) \text{ or } (5, 3, 1, 0, 0, \dots).$$

Definition 1.3.2. Following [12], we will use the following characteristic of a partition of a positive integer:

1. $l(\lambda)$ is called the *length* of λ, and is defined to be the number of nonzero parts in λ.
2. $m_i(\lambda)$ is called *multiplicity* of i in λ, and is defined to be the number of parts of λ that are equal to i.
3. $|\lambda|$ is called the *weight* of λ, and is defined to be the sum of its parts.

Notation 1.3.3. Another common way to write partition is $\lambda = (1^{n_1} 2^{n_2} \cdots)$. This notation emphasizes the multiplicities of each part of λ.

Definition 1.3.4. Let λ be a partition. If $|\lambda| = n$, we say that λ is a partition of n. The set of all partitions of n is denoted by $\mathcal{P}_n$ and the set of all partitions by $\mathcal{P}$. Its cardinality $p_n = |\mathcal{P}_n|$ can be computed by as the coefficient of x^n in the following generating series

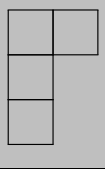
$$\sum_{i \geq 1} p_i x^i = \prod_{j \geq 1} \frac{1}{1 - x^j} \tag{1.61}$$

Definition 1.3.5. [12] Given a partition λ, we define its *conjugate* λ', by defining λ'_i to be the number of parts of λ that are greater than or equal to i. Since a partition is a nonincreasing sequence, it is clear that λ' is also a partition.

Observation: Another useful way to write the parts of λ' is by the formula $\lambda'_i = \text{Max}\{k \geq 1 \mid \lambda_k \geq i\}$. In fact, let $a = \text{Max}\{k \geq 1 \mid \lambda_k \geq i\}$, then $\lambda_a \geq i$ and $\lambda_j \geq \lambda_a \ \forall j \ (1 \leq j \leq a)$. But, since $\lambda_j < i$ for $j > a$, we have $\#\{k \geq 1 \mid \lambda_k \geq i\} = \#\{1, \ldots, a\} = a$.

To each partition λ we can associate a diagram as follows. The diagram of λ is by definition the set $T_\lambda = \{(i, j) | \ j \leq \lambda_i\}$ where each pair $(i, j) \in T_\lambda$ is called a *square*. Clearly, the ith row of T_λ consist of λ_i squares. To draw diagrams we use matrix convention, that is, rows going downward as i increases, and columns indexed by j, going to the right as j increases. With this convention, we claim that $T_{\lambda'} = T_\lambda^t$, where $(-)^t$ denotes transpose, as for matrices. This follows from $j \leq \lambda'_i \iff \lambda_j \leq i$ because $\lambda'_i = \text{Max}\{k \mid \lambda_k \geq i\}$. So λ'_i counts the number of squares in the ith column of the diagram T_λ.

Example 1.3.6. The diagram of the partition $\lambda = (2, 1^2)$ is



Definition 1.3.7. [12] Given two partitions λ and μ, we write $\mu \subset \lambda$ if $\lambda_i - \mu_i \geq 0$ for any $i \geq 1$. So $\subset$ gives a partial order in $\mathcal{P}$.

Definition 1.3.8. [12] Let $\lambda, \mu \in \mathcal{P}$ such that $\mu \subset \lambda$. We define the skew diagram $\theta = \lambda - \mu$ to be the set theoretical difference between the diagrams of λ and μ. In this case, we will write $\theta_i = \lambda_i - \mu_i$ to be the number of squares on the ith lines on the skew diagram θ. Also, we define the *weight* of the skew diagram θ to be the sum $\sum_{i\geq 1} \theta_i = |\lambda| - |\mu|$.

We can see that if $\mu \subset \lambda$, then $\mu' \subset \lambda'$. Define the skew diagram $\theta = \lambda - \mu$. In this case we write $\theta' = \lambda' - \mu'$. Another partial order on $\mathcal{P}$ that will be useful is the following.

Definition 1.3.9. Given $\mu, \lambda \in \mathcal{P}$, we define a partial order on $\mathcal{P}$ by $\mu \leq \lambda$ if

$$\mu_1 + \cdots + \mu_i \leq \lambda_1 + \cdots + \lambda_i \text{ for all } i \geq 1.$$

This is called the *dominance* partial order in $\mathcal{P}$ [12].

1.3.2 Discrete valuation rings

Definition 1.3.10. A *discrete valuation ring* R is a principal ideal domain which is local with maximal ideal $P \neq (0)$. The *residue field* of R is defined by the quotient R/P.

With this definition we can already see that there is just one simple $R-$module, namely the residue field R/P. In fact, if $M = Rm$ (any simple module must be cyclic), then we can define a surjective homomorphism $\varphi : R \to M : r \mapsto rm$. So $R/\mathrm{Ker}(\varphi) \cong M$. But again, M is simple, so $\mathrm{Ker}(\varphi)$ must be a maximal ideal of R. Therefore $\mathrm{Ker}(\varphi) = P$.

Next, an important property of discrete valuation rings.

Theorem 1.3.11. Let R be a discrete valuation ring, and $p \in R$ a fixed generator of the unique maximal ideal. The, every $x \in R$ is of the form $x = up^m$ for some $m \geq 0$ and unit $u \in R$.

Proof: Since R is noetherian, any non unit $a \in R$ is contained in (p). Therefore, the set $R-(p)$ is precisely the set of units in R. We claim that for any $x \in R$ there exists a maximal $m \geq 0$ such that $x \in (p^m)$. To show this, it's sufficient to prove that $\bigcap_{m\geq 0}(p^m) = (0)$. By contradiction, suppose that there exists a nonzero element $x \in \bigcap_{m\geq 0}(p^m)$. Then there is a

sequence $(u_1, u_2, \dots)$ such that $x = u_m p^m \ \forall m \geq 1 \Rightarrow u_m = p u_{m+1} \ \forall m \geq 1$. Therefore, we have a strictly increasing chain of ideals

$$(u_1) \subset (u_2) \subset \cdots \tag{1.62}$$

which contradicts the fact that R is noetherian (see theorem (1.1.33)).

Now, for $x \in R$, let $m \geq 0$ be such that $x \in (p^m)$ and $x \notin (p^{m+1})$. Then, in this case $x = up^m$ where $u \notin (p)$, so u is a unit.

Example 1.3.12. Let R be a PID. By theorem (1.2.9), R_P is a discrete valuation ring. In this case, the residue field is given by $R_P/S^{-1}P \cong S^{-1}(R/P)$. Moreover if $S_R \subset \mathrm{Aut}(R)$, then by the theorem (1.2.16), the localization map $f : R \to S^{-1}R$ is an isomorphism of $S^{-1}R-$modules. But this map is also a ring homomorphism. Therefore, the residue field of R_P is $S^{-1}(R/P) \cong R/P$.

Example 1.3.13. Another example of discrete valuation ring is given by the ring of Taylor series $\mathbb{F}[[t]]$. In this case, the unique maximal ideal is (t) and the residue field is $\mathbb{F}[[t]]/(t) \cong \mathbb{F}$.

1.3.3 Finite modules over a discrete valuation ring

In this section, all rings R considered are discrete valuation rings. Also, we will denote the maximal ideal by $P = (p)$ (so p is a fixed generator). Now, if M if a finitely generated $R-$module, then by theorem (1.1.50) we have

$$M \cong \left(\bigoplus_{i=1}^{r} \frac{R}{(p^{\lambda_i})} \right) \oplus F \tag{1.63}$$

where F is the free part of M and $\lambda = (\lambda_1, \lambda_2, \dots)$ is a partition. Now we will be concerned with finitely generated modules M over a discrete valuation ring R that have the additional property that $p^r M = (0)$ for some $r \geq 1$. In our context, these modules will be called *finite* $R-$modules.

Definition 1.3.14. Let M be a finitely generated module over a discrete valuation ring R with maximal ideal $P = (p)$. Then M is said to be a *finite* $R-$module if $p^r M = (0)$ for some $r \geq 1$.

In this case, the free part in the equation (1.63) must be $F = (0)$. So

$$M \cong \bigoplus_{i=1}^{r} \frac{R}{(p^{\lambda_i})} \tag{1.64}$$

Now, lets show that if $|R/(p)| = |\mathbb{F}| < \infty$, then these modules have a finite number of elements.

Lemma 1.3.15. Let M be a module and suppose that there exists a finite submodule $N \subset M$ such that M/N is of finite cardinality, then M is of finite cardinality.

Proof: Let $N = \{n_1, \ldots, n_l\}$, so that $[x] = \{x + n_1, \ldots, x + n_l\}$. Now, if $M/N = \{[x_1], \ldots, [x_k]\}$, then we have

$$M \subset \{x_i + n_j \mid i = 1, \ldots, k \text{ and } j = 1, \ldots, l\}.$$

Theorem 1.3.16. If the residue field $\mathbb{F}$ of R has q elements, then $|R/(p^r)| = q^r$.

Proof: By induction assume that the statement is true for $r-1$. Using the isomorphism theorem, we get

$$\frac{R/(p^r)}{(p^{r-1})/(p^r)} \cong \frac{R}{(p^{r-1})} \tag{1.65}$$

By lemma (1.3.15) $R/(p^r)$ is finite and

$$\left|\frac{R/(p^r)}{(p^{r-1})/(p^r)}\right| = \frac{|R/(p^r)|}{|(p^{r-1})/(p^r)|} = \left|\frac{R}{(p^{r-1})}\right| = q^{r-1} \tag{1.66}$$

But $(p^{r-1})/(p^r) \cong R/(p)$ and $|(p^{r-1})/(p^r)| = q$. So by the last equation $|R/(p^r)| = qq^{r-1} = q^r$.

Corollary 1.3.17. Let M be a finite $R-$module where the residue field of R has q elements. If M has the decomposition as in the equation (1.64), then

$$|M| = q^{\sum_j \lambda_j} = q^{|\lambda|}$$

1.3.4 Composition series

Definition 1.3.18. [3] Let M be a module (over any ring). A *series* for M is a strictly decreasing sequence of submodules

$$(0) = M_0 \subset \cdots \subset M_{n-1} \subset M_n = M \tag{1.67}$$

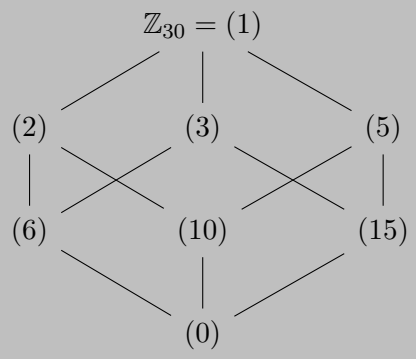
If the additional property that each M_i/M_{i-1} is simple, then we say that this series is a *composition series* for M. In this case n is called the *length* of the composition series.

Of course if $l(M) = 0$, then $M = (0)$ and $l(M) = 1$ if and only if $M \neq (0)$ is simple.

Definition 1.3.19. [3] If M has a composition series, we define $l(M)$ to be the least length of a composition series of M. If a module doesn't have a composition series, we put $l(M) = \infty$.

Example 1.3.20. If V is a finite dimensional vector space over $\mathbb{F}$, then $l(V) = \dim_{\mathbb{F}}(V)$.

Example 1.3.21. Consider the lattice of submodules of $\mathbb{Z}_{30}$.



Any path from the top to the bottom yields a composition series. For example

$$(0) \subset (6) \subset (3) \subset \mathbb{Z}_{30} \tag{1.68}$$

In fact, since $(6) = \mathbb{Z}_{30}/6\mathbb{Z}_{30} \cong \mathbb{Z}/6\mathbb{Z}$ and $(3) = \mathbb{Z}_{30}/3\mathbb{Z}_{30} \cong \mathbb{Z}/3\mathbb{Z}$. We have $\mathbb{Z}_{30}/(3) \cong \mathbb{Z}_{10}$ and $(3)/(2) \cong \mathbb{Z}_2$ which are simple submodules of $\mathbb{Z}$.

Theorem 1.3.22. (Jordan-Holder theorem, [3]) Suppose that M has a composition series of length n. Then, any other composition series of M has length n. Also, any series of M can be refined to yield a composition series.

Proof: See ([3], Chapter III).
An useful property of length, is that it is additive in the following sense.

Theorem 1.3.23. [3] Suppose that we have a short exact sequence of modules of finite length

$$0 \longrightarrow L \xrightarrow{u} M \xrightarrow{v} N \longrightarrow 0$$

then $l(M) = l(L) + l(N)$.

Proof: Let $l(L) = l$ and $l(N) = n$. By hypothesis, we have composition series

$$\begin{aligned} (0) = L_l \subset \cdots \subset L_1 = L \\ (0) = N_n \subset \cdots \subset N_1 = N \end{aligned} \tag{1.69}$$

Since u is injective, we also have another two series

$$\begin{aligned} (0) = u(L_l) \subset \cdots \subset u(L_1) = u(L) \\ v^{-1}(0) = v^{-1}(N_n) \subset \cdots \subset v^{-1}(N_1) = v^{-1}(N) \end{aligned} \tag{1.70}$$

But $u(L) = \mathrm{Ker}(v) = v^{-1}(0)$. And since u is injective, the first series in the above equation is a composition series for $u(L)$. So we build a new series

$$(0) = u(L_l) \subset \cdots \subset u(L_1) = u(L) = v^{-1}(0) = v^{-1}(N_n) \subset \cdots \subset v^{-1}(N_1) = v^{-1}(N) = M \tag{1.71}$$

and we claim that this is a composition series for M. It's clear that if this is the case, then the length of this series is $n + l$. Therefore, from the above considerations, we just need to show that

$$\frac{v^{-1}(N_i)}{v^{-1}(N_{i-1})} \tag{1.72}$$

are simple modules. To do this, we will use the isomorphism theorem to show that

$$\frac{v^{-1}(N_i)}{v^{-1}(N_{i-1})} \cong \frac{v^{-1}(N_i)/\mathrm{Ker}(v)}{v^{-1}(N_{i-1})/\mathrm{Ker}(v)} \cong \frac{N_i}{N_{i-1}} \tag{1.73}$$

To do this, note that $N_i \cong v^{-1}(N_i)/\ker(v)$ by the isomorphism theorem. So let $\varphi' : N_i \to v^{-1}(N_i)/\mathrm{Ker}(v) : x \mapsto y + \mathrm{Ker}(v)$ where $y \in v^{-1}(x)$ be an isomorphism. To ease notation, lets write $A_i := v^{-1}(N_i)/\mathrm{Ker}(v)$. Now compose φ' with the projection $A_i \to A_i/A_{i-1}$ to get

$$\varphi : N_i \to A_i/A_{i-1} : x \mapsto \overline{y} + A_{i-1} \tag{1.74}$$

where $\overline{y} = y + \mathrm{Ker}(v)$. Then $x \in \mathrm{Ker}(\varphi) \iff \varphi(x) = \overline{y} + A_{i-1} = 0 \Rightarrow \overline{y} \in A_{i-1}$. This means that $y + \mathrm{Ker}(v) \in v^{-1}(N_{i-1})/\mathrm{Ker}(v) \iff y \in v^{-1}(N_{i-1}) \iff x \in N_{i-1}$. So, by the isomorphism theorem, we have $N_i/N_{i-1} \cong A_i/A_{i-1}$.

Corollary 1.3.24. Let M be a module of finite length, if $N \subset M$ is of finite length, then $l(N) \leq l(M)$.

Now we will show that finite $R-$modules over discrete valuation rings have finite length.

Definition 1.3.25. [1] Let M be an $R-$module and $u \in M$. We write $\mathrm{Ann}_R(u)$ (called the *annihilator* of u) for the set of elements $r \in R$ such that $ru = 0$. It's clear from the definition that this set forms an ideal in R.

Lemma 1.3.26. Let $M \neq (0)$ be a module over a noetherian ring R. Then there exists a prime ideal $P \subset R$ of the form $P = \mathrm{Ann}_R(u)$ for some $u \in M$.

Proof: Let $\mathcal{P} = \{\mathrm{Ann}_R(u) \mid u \in M - \{0\}\}$. Since R is noetherian, $\mathcal{P}$ has a maximal element $P = \mathrm{Ann}_R(u)$, i.e., does not exist $Q \in \mathcal{P}$ such that $Q \supset P, Q \neq P$. Note that $u \neq 0 \Rightarrow P \neq M$ (for example $1 \notin P$). Now, $b \notin P \Rightarrow bu \neq 0$ and $\mathrm{Ann}_R(bu) \in \mathcal{P}$. But $\mathrm{Ann}_R(u) \subset \mathrm{Ann}_R(bu) \Rightarrow \mathrm{Ann}_R(u) = \mathrm{Ann}_R(bu) = P$ by maximality. So, if $ab \in P \Rightarrow abu = a(bu) = 0$. Therefore $a \in \mathrm{Ann}_R(bu) = P$. This shows that P is prime.

Lemma 1.3.27. Let $M \neq 0$, then there exists $M_1 \subset M$ such that $M_1 \cong R/P_1$ for some prime ideal P_1.

Proof: By lemma (1.3.26), there exists a prime ideal of the form $P_1 = \mathrm{Ann}_R(u)$ for some $u \neq 0$. Now, define the homomorphism $\varphi : R \to M : r \mapsto ru$. This map induces an injection $i : R/\mathrm{Ker}(\varphi) \hookrightarrow M$. But $\mathrm{Ker}(\varphi) = \mathrm{Ann}_R(u) = P_1$. Therefore we have an inclusion $M_1 \hookrightarrow M$ where $M_1 := i(R/P_1) \cong R/P_1$.

Theorem 1.3.28. Let $M \neq (0)$ be a noetherian module over a noetherian ring R. Then M has a series of the form

$$(0) \subset M_1 \subset M_2 \subset \cdots \subset M_n = M \tag{1.75}$$

such that $M_i/M_{i-1} \cong R/P_i$ for prime ideals $P_i \subset R$.

Proof: By the lemma (1.3.27), there exists $M_1 \subset M$ such that $M_1 \cong R/P_1$ for some prime ideal $P_1 \subset R$. If $M/M_1 \neq (0)$, we can apply lemma (1.3.27) again to the quotient module M/M_1 to get a module $M_2' \subset M/M_1$ such that $M_2' \subset R/P_2$ for some prime ideal $P_2 \subset R$. But $M_2' = M_2/M_1$ for some M_2 containing M_1. Repeating this procedure, we get a chain of submodules: $(0) \subset M_1 \subset M_2 \subset \ldots$ such that $M_i/M_{i-1} \cong R/P_i$. Now, since M is noetherian, the chain must terminate at some $M_n = M$ (if $M_n \neq M$, then $M/M_n \neq (0)$ and we can apply the lemma (1.3.26) again to get a new module M_{n+1} strictly bigger than M_n). So we get

$$(0) \subset M_1 \subset M_2 \subset \cdots \subset M_n = M \tag{1.76}$$

such that $M_i/M_{i-1} \cong R/P_i$ for prime ideals $P_i \subset R$.

Now, if we add the condition that R is a principal ideal domain, then each prime ideal P_i is also maximal. Therefore R/P_i are simple modules. So we get the following corollary.

Corollary 1.3.29. With the additional condition that R is a principal ideal domain we have that $l(M) < \infty$.

Now, let M be a finite $R-$module, where R is a discrete valuation ring with finite residue field. Then, R is noetherian because it's a principal ideal ring and M is noetherian because it's of finite cardinality.

Corollary 1.3.30. If M is a finite module over a discrete valuation ring, then $l(M) < \infty$.

1.3.5 Formula for the number of automorphisms

From now on, we will only be interested in the case that R is a discrete valuation ring with finite residue field $\mathbb{F} = R/(p)$. The number of elements of $\mathbb{F}$ will be denoted by q.

In the equation (1.64) it is not clear whether λ is uniquely defined by M. We will see that this is the case. In fact, let M be a module over a discrete valuation ring R with the property that $p^r M = (0)$ and $p^{r-1}M \neq (0)$ for some $r \geq 1$. We have a series

$$(0) = p^r M \subset \cdots \subset pM \subset M \tag{1.77}$$

Since $p\left(p^{i-1}M/p^iM = (0)\right)$, each quotient $p^{i-1}M/p^iM$ can be viewed as a vector space over the residue field $\mathbb{F} = R/(p)$.

Theorem 1.3.31. [12] Let

$$M \cong \bigoplus_{i=1}^{r} \frac{R}{(p^{\lambda_i})} \tag{1.78}$$

then we have $\dim_{\mathbb{F}}(p^{i-1}M/p^iM) = \lambda_i'$.

Proof: We have

$$\begin{aligned} p^{i-1}M &\cong \bigoplus_{\lambda_j \geq i} \frac{p^{i-1}R}{(p^{\lambda_j})} \\ &= \bigoplus_{\lambda_j = i} \frac{p^{i-1}R}{(p^{\lambda_j})} \oplus \bigoplus_{\lambda_j \geq i+1} \frac{p^{i-1}R}{(p^{\lambda_j})} \\ &= \bigoplus_{\lambda_j = i} \mathbb{F} \oplus \bigoplus_{\lambda_j \geq i+1} \frac{p^{i-1}R}{(p^{\lambda_j})} \end{aligned} \tag{1.79}$$

So

$$\begin{aligned} p^{i-1}M/p^iM &\cong \bigoplus_{\lambda_j=i} \mathbb{F} \oplus \bigoplus_{\lambda_j\geq i+1} \frac{p^{i-1}R/(p^{\lambda_j})}{p^iR/(p^{\lambda_j})} \\ &\cong \bigoplus_{\lambda_j=i} \mathbb{F} \oplus \bigoplus_{\lambda_j\geq i+1} \mathbb{F} \\ &= \bigoplus_{\lambda_j\geq i} \mathbb{F} \\ &= \mathbb{F}^{\lambda_i'} \end{aligned} \tag{1.80}$$

therefore $\dim_{\mathbb{F}}(p^{i-1}M/p^iM) = \lambda_i'$.

Since λ is uniquely defined by the module M we can formulate the following definition.

Definition 1.3.32. If M has a decomposition as in equation (1.64), then M is said to have *type* λ.

Of course, two modules have the same type if and only if they are isomorphic. By corollary (1.3.30) we see that finite $R-$modules have finite length. Now we claim that if M is a finite $R-$module of type λ, then $l(M) = |\lambda|$. In fact, take a composition series

$$(0) = M_0 \subset M_1 \subset \cdots \subset M_{l(M)} = M \tag{1.81}$$

then each $M_i/M_{i-1} \cong \mathbb{F}$. In particular, M is finite and

$$|M| = \prod_{i=1}^{l(M)} \left| \frac{M_i}{M_{i-1}} \right| = \prod_{i=1}^{l(M)} q = q^{l(M)} \tag{1.82}$$

By corollary (1.3.17) we have that $q^{l(M)} = q^{|\lambda|} \Rightarrow l(M) = |\lambda|$.

Notation 1.3.33. [12] For a partition λ, we denote $n(\lambda) = \sum_{i\geq 1}(i-1)\lambda_i$. Summing the columns of T_λ (with these numbers attached) we also see that

$$n(\lambda) = \sum_{i\geq 1} \binom{\lambda_i'}{2}$$

Now let's prove the main theorem of this section.

Theorem 1.3.34. [12] The number of automorphisms of a finite $R-$module of type λ is

$$a_\lambda(q) = q^{|\lambda|+2n(\lambda)} \prod_{i\geq 1} \varphi_{m_i(\lambda)}(q^{-1}) \tag{1.83}$$

where $\varphi_m(t) = (1-t)(1-t^2)\dots(1-t^m)$.

Definition 1.3.35. [12] Let $x \in M - \{0\}$, we shall say that x has *height* r if $p^r x = 0$ while $p^{r-1}x \neq 0$. So x has height r if and only if $Rx \cong R/(p^r)$.

Notation 1.3.36. Let M be a finite $R-$module. We will denote M_r the submodule of M generated by elements of height less than or equal to r.

To count the number of automorphisms of a finite $R-$module of type λ, we need to list the number of sequences $(x_1, x_2, \dots)$ such that each x_i has height λ_i. So we proceed as follows: to count the number of elements $x_1 \in M$ with height λ_1, we need to calculate the cardinality of the set $M_{\lambda_1} - M_{\lambda_1 - 1}$. Now, choose x_1 in this set, then we need to count the number of elements $x_2 \in M$ of height λ_2 that don't belong to the submodule Rx_1. Repeating this procedure, we see that at each step we need to choose a x_j such that $Rx_j \cap (Rx_{j-1} \oplus \dots \oplus Rx_1) = (0)$ where each x_i has height λ_i. To count this, the only property of the submodule $N = Rx_{j-1} \oplus \dots \oplus Rx_1$ that we need, is that it is a submodule generated by elements of height greater than or equal to λ_{j-1}.

Lemma 1.3.37. [12] Let M be a $R-$module of type λ. Then $l(M_r) = \lambda'_1 + \dots + \lambda'_r$.

Proof: By theorem (1.3.31), we have

$$\begin{aligned} \lambda'_1 + \dots + \lambda'_r &= \sum_{i=1}^{r} \dim_{\mathbb{F}} \left(\frac{p^{i-1}M}{p^i M} \right) \\ &= \sum_{i=1}^{r} l \left(\frac{p^{i-1}M}{p^i M} \right) \\ &= \sum_{i=1}^{r} (l(p^{i-1}M) - l(p^i M)) \\ &= l(M) - l(p^r M) \end{aligned} \tag{1.84}$$

But, by definition M_r is the kernel of the map $M \to p^r M : m \mapsto p^r m$. Therefore $M/M_r \cong p^r M$ and by the additive property of length we get $l(M_r) = l(M) - l(p^r M) = \lambda'_1 + \dots + \lambda'_r$.

Lemma 1.3.38. [12] Let M be a finite $R-$module and $N \subset M$ a submodule generated by elements of height greater than or equal to r. Then $N_1 \subset p^{r-1}N$.

Proof: Since M is a finite $R-$module, N is a finite $R-$module and $N = \bigoplus_i Rx_i$ where each $x_i \in M$ has height greater than or equal to r. Now, if $z \in N_1, z \neq 0$ we have $z = \sum_i r_i x_i$. Moreover $pz = 0 \Rightarrow pr_i x_i = 0$ for each i. Therefore $pr_i \in (p^r) \Rightarrow r_i = p^{r-1}s_i$ for each i. This shows that $z = p^{r-1}\sum_i s_i x_i \in p^{r-1}N$.

Lemma 1.3.39. [12] Let M be a finite $R-$module and $N \subset M$ a submodule generated by elements of height greater than or equal to r, and let $x \in M$. Then

$$x \text{ has height } r \text{ and } Rx \cap N = (0) \iff x \in M_r - (M_{r-1} + N_r) \tag{1.85}$$

Proof: Suppose that x has height r and $Rx \cap N = (0)$. Since x has height r, we have $x \in M_r$. Also, if $x \in M_{r-1} + N_r \Rightarrow p^{r-1}x \in N - \{0\}$ contradicting the hypothesis that $Rx \cap N = (0)$. Now if $x \in M_r - (M_{r-1} + N_r)$, we have that x has height r. But, if $Rx \cap N \neq (0) \Rightarrow$ there exists $m < r$ such that $p^m x \in N - \{0\}$ (see theorem (1.3.11)) $\Rightarrow$ $p^{r-1}x \in N_1 \subset N$. By the previous lemma, $p^{r-1}x = p^{r-1}y$ for some $y \in N$ and since $p^{r-1}x \neq 0$ we have that $y \in N_r$. Therefore $p^{r-1}(x-y) = 0 \Rightarrow x - y \in M_{r-1}$ while $y \in N_r \Rightarrow \in xM_{r-1} + N_r$.

Lemma 1.3.40. [12] Let M be a finite $R-$module of type λ. If $N \subset M$ is a submodule of type ν, then the number of elements $x \in M$ with height r and satisfying $Rx \cap N = (0)$ is

$$q^{\lambda'_1 + \cdots + \lambda'_r}(1 - q^{\nu'_r - \lambda'_r}) \tag{1.86}$$

Proof: By lemma (1.3.39) we just need to compute $|M_r - (M_{r-1} + N_r)|$. But

$$|M_r - (M_{r-1} + N_r)| = |M_r| - |M_{r-1} + N_r| = q^{l(M_r)} - q^{l(M_{r-1}+N_r)} \tag{1.87}$$

and

$$\begin{aligned} l(M_{r-1} + N_r) &= l(M_{r-1}) - l\left(\frac{M_{r-1} + N_r}{M_{r-1}}\right) \\ &= l(M_{r-1}) - l\left(\frac{N_r}{M_{r-1} \cap N_r}\right) \\ &= l(M_{r-1}) - (l(N_r) - l(N_{r-1})) \\ &= \lambda'_1 + \ldots \lambda'_{r-1} - \nu'_r \ \text{ (by lemma (1.3.37))} \end{aligned} \tag{1.88}$$

Using the last two equations, we get $|M_r - (M_{r-1} + N_r)| = q^{\lambda'_1 + \cdots + \lambda'_r}(1 - q^{\lambda'_r - \nu'_r})$

Going back to the theorem (1.3.34), we see that each submodule $Rx_{j-1} \oplus \cdots \oplus Rx_1$ has type $\nu = (\lambda_1, \ldots, \lambda_{j-1})$. Therefore, using lemma (1.3.40) we see that

$$\begin{aligned} |\mathrm{Aut}(M)| &= \prod_{i\geq 1} q^{\lambda'_1+\cdots+\lambda'_{\lambda_i}}(1-q^{\nu'_{\lambda_i}-\lambda'_{\lambda_i}}) \\ &= \prod_{i\geq 1} q^{\lambda'_1+\cdots+\lambda'_{\lambda_i}}(1-q^{i-1-\lambda'_{\lambda_i}}) \ \text{ (because } \nu'_{\lambda_i} = i-1) \end{aligned} \tag{1.89}$$

Now, for $a \in \mathrm{Im}(\lambda)$ (the set of parts of λ), define $j_a := \mathrm{Min}\{j \geq 1 \mid \lambda_j = a\}$. Then $j_a = 1 + \lambda'_a - m_a(\lambda)$. To verify this, just note that $m_a(\lambda) = \lambda'_a - \lambda'_{a+1}$ and $\lambda'_{a+1} = \mathrm{Max}\{j \geq 1 \mid \lambda_j \geq a+1\} = j_a - 1$. Using this notation, we can write

$$\begin{aligned} \prod_{i\geq 1} q^{\lambda'_1+\cdots+\lambda'_{\lambda_i}}(1-q^{i-1-\lambda'_{\lambda_i}}) &= \prod_{a\in\mathrm{Im}(\lambda)} \prod_{i=0}^{m_a(\lambda)-1} q^{\lambda'_1+\cdots+\lambda'_a}(1-q^{(i+j_a)-1-\lambda'_a}) \\ &= \prod_{a\in\mathrm{Im}(\lambda)} \prod_{i=0}^{m_a(\lambda)-1} q^{\lambda'_1+\cdots+\lambda'_a}(1-q^{i-m_a(\lambda)}) \\ &= \prod_{a\in\mathrm{Im}(\lambda)} q^{m_a(\lambda)(\lambda'_1+\cdots+\lambda'_a)}\varphi_{m_a(\lambda)}(q^{-1}) \\ &= q^{\sum_{a\geq 1} m_a(\lambda)(\lambda'_1+\cdots+\lambda'_a)} \prod_{a\geq 1} \varphi_{m_a(\lambda)}(q^{-1}) \\ &= q^{\sum_{a\geq 1}(\lambda'_a-\lambda'_{a+1})(\lambda'_1+\cdots+\lambda'_a)} \prod_{a\geq 1} \varphi_{m_a(\lambda)}(q^{-1}) \\ &= q^{{\lambda'_1}^2+\cdots+{\lambda'_{\lambda_1}}^2} \prod_{a\geq 1} \varphi_{m_a(\lambda)}(q^{-1}) \\ &= q^{|\lambda|+2n(\lambda)} \prod_{a\geq 1} \varphi_{m_a(\lambda)}(q^{-1}) \end{aligned} \tag{1.90}$$

1.4 Hall Algebras

1.4.1 Definition of the Hall algebra

Definition 1.4.1. [12] Let M be a finite $R-$module. Then $N \subset M$ is said to have *cotype* μ in M if M/N is of type μ.

Definition 1.4.2. [12] A finite $R-$module M is called *elementary* if $pM = (0)$. In this case we see that M is elementary if and only if it is of type (1^m) for some $m \geq 1$.

An important property of elementary submodules is that they are precisely vector spaces over $\mathbb{F}$. In fact, if $pM = (0)$, the $R-$action pass through the quotient $(x + (p)) \cdot m := xm$ and M becomes a vector space over $\mathbb{F}$. Conversely, if M is a vector space over F, then we can compose the $\mathbb{F}-$action with the projection $R \to R/(p) = \mathbb{F}$ to get a $R-$module.

Definition 1.4.3. [12] The *Hall algebra* $H(R)$ is defined to be the free $\mathbb{Z}-$module on a basis (u_λ) generated by all partitions λ. The product is defined by the equation

$$u_\mu u_\nu := \sum_\lambda G_{\mu\nu}^\lambda u_\lambda \tag{1.91}$$

where the structure constants $G_{\mu\nu}^\lambda$ are defined as follows: Fix a finite $R-$module M of type λ. The number $G_{\mu\nu}^\lambda$ is the number of chains

$$(0) \subset N \subset M \tag{1.92}$$

such that N is of type ν and cotype μ in M.

1.4.2 Calculantion of the structure constants

From the additive property of the length, we get that $G_{\mu\nu}^\lambda \neq 0$ only if $|\lambda| = |\mu| + |\nu|$. Before we prove the main properties of the Hall algebra, lets compute its most basics structure constants.

Lemma 1.4.4. Let V be a vector space over a finite field $\mathbb{F}$ of cardinality q. Then, for a fixed $i \leq d = \dim_{\mathbb{F}}(V)$, the number of sequences $(v_1, \ldots, v_i)$ of linearly independent vectors in V is

$$(q^d - 1)(q^d - q) \ldots (q^d - q^{i-1}) \tag{1.93}$$

Proof: If $i = 1$ we just need to choose a nonzero vector v_1. The number of such vectors is $q^d - 1$, so the result is true for $i = 1$. Now, assume by induction that the result is true for some $i \geq 1$. Then, there are $(q^d - 1)(q^d - q) \ldots (q^d - q^{i-1})$ possibilities to choose a linearly independent sequence $(v_1, \ldots, v_i)$ in V. Since the map $(a_1, \ldots, a_i) \in \mathbb{F}_q^i \mapsto \sum_{j=1}^i a_j v_j \in V$ is injective, the span of the sequence $(v_1, \ldots, v_i)$ have exactly q^i elements. So, there are $q^d - q^i$ ways to choose v_{i+1} such that $(v_1, \ldots, v_i, v_{i+1})$ is linearly independent. This shows that the number of linearly independent sequences of $i+1$ vectors is $(q^d - 1)(q^d - q) \ldots (q^d - q^{i-1})(q^d - q^i)$ which completes the induction argument.

Lemma 1.4.5. Let $N \subset M$ be such that $pM \subset N$. Then, M/N is a vector space of dimension $d = \dim_{\mathbb{F}}(M/N)$, where $\mathbb{F}$ is the residue field of R. In this case, the number of sequences $(x_1, \ldots, x_i)$ in M, for $i \leq d$, which are linearly independent modulo in M/N is

$$(q^{d+n} - q^n)(q^{d+n} - q^{n+1}) \ldots (q^{d+n} - q^{n+i-1}) \tag{1.94}$$

where $n = l(N)$.

Proof: Fix $i \leq 1$ A sequence $(x_1, \ldots, x_i)$ in M is linearly independent modulo N if and only if $(\overline{x_1}, \ldots, \overline{x_i})$ is linearly independent in M/N. By the previous lemma, there are $(q^d-1)(q^d-q)\ldots(q^d-q^{i-1})$ linearly independent sequences $(\overline{x_1}, \ldots, \overline{x_i})$ in $V = M/N$. Now, $(\overline{x_1}, \ldots, \overline{x_i}) = (\overline{y_1}, \ldots, \overline{y_i})$ if and only if $x_j - y_j \in N$. Since N has q^n elements, the number of sequences $(x_1, \ldots, x_i)$ that have the same image under the projection is q^{ni}. Therefore, the number of sequences $(x_1, \ldots, x_i)$ in M that are linearly independent modulo N is

$$q^{ni}(q^d - 1)(q^d - q)\ldots(q^d - q^{i-1}) = (q^{d+n} - q^n)(q^{d+n} - q^{n+1})\ldots(q^{d+n} - q^{n+i-1}). \tag{1.95}$$

Lets compute explicitly a few values of $G^{\lambda}_{\mu\nu}$. It's convenient to introduce the Gauss polynomials $\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right]_t = \prod_{i=1}^{b}(1 - t^{a+i-1})/(1 - t^i)$.

Example 1.4.6. $G^{(1^{r+s})}_{(1^r)(1^s)}$ counts the number of submodules $N \subset \mathbb{F}^{r+s}$ of type (1^s) and cotype (1^r). We claim that

$$G^{(1^{r+s})}_{(1^r)(1^s)} = q^{sr} \begin{bmatrix} r+s \\ s \end{bmatrix}_{q^{-1}} \tag{1.96}$$

In fact, these modules are precisely the subspaces of dimension s in $\mathbb{F}^{r+s}$. Let $d = r + s$, from lemma (1.4.4) we get that there exists

$$(q^d - 1)(q^d - q)\cdots(q^d - q^{s-1}) \tag{1.97}$$

linearly independent sequences in $(v_1, \ldots, v_s)$ in M. Each one of these sequences generates a subspace N of dimension s. But, given N, it has precisely

$$(q^s - 1)(q^s - q)\cdots(q^s - q^{s-1}) \tag{1.98}$$

basis, i.e. linearly independent sequences which generated the whole N. Therefore, the

number of subspaces of dimension s in M is the quotient

$$\begin{aligned}
&\frac{(q^d-1)(q^d-q)\cdots(q^d-q^{s-1})}{(q^s-1)(q^s-q)\cdots(q^s-q^{s-1})} \\
&= \frac{q^{ds}}{q^{ss}}\frac{(1-q^{-d})(1-q^{-d+1})\cdots(1-q^{-d+s-1})}{(1-q^{-s})(1-q^{-s+1})\cdots(1-q^{-s+s-1})} \\
&= q^{sr}\begin{bmatrix} r+s \\ s \end{bmatrix}_{q^{-1}}
\end{aligned} \tag{1.99}$$

Of course, each one of the subspaces N of dimension s, has codimension r in N, i.e. the quotient M/N is of type (1^r). In conclusion, we get

$$G^{(1^{r+s})}_{(1^r)(1^s)} = q^{sr}\begin{bmatrix} r+s \\ s \end{bmatrix}_{q^{-1}}. \tag{1.100}$$

Example 1.4.7. $G^{(r+s)}_{(r)(s)}$ counts the number of submodules N of type (s) and cotype (r) of $M = R/(p^{(r+s)})$. We claim that

$$G^{(r+s)}_{(r)(s)} = 1 \tag{1.101}$$

From the isomorphism theorem, each submodule is of the form $N = (p^m)/(p^{r+s})$ for some $m \in \{0, 1, \ldots, r+s\}$. But, N is of type (s) if and only if $m = r$. Therefore $N = (p^r)/(p^{r+s})$. We must check that N has cotype (r) in M. But again, by the isomorphism theorem, $M/N \cong R/(p^r)$ is of type (r). Therefore the unique possibility is $N = (p^r)/(p^{r+s})$ and we conclude that $G^{(r+s)}_{(r)(s)} = 1$.

From the previous examples we get that $G^{(2)}_{(1)(1)} = 1$ and

$$G^{(1^2)}_{(1)(1)} = q\begin{bmatrix} 2 \\ 1 \end{bmatrix}_{q^{-1}} = q(1-q^{-2})/(1-q^{-1}) = q(1+q^{-1}) = q+1 \tag{1.102}$$

Therefore, we can compute the product

$$u_{(1)}u_{(1)} = u_{(2)} + (q+1)u_{(1^2)}. \tag{1.103}$$

Definition 1.4.8. [12] Let M be a fixed $R-$module of type λ and partitions μ_i, $i = 1, \ldots n$. We will write $G^{\lambda}_{\mu_1 \ldots \mu_n}$ to denote the number of chains

$$(0) = M_n \subset M_{n-1} \subset \cdots \subset M_1 \subset M_0 = M \tag{1.104}$$

such that M_{i-1}/M_i is of type μ_i for each $i = 1, \ldots, n$.

Lemma 1.4.9. We have

$$\sum_{\lambda} G^{\lambda}_{\mu_1 \dots \mu_r} G^{\xi}_{\lambda \mu_{r+1} \dots \mu_{r+s}} = G^{\xi}_{\mu_1 \dots \mu_{r+s}} = \sum_{\lambda} G^{\xi}_{\mu_1 \dots \mu_{r-1} \lambda} G^{\lambda}_{\mu_r \dots \mu_{r+s}} \tag{1.105}$$

Proof: First, let's focus on the left equality. We will compute $G^{\xi}_{\mu_1 \dots \mu_{r+s}}$ by dividing in cases according to the type of M/M_r. Clearly, each case will produce a different chain of submodules. Now, fix λ. By definition, there are $G^{\xi}_{\lambda \mu_{r+1} \dots \mu_{r+s}}$ ways to construct a chain of the form

$$(0) = M_{r+s} \subset \cdots \subset M_r \subset M \tag{1.106}$$

such that M_{i-1}/M_i is of type μ_i for each $i = r+1, \dots, r+s$ while M/M_r is of type λ. Also, there are $G^{\lambda}_{\mu_1 \dots \mu_r}$ ways to construct a chain

$$(0) = M'_r \subset \cdots \subset M'_1 \subset M'_0 = M/M_r \tag{1.107}$$

where M'_{i-1}/M'_i is of type μ_i for each $i = 1, \dots, r$. But, by the isomorphism theorem, for each i, there is a unique submodule $M_i \subset M$ such that $M/M_i \cong M'_i$. So, for each chain (1.107) there is a unique chain of submodules

$$M_r \subset \dots M_1 \subset M_0 = M \tag{1.108}$$

such that $M_{i-1}/M_i \cong M'_{i-1}/M'_i$ is of type μ_i for each $i = 1, \dots, r$, while M/M_r is of type λ. Conversely, each chain (1.108) give a chain of the form (1.107) by taking the quotient by M_r. Therefore, there exists $G^{\lambda}_{\mu_1 \dots \mu_r}$ ways to complete the sequence (1.106) and we conclude that there exists $G^{\lambda}_{\mu_1 \dots \mu_r} G^{\xi}_{\mu_1 \dots \mu_{r+s}}$ ways to construct a chain $(0) = M_{r+s} \subset \cdots \subset M_1 \subset M_0 = M$ such that M_{i-1}/M_i is of type μ_i for each $i = 1, \dots, r+s$ while M/M_r is of type λ. Summing these numbers for all λ, we get $G^{\xi}_{\mu_1 \dots \mu_{r+s}}$. The same reasoning can be applied to obtain the right equation of the lemma.

Theorem 1.4.10. [12] The Hall algebra $H(R)$ is associative.

Proof: This follows from the previous lemma for $r = 2$ and $s = 1$.

Theorem 1.4.11. [12] The Hall algebra $H(R)$ is commutative.

Proof: See ([12], chapter II).

Definition 1.4.12. Let $n \geq 1$ be a positive integer and $\mathcal{P}_n$ be the set of partitions of n. The *reverse lexicographic ordering* L_n is the subset of $\mathcal{P}_n \times \mathcal{P}_n$ consisting of pairs (μ, λ) such that the first nonvanishing entry $\mu_i - \lambda_i$ is positive. We write $\mu > \lambda$ if $(\mu, \lambda) \in L_n$.

For example, for $n = 5$ we have

$$(5) > (4,1) > (3,2) > (3,1^2) > (2^2,1) > (2,1^3) > (1^5) \tag{1.109}$$

Using this, we put a total order on $\mathcal{P}$ by $\mu > \lambda$ if $\mu \in \mathcal{P}_n$ and $\lambda \in \mathcal{P}_m$ with $n > m$ or if $(\mu, \lambda) \in L_n$ in the case $\mu, \lambda \in \mathcal{P}_n$. With this total order, we are able to work with matrices $(A_{\mu\lambda})$ indexed by $\mathcal{P}$. In this case, multiplication is defined as usual $(AB)_{\mu\lambda} = \sum_\nu A_{\mu\nu} B_{\nu\lambda}$.

Theorem 1.4.13. [12] Let $\lambda = (\lambda_1, \ldots, \lambda_r)$ be a partition and $v_{\lambda'} := v_{\lambda_1} \cdots v_{\lambda_r}$ where $v_i := u_{(1^i)}$. Then, the set $\{v_\lambda |\ \lambda \in \mathcal{P}\}$ forms a $\mathbb{Z}-$basis of $H(R)$.

Proof: Fix a partition λ and let $\lambda' = (\lambda'_1, \ldots, \lambda'_s)$ be the conjugate of λ. By the lemma (1.4.9) we get that

$$v_{\lambda'} := v_{\lambda'_1} \cdots v_{\lambda'_r} = \sum_\xi G^{\xi}_{(1^{\lambda'_1})\ldots(1^{\lambda'_s})} u_\xi \tag{1.110}$$

To ease notation, let's write $a_{\lambda\xi} := G^{\xi}_{(1^{\lambda'_1})\ldots(1^{\lambda'_s})}$. Now, fix a module M of type ξ. By definition, $a_{\lambda\xi}$ is the number of chains

$$(0) = M_s \subset M_{s-1} \cdots \subset M_1 \subset M_0 = M \tag{1.111}$$

such that M_{i-1}/M_i is of type $(1^{\lambda'_i})$ for each $i = 1, \ldots, s$. Suppose that such a sequence exist. Since each M_{i-1}/M_i is elementary, we have $p(M_{i-1}/M_i) = (0) \iff pM_{i-1} \subset M_i$ for each $i = 1, \ldots, s$. Therefore $p^i M \subset M_i$ and $l(p^i M) \leq l(M_i)$. So

$$\begin{aligned}
&l(M/p^i M) \geq l(M/M_i) \iff \\
&l(M) - l(p^i M) \geq l(M) - l(M_i) \iff \\
&\sum_{j=1}^{i} l(p^{j-1}M/p^j M) \geq \sum_{j=1}^{i} l(M_{j-1}/M_j) \iff \\
&\sum_{j=1}^{i} \xi'_j \geq \sum_{j=1}^{i} \lambda'_j \quad \text{(by theorem (1.3.31))}
\end{aligned} \tag{1.112}$$

and the fact that $l(M_{j-1}/M_j) = |(1^{\lambda'_j})| = \lambda'_j$. This is true for each $i = 1, \ldots, s$. Therefore $\xi' \geq \lambda' \iff \xi \leq \lambda$ which implies that $a_{\lambda\xi} \neq 0 \iff \xi \leq \lambda$. Conversely, if $\xi = \lambda$, we see by the previous equation that $l(M/p^i M) = l(M/M_i)$, which happens only if $l(p^i M) = l(M_i) \Rightarrow p^i M = M$. So in this case, there is only one decreasing chain (M_i) satisfying that M_{i-1}/M_i is of type $(1^{\lambda'_i})$, which is the chain $(p^i M)$. This shows that $a_{\lambda\lambda} = 1$. So the change of basis matrix $(a_{\lambda\xi})$ is unitriangular, and therefore unimodular, which implies that we can solve equation (1.110) to express the u_ξ in term of the v_λ. This proves the theorem.

One of the consequences of this theorem is that the structure constants $G^{\lambda}_{\mu\nu}$ are completely determined by the structure constants of the form $G^{\lambda}_{\mu(1^m)}$. So the next task will be to compute these structure constants explicitly.

Lemma 1.4.14. [12] Let M be a finite $R-$module of type λ and $N \subset M$ a submodule of cotype μ in M. Then $\mu \subset \lambda$.

Proof: We need to show that $\mu_i \leq \lambda_i$ for all $i \geq 1$. Since $p^i(M/N)$ is a submodule of M/N, by the isomorphism theorem, there exists a submodule $L_i \subset M$ such that $M/L_i = p^i(M/N)$. In this case $L^i = \pi^{-1}(p^i(M/N)) = p^iM + N$ where π in the natural projection of M onto M/N. Therefore

$$\begin{aligned}\frac{p^{i-1}(M/N)}{p^i(M/N)} &= \frac{(p^{i-1}M+N)/N}{(p^iM+N)/N} \\ &\cong \frac{p^{i-1}M+N}{p^iM+N} \\ &= \frac{p^{i-1}M+p^iM+N}{p^iM+N} \quad \text{(because } p^iM \subset p^{i-1}M\text{)} \\ &\cong \frac{p^{i-1}M}{p^{i-1}M\cap(p^iM+N)}\end{aligned} \tag{1.113}$$

Therefore

$$l\left(\frac{p^{i-1}(M/N)}{p^i(M/N)}\right) = l(p^{i-1}M) - l(p^{i-1}M\cap(p^iM+N)) \tag{1.114}$$

But $p^iM \subset p^{i-1}M \cap (p^iM+N)$, so

$$\mu'_i = l\left(\frac{p^{i-1}(M/N)}{p^i(M/N)}\right) \leq l(p^{i-1}M) - l(p^iM) = l\left(\frac{p^{i-1}M}{p^iM}\right) = \lambda'_i \tag{1.115}$$

and we conclude that $\mu_i \leq \lambda_i$ for each $i \geq 1 \Rightarrow \mu \subset \lambda$.

Definition 1.4.15. [12] Let M be a $R-$module. The set $S = \{x \in M |\ px = 0\}$ is called the *socle* of M.

It's clear by the definition that $S \subset M$ is the unique maximal elementary submodule of M. Now, if M is of type $\lambda = (\lambda_1, \ldots, \lambda_r)$, i.e.,

$$M = \bigoplus_{i=1}^{r} \frac{R}{(p^{\lambda_i})} \tag{1.116}$$

then $S = \bigoplus_{i=1}^{r}(p^{\lambda_i-1})/(p^{\lambda_i})$ and $M/S \cong \bigoplus_{i=1}^{r}(R/(p^{\lambda_i-1}))$. So M/S is of type $\tilde{\lambda} = (\lambda_1 - 1, \ldots, \lambda_r - 1)$.

Lemma 1.4.16. [12] Let M be a finite $R-$module of type λ and $L \subset N \subset M$ where L (resp. N) is of cotype β (resp. α) in M. With these conditions $\alpha \subset \beta \subset \lambda$.

Proof: We have the isomorphism $M/N \cong (M/L)/(N/L)$. So N/L is a submodule of cotype α in M/L . But M/L is of type β, which implies that $\alpha \subset \beta$ by lemma (1.4.14). Moreover, the same lemma implies that $\beta \subset \lambda$ and the result follows.

Lemma 1.4.17. [12] Let M be a finite $R-$module of type λ and $N \subset M$ be an elementary submodule of cotype μ. Then $\lambda - \mu$ is a *vertical strip*, i.e. $\lambda_i - \mu_i \in \{0, 1\}$ for each $i \geq 1$.

Proof: Since N is elementary, we have $N \subset S$. Therefore, from the previous lemma $\tilde{\lambda} \subset \mu \subset \lambda \Rightarrow \lambda_i - 1 \leq \mu_i \leq \lambda_i$ for each $i \geq 1$. This shows that $\lambda_i - \mu_i \in \{0, 1\}$ for each $i \geq 1$.

Theorem 1.4.18. [12] Let M be a finite $R-$module of type λ and $N \subset M$ a submodule of cotype α in M. Let β be a partition such that $\alpha \subset \beta \subset \lambda$. The number of submodules $L \subset N$ of cotype β in M is given by

$$h_{\alpha\beta\lambda}(q) := q^{d(\alpha,\beta,\lambda)} \prod_{i\geq 1} \begin{bmatrix} \lambda'_i - \alpha'_i \\ \lambda'_i - \beta'_i \end{bmatrix}_{q^{-1}} \tag{1.117}$$

where $d(\alpha, \beta, \lambda) = \sum_{j\geq i}(\beta'_j - \alpha'_j)(\lambda'_i - \beta'_i)$.

Proof: Let $L \subset N$ be a submodule (not necessarily of cotype β in M). Also, define $N_i := N \cap p^i M,\ L_i := L \cap p^i M\ (= L \cap N_i),\ n_i := l(N_i)$ and $l_i := l(L_i)$. Then

$$\begin{aligned} l(L_{i-1}/L_i) &= l(L/L_i) - l(L/L_{i-1}) \\ &= l(L/(L \cap p^i M)) - l(L/L \cap p^{i-1}M) \\ &= l\left(\frac{L + p^i M}{p^i M}\right) - l\left(\frac{L + p^{i-1}M}{p^{i-1}M}\right) \\ &= l(p^{i-1}M/p^i M) - l((L + p^{i-1}M)/(L + p^i M)) \\ &= \lambda'_i - l\left(\frac{(L + p^{i-1}M)/L}{(L + p^i M)/L}\right) \\ &= \lambda'_i - l\left(\frac{p^{i-1}(M/L)}{p^i(M/L)}\right). \end{aligned} \tag{1.118}$$

Now, let $\theta := \lambda - \beta$. The previous equation shows that L is of cotype β in M if and only if $l(L_{i-1}/L_i) = \lambda'_i - \beta'_i = \theta'_i$. So we see that for each submodule L of cotype β in M, there is

a sequence $(L_i)_{i\geq 0}$ (with $L = L_0$) which satisfies the following conditions:

$$\textbf{(1)}\ L_i = L_{i-1} \cap N_i \quad \text{and} \quad \textbf{(2)}\ l(L_{i-1}/L_i) = \theta_i' \quad \forall i \tag{1.119}$$

Conversely, for each sequence $(L_i)_{i\geq 0}$ satisfying (1) and (2), we have $L_i = L_0 \cap N_i$ while $l(L_{i-1}/L_i) = \theta_i'$. So L_0 is a submodule of cotype β in M. This shows that, the map $(L_i)_{i\geq 0} \mapsto L_0$ gives a one-to-one correspondence between sequences satisfying (1) and (2) and submodules of cotype β in M. Therefore, we will focus on enumerating such sequences.

The strategy will be the following: For a given $i \geq 1$ and $L_i \subset N_i$, we will count the number of submodules $L_{i-1} \subset N_{i-1}$ such that (1) and (2) are satisfied for i. We will see that L_{i-1} is generated by L_i and some linearly independent set modulo N_i in N_{i-1}, with θ_i' elements.

Now we need to make an observation: Let $L_j \subset N_j$ $(j = i-1, i)$ be two submodules satisfying (1) and (2) for i. Since L_{i-1}/L_i is a vector space, by the property (1) and (2), we get that $\dim_{\mathbb{F}}(L_{i-1}/L_i) = l(L_{i-1}/L_i) = \theta_i'$. Similarly, N_{i-1}/N_i is a vector space of dimension $l(N_{i-1}/N_i) = n_{i-1} - n_i$. But, since $N_i = N \cap p^i M$, we get

$$\begin{aligned} & p^i M/N_i \cong (N + p^i M)/N \cong p^i(M/N) \\ \Rightarrow\ & l(N_i) = l(p^i M) - l(p^i(M/N)) \\ & \qquad = \sum_{j>i} l(p^{j-1}M/p^j M) - l(p^{j-1}(M/N)/p^j(M/N)) \\ & \qquad = \sum_{j>i} (\lambda_j' - \alpha_j') \quad \text{(because of the lemma (1.3.31))} \end{aligned} \tag{1.120}$$

The last equation implies that $n_{i-1} - n_i = \lambda_i' - \alpha_i'$. To ease notation, define $\varphi := \beta - \alpha$, so we can write $\theta_i' + \varphi_i' = \lambda_i' - \alpha_i'$ for the dimension of L_{i-1}/L_i.

With these observations, we proceed to count the submodules L_{i-1}. By Lemma (1.4.5), there exists

$$A_i = (q^{\theta_i'+\varphi_i'+n_i} - q^{n_i})(q^{\theta_i'+\varphi_i'+n_i} - q^{n_i+1}) \dots (q^{\theta_i'+\varphi_i'+n_i} - q^{n_i+\theta_i'-1}) \tag{1.121}$$

sequences $x = (x_1, \dots, x_{\theta_i'})$ in N_{i-1}, which are linearly independent modulo N_i. Writing Rx for the $R-$span of x, we claim that $L_{i-1} := Rx + L_i$ satisfies properties (1) and (2).

First, lets verify the first property. It's clear that $L_i \subset (Rx + L_i) \cap N_i$. Now, if $y \in (Rx + L_i) \cap N_i$, then $y = z + l \in N_i$, where $z \in Rx$ and $l \in N_i$. But this implies that $z \in N_i$, because $L_i \subset N_i$. But x is linearly independent modulo N_i. Therefore $z = 0$ and $y = l \in N_i$. This shows that $L_i = (Rx + L_i) \cap N_i = L_{i-1} \cap N_i$. For the second property, we just need to see that $L_{i-1}/L_i = (Rx + L_i)/L_i$ is a vector space of dimension θ_i', because $(x_1 + N_i, \dots, x_{\theta_i'} + N_i)$ is a basis of this space.

Conversely, if $L_{i-1} \subset L_i$ is such that (1) and (2) is true for i, then L_{i-1}/L_i is a vector space of dimension θ_i'. Let $(x_1 + L_i, \dots, x_{\theta_i'} + L_i)$ be a basis of L_{i-1}/L_i, so $L_{i-1} = Rx + L_i$ where $(x_1, \dots, x_{\theta_i'})$ is a sequence in $L_{i-1} \subset N_{i-1}$ which is linearly independent modulo L_i. By the lemma (1.4.5), there are

$$B_i = (q^{\theta_i'+l_i} - q^{l_i})(q^{\theta_i'+l_i} - q^{l_i+1}) \dots (q^{\theta_i'+l_i} - q^{l_i+\theta_i'-1}) \tag{1.122}$$

such sequences. This shows that, for each one of the A_i ways to construct L_{i-1}, there are exactly B_i that produces the same module. Therefore, for a fixed $i \geq 1$ and $L_i \subset N_i$, the number of modules $L_{i-1} \subset N_{i-1}$ such that (1) and (2) hold for i is

$$\begin{aligned}
\frac{A_i}{B_i} &= \frac{(q^{\theta'_i+\varphi'_i+n_i} - q^{n_i})(q^{\theta'_i+\varphi'_i+n_i} - q^{n_i+1}) \dots (q^{\theta'_i+\varphi'_i+n_i} - q^{n_i+\theta'_i-1})}{(q^{\theta'_i+l_i} - q^{l_i})(q^{\theta'_i+l_i} - q^{l_i+1}) \dots (q^{\theta'_i+l_i} - q^{l_i+\theta'_i-1})} \\
&= \frac{(1-q^{-(\theta'_i+\varphi'_i)})(1-q^{-(\theta'_i+\varphi'_i)+1}) \dots (1-q^{-(\theta'_i+\varphi'_i)+\theta'_i-1})}{(1-q^{-\theta'_i})(1-q^{-\theta'_i+1}) \dots (1-q^{-\theta'_i+\theta'_i-1})} \frac{q^{\theta'_i n_{i-1}}}{q^{\theta'_i l_{i-1}}} \\
&= q^{\theta'_i(n_{i-1}-l_{i-1})} \begin{bmatrix} \theta'_i + \varphi'_i \\ \theta'_i \end{bmatrix}_{q^{-1}} \\
&= q^{\theta'_i \sum_{j\geq i} \varphi'_j} \begin{bmatrix} \theta'_i + \varphi'_i \\ \theta'_i \end{bmatrix}_{q^{-1}} \quad (\text{because } \; l_{i-1} = \sum_{j\geq i} \theta'_i, \quad \text{which follows from (2)})
\end{aligned} \tag{1.123}$$

Therefore, the number of sequences $(L_i)_{i\geq 0}$, satisfying conditions (1) and (2), is

$$\begin{aligned}
&\prod_{i\geq 1} q^{\theta'_i \sum_{j\geq i} \varphi'_j} \begin{bmatrix} \theta'_i + \varphi'_i \\ \theta'_i \end{bmatrix}_{q^{-1}} \\
&= q^{\sum_{i\geq 1} \theta'_i \sum_{j\geq i} \varphi'_j} \prod_{i\geq 1} \begin{bmatrix} \theta'_i + \varphi'_i \\ \theta'_i \end{bmatrix}_{q^{-1}} \\
&= q^{\sum_{j\geq i} \theta'_i \varphi'_j} \prod_{i\geq 1} \begin{bmatrix} \theta'_i + \varphi'_i \\ \theta'_i \end{bmatrix}_{q^{-1}} \\
&= q^{d(\alpha,\beta,\lambda)} \prod_{i\geq 1} \begin{bmatrix} \lambda'_i - \alpha'_i \\ \lambda'_i - \beta'_i \end{bmatrix}_{q^{-1}} \quad (\text{because } \theta'_i = \lambda'_i - \beta'_i \text{ and } \varphi'_i = \beta'_i - \alpha'_i).
\end{aligned} \tag{1.124}$$

This completes the proof of the theorem.

Now, we will use this theorem to compute structure constants of the form $G^{\lambda}_{\beta(1^m)}$. To do this, we need to make some observations about the previous theorem. First, the condition $\alpha \subset \beta$ is necessary, because of the lemma (1.4.16). Also, if we take $N = S$, the socle of M, we get that $L \subset S$ if and only if L is elementary. Then $L \subset S$ implies that L is elementary and it is of type (1^m) for some $m \geq 1$. In this case, L exists only if $|\lambda| = |\beta| + |(1^m)| \iff m = |\lambda| - |\beta|$. Also, since S is the socle of M, we get that $\alpha = \tilde{\lambda} = (\lambda_1 - 1, \lambda_2 - 1, \dots)$. But also

$$\begin{aligned}
\tilde{\lambda}'_i &= \text{Max}\{k \mid \tilde{\lambda}_k \geq i\} \\
&= \text{Max}\{k \mid \lambda_k - 1 \geq i\} \\
&= \text{Max}\{k \mid \lambda_k \geq i+1\} \\
&= \lambda'_{i+1}
\end{aligned} \tag{1.125}$$

So we have

$$\prod_{i\geq 1} \begin{bmatrix} \lambda'_i - \alpha'_i \\ \lambda'_i - \beta'_i \end{bmatrix}_{q^{-1}} = \prod_{i\geq 1} \begin{bmatrix} \lambda'_i - \lambda'_{i+1} \\ \lambda'_i - \beta'_i \end{bmatrix}_{q^{-1}} \tag{1.126}$$

Now, it remains to compute $d(\alpha,\beta,\lambda)$. Recall the notation $\theta=\lambda-\beta$ and $\varphi=\beta-\alpha=\beta-\tilde{\lambda}$. So $\varphi_i=\beta_i-\lambda_i+1=1-\theta_i$. Now, we need to calculate the sum $d(\alpha,\beta,\lambda)=\sum_{j\geq i}\varphi'_j\theta'_i=\sum_{j\geq i}\theta_j\varphi_i$. We have

$$\begin{aligned}\sum_{j\geq i}\theta_j\varphi_i &= \sum_{j\geq i}\theta_j(1-\theta_i)\\ &= \sum_{j>i}\theta_j(1-\theta_i) \quad \text{(because } \theta_i\in\{0,1\} \text{ by lemma (1.4.17))}\\ &= \sum_{j>i}\theta_j-\sum_{j>i}\theta_j\theta_i\\ &= \sum_{i\geq 1}(i-1)\theta_i-\frac{(\sum_i\theta_i)^2-\sum_i\theta_i^2}{2}\\ &= n(\lambda)-n(\beta)-\frac{m^2-m}{2} \quad \text{(because } m=|\theta| \text{ and } \theta_i^2=\theta_i)\\ &= n(\lambda)-n(\beta)-n((1^m)).\end{aligned} \tag{1.127}$$

So we get the next result.

Theorem 1.4.19. [12] Let M be a finite $R-$module of type λ. Then, for any elementary submodule $L\subset M$ of cotype β, the skew diagram $\lambda-\beta$ is a vertical strip and the number of such modules which are of type (1^m) is

$$G^{\lambda}_{\beta(1^m)}=q^{n(\lambda)-n(\beta)-n((1^m))}\prod_{i\geq 1}\begin{bmatrix}\lambda'_i-\lambda'_{i+1}\\ \lambda'_i-\beta'_i\end{bmatrix}_{q^{-1}}.$$

Theorem (1.4.19) implies that the structure constants $G^{\lambda}_{\beta(1^m)}$ are determined by the following "universal" polynomial

$$f^{\lambda}_{\beta(1^m)}(t):=t^{n(\lambda)-n(\beta)-n((1^m))}\prod_{i\geq 1}\begin{bmatrix}\lambda'_i-\lambda'_{i+1}\\ \lambda'_i-\beta'_i\end{bmatrix}_{t^{-1}}. \tag{1.128}$$

That is, $G^{\lambda}_{\beta(1^m)}=f^{\lambda}_{\beta(1^m)}(q)$ for any ring R with residue field of cardinality q. This, and theorem (1.4.13) show that the structure constants depend only on the cardinality of the residue field of R. Also, it is possible to explicitly determine all structure constants $G^{\lambda}_{\mu\nu}$ (see [12] chapter II).

Theorem (1.4.19) will be useful to show that the space of characters of $GL_n(\mathbb{F}_q)$ is a certain infinite tensor product of Hall algebras, which we do in the next chapter.

Chapter 2

Conjugacy classes of $GL_n(\mathbb{F}_q)$ and its space of characters

2.1 Finite fields

2.1.1 Basic definitions and theorems

Definition 2.1.1. [11] Let R be a ring. So $(R, +)$ is an abelian group, and hence a $\mathbb{Z}-$module. If there exists $n \in \mathbb{Z}$ such that $nx = 0$ for all $x \in R$ we say that R is of *positive characteristic*. In this case, the smallest positive integer n such that $nx = 0$ for all $x \in R$ is called the *characteristic* of R.

We make an observation that $nx = 0$ for all $x \in R$ if and only if $n1 = 0$.

Theorem 2.1.2. [11] A ring R of positive characteristic having no zero divisors must have prime characteristic.

Proof: By hypothesis there exists a smallest $n \geq 1$ such that $nx = 0$ for all $x \in R$. If n is not prime, then $n = md$ for some $m, d > 1$. In this case we have $(m1)(d1) = md1 = n1 = 0$. Since R has no zero divisors, we have $m1 = 0$ or $d1 = 0$. In any case we have a contradiction with the minimality of n.

Theorem 2.1.3. For every prime $q > 1$, there is an unique field (up to isomorphism) $\mathbb{F}_q$ of cardinality q.

Proof: For every prime $q > 1$, (q) is a prime ideal and $\mathbb{Z}/(q)$ is a finite integral domain with q elements. By theorem (1.1.31) $\mathbb{Z}/(q)$ is a field. For uniqueness, let $\mathbb{F}$ be a field of prime cardinality q. Now, the homomorphism $f : \mathbb{Z} \to \mathbb{F} : m \mapsto m1$ has kernel (q), by

the definition of the characteristic of $\mathbb{F}$. Therefore, we have an injective homomorphism $\mathbb{Z}/(q) \hookrightarrow F$. Since $|\mathbb{F}| = q$, this homomorphism is also a surjection, so it is in fact an isomorphism. Therefore $\mathbb{F} \cong \mathbb{Z}/(q)$.

Notation 2.1.4. For each prime q we will write $\mathbb{F}_q := \mathbb{Z}/(q)$ to be the unique field (up to isomorphism) of cardinality q.

Theorem 2.1.5. [11] Let F be a finite field. So it's of prime characteristic $q > 1$ and there is an injection $\mathbb{F}_q \hookrightarrow F$. In this case $|F| = q^{[F:\mathbb{F}_q]}$.

Proof: Let $f : \mathbb{Z} \to F : m \mapsto m1$, then clearly $\mathrm{Ker}(f) = (q)$. So we have an injection $\mathbb{F}_q = \mathbb{Z}/(q) \hookrightarrow F$. For this reason, we can think of $\mathbb{F}_q \subset F$. So F is a vector space over $\mathbb{F}_q$ of dimension $\dim_{\mathbb{F}_q}(F) = [F : \mathbb{F}_q] \Rightarrow F \cong \mathbb{F}_q^{[F:\mathbb{F}_q]}$. Therefore $|F| = q^{[F:\mathbb{F}_q]}$.

Theorem 2.1.6. Let F be a field and $f \in F[x]$ a polynomial of degree n. Then there exists a unique field E (up to isomorphism), such that $f(x) = \prod_{i=1}^{n}(x - r_i)$ in E while $E = F(r_1, \ldots, r_n)$. In this case, we write $E = \mathrm{Split}_F(f)$ and call it the *splitting* field of f over F.

Proof: See ([6], Chapter IV).

Theorem 2.1.7. [11] For every prime q and integer $n > 0$ there is a unique (up to isomorphism) field $\mathbb{F}$ of cardinality q^n.

Proof: (Existence) Let $f = x^{q^n} - x \in \mathbb{F}_q[x]$ and $E = \mathrm{Split}_{\mathbb{F}_q}(f)$ be the splitting field of f over $\mathbb{F}_q$. Since $f' = -1 \Rightarrow \gcd(f, f') = 1$. Therefore, all the roots of f in E are distinct. So $F := \{\alpha \in E \mid \alpha^{q^n} = \alpha\}$ is a subfield of E with q^n elements (it is clear that F is closed under multiplication, but is also closed under sum because $(a + b)^q = a^q + b^q$. Clearly, F if obtained from $\mathbb{F}_q$ by adjoining all the roots of $x^{q^n} - x \in \mathbb{F}_q[x]$. So $F \cong E \Rightarrow F = E$, by the uniqueness (up to isomorphisms) of splitting fields. So, the field that we are looking is exactly the splitting field of the polynomial $x^{q^n} - x$ over $\mathbb{F}_q$.

(Uniqueness) Let F be a field of cardinality q^n, so it is of prime characteristic, say p. By the theorem (2.1.5), we have $\mathbb{F}_p \subset F$ and $p^d = q^n$ where $d = [F : \mathbb{F}_p]$. Since both q and p are prime, we have $q = p$ and $n = [F : \mathbb{F}_q]$. Now, the polynomial $x^{q^n} - x \in \mathbb{F}_q[x]$ factors in F, because $\alpha^{q^n} = \alpha$ for all $\alpha \in F$. Therefore, we have the decomposition

$$x^{q^n} - x = \prod_{\alpha \in F} (x - \alpha). \tag{2.1}$$

Clearly F is obtained from $\mathbb{F}_q$ by adjoining all the roots of $x^{q^n} - x \in \mathbb{F}_q[x]$. So F is a

splitting field of $x^{q^n} - x$ over $\mathbb{F}_q$.

Notation 2.1.8. Let q be a prime number and $n \geq 1$ a positive integer. Then we saw that $\mathbb{F}_q := \mathbb{Z}/(q)$ and $\text{Split}_{\mathbb{F}_q}(x^{q^n} - x)$ is the unique (up to isomorphisms) field extension of $\mathbb{F}_q$ of cardinality q^n (or equivalently, the unique field extension of degree n of $\mathbb{F}_q$). So we write $\mathbb{F}_{q^n} = \text{Split}_{\mathbb{F}_q}(x^{q^n} - x)$.

2.1.2 Orbits for the Frobenius map

In this subsection we will set the stage to understand how to parametrize conjugacy classes on $GL_n(\mathbb{F}_q)$.

Definition 2.1.9. [11] For any prime q and positive integer $n \geq 1$, denote the multiplicative group of $\mathbb{F}_{q^n}$ by M_n. We define the *Frobenius automorphism* of $\mathbb{F}_{q^m}$ to be the map $F : M_n \to M_n : x \mapsto x^q$.

Theorem 2.1.10. [11] Let $n, m \geq 1$ be positive integers such that $m|n$ and $q > 1$ a prime number. Then $\mathbb{F}_{q^n}$ is the unique field extension of $\mathbb{F}_{q^m}$ of degree $d = n/m$.

Proof: Since $m|n$, we have that $(q^m - 1)|(q^n - 1)$ and $(x^{q^m-1} - 1)|(x^{q^n-1} - 1) \iff (x^{q^m} - x)|(x^{q^n} - x)$. Therefore $\mathbb{F}_q \subset \text{Split}_{\mathbb{F}_q}(x^{q^m} - x) \subset \text{Split}_{\mathbb{F}_q}(x^{q^n} - x)$, i.e. $\mathbb{F}_{q^m} \subset \mathbb{F}_{q^n}$. Moreover $[\mathbb{F}_{q^n} : \mathbb{F}_{q^m}] = [\mathbb{F}_{q^n} : \mathbb{F}_q]/[\mathbb{F}_{q^m} : \mathbb{F}_q] = d$. Now, if $F|\mathbb{F}_{q^m}$ is a field extension of degree $d = n/m$, then by lemma (2.1.5), $|F| = (q^m)^d = q^n \Rightarrow F = \mathbb{F}_{q^n}$.

It's important to note in the previous proof that $\mathbb{F}_{q^m}$ is the *unique* subfield of $\mathbb{F}_{q^n}$ with q^m elements. Otherwise $x^{q^m} - x$ would have more than q^m roots in $\mathbb{F}_{q^n}$, which is a contradiction. Therefore, the subfield $\mathbb{F}_{q^m} \subset \mathbb{F}_{q^n}$ is characterized as being the set of fixed point of $F^m : x \mapsto x^{q^m}$ in $\mathbb{F}_{q^m}$. So we get the following corollary.

Corollary 2.1.11. Let q be a prime and n a positive integer. Then, for any positive integer m that divides n, the subfield $\mathbb{F}_{q^m} \subset \mathbb{F}_{q^n}$ is characterized as the set of fixed points of F^m, where F is the Frobenius automorphism of $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$.

Notation 2.1.12. We saw that a finite field necessarily have a power of a prime as its number of elements. So, from now on, q will denote a power of a prime.

Lemma 2.1.13. Let $f \in \mathbb{F}_q[x]$ be an irreducible polynomial over $\mathbb{F}_q$ of degree m. Then f divides $x^{q^n} - x \in \mathbb{F}_q[x]$ if and only if m divides n.

Proof: Suppose that f divides $x^{q^n} - x$. Then $\text{Split}_{\mathbb{F}_q}(f) \subset \text{Split}_{\mathbb{F}_q}(x^{q^n} - x) = \mathbb{F}_{q^n}$. But, for any root α of f in its splitting field over $\mathbb{F}_q$ we have $\mathbb{F}_q(\alpha) \subset \mathbb{F}_{q^n}$. Since f is irreducible of degree m, $\mathbb{F}_q(\alpha)|\mathbb{F}_q$ is a field extension of degree m and $\mathbb{F}_q(\alpha) = \mathbb{F}_{q^m}$ by lemma (2.1.5). Hence $\mathbb{F} \subset \mathbb{F}_{q^m} \subset \mathbb{F}_{q^n} \Rightarrow n = [\mathbb{F}_{q^n} : \mathbb{F}_q] = [\mathbb{F}_{q^n} : \mathbb{F}_{q^m}][\mathbb{F}_{q^m} : \mathbb{F}_q] = [\mathbb{F}_{q^n} : \mathbb{F}_{q^m}]m$, which implies that m divides n.

Conversely, suppose that m divides n. Then, if $\alpha \in \text{Split}_{\mathbb{F}_q}(f)$ is a root of f, then $\mathbb{F}_q(\alpha) = \mathbb{F}_{q^m}$. But, by theorem (2.1.10) $\mathbb{F}_{q^m} \subset \mathbb{F}_{q^n} \Rightarrow \alpha \in \mathbb{F}_{q^n}$ and α must be a root of $x^{q^n} - x \in \mathbb{F}_q[x]$. Since f is irreducible over $\mathbb{F}_q$, f must divide $x^{q^n} - x$ in $\mathbb{F}_q[x]$.

Now we will show that if $f \in \mathbb{F}_q[x]$ is irreducible of degree m, then the splitting field of f is exactly $\mathbb{F}_{q^m}$.

Theorem 2.1.14. Let $f \in \mathbb{F}_q[x]$ be an irreducible polynomial over $\mathbb{F}_q$ of degree m. Then, if α is a root in the splitting field of f over $\mathbb{F}_q$, we get that

$$f = \prod_{i=0}^{m-1} (x - \alpha^{q^i}) \tag{2.2}$$

in its splitting field. Moreover, all the roots $\{\alpha, \ldots, \alpha^{q^m-1}\}$ are distinct (with $\alpha^{q^m} = \alpha$), and the splitting field of f over $\mathbb{F}_q$ is $\mathbb{F}_{q^m}$.

Proof: Let α be a root of f in the splitting field of f over $\mathbb{F}_q$. Then $\mathbb{F}_q(\alpha) = \mathbb{F}_{q^m} \subset \text{Split}_{\mathbb{F}_q}(f)$. But we have that $f(\beta^q) = f(\beta)^q$ for any $\beta \in \text{Split}_{\mathbb{F}_q}(f)$.

Therefore $\{\alpha, \alpha^q, \ldots, \alpha^{q^{m-1}}\}$ are all roots of f. We claim that these roots are distinct. Suppose that $\alpha^{q^i} = \alpha^{q^j}$ for some i and j such that $1 \leq i < j < m$, then $\alpha^{q^{m+i-j}} = (\alpha^{q^i})^{q^{m-j}} = (\alpha^{q^j})^{q^{m-j}} = \alpha^{q^m} = \alpha$ because $\alpha \in \mathbb{F}_{q^m}$. But, since f is irreducible, f must divide $x^{q^{m+i-j}} - x$. By lemma (2.1.13), this happens only if $m|(m+i-j)$, which is impossible because $m + i - j < m$. Therefore, since $\mathbb{F}_q(\alpha) = \mathbb{F}_q(\alpha, \ldots, \alpha^{q^{n-1}})$, we see that $\mathbb{F}_{q^m} = \mathbb{F}_q(\alpha) = \text{Split}_{\mathbb{F}_q}(f)$.

Lets summarize what we have done so far. We know that for each prime q and positive integer $n \geq 1$, there is a field $\mathbb{F}_{q^n}$ of cardinality q^n. These fields are uniquely defined (up to isomorphism). Moreover, for each $m|n$ we have an inclusion $\mathbb{F}_{q^m} \subset \mathbb{F}_{q^n}$. In particular, this shows that

$$F := \bigcup_{n \geq 1} \mathbb{F}_{q^{n!}} = \bigcup_{n \geq 1} \mathbb{F}_{q^n} \tag{2.3}$$

is a field. Moreover, by the theorem (2.1.14), for each irreducible polynomial $f \in \mathbb{F}_q[x]$ of degree m, the splitting field of f is exactly $\mathbb{F}_{q^m}$. This shows that each polynomial over $\mathbb{F}_q$ has a root in F and $F = \overline{\mathbb{F}}_q$ must be the algebraic closure of $\mathbb{F}_q$.

Notation 2.1.15. The multiplicative group of $\overline{\mathbb{F}}_q$ will be denoted by M.

It follows from the theorem (2.1.14) that each irreducible polynomial over $\mathbb{F}_q$ gives an orbit of the action of the Frobenius automorphism on M. Conversely, let $\{\alpha, \ldots, \alpha^{q^{n-1}}\}$ (where n is the smallest positive integer such that $\alpha^{q^n} = \alpha$) be an orbit of the Frobenius automorphism on M. We claim that the polynomial

$$f := \prod_{i=0}^{n-1} (x - \alpha^{q^i}) \in \mathbb{F}_{q^n}[x] \tag{2.4}$$

has coefficients in $\mathbb{F}_q$. In fact, extend the Frobenius automorphism F over $\mathbb{F}_{q^n}$ to $\mathbb{F}_{q^n}[x]$ in the natural way, i.e.

$$F\left(\sum_{i=0}^{n} a_i x^i\right) := \sum_{i=0}^{n} F(a_i) x^i. \tag{2.5}$$

Then

$$\begin{aligned} F(f) &= \prod_{i=0}^{n-1} (x - F(\alpha^{q^i})) \\ &= \prod_{i=0}^{n-1} (x - \alpha^{q^{i+1}}) \\ &= \prod_{i=0}^{n-1} (x - \alpha^{q^i}) \quad \text{because } \alpha^{q^n} = \alpha. \end{aligned} \tag{2.6}$$

So, all the coefficients of f are fixed by F. By corollary (2.1.11), we see that all the coefficients of f must belong to $\mathbb{F}_q$ and $f \in \mathbb{F}_q[x]$. Now, suppose that $f_1 \in \mathbb{F}_q[x]$ is an irreducible component of f with α as one of its roots. Then, by theorem (2.1.14), f_1 has $d(f_1)$ distinct roots $\{\alpha, \ldots, \alpha^{q^{d(f_1)-1}}\} \subset \mathbb{F}_{q^n}$ where $\alpha^{q^{d(f_1)}} = \alpha$. So $d(f_1)$ is the smallest positive integer such that $\alpha^{q^{d(f_1)}} = \alpha \Rightarrow d(f_1) = n$ and $f = f_1$. Therefore, f is irreducible and we deduce the next theorem.

Theorem 2.1.16. Let Φ denote the set of $F-$orbits in M. Then, there is a one to one correspondence between irreducible polynomials over $\mathbb{F}_q$ and elements of Φ. Moreover, for each irreducible polynomial f of degree m, the corresponding orbit induced by f has m elements. For this reason, we will identify Φ with the set of irreducible polynomials over $\mathbb{F}_q$.

Corollary 2.1.17. [12] Let Φ_n be the set of $F-$orbits in M_n, then $\Phi_n = \{f \in \Phi \mid d(f)|n\}$.

Proof: This follows directly from the fact that $\mathbb{F}_{q^{d(f)}}$ is the splitting field of f (by theorem (2.1.14)) and $\mathbb{F}_{q^{d(f)}} \subset \mathbb{F}_{q^n}$ if and only if $d(f)|n$ (by the theorem (2.1.10)).

Theorem 2.1.18. Let $\mathbb{F}$ be a finite field. Then its multiplicative group is cyclic.

Proof: Since $G := \mathbb{F}^\times$ is a finite abelian group, it is a finite $\mathbb{Z}-$module. By corollary (1.1.53) we have

$$G \cong \frac{\mathbb{Z}}{(p_1^{m_1})} \oplus \cdots \oplus \frac{\mathbb{Z}}{(p_k^{m_k})} \tag{2.7}$$

for (not necessarily distinct) primes $p_i,\ i = 1, \ldots, k$. We claim that all p_i's are distinct. Let $L := \text{lcm}(p_1^{m_1}, \ldots, p_k^{m_k})$, then $a^L = 1$ for all $a \in G \Rightarrow G \subset \{\text{roots of } x^L - 1 \in \mathbb{F}[x]\}$. But $x^L - 1$ is of degree L. So $|G| \leq L \leq p_1^{m_1} \cdots p_k^{m_k} = |G|$. This shows that $L = p_1^{m_1} \cdots p_k^{m_k}$ and thus all p_i's must be distinct. By lemma (1.1.51), we get that $G \cong \mathbb{Z}/(L)$ where $L = |G| = |\mathbb{F}| - 1$.

Lemma 2.1.19. [12] Let n, m be positive integers such that $m|n$. Then the norm map $N_{n,m} : M_n \to M_m$ is defined by $x \mapsto \prod_{i=0}^{d-1} F^{mi}x$, where $d = n/m$. The norm map is a surjective homomorphism.

Proof: By theorem (2.1.18), we have $M_n = \langle\alpha\rangle$ for some $\alpha \in \mathbb{F}_{q^n}$. Let $\beta = \alpha^{(q^n-1)/(q^m-1)}$, then β has order $q^m - 1$ which is equivalent to $F^m(\beta) = \beta$. So $\beta \in M_m$ by corollary (2.1.11) and $M_m = \langle\beta\rangle$. But

$$N_{n,m}(\alpha) = \prod_{i=0}^{d-1} F^{mi}(\alpha) = \alpha\alpha^{q^m} \cdots \alpha^{q^{m(d-1)}} = \alpha^{1+q^m+\cdots+q^{(d-1)m}} = \beta, \tag{2.8}$$

which shows that $N_{n,m}$ maps a generator of M_n to a generator of M_m, so its must be surjective.

Let $M_n^* := \text{Hom}(M_n, \mathbb{C})$. By the previous lemma we have an embedding $N_{n,m}^* : M_m^* \hookrightarrow M_n^*$ whenever $m|n$. So the family $\{M_n^*\}_{n\geq 1}$ together with the monomorphisms $N_{n,m}^*$ for $m|n$, forms a direct system. So we have the direct limit $L := \varinjlim M_n^*$, with canonical embeddings $\varphi_n : M_n^* \hookrightarrow L : \xi \mapsto [\xi]$. Remember

$$L := \left(\bigcup_{n\geq 1} M_n^*\right) \Big/ \sim, \tag{2.9}$$

where if $\eta \in M_i^*$ and $\xi \in M_j^*$, then $\eta \sim \xi \iff N_{ki}^*(\eta) = N_{kj}^*(\xi)$ for some k such that $\text{lcm}(i, j)|k$. Also, the Frobenius map acts on M_n^* by $F\xi := \xi \circ F \iff F\xi = \xi^q$. Therefore, we see that $\eta \sim \xi \Rightarrow F(\eta) \sim F(\xi)$, so the action $F[\xi] := [F(\xi)]$ is well defined. We claim that M_n^* can be identified with L_n, the F^n-invariant space in L, through the embedding φ_n. Let $[\xi] \in M^*$ be such that $F^n[\xi] = [\xi] \Rightarrow [\xi^{q^n}] = [\xi]$. Let $i = o(\xi)$, the order of ξ, then $\langle\xi\rangle = M_i^*$.

By definition there exists $j \geq 1$ such that $N^*_{j,i}(\xi^{q^n}) = N^*_{j,i}(\xi) \Rightarrow \xi^{q^n} = \xi \Rightarrow \xi^{q^n-1} = 1$ and hence $(q^i - 1)|(q^n - 1) \Rightarrow i|n$. Therefore $N^*_{n,i}(\xi) \in M^*_n$ and $\varphi_n(N^*_{n,i}(\xi)) = [\xi]$. Moreover $L_m \subset L_n$ if and only if $m|n$. In fact, if $m|n$, then we have the diagram

$$\begin{array}{ccc} M_m & \xrightarrow{N^*_{n,m}} & M_n \\ \downarrow{\scriptstyle \varphi_m} & & \downarrow{\scriptstyle \varphi_n} \\ L_m & \dashrightarrow & L_n \end{array}$$

Since $N^*_{n,m}(\xi) \sim \xi \ \forall \xi \in M^*_m$, we get an inclusion $\varphi_n N^*_{n,m} \varphi_m^{-1} : L_m \hookrightarrow L_n$. Conversely, if $L_m \subset L_n \Rightarrow |L_m|$ divides $|L_n|$, i.e. $(q^m - 1)|(q^n - 1) \Rightarrow m|n$. Then we get the next theorem.

Theorem 2.1.20. [12] Let L be the direct limit of the directed system $(M^*_n, N^*_{n,m})$. Then, the Frobenius automorphism defines an action on L through the map $[\xi] \mapsto [F(\xi)]$. Also denote by L_n the F^n-invariant subgroup of L, then

(1) $L = \bigcup_{n\geq 1} L_n$ where $L_n \cong M^*_n$ for each $n \geq 1$.

(2) $L_m \subset L_n \iff m|n$.

Let G be a group acting on a set S. For $s \in S$, $G_s := \{g \in G \,|g * s = s\}$ is a subgroup of G called the *stabilizer* of s. Also, the orbit of $s \in S$ by the action of G is defined to be the set $O_s := \{s' \in S \mid s' = gs \text{ for some } g \in G\}$. With these definitions, the orbit-stabilizer theorem asserts

$$G/G_s \to O_s : gG_s \mapsto g * s \tag{2.10}$$

Particularly, if G is a finite group, then $|O_s| = [G : G_s] = |G|/|G_s| \Rightarrow |G| = |O_s||G_s|$. This last expression is called the *counting formula* for the action of G on S.

Theorem 2.1.21. [12] Let Φ^* (resp. Φ^*_n) be the set of $F-$orbits in L (resp. L_n). If $d(\phi)$ denotes the cardinality of $\phi \in \Phi^*$, then $\Phi^*_n = \{\phi \in \Phi^* \mid d(\phi)|n\}$.

Proof: For each $n \geq 1$, the cyclic group $G := \langle F|_{L_n} \rangle$ is of cardinality n and we have an action $G \times L_n \to L_n$. The set of orbits induced by this action is Φ^*_n and, by the counting formula for the action of G on L_n, $d(\phi)|n$ for each orbit $\phi \in \Phi^*_n$. Conversely, if $\phi \in \Phi^*$ and $d(\phi)|n$, then since each $[\xi] \in \phi$ is $F^{d(\phi)}-$invariant, we have $[\xi] \in L_{d(\phi)} \subset L_n \Rightarrow [\xi] \in L_n$ and $\phi \in \Phi^*_n$.

Definition 2.1.22. [12] Define a pairing between M^*_n and M_n by $\langle \xi, x \rangle_n = \xi(x)$.

It's important to note that if $m|n$, $x \in M_m$ and $\xi \in M^*_m$, then $\langle N^*_{n,m}(\xi), x \rangle_n =$

$(\langle \xi, x\rangle_m)^{\frac{n}{m}}$. So, the pairings $\langle \cdot, \cdot \rangle_n$ are not compatible with the embeddings

$$N^*_{n,m} \times \iota : M^*_m \times M_m \hookrightarrow M^*_n \times M_n : (\xi, x) \mapsto (N^*_{n,m}(\xi), x). \tag{2.11}$$

Definition 2.1.23. [9] Let $n \geq 1$, $\phi \in \Phi^*$ and $f \in \Phi$. If $d(\phi), d(f)|n$, we define

$$(\phi, f)_n = \frac{1}{d(\phi)} \sum_{\xi \in \phi} \langle \xi, x\rangle_n$$

where $x \in f$ is fixed, and

$$(\phi, f)_n = 0$$

otherwise. In the first case, the expression does not depend on the choice of $x \in f$.

If both $d(\phi), d(f)|n \Rightarrow \phi \in \Phi^*_n$ and $f \in \Phi_n$. So $\langle \xi, F^k x\rangle_n = \xi(F^k x) = (F^k \xi)(x)$. Therefore $\langle \xi, F^k x\rangle_n = \langle F^k \xi, x\rangle_n$ and $\sum_{\xi \in \phi} \langle F^k \xi, x\rangle_n = \sum_{\xi \in \phi} \langle \xi, x\rangle_n$. This proves the independence of the choice of $x \in f$.

Lemma 2.1.24. [9] In the context of definition (2.1.23). If $d(\phi), d(f)|n$, then

$$(\phi, f)_n = \frac{1}{d(f)} \sum_{x \in f} \langle \xi, x\rangle_n$$

where $\xi \in \phi$ is fixed. Futhermore this expression does not depend on the choice of ξ.

Proof: Let $g(i) = \langle \xi^{q^i}, x\rangle_n$. Then g is $d([\xi])-$periodic. So

$$\sum_{i=0}^{n-1} g(i) = \sum_{j=0}^{\frac{n}{d([\xi])}-1} \sum_{i=0}^{d([\xi])-1} g(i + jd([\xi])) = \frac{n}{d([\xi])} \sum_{i=0}^{d([\xi])-1} g(i) \tag{2.12}$$

which is equivalent to

$$\frac{1}{n} \sum_{i=0}^{n-1} \langle \xi^{q^i}, x\rangle_n = \frac{1}{d([\xi])} \sum_{i=0}^{d([\xi])-1} \langle \xi^{q^i}, x\rangle_n = ([\xi], [x])_n \tag{2.13}$$

In the same way we can show that

$$\frac{1}{n} \sum_{i=0}^{n-1} \langle \xi^{q^i}, x\rangle_n = \frac{1}{d(f)} \sum_{i=0}^{d(f)-1} \langle \xi, x^{q^i}\rangle_n. \tag{2.14}$$

Comparing the last two equations, we get the result.

2.1.3 Irreducible polynomials over $\mathbb{F}_q$

We have seen in the proof of theorem (2.1.14) that if $f \in \mathbb{F}_q[x]$ is irreducible of degree m, then $\mathbb{F}_{q^m}$ is obtained from $\mathbb{F}_q$ by adjunction of one of the roots of f. In this section, we will prove that for each $m \geq 1$ there is an irreducible polynomial f of degree m, so we can construct $\mathbb{F}_{q^m}$ explicitly, provided we have an algorithm to find these polynomials.

Theorem 2.1.25. For each $m \geq 1$ there exists an irreducible polynomial f of degree m over $\mathbb{F}_q$.

Proof: Let $\mathbb{F}_{q^m} := \text{Split}_{\mathbb{F}_q}(x^{q^m} - x)$. We know that $M_m = \mathbb{F}_{q^m}^{\times}$ is cyclic because of lemma (2.1.18). Therefore, there is a generator $\alpha \in M_n$ which is a root of $x^{q^m} - x \in \mathbb{F}_q[x]$, so it must be a root of some irreducible polynomial $f \in \mathbb{F}_q[x]$ (by unique factorization in $\mathbb{F}_q[x]$). We claim that $d(f) = m$, in fact $\mathbb{F}_q(\alpha) = \mathbb{F}_{q^m}$ (because $\mathbb{F}_{q^m} = \{0\} \cup \langle\alpha\rangle$) and $\mathbb{F}_q(\alpha) = \mathbb{F}_{q^m}$. By theorem (2.1.10), $\mathbb{F}_q(\alpha)|\mathbb{F}_q$ is an extension of degree m, so f must have degree m.

Now we know that for each integer $m \geq 1$, there is a irreducible polynomial f of degree m over $\mathbb{F}_q$. But the proof of the previous theorem is somewhat nonconstructive and it would be interesting to find f, so $\mathbb{F}_{q^m}$ can be realized as $\mathbb{F}_q[x]/(f)$, where calculations can be carried out explicitly. Said this, we must find a criteria to determine if some polynomial over $\mathbb{F}_q$ is irreducible or not. To do this observe that if $f \in \mathbb{F}_q[x]$ and $\gcd(f, f') \neq 1$, i.e., is non-constant, then f must be reducible. So we will just consider the cases where $\gcd(f, f') = 1$, i.e. all the irreducible components of f are distinct.

Lemma 2.1.26. [5] Let f be a polynomial over a field $\mathbb{F}$. Suppose that f factors in $\mathbb{F}$ as a product of pairwise coprime factors $g_1, \ldots, g_m$. Then

$$\theta : \frac{\mathbb{F}[x]}{(f)} \to \bigoplus_{j=1}^{m} \frac{\mathbb{F}[x]}{(g_j)} : h + (f) \mapsto (h + (g_1), \ldots, h + (g_m)) \tag{2.15}$$

is an isomorphism or rings.

Proof: Lets write $R = \mathbb{F}[x]/(f)$, $R_j = \mathbb{F}[x]/(g_j)$ and $\overline{h} = h + (f) \in R$. Since the product on $\bigoplus_{j=1}^m R_j$ is given pointwise on each component, θ is clearly a homomorphism. Now, $\theta(\overline{h}) = 0 \Rightarrow h \in (g_j)\ \forall j$ and since the factors g_j are pairwise coprime, we conclude that $h \in (f) \Rightarrow \overline{h} = 0$, so θ is injective. Therefore $\theta(R)$ (viewed as a vector space over $\mathbb{F}$) is of dimension $\dim_{\mathbb{F}}(R) = d(f)$. But, viewing $\bigoplus_{j=1}^m R_j$ as a vector space over $\mathbb{F}$ we get $\dim_{\mathbb{F}}(\bigoplus_{j=1}^m R_j) = \sum_{j=1}^m \dim_{\mathbb{F}}(R_j) = \sum_{j=1}^m d(g_j) = d(f)$, the same dimension as the image of θ, proving the surjectivity of θ. So we conclude that θ is an isomorphism of rings.

Definition 2.1.27. [5] Let f be a polynomial over $\mathbb{F}_q$. We say that $h \in \mathbb{F}_q[x]$ is a factorization polynomial for f if f divides the polynomial $\prod_{a\in\mathbb{F}_q}(h-a)$.

For example, for any polynomial $f \in \mathbb{F}_q[x]$, a degree zero polynomial is always a factorization polynomials for f. Also, f is clearly a factorization polynomial for f. Much more interesting, is the case where we have a factorization polynomial h for f such that $0 < d(h) < d(f)$. We will see in the next theorem, an algorithm to produce such polynomials. Before we do that, a little observation: Let $f \in \mathbb{F}_q[x]$ be a polynomial with distinct irreducible factors $g_j,\ j = 1,\dots,m$ and h a factorization polynomial for f. Since f divide the product $\prod_{a\in\mathbb{F}_q}(h-a)$, each irreducible factor g_j must divide $h-a$ for some $a \in \mathbb{F}_q$. But the polynomials $h-a,\ a \in \mathbb{F}_q$ are pairwise coprime and hence, for each j there is a unique $a_j \in \mathbb{F}_q$ such that $g_j|(h-a_j)$. Hence each factorization polynomial h uniquely determine a list $(a_1,\dots,a_m) \in \mathbb{F}_q^m$ with the property that $g_j|(h-a_j)\ \ \forall j = 1,\dots,m$.

Therefore we have a map

$$\rho : h \mapsto (a_1,\dots,a_m) \tag{2.16}$$

which maps factorization polynomials for f to vectors in $\mathbb{F}_q^m$.

Theorem 2.1.28. [5] Suppose that $f \in \mathbb{F}_q[x]$ is a polynomial with distinct irreducible factors $g_j,\ j = 1,\dots,m$. Let $R = \mathbb{F}_q[x]/(f),\ L_j = \mathbb{F}_q[x]/(g_j),\ \psi : R \to R : v \mapsto v^q$ (a linear map over $\mathbb{F}_q$) and $U := \mathrm{Ker}(\psi - I_R)$. We claim that if $\overline{h} \in U$, then h is a factorization polynomial for f and the map $\varphi : U \to \mathbb{F}_q^m : \overline{h} \mapsto \rho(h)$ is a well defined isomorphism. In particular, the number of irreducible factors of f is $m = \dim_{\mathbb{F}_q}(U)$.

Proof: By theorem (2.1.26), we have a (ring) isomorphism

$$\theta : R \to L_1 \oplus \cdots \oplus L_m \tag{2.17}$$

Write $L = \bigoplus_{j=1}^m L_j$ and consider the projections $p_j : L \to L_j$. Now, since $p_j \circ \theta|_{\mathbb{F}_q} : \mathbb{F}_q \to L_j$ is a nonzero ring homomorphism and $\mathbb{F}_q$ is field, we get that $E_j := p_j \circ \theta(\mathbb{F}_q)$ is a field isomorphic to $\mathbb{F}_q$. Therefore, by corollary (2.1.11), $E_j \subset L_j$ is characterized as being the subset fixed by the action of the Frobenius map $x \mapsto x^q$. Since $\overline{h}^q = \overline{h}$ whenever $\overline{h} \in U$, we have $p_j \circ \theta(\overline{h}) = h + (g_j) \in E_j$. But we also have that $p_j \circ \theta(\mathbb{F}_q) = (\mathbb{F}_q + (g_j))/(g_j)$. So, for each j, there is a unique $a_j \in \mathbb{F}_q$ such that $h + (g_j) = a_j + (g_j) \Rightarrow g_j|(h-a_j)$, which implies that h is a factorization polynomial for f and $\rho(h) = (a_1,\dots,a_m)$.

From its definition, it is clear that the map $\varphi : \overline{h} \mapsto \rho(h)$ is a vector space homomorphism. Let's show that φ is in fact a vector space isomorphism. To see injectivity, suppose that $\rho(h) = 0$, then $h + (g_j) = 0 + (g_j)$ ($\iff\ h \in (g_j)$) for each $j = 1,\dots,m$. Since, the factors g_j are pairwise coprime, we get that their product divides h, i.e. $f|h \Rightarrow \overline{h} = 0$.

Now let $(a_1,\dots,a_m) \in \mathbb{F}_q^m$ and $v = (a_1 + (g_1),\dots,a_m + (g_m)) \in L$. Since θ is surjective, there exists $\overline{h} \in R$ such that $\theta(\overline{h}) = v$. But, we know that

$$v^q = (a_1^q + (g_1),\dots,a_m^q + (g_m)) = (a_1 + (g_1),\dots,a_m + (g_m)) = v \tag{2.18}$$

so $\theta(\bar{h}^q) = \theta(\overline{h}) \Rightarrow \bar{h}^q = \overline{h}$ and $\overline{h} \in U$. This shows that φ is surjective and we conclude the proof.

From this result, we can already decide if a polynomial with distinct factors is irreducible or not, we just need to compute the dimension of U.

Example 2.1.29. The polynomial $x^4+2 \in \mathbb{F}_3[x]$ has two irreducible factors, therefore is not irreducible.

In this case $R = \mathbb{F}_3[x]/(x^4+2)$ is a vector space of dimension 4 over $\mathbb{F}_3$. It has a basis $\{\overline{1}, \overline{x}, \overline{x}^2, \overline{x}^3\}$. So $\psi(\overline{1}) = \overline{1}$ and $\psi(\overline{x}) = \overline{x}^3$. To compute $\psi(\overline{x}^2) = \overline{x}^6$, we just divide x^6 by x^4+2 using long division, to get the remainder x^2. So $\psi(\overline{x}^2) = \overline{x}^6 = \overline{x}^2$ and similarly for $\overline{x}^3$ we compute $\psi(\overline{x}^3) = \overline{x}^9 = \overline{x}$. Therefore, the matrix of ψ is

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \Rightarrow M - I = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \end{pmatrix} \tag{2.19}$$

So, the kernel of $M - I$ is the span of the set $\{(a, b, c, b) \mid a, b, c \in \mathbb{F}_3\}$ which corresponds to $U = \text{Span}\{\overline{1}, \overline{x}^2, \overline{x}^3 + \overline{x}\}$ in R. By the previous theorem, all the polynomials $1, x^2$ and $x^3 + x$ are factorization polynomials for $x^4 + 2$ over $\mathbb{F}_3$. Since $\dim_{\mathbb{F}_3}(U) = 3$, the polynomial $x^4 + 2 \in \mathbb{F}_3[x]$ must have three irreducible factors, particularly it isn't irreducible.

2.2 Structure of $GL_n(\mathbb{F}_q)$

2.2.1 Parametrization of conjugacy classes

The purpose of this section is to parametrize the conjugacy classes of $GL_n(\mathbb{F}_q)$. To do this, given $g \in GL_n(\mathbb{F}_q)$, we define an $\mathbb{F}_q[t]-$module V_g as follows: $V_g = \mathbb{F}_q^n$ with the action $t \cdot v := gv$ for $v \in \mathbb{F}_q^n$. Therefore, V_g is a finitely generated module over the principal ideal domain $\mathbb{F}_q[t]$, and we can apply Corollary (1.1.53) to get

$$V_g \cong \bigoplus_{f \in \Phi} \bigoplus_{i \geq 1} \mathbb{F}_q[t]/(f)^{\mu_i(f)} \tag{2.20}$$

Here we have used the fact that each element $h \in \mathbb{F}_q[t]$ decomposes uniquely (up to $\mathbb{F}_q$) as a product of irreducible factors $h_1, h_2, \cdots h_n$, which is a consequence of the euclidean division algorithm. Observe that

$$n = \dim_{\mathbb{F}_q}(V_g) = \sum_{f \in \Phi} \sum_{i \geq 1} d(f)\mu_i(f) = \sum_{f \in \Phi} d(f)|\mu(f)| \tag{2.21}$$

Therefore μ is a *weighted* partition valued function, i.e., it's a map

$$\mu : \Phi \to \mathcal{P} \tag{2.22}$$

satisfying $||\mu|| < \infty$ where $||\mu|| := \sum_{f\in\Phi} d(f)|\mu(f)|$ is called the *weight* of μ. Also, since t acts on V_g as an isomorphism we necessarily have $\mu(t) = 0$.

Theorem 2.2.1. $V_g \cong V_h \iff g$ and h belongs to the same conjugacy class.

Proof: We have $V_g \cong V_h$ if and only if there exists an isomorphism $T : V_g \to V_h$ such that $T(t \cdot v) = t \cdot (Tv) \iff T(gv) = h(Tv)$ for all $v \in \mathbb{F}_q^n$. Since at the level of vector spaces $V_g = V_h = \mathbb{F}_q^n$, we have that T has a matrix representation $x \in GL_n(\mathbb{F}_q)$. Therefore $x(gv) = h(xv)$ for all $v \in \mathbb{F}_q^n \Rightarrow xg = hx$, $x \in GL_n(\mathbb{F}_q) \Rightarrow g$ and h are conjugates. Conversely, if there exists $x \in GL_n(\mathbb{F}_q)$ such that $xg = hx$, then $x : V_g \to V_h : v \mapsto xv$ is an isomorphism of $\mathbb{F}_q[t]-$modules.

Lemma 2.2.2. Let $f, h \in \Phi$, $f \neq h$ and $m \geq 1$ a positive integer. Then the map

$$\varphi_h : \mathbb{F}_q[t]/(f^m) \to \mathbb{F}_q[t]/(f^m) : \overline{u} \mapsto h\overline{u} \tag{2.23}$$

is an automorphism of $\mathbb{F}_q[t]/(f^m)$.

Proof: Suppose that $h\overline{u} = 0$, then $hu \in (f^m) \Rightarrow f^m|hu \Rightarrow f^m|u$ because $\gcd(f, h) = 1$. So $\overline{u} = 0$ and φ_h is an injective endomorphism of $\mathbb{F}_q[t]/(f^m)$, a finite dimensional vector space, then φ_h is also automorphism of $\mathbb{F}_q[t]/(f^m)$.

Definition 2.2.3. Let μ be a weighted partition valued function. We will define the set $\mathrm{Supp}(f)$, called the *support* of μ, to be the set of $f \in \Phi$ such that $\mu(f) \neq 0$.

Theorem 2.2.4. There is a one-to-one correspondence between weighted partition valued functions $\mu : \Phi \to \mathcal{P}$ such that $||\mu|| = n$ and $\mu(t) = 0$, and conjugacy classes c_μ in $GL_n(\mathbb{F}_q)$.

Proof: A direct consequence of the Theorem (2.2.1), is that there is a one to one correspondence between conjugacy classes c of $GL_n(\mathbb{F}_q)$ and $\mathbb{F}_q[t]-$modules V_c such that t acts on V_c as an isomorphism. By equation (2.21) we see that each conjugacy class gives a weighted partition function $\mu : \Phi \to \mathcal{P}$ such that $||\mu|| = n$ and $\mu(t) = 0$.

Conversely, each weighted partition valued function $\mu : \Phi \to \mathcal{P}$ such that $||\mu|| = n$ and $\mu(t) = 0$ determines a $\mathbb{F}_q[t]-$module $V_\mu = \bigoplus_{f\in\Phi} \bigoplus_{i\geq 1} \mathbb{F}_q[t]/(f)^{\mu_i(f)}$. Furthermore, the action of t on V_μ is defined by the map $\bigoplus_{f\in\mathrm{Supp}(\mu)} \varphi_t$. So, by the lemma (2.2.2), t acts on V_μ as an isomorphism of vector spaces. Therefore μ can be identified with a conjugacy class c_μ on $GL_n(\mathbb{F}_q)$.

Notation 2.2.5. Because of the previous theorem, we will write V_μ for the isomorphism class of modules V_g, for $g \in c_\mu$. Of course, by construction V_μ has decomposition as in the equation (2.20).

Now is a good time to give some examples. Let's parametrize the conjugacy classes of $GL_2(\mathbb{F}_2)$ using the machinery built up so far.

Example 2.2.6. Let $n = 2$ and $q = 2$. Since $n = 2 \Rightarrow \mu(f) = 0$ if $d(f) \geq 3$, and the only irreducible polynomials over $\mathbb{F}_2$ of degree at most two are $t, t+1$ and t^2+t+1 (see theorem (2.1.28) for a criterion for irreducibility). Therefore, we have three cases for the weighted partition function μ and the corresponding representatives for conjugacy classes in $GL_2(\mathbb{F}_2)$:

$$(1) \quad \begin{cases} \mu(t^2+t+1) = (1,0,\dots) \\ \mu(f) = 0 \text{ if } f \neq t^2+t+1 \end{cases} \longleftrightarrow \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

$$(2) \quad \begin{cases} \mu(t+1) = (2,0,\dots) \\ \mu(f) = 0 \text{ if } f \neq t+1 \end{cases} \longleftrightarrow \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$(3) \quad \begin{cases} \mu(t+1) = (1,1,0,\dots) \\ \mu(f) = 0 \text{ if } f \neq t+1 \end{cases} \longleftrightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

In the first case, we have the vector space isomorphism $V_\mu \cong 1 \cdot \mathbb{F}_2 \oplus t \cdot \mathbb{F}_2$. So, in this case, the action of t on the basis $\{1, t\}$ of V_μ is as follows:

$$\varphi_t : 1 \mapsto t \text{ and } t \mapsto t^2 = t+1 \tag{2.24}$$

Therefore, the corresponding conjugacy class in $GL_2(\mathbb{F}_2)$ is the one containing the matrix representation of φ_t, that is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$.

Similarly, in the second case we have the vector space isomorphism $V_\mu \cong 1 \cdot \mathbb{F}_2 \oplus t \cdot \mathbb{F}_2$, but the action of t on the basis $\{1, t\}$ of V_μ is

$$\varphi_t : 1 \mapsto t \text{ and } t \mapsto t^2 = 1 \tag{2.25}$$

Therefore, the corresponding conjugacy class, is the one containing the matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$.

In the last case, t acts diagonally like the identity and the corresponding conjugacy class is the singleton containing the identity matrix. Evidently, this last case can be generalized to

$$V_{\{I_n\}} \cong \left(\frac{\mathbb{F}_q[t]}{(t-1)}\right)^{\oplus n} \tag{2.26}$$

2.2.2 Formula for the order of the centralizer

Now, let's compute the cardinality of centralizers in $GL_n(\mathbb{F}_q)$. To do this, observe that there is a one-to-one correspondence between elements of the centralizer of $g \in GL_n(\mathbb{F}_q)$ and elements of $\mathrm{Aut}(V_g)$. Now, if V has decomposition as in equation (2.20) and $T \in \mathrm{Aut}(V)$, then $T(V_{(f)}) \subset V_{(f)}$ where $V_{(f)} := \bigoplus_{i\geq 1} \mathbb{F}_q[t]/(f)^{\mu_i(f)}$. In fact, let $m \geq 1$ and $h \in \mathbb{F}_q[t]$ be such that $\gcd(f,h) = 1$ (which is true whenever $h \in \Phi - \{f\}$). Then, if $\overline{u} \in \mathbb{F}_q[t]/(h)^m$ and $f^n\overline{u} = 0$ for some $n \geq 1$ then $f^n u \in (h)^m \Rightarrow h^m | f^n u \Rightarrow h^m | u$ and $\overline{u} = 0$. Since $f^{|\mu(f)|}T(v) = 0$ for any $v \in V_{(f)}$, we see that for each $h \in \Phi - \{f\}$, the projection of $T(v)$ in V_h is zero. Now, since $T|_{V_{(f)}} : V_{(f)} \to V_{(f)}$ is an injective map between vector spaces over $\mathbb{F}_q$, we obtain that $T(V_{(f)}) = V_{(f)}$.

The modules $V_{(f)}$ are called *$f-$primary components* of V and, by the above discussion, they are characterized by the property of being the submodule of V generated by all elements that are annihilated by some power of f. So we get the next theorem.

Theorem 2.2.7. [12] Let $V = V_\mu$ be a $\mathbb{F}_q[t]-$module as in equation (2.20). Then if T is an automorphism of V, its restriction $T|_{V_{(f)}}$ is an automorphism of $V_{(f)}$. In particular, we have an isomorphism of groups

$$\begin{aligned} \mathrm{Aut}(V) &\to \prod_{i\geq 1} \mathrm{Aut}(V_{(f)}) \\ T &\longmapsto \left(T|_{V_{(f)}}\right)_{f\in\Phi} \end{aligned} \tag{2.27}$$

We will show that each $f-$primary component $V_{(f)} \subset V_\mu$ is a $\mathbb{F}_q[t]_{(f)}-$module of type $\mu(f)$. For each $h \in \mathrm{Supp}(\mu) - \{f\}$ the map $\bigoplus_{1\leq i\leq l(\mu(f))} \varphi_h : V_{(f)} \to V_{(f)}$ is an isomorphism by Lemma (2.2.2). Therefore, multiplication by h defines an automorphism of $V_{(f)}$. So if $S := \mathbb{F}_q[t] - (f)$, then

$$S_{V_{(f)}} \subset \mathrm{Aut}(V_{(f)}) \tag{2.28}$$

(remember the notation (1.2.12)). By the functorial property of localization (1.2.13) we have that $V_{(f)}$ is an $\mathbb{F}_q[t]_{(f)}-$module. Moreover, by Theorem (1.2.9), $\mathbb{F}_q[t]_{(f)}$ is a discrete valuation ring with maximal ideal given by $S^{-1}(f) = (f/1)$ and residue field $\mathbb{F}_q[t]/(f) = \mathbb{F}_{q^{d(f)}}$ (see example (1.3.12)). This shows that $V_{(f)}$ is a module over a discrete valuation ring $\mathbb{F}_q[t]_{(f)}$, that is killed by some power of $f/1$.

In this context, $V_{(f)}$ is a finite $\mathbb{F}_q[t]_{(f)}-$module and must have some type. We will show that this type is exactly $\mu(f)$. In fact, since $S_{V_{(f)}} \subset \mathrm{Aut}(V_{(f)})$, we have the isomorphisms

$$\begin{aligned} V_{(f)} &= \bigoplus_{i\geq 1} \mathbb{F}_q[t]/(f^{\mu_i(f)}) \\ &\cong \bigoplus_{i\geq 1} S^{-1}\left(\mathbb{F}_q[t]/(f^{\mu_i(f)})\right) \quad \text{(by Theorem (1.2.16))} \\ &\cong \bigoplus_{i\geq 1} \mathbb{F}_q[t]_{(f)}/((f/1)^{\mu_i(f)}) \quad \text{(by the Corollary (1.2.15))} \end{aligned} \tag{2.29}$$

The isomorphism $V_{(f)} \cong \bigoplus_{i\geq 1} \mathbb{F}_q[t]_{(f)}/((f/1)^{\mu_i(f)})$ is an $\mathbb{F}_q[t]_{(f)}-$module isomorphism, which shows that $V_{(f)}$ is of type $\mu(f)$. Now, we can apply theorem (1.3.34) to deduce that

$$|\mathrm{Aut}(V_{(f)})| = q_f^{|\mu(f)|+2n(\mu(f))} \prod_{i\geq 1} \varphi_{m_i(\mu(f))}(q_f^{-1}) \tag{2.30}$$

where $q_f := q^{d(f)}$.

Notation 2.2.8. We will write $b_\lambda(q) := \prod_{i\geq 1} \varphi_{m_i(\lambda)}(q)$ and $a_\lambda(q) = q^{|\lambda|+2n(\lambda)} b_\lambda(q^{-1})$.

By theorem (2.2.7) and equation (2.30), we get the next result.

Theorem 2.2.9. [12] Suppose that $g \in GL_n(\mathbb{F}_q)$ belongs to a conjugacy class c_μ. Then, the order of the centralizer of g is

$$a_\mu := |\mathrm{Aut}(V_\mu)| = \prod_{f\in\Phi} a_{\mu(f)}(q_f) \tag{2.31}$$

As an example, we get another way to evaluate the cardinality of $GL_n(\mathbb{F}_q)$.

Corollary 2.2.10. $|GL_n(\mathbb{F}_q)| = (q^n-1)(q^n-q)\cdots(q^n-q^{n-1})$

Proof: The centralizer of the identity $I_n \in GL_n(\mathbb{F}_q)$ is the whole group. Moreover its conjugacy class is in correspondence with the weighted valued partition function with support in $t-1$, with (1^n) as its unique nonzero value. So we can apply formula (2.2.9) to get

$$\begin{aligned} |GL_n(\mathbb{F}_q)| &= a_{(1^n)}(q) \\ &= q^{n+n(n-1)}\varphi_n(q^{-1}) \\ &= q^{n^2}(1-q^{-1})(1-q^{-2})\cdots(1-q^{-n}) \\ &= (q^n-1)(q^n-q)\cdots(q^n-q^{n-1}). \end{aligned} \tag{2.32}$$

2.2.3 The space of characters $\mathcal{R}(q)$

Let $\mathcal{R}_n(q)$ be the vector space of complex valued class functions on $GL_n(\mathbb{F}_q)$ (To ease notation we will write G_n for $GL_n(\mathbb{F}_q)$). Then, $\mathcal{R}_n(q)$ is equipped with a canonical inner product defined by

$$\langle u, v\rangle_{G_n} = \frac{1}{|G_n|} \sum_{g\in G_n} u(g)\overline{v(g)} \quad u, v \in \mathcal{R}_n(q). \tag{2.33}$$

The dimension of $\mathcal{R}_n(q)$ equals the number of conjugacy classes of G_n, i.e. the number of weighted partition valued functions $\mu : \Phi \to \mathcal{P}$ such that $||\mu|| = n$. Define the graded vector space

$$\mathcal{R}(q) = \bigoplus_{n \geq 0} \mathcal{R}_n(q) \tag{2.34}$$

where $R_0 = \mathbb{C}$. Extending $\langle \cdot, \cdot \rangle$ to $\mathcal{R}(q)$ by putting $\langle \mathcal{R}_n(q), \mathcal{R}_m(q) \rangle = 0$ for $n \neq m$, we get an inner product on $\mathcal{R}(q)$ such that the irreducible characters of G_n for different n are all mutually orthogonal.

The first objective of this subsection, is to prove that $\mathcal{R}(q)$ has a structure of an associative algebra. To show this, we first need to define the so called "parabolic induction" functor. Let $n = n_1 + \cdots + n_r$ and let V_i be the subspace of $V = \mathbb{F}_q^n$ spanned by the first $n_1 + \cdots + n_i$ basis vectors, i.e.

$$V_i = \text{Span}\{v_1, \ldots, v_{n_1 + \cdots + v_{n_i}}\} \tag{2.35}$$

where $\{v_i\}$ denotes the canonical basis of $V = \mathbb{F}_q^n$. Let F denote the flag

$$(0) = V_0 \subset V_1 \subset \cdots \subset V_r = V. \tag{2.36}$$

The subgroup $P_{(n_1,\ldots,n_r)}$ denotes those matrices in G_n that fix the flag F. So we have

$$P_{(n_1,n_2,\ldots,n_r)} = \left\{ \begin{pmatrix} A_1 & B_{12} & B_{1r} \\ 0 & A_2 & B_{2r} \\ \cdots & & \cdots \\ 0 & 0 & A_r \end{pmatrix} \right\} \tag{2.37}$$

where $A_i \in G_{n_i}$. Then we have the decomposition $P_{(n_1,\ldots,n_r)} = MU$ where

$$M = \left\{ \begin{pmatrix} A_1 & 0 & 0 \\ 0 & A_2 & 0 \\ \cdots & & \cdots \\ 0 & 0 & A_r \end{pmatrix} \right\} \cong G_{n_1} \times \cdots \times G_{n_r} \tag{2.38}$$

and

$$U = \left\{ \begin{pmatrix} I_{n_1} & B_{12} & B_{1r} \\ 0 & I_{n_2} & B_{2r} \\ \cdots & & \cdots \\ 0 & 0 & I_{n_r} \end{pmatrix} \right\} \tag{2.39}$$

is a normal subgroup of $P_{(n_1,\ldots,n_r)}$. We can see that the map

$$M \to P_{(n_1,\ldots,n_r)}/U : \begin{pmatrix} A_1 & 0 & 0 \\ 0 & A_2 & 0 \\ \cdots & & \cdots \\ 0 & 0 & A_r \end{pmatrix} \mapsto \begin{pmatrix} A_1 & 0 & 0 \\ 0 & A_2 & 0 \\ \cdots & & \cdots \\ 0 & 0 & A_r \end{pmatrix} U \tag{2.40}$$

is an isomorphism. In fact, surjectivity follows from

$$\begin{aligned} \begin{pmatrix} A_1 & B_{12} & B_{1r} \\ 0 & A_2 & B_{2r} \\ \cdots & & \cdots \\ 0 & 0 & A_r \end{pmatrix} U &= \begin{pmatrix} A_1 & 0 & 0 \\ 0 & A_2 & 0 \\ \cdots & & \cdots \\ 0 & 0 & A_r \end{pmatrix} \begin{pmatrix} I_{n_1} & A_1^{-1}B_{12} & A_1^{-1}B_{1r} \\ 0 & I_{n_2} & A_2^{-1}B_{2r} \\ \cdots & & \cdots \\ 0 & 0 & I_{n_r} \end{pmatrix} U \\ &= \begin{pmatrix} A_1 & 0 & 0 \\ 0 & A_2 & 0 \\ \cdots & & \cdots \\ 0 & 0 & A_r \end{pmatrix} U, \end{aligned} \tag{2.41}$$

and injectivity follows from the fact that $U \cap M = \{1\}$. Now, if

$$p_{(n_1,\ldots,n_r)} : P_{(n_1,\ldots,n_r)} \to P_{(n_1,\ldots,n_r)}/U \cong M \tag{2.42}$$

is the canonical projection, then each representation of $M \cong G_{n_1 \times \cdots \times G_{n_r}}$ can be pulled back to $P_{(n_1,\ldots,n_r)}$ through $p_{(n_1,\ldots,n_r)}$.

Notation 2.2.11. Let G be a finite group, then $\mathcal{A}(G)$ will denote the category of finite dimensional complex representations of G, or equivalently the category of $\mathbb{C}G-$modules V such that $\dim_{\mathbb{C}} V < \infty$.

Recall that if G is a group and $H \subset G$ is a subgroup, then we have a functor

$$\mathrm{Ind}_H^G : \mathcal{A}(H) \to \mathcal{A}(G) : V \mapsto V^G = \mathbb{C}G \otimes_{\mathbb{C}H} V \tag{2.43}$$

called *induction* from H to G.

Theorem 2.2.12. The induction functor has the following properties:

(1) It is transitive, i.e., if we have subgroups $K \subset H \subset G$ and (V, ρ) a $\mathbb{C}K-$module, then $\mathrm{Ind}_H^G \mathrm{Ind}_K^H \rho \cong \mathrm{Ind}_K^G \rho$.

(2) If (V, ρ) is a $\mathbb{C}H-$module, where $H \subset G$ is a subgroup, then

$$\dim_{\mathbb{C}} V^G = [G : H] \dim_{\mathbb{C}} V. \tag{2.44}$$

The first property is a direct consequence of Lemma (1.1.18). Now, for the second property let $G = \bigcup_{i=1}^k s_i H$ be a decomposition of G into left cosets, where $k = [G : H]$. So, for each $g \in G$ there exist a unique i, $1 \leq i \leq t$ and $h \in H$ such that $g = s_i h$. This shows that each element of $\mathbb{C}G$ can be written uniquely as a sum $\sum_{i=1}^k s_i b_i$ where $b_i \in \mathbb{C}H$. Therefore we have the vector space decomposition

$$\mathbb{C}G \cong \bigoplus_{i=1}^k s_i \mathbb{C}H. \tag{2.45}$$

So, by applying theorem (1.1.23), we obtain the vector space decomposition

$$\begin{aligned} \mathbb{C}G \otimes_{\mathbb{C}H} V &\cong \bigoplus_{i=1}^k s_i \mathbb{C}H \otimes_{\mathbb{C}H} V \\ &\cong \bigoplus_{i=1}^k \mathbb{C}H \otimes_{\mathbb{C}H} V \quad (\text{because } s_i \mathbb{C}H \cong \mathbb{C}H \text{ as right } \mathbb{C}H - \text{modules}) \\ &\cong \bigoplus_{i=1}^k V \quad (\text{because of lemma (1.1.20)}) \end{aligned} \tag{2.46}$$

By the last equation we get $\dim_{\mathbb{C}} V^G = k \cdot \dim_{\mathbb{C}} V = [G:H]\dim_{\mathbb{C}} V$. In particular, if $\{v_j\}$ is a basis of V, then $\{s_i \otimes v_j\}$ is a spanning set for V^G of the same cardinality as the dimension of V^G, so it's a basis for V^G.

Now we are ready to describe the parabolic induction functor. Let (ρ_j, V_j) be representations for the the groups $G_{n_j},\ j = 1, 2$. Then $(\rho_1 \otimes \rho_2, V_1 \otimes V_2)$ is a representation of $G_{n_1} \times G_{n_2}$ and we can pull it back to a representation $(p^*_{(n,m)}(\rho_1 \otimes \rho_2), V_1 \otimes V_2)$ of $P_{(n_1,n_2)}$. Finally, we induce this representation to the full linear group $G_{n_1+n_2}$.

Definition 2.2.13. (Parabolic induction) For each $j = 1, 2$, let (ρ_j, V_j) be a representation for the the group G_{n_j}. We define the representation $(\rho_1 \circ \rho_2, V_1 \otimes V_2)$ of $G_{n_1+n_2}$ by setting

$$\rho_1 \circ \rho_2 = \mathrm{Ind}_{P_{(n_1,n_2)}}^{G_{n_1+n_2}} p^*_{(n_1,n_2)}(\rho_1 \otimes \rho_2). \tag{2.47}$$

Moreover, if χ_j is the character associated to (ρ_j, V_j), we write

$$\chi_1 \circ \chi_2 = \mathrm{Ind}_{P_{(n_1,n_2)}}^{G_{n_1+n_2}} p^*_{(n_1,n_2)}(\chi_1 \otimes \chi_2) \tag{2.48}$$

for the character associated to the representation $(\rho_1 \circ \rho_2, V_1 \otimes V_2)$.

The effect of pulling back a representation (ρ, V) of $G_{n_1} \times G_{n_2}$ is the following. Let $v \in V$, then

$$p^*_{(n_1,n_2)}\rho \begin{pmatrix} A_1 & B_{12} \\ & A_2 \end{pmatrix}(v) = \rho \begin{pmatrix} A_1 & 0 \\ & A_2 \end{pmatrix}(v) \tag{2.49}$$

So, to ease notation, we will drop the symbol for the pullback of the projection in the parabolic induction. We just need to have in mind that after the pullback, the action is defined only by the block diagonal elements A_1 and A_2 as in the previous equation. For example, let $A = \begin{pmatrix} A_1 & B_{12} \\ & A_2 \end{pmatrix} \in P_{(n_1,n_2)}$ and (ρ_j, V_j) representations of G_{n_j} for $j = 1, 2$. Then, viewing the vector space $V_1 \otimes V_2$ as a $\mathbb{C}P_{(n,m)}-$module, we get the following action

$$A(v_1 \otimes v_2) = \rho_1(A_1)v_1 \otimes \rho_2(A_2)v_2. \tag{2.50}$$

This point of view will be useful to avoid a cumbersome notation by writing $p^*_{(n,m)}(\rho_1 \otimes \rho_2)$ everywhere.

Theorem 2.2.14. For each $j = 1, 2, 3$, let (ρ_j, V_j) be a representation of the group G_{n_j}. Then there is an isomorphism $\rho_1 \circ (\rho_2 \circ \rho_2) \cong (\rho_1 \circ \rho_2) \circ \rho_3$ of representations of $G_{n_1+n_2+n_3}$.

Proof: Let

$$P_{(n_1,n_2,n_3)} = \left\{ \begin{pmatrix} A_1 & B_{12} & B_{13} \\ & A_2 & B_{23} \\ & & A_3 \end{pmatrix} \right\} \tag{2.51}$$

where $A_j \in G_{n_j}$ for $j = 1, 2, 3$. We define $\rho_1 \circ \rho_2 \circ \rho_3$ (without parentheses) to be the representation $\mathrm{Ind}_{P_{(n_1,n_2,n_3)}}^{G_{n_1+n_2+n_3}}(\rho_1 \otimes \rho_2 \otimes \rho_3)$ (remember that we are dropping the pullback of

the projection). We will show that $\rho_1 \circ \rho_2 \circ \rho_3 \cong (\rho_1 \circ \rho_2) \circ \rho_3$. By definition

$$\begin{aligned}(\rho_1 \circ \rho_2) \circ \rho_3 &= \mathrm{Ind}_{P_{(n_1+n_2,n_3)}}^{G_{n_1+n_2+n_3}} (\rho_1 \circ \rho_2) \otimes \rho_3 \\ &= \mathrm{Ind}_{P_{(n_1+n_2,n_3)}}^{G_{n_1+n_2+n_3}} (\mathrm{Ind}_{P_{(n_1+n_2,n_3)}}^{G_{n_1+n_2+n_3}} \rho_1 \otimes \rho_2) \otimes \rho_3.\end{aligned} \tag{2.52}$$

So, if we show that

$$(\mathrm{Ind}_{P_{(n_1+n_2,n_3)}}^{G_{n_1+n_2+n_3}} \rho_1 \otimes \rho_2) \otimes \rho_3 \cong \mathrm{Ind}_{P_{(n_1,n_2,n_3)}}^{P_{(n_1+n_2,n_3)}} \rho_1 \otimes \rho_2 \otimes \rho_3, \tag{2.53}$$

we get the result by the transitivity property of induction (see theorem (1.1.18). Therefore, we need to construct an isomorphism

$$\mathbb{C}P_{(n_1+n_2,n_3)} \otimes_{\mathbb{C}P_{n_1,n_2,n_3}} V_1 \otimes V_2 \otimes V_3 \to (\mathbb{C}G_{n_1+n_2} \otimes_{\mathbb{C}P_{(n_1,n_2)}} V_1 \otimes V_2) \otimes V_3 \tag{2.54}$$

First, lets show that both of the vector spaces in the last equation have the same dimension. Choose a decomposition into left cosets $G_{n_1+n_n} = \bigcup_i s_i P_{(n_1,n_2)}$. This decomposition induces another decomposition into left cosets

$$P_{(n_1+n_2,n_3)} = \bigcup_i \begin{pmatrix} s_i & \\ & I_{n_3} \end{pmatrix} P_{(n_1,n_2,n_3)} \tag{2.55}$$

so we get that $[G_{n_1+n_2} : P_{(n_1,n_2)}] = [P_{(n_1+n_2,n_3)} : P_{(n_1,n_2,n_3)}]$, and by the Theorem (2.2.12) we obtain the equality of dimensions. Writing g_i for the block diagonal matrix $\mathrm{diag}(s_i, I_{n_3})$ appearing in the decomposition of $P_{(n_1+n_2,n_3)}$ into left cosets, and $\{v_i^j \mid i = 1, \ldots, \dim_{\mathbb{C}} V_j\}$ for the basis of V_j for $j = 1, 2, 3$ we get that $\{g_i \otimes v_j \otimes v_k \otimes v_l\}$ is a basis for

$$\mathbb{C}P_{(n_1+n_2,n_3)} \otimes_{\mathbb{C}P_{n_1,n_2,n_3}} V_1 \otimes V_2 \otimes V_3. \tag{2.56}$$

Therefore, we have a well defined homomorphism of vector spaces

$$\begin{aligned}\varphi : \mathbb{C}P_{(n_1+n_2,n_3)} \otimes_{\mathbb{C}P_{(n_1,n_2,n_3)}} V_1 \otimes V_2 \otimes V_3 &\to (\mathbb{C}G_{n_1+n_2} \otimes_{\mathbb{C}P_{(n_1,n_2)}} V_1 \otimes V_2) \otimes V_3 \\ g_i \otimes v_j \otimes v_k \otimes v_l &\longmapsto (s_i \otimes (v_j \otimes v_k)) \otimes v_l\end{aligned} \tag{2.57}$$

Observe that by multilinearity we have $\varphi(g_i \otimes u \otimes v \otimes w) = (s_i \otimes (u \otimes v)) \otimes w \; \forall u \in V_1, v \in V_2, w \in V_3$. This homomorphism is clearly seen to be surjective. Moreover, since both the domain and its image have the same dimension, we get that φ is in fact a vector space isomorphism. We claim that φ is an isomorphism of representations of $P_{(n_1+n_2,n_3)}$. Let

$$A = \begin{pmatrix} A_1 & B_{12} \\ & A_2 \end{pmatrix} \in P_{(n_1+n_2,n_3)}, \tag{2.58}$$

then, if $A_1 s_i = s_m C$ where $C = \left(\begin{smallmatrix} C_1 & D_{12} \\ & C_2 \end{smallmatrix}\right) \in P_{(n_1,n_2)}$, we get

$$\begin{aligned} A(g_i \otimes v_j \otimes v_k \otimes v_l) &= Ag_i \otimes v_j \otimes v_k \otimes v_l \\ &= \begin{pmatrix} A_1 s_i & B_{12} \\ & A_2 \end{pmatrix} \otimes v_j \otimes v_k \otimes v_l \\ &= \begin{pmatrix} s_m C & B_{12} \\ & A_2 \end{pmatrix} \otimes v_j \otimes v_k \otimes v_l \\ &= g_m \otimes \begin{pmatrix} C & B_{12} \\ & A_2 \end{pmatrix} (v_j \otimes v_k \otimes v_l) \\ &= g_m \otimes C_1 v_j \otimes C_2 v_k \otimes A_2 v_l. \end{aligned} \tag{2.59}$$

But

$$\begin{aligned} \varphi((g_m \otimes C_1 v_j \otimes C_2 v_k) \otimes A_2 v_l) &= (s_m \otimes C_1 v_j \otimes C_2 v_k) \otimes A_2 v_l \\ &= (s_m \otimes C(v_j \otimes v_k)) \otimes A_2 v_l \\ &= (s_m C \otimes v_j \otimes v_k) \otimes A_2 v_l \\ &= (A_1 s_i \otimes v_j \otimes v_k) \otimes A_2 v_l \\ &= A(s_i \otimes v_j \otimes v_k) \otimes v_l. \end{aligned} \tag{2.60}$$

So we conclude that φ is in fact an isomorphism of representations and the theorem is proved.

As a direct consequence of this theorem, we see that if χ_j is the character associated to the representation $(\rho_j, V_j),\ j = 1, 2, 3$, then $(\chi_1 \circ \chi_2) \circ \chi_3 = \chi_1 \circ (\chi_2 \circ \chi_3)$. Moreover, the proof of Theorem (2.2.14) can be generalized to show that if χ_j are characters for the groups G_{n_j} for $j = 1, \ldots, r$ and $n = n_1 + \cdots + n_r$. Then

$$\chi_1 \circ \cdots \circ \chi_r := \mathrm{Ind}_{P_{(n_1,\ldots,n_r)}}^{G_n} \chi_1 \otimes \cdots \otimes \chi_r \tag{2.61}$$

is exactly the product of the of the characters $\{\chi_j\}$ in $\mathcal{R}(q)$ in any order.

Theorem 2.2.15. [9] The product $\circ : \mathcal{R}_n(q) \times \mathcal{R}_m(q) \to \mathcal{R}_{n+m}(q)\ \forall n, m$ can be extended to the whole space $\mathcal{R}(q)$ turning it into a commutative and associative $\mathbb{C}-$algebra.

Proof: Associativity follows from Theorem (2.2.14), so it only remains to prove the commutativity of $\circ$, which follows from the isomorphism of representations

$$p^*_{(n_1,n_2)}(\rho_1 \otimes \rho_2) \cong p^*_{(n_2,n_1)}(\rho_2 \otimes \rho_1). \tag{2.62}$$

The next objective of this subsection is to show that $\mathcal{R}(q)$ is a certain tensor product of Hall algebras. Let G be a finite group and $H \subset G$ a subgroup. We will show that if (ρ, V)

is a representation of H with associated character χ, and $G = \bigcup_{i=1}^{k} s_i H$ is a decomposition of G into left cosets, then the associated character of V^G is

$$\mathrm{Ind}_H^G \chi(g) = \sum_{i=1}^{k} \chi(s_i^{-1} g s_i), \tag{2.63}$$

where we set $\chi(s_i^{-1} g s_i) = 0$ if $s_i^{-1} g s_i \notin H$. To show this, we use the vector space decomposition

$$V^G := \mathbb{C}G \otimes_{\mathbb{C}H} V \cong \bigoplus_{i=1}^{k} s_i \otimes_{\mathbb{C}H} V. \tag{2.64}$$

Let $g \in G$ and let i be fixed, then $g s_i = s_j h$ for unique j and $h \in H$. Therefore $g(s_i \otimes v) = s_j h \otimes v = s_j \otimes \rho(h) v = s_j \otimes \rho(s_j^{-1} g s_i) v$ and those i's that give a nonzero contribution to $\mathrm{Ind}_H^G(g)$ are those such that $g s_i = s_i h$ for some $h \in H \iff s_i^{-1} g s_i \in H$. In particular, if $\{v_j \mid j = 1, \ldots, n\}$ is a basis of V, then for $h \in H$ we have $\chi(h) = \mathrm{Tr}\rho(h) = \sum_i (\rho(h) v_i)_i$ where $(\cdot)_i : V \to \mathbb{C}$ denotes the projection taking the ith coordinate. Therefore $(s_j \otimes \rho(h) v_i)_{ij} = (\rho(h) v_i)_i$ where $(\cdot)_{ij} : V^G \to \mathbb{C}$ is the projection of the (i,j)th coordinate. So we conclude that

$$\begin{aligned} \mathrm{Ind}_H^G \chi(g) &= \sum_{i : s_i^{-1} g s_i \in H} \sum_{j=1}^{n} (s_i \otimes \rho(s_i^{-1} g s_i) v_j)_{ij} \\ &= \sum_{i : s_i^{-1} g s_i \in H} \sum_{j=1}^{n} (\rho(s_i^{-1} g s_i) v_j)_j \\ &= \sum_{i : s_i^{-1} g s_i \in H} \chi(s_i^{-1} g s_i). \end{aligned} \tag{2.65}$$

If we put $\chi(s_i^{-1} g s_i) = 0$ if $s_i^{-1} g s_i \notin H$ we get the result.

Now, let $n = n_1 + \cdots + n_r$ and $G_n = \bigcup_{s \in S} s P_{(n_1, \ldots, n_r)}$ a left coset decomposition. By the discussion at the beginning of this section, if u_j $(j = 1, \ldots, r)$ are characters (or more generally class functions), then the class function $u = u_1 \otimes \cdots \otimes u_r$ of $G_{n_1} \times \cdots \times G_{n_r}$ can be pulled back to $P_{(n_1, \ldots, n_r)}$ in such a way that

$$\begin{aligned} p^*_{(n_1,\ldots,n_r)} u \begin{pmatrix} A_1 & B_{12} & \cdots & B_{1r} \\ 0 & A_2 & \cdots & B_{2r} \\ \vdots & 0 & \ddots & \vdots \\ 0 & 0 & \cdots & A_r \end{pmatrix} &= u_1 \otimes \cdots \otimes u_r \begin{pmatrix} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \vdots & 0 & \ddots & \vdots \\ 0 & 0 & \cdots & A_r \end{pmatrix} \\ &= \prod_{i=1}^{r} u_i(A_i) \end{aligned} \tag{2.66}$$

Of course for simplicity of notation, we will continue to omit the pullback, and use the same notation "u" for the character $p^*_{(n_1,\ldots,n_r)} u$. Now, by equation (2.61) we get

$$u_1 \circ \cdots \circ u_r(x) = \sum_{s \in S} u(s^{-1} x s). \tag{2.67}$$

So $u(s^{-1}xs)$ contributes to $u_1 \circ \cdots \circ u_r(x)$ if and only if $s^{-1}xs \in P_{(n_1,\ldots,n_r)}$, i.e. if and only if $s^{-1}xs$ preserves the flag F

$$(0) = V_0 \subset V_1 \subset \cdots V_r = V \tag{2.68}$$

as defined in equation (2.35). Moreover, $s^{-1}xs$ preserves F if and only if x preserves the flag sF

$$(0) = W_0 \subset W_1 \subset \cdots \subset W_r = V. \tag{2.69}$$

where $W_i = sV_i$. This means that the chain $(W_i)_{i=0}^r$ (with $W_r = V_x$ and $W_0 = (0)$), is a chain of $\mathbb{F}_q[t]-$submodules. Now, if we put

$$s^{-1}xs = \begin{pmatrix} A_1 & B_{12} & & B_{1r} \\ 0 & A_2 & & B_{2r} \\ \cdots & & & \cdots \\ 0 & 0 & & A_r \end{pmatrix} \tag{2.70}$$

then $u(s^{-1}xs) = \prod_{i=1}^r u_i(A_i)$ and for $n_{i-1} < k \leq n_i$

$$\begin{aligned} s^{-1}xs(v_k) - \sum_{l=n_1+\cdots+n_{i-1}+1}^{n_1+\cdots+n_i} (A_i)_{lk} v_l \quad &\in V_{i-1} \iff \\ x(sv_k) - \sum_{l=n_1+\cdots+n_{i-1}+1}^{n_1+\cdots+n_i} (A_i)_{lk}(sv_l) \quad &\in W_{i-1} \end{aligned} \tag{2.71}$$

So x (and hence t) acts on W_i/W_{i-1} as the matrix A_i, which implies that $W_i/W_{i-1} \cong V_{A_i}$ as $\mathbb{F}_q[t]-$modules. Conversely, let

$$(0) = W_0 \subset W_1 \subset \cdots \subset W_r = V_x \tag{2.72}$$

be a chain of $\mathbb{F}_q[t]-$submodules, such that $W_i/W_{i-1} \cong V_{A_i}$. This means that there is a choice of basis of $V_x = \mathbb{F}_q^n$ such that x has matrix representation

$$A = \begin{pmatrix} A_1 & B_{12} & & B_{1r} \\ 0 & A_2 & & B_{2r} \\ \cdots & & & \cdots \\ 0 & 0 & & A_r \end{pmatrix} \tag{2.73}$$

which implies that A is conjugate in $P_{(n_1,\ldots,n_r)}$ to $s^{-1}xs$ for some $s \in S$. So, the contributions for the summand $\prod_{i=1}^r u_i(A_i)$ are in one to one correspondence with chains of $\mathbb{F}_q[t]-$submodules

$$(0) = W_0 \subset W_1 \subset \cdots \subset W_r = V_x \tag{2.74}$$

such that $W_i/W_{i-1} \cong V_{A_i}$. From this discussion we have proved the following theorem.

Theorem 2.2.16. [12] Let u_i be a class function on G_{n_i} for each $i = 1, \dots, r$ and write $n = n_1 + \cdots + n_r$. Then if c_μ is a conjugacy class in G_n, we get

$$u_1 \circ \cdots \circ u_r(c_\mu) = \sum g^\mu_{\mu^1 \dots \mu^r} \prod_{i=1}^{r} u_i(c_{\mu^i}) \tag{2.75}$$

where the sum runs over all sequences $(\mu^1, \dots, \mu^r)$ such that c_{μ^i} is a conjugacy class in G_{n_i} for $i = 1, \dots, r$, and $g^\mu_{\mu^1 \dots \mu^r}$ is the number of chains of submodules

$$(0) = W_0 \subset W_1 \subset \cdots \subset W_r = V_\mu \tag{2.76}$$

such that $W_i / W_{i-1} \cong V_{\mu^i}$.

Now, for c_μ a conjugacy class in G_n, define the class function π_μ on G_n by

$$\pi_\mu(c_\lambda) = \begin{cases} 1 & \text{if } \lambda = \mu \\ 0 & \text{if } \lambda \neq \mu \end{cases} \tag{2.77}$$

Now fix conjugacy classes c_{μ^i} in G_{n_i} for each $i = 1, \dots, r$. Then by the previous theorem, we get that

$$\pi_{\mu^1} \circ \cdots \circ \pi_{\mu^r}(c_\mu) = g^\mu_{\mu^1 \dots \mu^r} = g^\mu_{\mu^1 \dots \mu^r} \pi_\mu(c_\mu) \tag{2.78}$$

for all conjugacy classes c_μ on G_n. Therefore

$$\pi_{\mu^1} \circ \cdots \circ \pi_{\mu^r} = \sum_\mu g^\mu_{\mu^1 \dots \mu^r} \pi_\mu. \tag{2.79}$$

Now we com to the link with Hall algebras. We claim that $g^\mu_{\mu^1 \dots \mu^r} = \prod_{f \in \Phi} G^{\mu(f)}_{\mu^1(f) \dots \mu^r(f)}(q_f)$ where $q_f := q^{d(f)}$. Indeed, if

$$(0) = W_0 \subset W_1 \subset \cdots \subset W_r = V_\mu, \tag{2.80}$$

then

$$\begin{aligned} &W_i / W_{i-1} \cong V_{\mu^i} \iff \\ &\bigoplus_{f \in \Phi} (W_i)_{(f)} / (W_{i-1})_{(f)} \cong \bigoplus_{f \in \Phi} \bigoplus_{j \geq 1} \mathbb{F}_q[t] / (f^{\mu^i_j(f)}) \ \ \forall i \iff \\ &(W_i)_{(f)} / (W_{i-1})_{(f)} \text{ is of type } \mu_i(f) \ \forall f \in \Phi \end{aligned} \tag{2.81}$$

Therefore, to build a chain of submodules $(W_i)_{i=0}^r$ (with $W_r = V_\mu$ and $W_0 = (0)$) such that $W_i / W_{i-1} \cong V_{\mu^i}$ for each $i = 1, \dots, r$, we just need to construct, for each $f \in \Phi$, a chain

$$(0) = (W_0)_{(f)} \subset (W_1)_{(f)} \subset \cdots \subset (W_r)_{(f)} = V_{\mu(f)} = (V_\mu)_{(f)} \tag{2.82}$$

such that the quotient module $(W_i)_{(f)}/(W_{i-1})_{(f)}$ is of type $\mu^i(f)$ for each $i = 1, \ldots, r$. The number of these chains is $G^{\mu(f)}_{\mu^1(f)\ldots\mu^r(f)}(q_f)$ by definition and we get the result.

Theorem 2.2.17. [9] We have an isomorphism of associative algebras:

$$\begin{aligned} \varphi : \mathcal{R}(q) &\to \bigotimes_{f\in\Phi} H(\mathbb{F}_q[t]_{(f)}) \otimes_{\mathbb{Z}} \mathbb{C} \\ \pi_\mu &\longmapsto \otimes_{f\in\Phi} u_{\mu(f)} \end{aligned} \tag{2.83}$$

Before we proceed to the proof we need to define carefully what we mean by the infinite tensor product $\bigotimes_{f\in\Phi} H(\mathbb{F}_q[t]_{(f)})$. For each $n \geq 1$ we can form the finite tensor product $\bigotimes_{f\in\Phi_n} H(\mathbb{F}_q[t]_{(f)})$. Moreover, by corollary (1.1.28) we have

$$\bigotimes_{f\in\Phi_n} H(\mathbb{F}_q[t]_{(f)}) = \bigoplus_{\mu:\mathrm{Supp}(\mu)\subset\Phi_n} \mathbb{Z} \bigotimes_{f\in\Phi_n} u_{\mu(f)}. \tag{2.84}$$

Also, if $m|n \Rightarrow \Phi_m \subset \Phi_n$ and we have a canonical monomorphism

$$\theta_{m,n} : \bigotimes_{f\in\Phi_m} H(\mathbb{F}_q[t]_{(f)}) \to \bigotimes_{f\in\Phi_n} H(\mathbb{F}_q[t]_{(f)}) : \bigotimes_{f\in\Phi_m} u_{\mu(f)} \mapsto \bigotimes_{f\in\Phi_n} u_{\mu(f)} \tag{2.85}$$

where $\mathrm{Supp}(\mu) \subset \Phi_m$. Therefore, $(\bigotimes_{f\in\Phi_n} H(\mathbb{F}_q[t]_{(f)}), \theta_{m,n})$ is a directed system and we define

$$\bigotimes_{f\in\Phi} H(\mathbb{F}_q[t]_{(f)}) := \varinjlim \bigotimes_{f\in\Phi_n} H(\mathbb{F}_q[t]_{(f)}). \tag{2.86}$$

Since each $\theta_{m,n}$ is injective, the canonical homomorphisms

$$\bigotimes_{f\in\Phi_n} H(\mathbb{F}_q[t]_{(f)}) \to \bigotimes_{f\in\Phi} H(\mathbb{F}_q[t]_{(f)}) : \bigotimes_{f\in\Phi_n} u_{\mu(f)} \mapsto [\bigotimes_{f\in\Phi_n} u_{\mu(f)}] \tag{2.87}$$

are also injective. By theorem (1.1.27) we get an isomorphism $\bigotimes_{f\in\Phi} H(\mathbb{F}_q[t]_{(f)}) \otimes_{\mathbb{Z}} \mathbb{C} \cong \varinjlim \bigotimes_{f\in\Phi_n} H(\mathbb{F}_q[t]_{(f)}) \otimes_{\mathbb{Z}} \mathbb{C}$. Moreover, if $n \geq 1$,

$$\begin{aligned} \bigotimes_{f\in\Phi_n} H(\mathbb{F}_q[t]_{(f)}) \otimes_{\mathbb{Z}} \mathbb{C} &\cong \bigoplus_{\mu\,:\,\mathrm{Supp}(\mu)\subset\Phi_n} \mathbb{C} \bigotimes_{f\in\Phi_n} u_{\mu(f)} \\ &\cong \bigoplus_{m\geq 1} \bigoplus_{\substack{\mathrm{Supp}(\mu)\subset\Phi_n \\ ||\mu||\leq m}} \mathbb{C} \bigotimes_{f\in\Phi_n} u_{\mu(f)} \end{aligned} \tag{2.88}$$

is a infinite direct sum, which can be seen as a direct limit of finite dimensional algebras over $\mathbb{C}$ (see the proof of the corollary (1.1.28)). But, by theorem (1.1.26) we see that $\bigotimes_{f\in\Phi_n} H(\mathbb{F}_q[t]_{(f)}) \otimes_{\mathbb{Z}} \mathbb{C}$ is a $\mathbb{C}-$algebra with multiplication defined componentwise. In this context it's clear that the group monomorphisms $\theta_{m,n}$ are also $\mathbb{C}-$algebra monomorphisms. Therefore $\bigotimes_{f\in\Phi} H(\mathbb{F}_q[t]_{(f)}) \otimes_{\mathbb{Z}} \mathbb{C}$, as a direct limit of $\mathbb{C}-$algebras, is a $\mathbb{C}-$algebra.

Moreover, for $\mathrm{Supp}(\mu) \subset \Phi_n$ write $\bigotimes_{f\in\Phi} u_{\mu(f)} = [\bigotimes_{f\in\Phi_n} u_{\mu(f)}]$, then since the canonical map $\bigotimes_{f\in\Phi_n} u_{\mu(f)} \mapsto \bigotimes_{f\in\Phi} u_{\mu(f)}$ is injective, the set $\{\bigotimes_{f\in\Phi} u_{\mu(f)} \mid \mu\}$ must be linearly independent.

Now we can proceed to the proof of the theorem (2.2.17).

Proof: The map φ is clearly surjective, by the previous construction of the tensor product $\bigotimes_{f\in\Phi} H(\mathbb{F}_q[t]_{(f)})$, and injectivity comes from the fact that $\{\bigotimes_{f\in\Phi} u_{\mu(f)} \mid \mu\}$ is a basis of $\bigotimes_{f\in\Phi} H(\mathbb{F}_q[t]_{(f)}) \otimes_{\mathbb{Z}} \mathbb{C}$. It remains to show that φ is a homomorphism of algebras. But

$$\begin{aligned}
\varphi(\pi_\mu)\varphi(\pi_\nu) &= \bigotimes_{f\in\Phi} u_{\mu(f)} \bigotimes_{f\in\Phi} u_{\nu(f)} \\
&= \bigotimes_{f\in\Phi} u_{\mu(f)} u_{\nu(f)} \\
&= \bigotimes_{f\in\Phi} \sum_{\lambda(f)} G^{\lambda(f)}_{\mu(f)\nu(f)}(q_f) u_{\lambda(f)} \\
&= \sum_{\lambda} \prod_{f\in\Phi} G^{\lambda(f)}_{\mu(f)\nu(f)}(q_f) \bigotimes_{f\in\Phi} u_{\lambda(f)} \\
&= \sum_{\lambda} g^{\lambda}_{\mu\nu} \bigotimes_{f\in\Phi} u_{\lambda(f)} \\
&= \varphi(\pi_\mu \circ \pi_\nu)
\end{aligned} \tag{2.89}$$

and we conclude the proof.

Chapter 3

Vertex algebras

3.1 Vector superspaces

Definition 3.1.1. [2] A vector superspace V is a $\mathbb{Z}/(2)-$graded vector space. We write $V = V_{\bar{0}} \oplus V_{\bar{1}}$ and if $v \in V_\alpha$ is a homogeneous vector of V, we define its *parity* by $p(v) = \alpha$. A subspace W of a vector superspace V is a subspace respecting the $\mathbb{Z}/(2)-$gradation of V, i.e. $W = W_{\bar{0}} \oplus W_{\bar{1}}$ with $W_\alpha \subset V_\alpha \ \ \forall \alpha \in \mathbb{Z}/(2)$.

It's important to note that if V, W are two vector superspaces, then $V \otimes W$ can be given a natural structure of a vector superspace by defining its $\mathbb{Z}/(2)-$gradation as follows:

$$(V \otimes W)_\alpha = \bigoplus_{p+q=\alpha} V_p \otimes W_q. \tag{3.1}$$

So for homogeneous vectors $v \in V, w \in W$ we have $p(v \otimes w) = p(v) + p(w)$.

Definition 3.1.2. [10] An associative superalgebra A is a $\mathbb{Z}/(2)-$graded associative algebra with the additional property that for $a \in A_\alpha,\ b \in A_\beta \Rightarrow ab \in A_{\alpha+\beta}$.

Definition 3.1.3. [10] A Lie superalgebra $\mathfrak{g} = \mathfrak{g}_{\bar{0}} \oplus \mathfrak{g}_{\bar{1}}$ is a $\mathbb{Z}/(2)-$graded vector superspace equipped with a bilinear map $[\ ,\ ] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ such that the following properties are satisfied for homogeneous elements $x, y \in \mathfrak{g}$:

1. (super-skewsymmetry) $[y, x] = -p(x, y)[x, y]$
2. (super-Jacobi identity) $[x, [y, z]] = [[x, y], z] + p(x, y)[y, [x, z]]$

where $p(x, y) = (-1)^{p(x)p(y)}$. We also impose the condition $p([x, y]) = p(x) + p(y)$.

As a consequence of the definition, we see that the restriction $[\ ,\ ] : \mathfrak{g}_{\bar{1}} \times \mathfrak{g}_{\bar{1}} \to \mathfrak{g}_{\bar{0}}$ is symmetric while $[\ ,\ ] : \mathfrak{g}_{\bar{0}} \times \mathfrak{g}_{\bar{0}} \to \mathfrak{g}_{\bar{0}}$ is skewsymmetric. Indeed $\mathfrak{g}_0$ is a Lie algebra in the usual sense. For Lie superalgebras we have an analogous definition for the universal enveloping algebra

$$\mathcal{U}(\mathfrak{g}) = \frac{T(\mathfrak{g})}{\langle x \otimes y - p(x,y) y \otimes x - [x,y] \rangle} \tag{3.2}$$

where $T(\mathfrak{g}) = \bigoplus_{k \geq 0} T^k(\mathfrak{g})$ is the tensor algebra of $\mathfrak{g}$. It's important to note that $T(\mathfrak{g})$ is a vector superspace according to the grading (3.1) and that the ideal $\langle x \otimes y - p(x,y) y \otimes x - [x,y] \rangle$ is generated by homogeneous elements, so it's a subspace of the vector superspace $T(\mathfrak{g})$. Therefore, the quotient $\mathcal{U}(\mathfrak{g})$ is also a vector superspace with grading

$$p(x_1 \cdots x_k) = p(x_1) + \cdots + p(x_k) \quad x_i \in \mathfrak{g}. \tag{3.3}$$

In particular this proves that $\mathcal{U}(\mathfrak{g})$ is an associative superalgebra.

There is an analogous version of the PBW-theorem for Lie superalgebras.

Theorem 3.1.4. [10] Let $\mathfrak{g} = \mathfrak{g}_{\bar{0}} \oplus \mathfrak{g}_{\bar{1}}$ be a Lie superalgebra, $\{x_1, \ldots, x_p\}$ be a basis for $\mathfrak{g}_{\bar{0}}$ and let $\{y_1, \ldots, y_q\}$ be a basis for $\mathfrak{g}_{\bar{1}}$. Then $\mathcal{U}(\mathfrak{g})$ is spanned by

$$x_1^{r_1} \cdots x_p^{r_p} y_1^{s_1} \cdots y_q^{s_q} \quad r_i \geq 0,\ s_i \leq 1. \tag{3.4}$$

Given an associative superalgebra A, one can endow it with a Lie superalgebra structure by defining the bracket $[a,b] = ab - p(a,b)ba$. As an example, let $V = V_{\bar{0}} \oplus V_{\bar{1}}$ be a vector superspace. Then $\mathrm{End}V$ has an associative superalgebra structure by defining the grading $\mathrm{End}V = (\mathrm{End}(V))_{\bar{0}} \oplus (\mathrm{End}(V))_{\bar{1}}$ where $(\mathrm{End}(V))_\alpha = \{T \in \mathrm{End}(V) \ | TV_\beta \subset V_{\alpha+\beta}\}$. This can be seen by choosing a basis for V, so we can identify

$$(\mathrm{End}(V))_{\bar{0}} = \left\{ \begin{pmatrix} * & 0 \\ 0 & * \end{pmatrix} \right\} \quad \text{and} \quad (\mathrm{End}(V))_{\bar{1}} = \left\{ \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \right\}. \tag{3.5}$$

3.2 The first definition

Definition 3.2.1. Let U be a vector superspace space, a $U-$valued formal distribution in l variables is a formal series $a(z_1, \ldots, z_l) = \sum_{n \in \mathbb{Z}} a_{n_1,\ldots,n_l} z_1^{-n_1-1} \cdots z_l^{-n_l-1}$ with coefficients $a_{n_1,\ldots,n_l} \in U$. We use the notation $U[[z_1^{\pm}, \ldots, z_l^{\pm}]]$ for the *vector superspace space* of $U-$valued formal distributions in l variables.

Definition 3.2.2. [2] Given a vector superspace V, a *quantum field* on V is an $\mathrm{End}V-$valued formal distribution $a(z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}$ with the property that for each $v \in V$ there is a positive integer $N = N(v) \geq 0$ such that $a_{(n)} v = 0\ \forall n \geq N$.

Definition 3.2.3. [2] Let V be a vector superspace. We say that a pair $(a(z), b(z))$ of End$V-$valued quantum field is *local* if

$$(z-w)^N[a(z), b(w)] = 0 \tag{3.6}$$

for some $N \geq 0$.

Definition 3.2.4. A vertex algebra is a quadruple $(V, T, |0\rangle, \mathcal{F})$ where V is a vector superspace, T is an endomorphism of V called the *infinitesimal translation operator* (or just translation operator), $|0\rangle$ is an element of $V_{\bar{0}}$ called *vacuum* and $\mathcal{F} = \{a^j(z)\}_{j\in J}$ is a collection of End$V-$valued quantum fields with the property that all endomorphisms a^j_n have the same parity for a fixed $j \in J$. This data satisfies the following axioms:

(1) (Vacuum and translation covariance) $T|0\rangle = 0$ and $[T, a^j(z)] = \partial_z a^j(z)\ \forall j \in J$.

(2) (Locality) $(z-w)^{N_{jk}}[a^j(z), a^k(w)] = 0$ for some $N_{jk} \geq 0$.

(3) (Completeness) V is spanned by element of the form $a^{j_1}_{(n_1)} \cdots a^{j_k}_{(n_k)}|0\rangle$

It's important to note that the translation operator is uniquely defined by axioms (1) and (3). In fact, by noting that $[T, a(z)] = \partial_z a(z) \iff [T, a_{(n)}] = -na_{(n-1)}$, we get

$$Ta^{j_1}_{n_1} \cdots a^{j_k}_{n_k}|0\rangle = \sum_{i=1}^{k} -n_i a^{j_1}_{n_1} \cdots a^{j_{i-1}}_{(n_{i-1})} a^{j_i}_{(n_i-1)} a^{j_{i+1}}_{(n_{i+1})} \cdots a^{j_k}_{n_k}|0\rangle \tag{3.7}$$

because of the axiom (1). Now, completeness (axiom (3)) implies that T is fully determined in V by the formula (3.7).

Lemma 3.2.5. [2] Let V be a vector superspace and let $|0\rangle \in V_{\bar{0}}, T \in (\text{End}V)_{\bar{0}}$ be such that $T|0\rangle = 0$. Then for any $T-$covariant End$V-$valued quantum field $a(z)$, we have $a(z)|0\rangle \in V[[z]]$.

Proof: By $T-$covariance we have $[T, a_n] = -na_{n-1}\ \forall n \in \mathbb{Z}$. So we can prove by induction that $T^k a_n|0\rangle = (-1)^k n!/(n-k)! a_{n-k}|0\rangle$. Now, since $a(z)$ is a quantum field, we have $a_n|0\rangle = 0$ whenever $n \geq N$ for a fixed $N \geq 0$ depending on $|0\rangle$. In particular, this means that for any $k, 1 \leq k \leq N$:

$$0 = T^k a_N|0\rangle = (-1)^k \frac{N!}{(N-k)!} a_{N-k}|0\rangle \tag{3.8}$$

and $a_{N-k}|0\rangle = 0$ $(1 \leq k \leq N)$. Therefore $a_n|0\rangle = 0\ \forall n \geq 0$ which is equivalent to $a(z)|0\rangle \in V[[z]]$.

Notation 3.2.6. From now on we will fix a vertex algebra $(V, |0\rangle, T, \mathcal{F})$. In this context we will write $\widetilde{\mathcal{F}}$ for the vector space of all T-covariant $\mathrm{End}V-$valued quantum fields. By the lemma (3.2.5), we are allowed to define a linear map $s : \widetilde{\mathcal{F}} \to V : a(z) \mapsto a(z)|0\rangle|_{z=0}$.

Lemma 3.2.7. [2] Let $a(z)$ be a $T-$covariant quantum field and write $s(a(z)) = a$. Then $a(z)|0\rangle = e^{zT}a$. In particular, this shows that if $a = 0$, then $a(z)|0\rangle = 0$.

Proof: By $T-$covariance we have $Ta_{(n)}|0\rangle = [T, a_{(n)}]|0\rangle = -na_{n-1}|0\rangle$ and by induction

$$T^k a_{(n)}|0\rangle = (-1)^k n(n-1)\cdots(n-k+1)a_{(n-k)}|0\rangle. \tag{3.9}$$

In particular $T^k a_{(-1)}|0\rangle = k!a_{(-k-1)}|0\rangle \Rightarrow a_{(-k-1)}|0\rangle = T^k a_{(-1)}|0\rangle/k! = T^k a/k!$. By lemma (3.2.5) we already know that $a_{(n)}|0\rangle = 0\ \forall n \geq 0$. Therefore $a(z)|0\rangle = \sum_{k\geq 0} T^k a/k! = e^{zT}a$.

Let $\mathbb{C}[[z,w]]$ be the ring of Taylor series in two variables and $\mathbb{C}((z,w))$ its fraction field. We have the following natural homomorphism

$$r_{z,w} : \mathbb{C}[[z,w]] \to \mathbb{C}((z))((w)) : \sum_{m,n\geq 0} a_{m,n} z^m w^n \mapsto \sum_{n\geq 0} \left(\sum_{m\geq 0} a_{m,n} z^m \right) w^n. \tag{3.10}$$

Then, by the universal property of the fraction field, we have a homomorphism $i_{z,w}$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathbb{C}[[z,w]] & \xrightarrow{r_{z,w}} & \mathbb{C}((z))((w)) \\ \downarrow & \nearrow_{i_{z,w}} & \\ \mathbb{C}((z,w)) & & \end{array}$$

Now, suppose that $D : \mathbb{C}[[z,w]] \to \mathbb{C}[[z,w]]$ is a derivation. It can be shown that there exists a unique extension of D to a derivation of $\mathbb{C}((z,w))$ (which will denote by D again). The formula is the usual one:

$$D\left(\frac{f}{g}\right) = \frac{(Df)g - f(Dg)}{g^2}. \tag{3.11}$$

Now, let $D = \partial_z$. Since $\partial_z \circ r_{z,w} = r_{z,w} \circ \partial_z$, we see that $\partial_z \circ i_{z,w} = i_{z,w} \circ \partial_z$ by the definition of the extension of ∂_z to $\mathbb{C}((z,w))$. In the same way we can see that $i_{z,w}$ commutes with multiplication by elements of $\mathbb{C}[[z,w]]$.

As an example, we have

$$i_{z,w}\frac{1}{z-w} = \sum_{k\geq 0} z^{-k-1}w^k \tag{3.12}$$

because $(z-w)\sum_{k\geq 0} z^{-k-1}w^k = 1$ in $\mathbb{C}((z))((w))$. Applying ∂_z^l to both sides of this equation we obtain

$$\begin{aligned}(-1)^l l! i_{z,w}\frac{1}{(z-w)^{l+1}} &= \sum_{k\geq 0}(-1)^l \frac{(k+l)!}{k!} z^{-k-l-1}w^k \Rightarrow \\ i_{z,w}\frac{1}{(z-w)^{l+1}} &= \sum_{k\geq 0}\binom{k+l}{k} z^{-l-1-k}w^k.\end{aligned} \tag{3.13}$$

Analytically, we can interpret $i_{z,w}(1/(z-w)^l)$ as the expansion of the series $1/(z-w)^l$ in the domain $|z|>|w|$. It's important to note that the diagram

$$\begin{array}{ccc} \mathbb{C}((z,w)) & \xrightarrow{i_{z,w}} & \mathbb{C}((z))((w)) \\ \downarrow{\scriptstyle i_{w,z}} & & \downarrow \\ \mathbb{C}((w))((z)) & \hookrightarrow & \mathbb{C}[[z^{\pm}, w^{\pm}]] \end{array}$$

is not commutative. In fact, we have the following important identity

$$i_{z,w}\frac{1}{z-w} - i_{w,z}\frac{1}{z-w} = \sum_{j\in\mathbb{Z}} z^{-j-1}w^j. \tag{3.14}$$

There is a special notation for the series appearing on the right side of this equation.

Notation 3.2.8. We write $\delta(z,w) = \sum_{z,w} z^{-j-1}w^j$ and we call it the *formal delta distribution.*

Lemma 3.2.9. [10] The formal delta distribution satisfies the following identities:

(1) $(z-w)^k\partial_w^j\delta(z,w) = 0$ for $k > j$.

(2) $(z-w)\partial_w^j\delta(z,w) = j\partial_w^{j-1}\delta(z,w)$.

Proof: Using equation (3.14) we see that

$$\partial_w^j\delta(z,w) = j!\left(i_{z,w}\frac{1}{(z-w)^{j+1}} - i_{w,z}\frac{1}{(z-w)^{j+1}}\right). \tag{3.15}$$

In particular, it's clear from this expression that $(z-w)^k\partial_w^j\delta(z,w) = 0$ for $k > j$. Now

$$\begin{aligned}(z-w)\partial_w^j\delta(z,w) &= j!\left(i_{z,w}\frac{z-w}{(z-w)^{j+1}} - i_{w,z}\frac{z-w}{(z-w)^{j+1}}\right) \\ &= j\partial_w^{j-1}\delta(z,w)\end{aligned} \tag{3.16}$$

proving the second identity.

As a consequence of the previous lemma, we have $z\delta(z,w) = w\delta(z,w) \Rightarrow f(z)\delta(z,w) = f(w)\delta(z,w)$ for any $U-$valued formal distribution f.

Now we can introduce the notion of nth product of $\mathrm{End}V-$valued quantum fields. Let U be a vector superspace and define the linear map $\mathrm{Res}_z : U[[z^{\pm}]] \to U : a(z) \mapsto a_0$, i.e., extraction of the coefficient of z^{-1}. Given two quantum fields $a(z), b(z)$, their nth product is defined to be

$$a(w)_{(n)}b(w) = \mathrm{Res}_z(i_{z,w}(z-w)^n a(z)b(w) - p(a,b)i_{w,z}(z-w)^n b(w)a(z)). \tag{3.17}$$

Lemma 3.2.10. [2] Let $a(z), b(z)$ be two quantum fields on V. Then, their nth product $a(z)_{(n)}b(z)$ is again a quantum field $\forall n \in \mathbb{Z}$.

Proof: Let $v \in V$, since $a(z)$ and $b(w)$ are quantum fields, we have $b(w)v \in V((w))$ and $a(z)b(w)v \in V((z))((w))$. But $i_{z,w}(z-w)^n \in \mathbb{C}((z))[[w]]$, so the product $i_{z,w}(z-w)^n a(z)b(w)v$ is well defined as an element of $V((z))((w))$, which implies that

$$\mathrm{Res}_z i_{z,w}(z-w)^n a(z)b(w)v \in V((w)).$$

Here we have used the inclusion

$$V((z))((w)) \hookrightarrow V[[z^{\pm}, w^{\pm}]] \cong V[[w^{\pm}]][[z^{\pm}]]$$

to apply Res_z. Similarly with the second term, we get $\mathrm{Res}_z i_{w,z}(z-w)^n b(w)a(z) \in V((w))$. Therefore $a(w)_{(n)}b(w)v \in V((w))\ \forall v \in V$ and $a(w)_{(n)}b(w)$ must be a quantum field.

Lemma 3.2.11. [2] Given two $T-$covariant quantum fields $a(z), b(z)$, their nth product $a(z)_{(n)}b(z)$ is also a $T-$covariant quantum field.

Proof: By lemma (3.2.10), we just need to verify that $a(z)_{(n)}b(z)$ is $T-$covariant. Now, using the fact that $a(z), b(z)$ are $T-$covariant quantum fields, we get

$$Ta(z)b(w) = a(z)b(w)T + \partial_z a(z)b(w) + a(z)\partial_w b(w). \tag{3.18}$$

Therefore

$$\begin{aligned}
&[T, a(w)_{(n)}b(w)] \\
&\quad = \mathrm{Res}_z(i_{z,w}(z-w)^n \partial_z a(z)b(w) + i_{z,w}(z-w)^n a(z)\partial_w b(w) \\
&\qquad - p(a,b)i_{w,z}(z-w)^n \partial_w b(w)a(z) - p(a,b)i_{w,z}(z-w)^n b(w)\partial_z a(z)).
\end{aligned} \tag{3.19}$$

But

$$\begin{aligned}
&\partial_z i_{z,w}(z-w)^n a(z)b(w) + \partial_w i_{w,z}(z-w)^n a(z)b(w) \\
&= i_{z,w}(z-w)^n \partial_z a(z)b(w) + i_{z,w}(z-w)^n a(z)\partial_w b(w)
\end{aligned} \tag{3.20}$$

and

$$\begin{aligned}&\partial_z i_{w,z}(z-w)^n b(w)a(z) + \partial_w i_{w,z}(z-w)^n b(w)a(z)\\ &= i_{w,z}(z-w)^n b(w)\partial_z a(z) + i_{w,z}(z-w)^n \partial_w b(w)a(z).\end{aligned} \tag{3.21}$$

Using the fact that $\mathrm{Res}_z \circ \partial_w = \partial_w \circ \mathrm{Res}_z$, $\mathrm{Res}_z \circ \partial_z = 0$, and the two previous equations, we get $[T, a(w)_{(n)}b(w)] = \partial_w(a(w)_{(n)}b(w))$.

Another important property of nth products is the following.

Lemma 3.2.12. $\partial_w(a(w)_{(n)}b(w)) = (\partial_w a(w))_{(n)}b(w) + a(w)_{(n)}\partial_w(b(w))$.

Proof: First we have

$$\begin{aligned}\partial_w(a(w)_{(n)}b(w)) = \mathrm{Res}_z(&-ni_{z,w}(z-w)^{n-1}a(z)b(w) + i_{z,w}(z-w)^n a(z)\partial_w b(w)\\ &+p(a,b)ni_{w,z}(z-w)^{n-1}b(w)a(z) - p(a,b)i_{w,z}(z-w)^n\partial_w b(w)a(z)).\end{aligned} \tag{3.22}$$

But

$$\begin{aligned}-ni_{z,w}(z-w)^{n-1}a(z)b(w) &= -\partial_z i_{z,w}(z-w)^n a(z)b(w) + i_{z,w}(z-w)^n\partial_z a(z)b(w) \quad \text{and}\\ ni_{w,z}(z-w)^{n-1}b(w)a(z) &= \partial_z i_{w,z}(z-w)^n b(w)a(z) - i_{w,z}(z-w)^n b(w)\partial_z a(z).\end{aligned} \tag{3.23}$$

So using that $\mathrm{Res}_z \circ \partial_z = 0$ we get the result.

Now we enunciate "Dong's lemma".

Lemma 3.2.13. [2] Given a triple $(a(z), b(z), c(z))$ of pairwise local quantum fields , then $(a(z)_{(n)}b(z), c(z))$ is a local pair of quantum field, for any $n \in \mathbb{Z}$.

Proof: Fix $n \in \mathbb{Z}$. By hypothesis, there is an integer $r \geq |n|$ such that

$$\begin{aligned}&(z-w)^r[a(z), b(w)] = 0,\\ &(z-x)^r[a(z), c(x)] = 0,\\ &(w-x)^r[b(w), c(x)] = 0.\end{aligned} \tag{3.24}$$

Now let $N > 4r$. By the binomial formula we have

$$(x-w)^N = (x-w)^r \sum_{s\geq 0}\binom{N-r}{s}(x-z)^{N-r-s}(z-w)^s. \tag{3.25}$$

So

$$\begin{aligned}&(x-w)^N[c(x), a(w)_{(n)}b(w)]\\ &= \mathrm{Res}_z(x-w)^r \sum_{s\geq 0}\binom{N-r}{s}(x-z)^{N-r-s}(z-w)^s \times\\ &\quad \times (i_{z,w}(z-w)^n[c(x), a(z)b(w)] - p(a,b)i_{w,z}(z-w)^n[c(x), b(w)a(z)]).\end{aligned} \tag{3.26}$$

But

$$[c(x), a(z)b(w)] - p(a,b)[c(x), b(w)a(z)] = c(x)[a(z), b(w)] + (-1)^{p(c)}[b(w), a(z)]c(x) \quad (3.27)$$

and consequently

$$\begin{aligned}(z-w)^s(i_{z,w}(z-w)^n[c(x), a(z)b(w)] - p(a,b)i_{w,z}(z-w)^n[c(x), b(w)a(z)]) \\ = (z-w)^{n+r}(z-w)^{s-r}(c(x)[a(z), b(w)] + (-1)^{p(c)}[b(w), a(z)]c(x)) \\ = 0\end{aligned} \quad (3.28)$$

whenever $s > 2r$. Moreover, we can write

$$\begin{aligned}[c(x), a(z)b(w)] = [c(x), a(z)]b(w) + p(a,c)a(z)[c(x), b(w)] \quad \text{and} \\ [c(x), b(w)a(z)] = [c(x), b(w)]a(z) + p(b,c)b(w)[c(x), a(z)].\end{aligned} \quad (3.29)$$

Therefore, if $s \leq 2r \Rightarrow N - r - s > r$ and

$$(x-w)^r(x-z)^{N-r-s}[c(x), a(z)b(w)] = 0 = (x-w)^r(x-z)^{N-r-s}[c(x), b(w)a(z)]. \quad (3.30)$$

So, we conclude that all the terms in the equation (3.26) are zero, proving the locality of the pair $(c(x), a(z)_{(n)}b(z))$.

Lemma 3.2.14. [2] Let $a(z), b(z)$ be quantum fields and $s(a(z)) = a,\ s(b(z)) = b$. Then $s(a(z)_{(n)}b(z)) = a_{(n)}b\ \forall n \in \mathbb{Z}$.

Proof: By the equation (3.17), we have

$$\begin{aligned}a(w)_{(n)}b(w)|0\rangle = \\ \mathrm{Res}_z(i_{z,w}(z-w)^n a(z)b(w)|0\rangle - p(a,b)i_{w,z}(z-w)^n b(w)a(z)|0\rangle).\end{aligned} \quad (3.31)$$

But $a(z)|0\rangle \in V[[z]]$ and $i_{w,z}(z-w)^n \in V((w))[[z]] \Rightarrow \mathrm{Res}_z i_{w,z}(z-w)^n b(w)a(z)|0\rangle = 0$. Therefore

$$\begin{aligned}a(w)_{(n)}b(w)|0\rangle\big|_{w=0} &= \mathrm{Res}_z i_{z,w}(z-w)^n a(z)b(w)|0\rangle\big|_{w=0} \\ &= \mathrm{Res}_z z^n a(z)b \\ &= a_{(n)}b.\end{aligned} \quad (3.32)$$

The next task will be to prove the *state-field correspondence*. Before we proceed, let's make some observations. The first one is that $\partial_z a(z)$ is $T-$covariant whenever $a(z)$ is $T-$covariant. Moreover, for any local pair $(a(z), b(z))$, $(\partial_z a(z), b(z))$ is also a local pair. Now, let $\mathcal{F}'$ be the smallest subspace of $\tilde{\mathcal{F}}$ that contains I_V, is local with each element of $\mathcal{F}$, is closed under ∂_z and nth products. We can construct $\mathcal{F}'$ inductively as follows. Let $\mathcal{F}_0 = \mathrm{Span}\{\mathcal{F}, I_V\}$ and for $k \geq 0$

$$\mathcal{F}_{k+1} = \mathcal{F}_k + \sum_{n\geq 0} \partial_z^n \mathcal{F}_k + \sum_{n\geq 0} (\mathcal{F}_k)_{(n)}\mathcal{F}_k. \quad (3.33)$$

The fact that the elements of $(\mathcal{F}_k)_{(n)}\mathcal{F}_k$ $(k \geq 0)$ are local with $\mathcal{F}$ is a consequence of the Dong's lemma (3.2.13). So we put $\mathcal{F}' = \bigcup_{k\geq 0}\mathcal{F}_k$. Now, let $\bar{\mathcal{F}}$ be the subspace of $\tilde{\mathcal{F}}$ consisting of End$V-$quantum fields that are local with $\mathcal{F}$. By definition, we have the sequence of inclusions

$$\mathcal{F} \subset \mathcal{F}' \subset \bar{\mathcal{F}} \subset \tilde{\mathcal{F}}. \tag{3.34}$$

Theorem 3.2.15. [2] Let $(V, T, |0\rangle, \mathcal{F})$ be a vertex algebra. Then the linear map $s : \mathcal{F}' \to V$ is bijective.

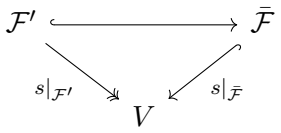
Proof: Suppose that $a(z)$ is a $T-$covariant quantum field that is local with each element of $\mathcal{F}$. Then by Dong's lemma (3.2.13), $a(z)$ is local with nth products of elements of $\mathcal{F}$ and its derivatives. This proves that the pair $(a(z), b(z))$, for $b(z) \in \mathcal{F}', a(z) \in \bar{\mathcal{F}}$, is local. Now, by the completeness axiom, the map $s : \mathcal{F}' \to V$ is surjective. This follows by the identity

$$\begin{aligned} s(a^{j_1}(z)_{(n_1)}(a^{j_2}(z)_{(n_2)}\cdots(a^{j_k}(z)_{(n_k)}I_V)\cdots) &= a^{j_1}_{n_1}\cdots a^{j_k}_{n_k}s(I_V) \quad \text{(by lemma (3.2.14))} \\ &= a^{j_1}_{n_1}\cdots a^{j_k}_{n_k}|0\rangle. \end{aligned} \tag{3.35}$$

In addition, we claim that the map $s : \bar{\mathcal{F}} \to V$ is injective. In fact, let and $a(z) \in \bar{\mathcal{F}}$ and suppose that $s(a(z)) = 0$. Now, suppose that $b(z) \in \mathcal{F}'$. Previously, we have shown that the pair $(a(z), b(z))$ is local. So there is an $N \geq 0$ such that

$$\begin{aligned} &(z-w)^N a(z)b(w) = p(a,b)(z-w)^N b(w)a(z) \Rightarrow \\ &(z-w)^N a(z)b(w)|0\rangle = p(a,b)(z-w)^N b(w)a(z)|0\rangle \Rightarrow \\ &(z-w)^N a(z)b(w)|0\rangle = 0 \quad \text{(because of lemma (3.2.7))} \Rightarrow \\ &(z-w)^N a(z)b(w)|0\rangle\big|_{w=0} = z^N a(z)b = 0. \end{aligned} \tag{3.36}$$

Since $a(z)b \in V((z))$, we can cancel out z^N in the last equation, to get $a(z)b = 0 \Rightarrow a_{(n)}b = 0\ \forall n \in \mathbb{Z}$. Since $s : \mathcal{F}' \to V$ is surjective, this shows that $a(z) = 0$. Now we have the commutative diagram



which implies that $\mathcal{F}' = \bar{\mathcal{F}}$ and that $s : \mathcal{F}' \to V$ is bijective.

Theorem (3.2.15) implies the existence of a linear map that associates to each $a \in V$ a quantum field $Y(a, z) \in \mathcal{F}'$ such that $Y(a, z)|0\rangle\big|_{z=0} = a$. Although the definition of $Y(a, z)$ is constructive, we can develop techniques to do calculations with these quantum fields. For example, we have the *Borcherds identity*:

$$\begin{aligned} i_{z,w}(z-w)^n Y(a,z)Y(b,w) - p(a,b)i_{w,z}(z-w)^n Y(b,w)Y(a,z) \\ = \sum_{j\geq 0} Y(a_{(n+j)}b, w)\partial_z^j \delta(z,w)/j! \end{aligned} \tag{3.37}$$

where $Y(a_{(n+j)}b,w) = Y(a,w)_{(n+j)}Y(b,w)$. To show this identity we need a lemma.

Lemma 3.2.16. [10] Let $D = D(z,w)$ be a local distribution, i.e. $(z-w)^N D = 0$ for some $N \geq 0$. Then

$$D = \sum_{j=0}^{N-1} c_j(w)\partial_z^j \delta(z,w)/j! \tag{3.38}$$

where $c_j(w) = \mathrm{Res}_z(z-w)^j D$. This equation is called *operator product expansion* of the formal distribution D.

Proof: The proof follows by induction on N. Let $D = \sum_{n,m\in\mathbb{Z}} D_{n,m} z^n w^m$. Suppose that $(z-w)D = 0 \Rightarrow D_{n,m} = D_{n+1,m-1}$ and

$$\begin{aligned} D &= \sum_{k\in\mathbb{Z}} \sum_{n+m=k} D_{n,m} z^n w^m \\ &= \sum_{k\in\mathbb{Z}} D_{k,0} \sum_{n\in\mathbb{Z}} z^n w^{k-n} \\ &= \sum_{k\in\mathbb{Z}} D_{k,0} w^{k+1} \delta(z,w) \\ &= c_0(w)\delta(z,w). \end{aligned} \tag{3.39}$$

Also, we have $\mathrm{Res}_z D = c_0(w)\mathrm{Res}_z\delta(z,w) = c_0(w)$. This proves the base case $N = 1$. Now, suppose that the statement of the lemma is true for some $N > 0$ and suppose that $(z-w)^{N+1}D = 0$. Define $D' = (z-w)^N D \Rightarrow (z-w)D' = 0$ and $D' = c_N(w)\delta(z,w)$ by the first part of the proof. Now, define $D'' = D - c_N(w)\partial_w^N \delta(z,w)/N!$. Then, by lemma (3.2.9) we get

$$\begin{aligned} (z-w)^N D'' &= (z-w)^N D - (z-w)^N c_N(w)\partial_w^N \delta(z,w)/N! \\ &= c_N(w)\delta(z,w) - c_N(w)\delta(z,w) \\ &= 0. \end{aligned} \tag{3.40}$$

By the induction hypothesis, we have $D'' = \sum_{j=0}^{N-1} c_j(w)\partial_w^j\delta(z,w)/j!$ where $c_j(w) = \mathrm{Res}_z(z-w)^j D'' = \mathrm{Res}_z(z-w)^j D$ because of the lemma (3.2.9) and the fact $\mathrm{Res}_z\partial_w^k\delta(z,w) = 0\ \forall k > 0$. Therefore, we get $D = \sum_{j=0}^{N} c_j(w)\partial_w^j\delta(z,w)/j!$ where $c_j(w) = \mathrm{Res}_z(z-w)^j D$.

Now, we are ready to prove the Borcherds identity. First, the pair $(Y(a,z), Y(b,z))$ is local, so there is a nonnegative integer $N \geq 0$ such that $(z-w)^N[Y(a,z),Y(b,w)] = 0$. So, multiplying the left hand side of the equation (3.37) by $(z-w)^{N+|n|}$ we get zero. Therefore, the expression

$$i_{z,w}(z-w)^n Y(a,z)Y(b,w) - p(a,b)i_{w,z}(z-w)^n Y(b,w)Y(a,z) \tag{3.41}$$

is a local distribution. Moreover, for $j \geq 0$

$$\begin{aligned} \mathrm{Res}_z(z-w)^j(i_{z,w}(z-w)^n Y(a,z)Y(b,w) - p(a,b)i_{w,z}(z-w)^n Y(b,w)Y(a,z)) \\ = Y(a,w)_{(n+j)}Y(b,w) \end{aligned} \tag{3.42}$$

and by applying the lemma (3.2.16) we get equation (3.37).

Definition 3.2.17. [2] Given two End$V-$valued quantum fields $a(z), b(z)$ we define the *normal ordered product* $: a(z)b(z) := a(z)_+ b(z) + a(z)b(z)_-$ where

$$a(z)_+ = \sum_{n<0} a_{(n)} z^{-n-1} \text{ and } a(z)_- = \sum_{n\geq 0} a_{(n)} z^{-n-1}. \tag{3.43}$$

Concerning the operators $(\cdot)_\pm : \mathrm{End}V[[z^\pm]] \to \mathrm{End}V[[z^\pm]]$, we have the useful identities:

$$\mathrm{Res}_z a(z) i_{z,w} \frac{1}{z-w} = a(w)_+ \text{ and } \mathrm{Res}_z a(z) i_{w,z} \frac{1}{z-w} = -a(w)_-. \tag{3.44}$$

Both identities can be verified by direct calculation. For example

$$\begin{aligned} \mathrm{Res}_z a(z) i_{z,w} \frac{1}{z-w} &= \mathrm{Res}_z \sum_{n\in\mathbb{Z}, k\geq 0} a_{(n)} z^{-k-1} w^k \\ &= \sum_{k\geq 0, n\in\mathbb{Z}} \sum_{n+k=-1} a_{(n)} w^k = \sum_{n<0} a_{(n)} w^{-n-1}. \end{aligned} \tag{3.45}$$

Now, using the definition of nth products of fields (3.17) for $n = -1$ we get the important identity : $Y(a,z)Y(b,z) := Y(a,z)_{(-1)}Y(b,z) = Y(a_{(-1)}b, z)$. So we define the normal ordered product in V as the bilinear map $V \otimes V \to V : (a,b) \mapsto : ab := a_{(-1)}b$. Sometimes it can be useful to explicitly open the definition of the normal ordered product between quantum fields. In general we have the identity

$$: a(z)b(z) := \sum_{n\in\mathbb{Z}} \sum_{m<0} (a_{(m)} b_{(n-m-1)} + b_{(n+m)} a_{(-m-1)}) z^{-n-1}. \tag{3.46}$$

3.3 Calculating in vertex algebra

Let $a(z) \in U((z))$ be a $U-$valued formal distribution. We define a map $U((z)) \to U[[\lambda]]$ by the formula $F_z^\lambda a(z) = \mathrm{Res}_z e^{\lambda z} a(z)$, which is called the *formal Fourier transform*. We can verify by direct computation that

$$F_z^\lambda a(z) = \mathrm{Res}_z \sum_{j\geq 0} \sum_{n\in\mathbb{Z}} \frac{\lambda^j}{j!} a_{(n)} z^{j-n-1} = \sum_{j\geq 0} \frac{\lambda^j}{j!} a_{(j)}. \tag{3.47}$$

Lemma 3.3.1. [2] Let $a(z) \in U((z))$. The formal Fourier transform satisfies the following identities:

(1) $F_z^\lambda(e^{zT}a(z)) = F_z^{\lambda+T}a(z)$.

(2) $F_z^\lambda a(-z) = -F_z^{-\lambda}a(z)$.

(3) $F_z^\lambda \partial_w^n \delta(z,w) = e^{\lambda w}\lambda^n$.

Proof: The first identity is immediate by the definition. For the second identity

$$\begin{aligned} F_z^\lambda a(z) &= \mathrm{Res}_z \sum_{j\geq 0}\sum_{n\in\mathbb{Z}} \frac{\lambda^j}{j!} a_{(n)}(-1)^{-n-1} z^{j-n-1} \\ &= \sum_{j\geq 0} \frac{\lambda^j}{j!} a_{(j)}(-1)^{-j-1} \\ &= -\sum_{j\geq 0} \frac{(-\lambda)^j}{j!} a_{(j)} \\ &= -F_z^{-\lambda}a(z). \end{aligned} \tag{3.48}$$

Now, for the third identity

$$\begin{aligned} F_z^\lambda \partial_w^n \delta(z,w) &= \mathrm{Res}_z e^{\lambda z}\partial_w^n \delta(z,w) \\ &= (-1)^n \mathrm{Res}_z e^{\lambda z}\partial_z^n \delta(z,w) \\ &= \mathrm{Res}_z (\partial_w^n e^{\lambda z})\delta(z,w) \\ &= \mathrm{Res}_z (\partial_w^n e^{\lambda w})\delta(z,w) \\ &= \partial_w^n e^{\lambda w}\mathrm{Res}_z \delta(z,w) \\ &= \partial_w^n e^{\lambda w}\cdot 1 = e^{\lambda w}\lambda^n. \end{aligned} \tag{3.49}$$

Now, let $(V, T, |0\rangle, \mathcal{F})$ be a vertex algebra. We define a bilinear map $[\cdot_\lambda \cdot] : V\otimes V \to \mathbb{C}[[\lambda]]\otimes V$ called *lambda-bracket* as follows. For $a, b \in V$ we put $[a_\lambda b] = F_z^\lambda Y(a,z)b$.

Theorem 3.3.2. [2] Let $(V, T, |0\rangle, \mathcal{F})$ be a vertex algebra. The lambda bracket satisfies the following identities:

(1) (sesquilinearity) $[Ta_\lambda b] = -\lambda[a_\lambda b]$ and $T[a_\lambda b] = [Ta_\lambda b] + [a_\lambda Tb]$.

(2) (skewsymmetry) $[b_\lambda a] = -p(a,b)[a_{-\lambda-T}b]$.

(3) (noncommutative Wick formula)

$$[a_\lambda : bc :] =: [a_\lambda b]c : + p(a,b) : b[a_\lambda c] : + \int_0^\lambda [[a_\lambda b]_\mu c] d\mu. \tag{3.50}$$

Before we prove this theorem, we need a few lemmas.

Lemma 3.3.3. [2] Let $(V, T, |0\rangle, \mathcal{F})$ be a vertex algebra. Then we have the identities $(Ta)_{(n)} = -na_{n-1}$ and $T(a_{(n)}b) = (Ta)_{(n)}b + a_{(n)}(Tb)$.

Proof: Since $\partial_z Y(a,z)|0\rangle\big|_{z=0} = TY(a,z)|0\rangle\big|_{z=0} = Ta$, we have $Y(Ta,z) = \partial_z Y(a,z)$ because of the state-field correspondence. This implies the identity $(Ta)_{(n)} = -na_{n-1}$. Now, by lemma (3.2.7) we have $e^{zT}(a_{(n)}b) = Y(a,z)_{(n)}Y(b,z)|0\rangle$ and

$$\begin{aligned} Te^{zT}(a_{(n)}b) &= (\partial_z Y(a,z))_{(n)}Y(b,z)|0\rangle + Y(a,z)_{(n)}\partial_z Y(b,z)|0\rangle \\ &= Y(Ta,z)_{(n)}Y(b,z)|0\rangle + Y(a,z)_{(n)}Y(Tb,z)|0\rangle \end{aligned} \tag{3.51}$$

by lemma (3.2.12). By putting $z = 0$ in the previous equation we get the result.

Corollary 3.3.4. [2] Let $(V, |0\rangle, T, \mathcal{F})$ be a vertex algebra. Then $T(: ab :) =: (Ta)b : + : a(Tb) :$.

Lemma 3.3.5. Let $X \in U[[z^{\pm}]]$ and $T \in \mathrm{End}U$. Then we have the identity $e^{wT}Xe^{-Tw} = e^{\mathrm{ad}(wT)}X$.

Proof: By induction, it can be verified that

$$\mathrm{ad}^k(T) = \sum_{j=0}^{k}(-1)^j \binom{k}{j} T^{k-j}XT^j. \tag{3.52}$$

Therefore

$$\begin{aligned} e^{wT}Xe^{-Tw} &= \sum_{k\geq 0}\sum_{j=0}^{k}\frac{w^{k-j}}{(k-j)!}T^{k-j}XT^{j}\frac{(-w)^{j}}{j!} \\ &= \sum_{k\geq 0}\sum_{j=0}^{k}\frac{(-1)^{j}}{(k-j)!j!}T^{k-j}XT^{j}w^{k} \\ &= \sum_{k\geq 0}\frac{1}{k!}\sum_{j=0}^{k}(-1)^{j}\binom{k}{j}T^{k-j}XT^{j}w^{k} \\ &= \sum_{k\geq 0}\frac{w^{k}}{k!}\mathrm{ad}^{k}(T)X. \end{aligned} \tag{3.53}$$

Lemma 3.3.6. Let $f(z)\in U[[z^{\pm}]]$ be a formal distribution. Then $e^{w\partial_z}f(z)=f(z+w)$, where the terms $(z+w)^{-1}$ are expanded in $\mathbb{C}((z))((w))$, i.e., with positive powers of w.

Proof: Let $f(z)=\sum_{n\in\mathbb{Z}}f_nz^n$. Then

$$\partial_z^k f(z)=\sum_{n\in\mathbb{Z}}f_n n(n-1)\cdots(n-k+1)z^{n-k} \tag{3.54}$$

and

$$\begin{aligned} e^{w\partial_z}f(z) &= \sum_{k\geq 0}\frac{w^{k}}{k!}\partial_z^k f(z) \\ &= \sum_{n\in\mathbb{Z}}f_n\sum_{k\geq 0}\frac{n(n-1)\cdots(n-k+1)}{k!}z^{n-k}w^{k} \\ &= \sum_{n\in\mathbb{Z}}f_n i_{z,w}(z+w)^{n} \\ &= f(z+w). \end{aligned} \tag{3.55}$$

Lemma 3.3.7. [2] Let $(V,T,|0\rangle,\mathcal{F})$ be a vertex algebra. Then we have the *skewsymmetry* relation $Y(a,z)b=p(a,b)e^{zT}Y(b,-z)a\ \ \forall a,b\in V$.

Proof: First, for $a\in V$ we have the sequence of identities

$$\begin{aligned} e^{wT}Y(a,z)e^{-Tw} &= e^{\mathrm{ad}(wT)}Y(a,z) \quad \text{(by lemma (3.3.5))} \\ &= e^{w\partial_z}Y(a,z) \quad \text{(by translation covariance)} \\ &= Y(a,z+w) \quad \text{(by lemma (3.3.6)).} \end{aligned} \tag{3.56}$$

Now let $b \in V$, by locality there is a integer $N \geq 0$ such that $(z-w)^N Y(a,z)Y(b,w) = p(a,b)(z-w)^N Y(b,w)Y(a,z)$. Applying both sides we to the vacuum, we get

$$\begin{aligned}(z-w)^N Y(a,z)e^{wT}b &= p(a,b)(z-w)^N Y(b,w)e^{zT}a \\ &= p(a,b)(z-w)^N e^{-zT}Y(b,w-z)a \quad \text{(by equation (3.56))},\end{aligned} \tag{3.57}$$

because of Lemma (3.2.7). We remark that the terms $(z+w)^{-1}$ appearing in $Y(a,w-z)$ are expanded in $\mathbb{C}((z))((w))$. Now, with N big enough, there is no negative power of $(z-w)$ in the equation (3.57), so we get an identity in $V[[z]][[w]]$. Therefore, we can put $w=0$ in the equation (3.57) and cancel z^N on both sides to get the skewsymmetry relation.

Now we can give the proof of Theorem (3.3.2). Lemma (3.3.3) together with the relation $[a_\lambda b] = \sum_{j\geq 0} \lambda^j a_{(j)}b/j!$ immediately implies the sesquilinearity identities of the theorem (3.3.2). Now, for the skewsymmetry we use Lemma (3.3.7) to obtain

$$\begin{aligned}F_z^\lambda Y(a,z)b &= p(a,b)F_z^\lambda e^{zT}Y(b,-z)a \\ &= -p(a,b)F_z^{-\lambda}e^{-zT}Y(b,z)a \quad \text{(by lemma (3.3.1))} \\ &= -p(a,b)F_z^{-\lambda-T}Y(b,z)a \\ &= -p(a,b)[b_{-\lambda-T}a].\end{aligned} \tag{3.58}$$

It remains to prove the noncommutative Wick formula. The Borcherds identity (3.37) for $n=0$ yields the commutator formula

$$\begin{aligned}&[Y(a,z),Y(b,w)] = \sum_{j\geq 0} Y(a_{(j)}b,w)\partial_w^j \delta(z,w)/j! \Rightarrow \\ &Y(a,z)Y(b,w)c - p(a,b)Y(b,w)Y(a,z)c = \sum_{j\geq 0} Y(a_{(j)}b,w)c\partial_w^j\delta(z,w)/j! \ \ \forall c \in V.\end{aligned} \tag{3.59}$$

Applying F_z^λ to both sides of this equation and using the third identity of lemma (3.3.1), we get

$$\begin{aligned}[a_\lambda Y(b,w)c] - p(a,b)Y(b,w)[a_\lambda c] &= \sum_{j\geq 0} Y(a_{(j)}b,w)c\lambda^j e^{w\lambda}/j! \\ &= e^{\lambda w}Y\left(\sum_{j\geq 0}\frac{\lambda^j}{j!}a_{(j)}b\ ,w\right)c \\ &= e^{w\lambda}Y([a_\lambda b],w)c.\end{aligned} \tag{3.60}$$

Now, by applying $\mathrm{Res}_z w^{-1}$ on both sides of this equation we obtain

$$\begin{aligned}[a_\lambda : bc :] - p(a,b) : b[a_\lambda c] : &= \mathrm{Res}_w w^{-1}e^{\lambda w}Y([a_\lambda b],w)c \\ &= \mathrm{Res}_w\left(\int_0^\lambda e^{\mu w}d\mu + w^{-1}\right)Y([a_\lambda b],w)c \\ &= \int_0^\lambda \mathrm{Res}_w e^{\mu w}Y([a_\lambda b],w)c\ d\mu + \mathrm{Res}_w w^{-1}Y([a_\lambda b],w)c \\ &= \int_0^\lambda [[a_\lambda b]_\mu c]d\mu + : [a_\lambda b]c :\end{aligned} \tag{3.61}$$

which finishes the proof of the noncommutative Wick formula.

3.4 Fermionic vertex algebras

Definition 3.4.1. [10] Let $V = V_{\bar{0}} \oplus V_{\bar{1}}$ be a vector superspace. A *skew-supersymmetric* bilinear form on V is a bilinear map $\langle \cdot, \cdot \rangle : V \otimes V \to \mathbb{C}$ satisfying $\langle a, b \rangle = -p(a,b)\langle b, a \rangle$ for homogeneous elements $a, b \in V$.

In this section we will give our first example of a vertex algebras. Let $A = A_{\bar{0}} \oplus A_{\bar{1}}$ be a vector superspace with a skew-supersymmetric bilinear form $\langle \cdot, \cdot \rangle$, and write $A[t^{\pm}] = \mathbb{C}[t^{\pm}] \otimes A$. Now we define the *Clifford affinization* of A to be the Lie superalgebra $\hat{A} = A[t^{\pm}] \oplus \mathbb{C}K$ with commutator given by

$$[at^m, bt^n] = \langle a, b \rangle \delta_{m,-n-1} K \quad \text{and} \quad [K, \cdot\,] = 0. \tag{3.62}$$

The parity on $\hat{A}$ is defined by putting $p(t) = 0$ (i.e $\mathbb{C}[t^{\pm}]$ is defined to be a purely even superalgebra) and $p(K) = 0$. So if a homogeneous in A, we have $p(at^m) = p(a)\ \forall m \in \mathbb{Z}$. Define a derivation $D = -d/dt$ of $\hat{A}$ by putting $D(at^n) = -nat^{n-1}$ and $D(K) = 0$. We claim that this in fact defines a derivation of $\hat{A}$. The only nontrivial case that we need to check is that one $D([at^m, bt^n]) = [D(at^m), bt^n] + [at^m, D(bt^n)]$. But

$$[D(at^m), bt^n] + [at^m, D(bt^n)] = -\langle a, b \rangle (m+n) \delta_{m,-n} K = 0 \tag{3.63}$$

and $D([at^m, bt^n]) = \langle a, b \rangle \delta_{m,-n-1} D(K) = 0$. Since D is a derivation of the Lie superalgebra $\hat{A}$, it can be extended to a derivation of its universal enveloping algebra $\mathcal{U}(\hat{A})$. Its expression is the usual one:

$$D(x_1 \cdots x_k) = \sum_{i=1}^{k} x_1 \cdots D(x_i) \cdots x_k, \quad x_i \in \hat{A}\ (1 \leq i \leq k) \tag{3.64}$$

and it's well defined because D is a derivation of $\hat{A}$, i.e. $D(x \otimes y - p(x,y) y \otimes x - [x, y]) = 0\ \forall x, y \in \hat{A}$.

We have the subalgebra $\hat{A}_+ = A[t^+] \oplus \mathbb{C}K \subset \hat{A}$. Define the $\hat{A}_+-$module $\mathbb{C}C$ with the following action $at^m C = 0$ for $m \geq 0$ and $KC = C$ (since $K \in Z(\hat{A})$, this implies that K is the identity map). Now, we induce it to a representation of $\hat{A}$ by defining the $\hat{A}-$module $F(A) = \mathcal{U}(\hat{A}) \otimes_{\mathcal{U}(\hat{A}_+)} \mathbb{C}C$. We claim that $F(A)$ has the structure of a vertex algebra with vacuum $|0\rangle = 1 \otimes C$, translation operator $T = D \otimes \mathbb{1}_{\mathbb{C}C}$ and quantum fields

$$\mathcal{F} = \left\{ a(z) = \sum_{n \in \mathbb{Z}} (at^n) z^{-n-1} \,\middle|\, a \in A \right\}. \tag{3.65}$$

Now, lets verify the vertex algebras axioms for $F(A)$. For the vacuum axiom, just note that $T|0\rangle = D(1) \otimes C = 0$ and for $\mu \in \mathcal{U}(\hat{A})$ we have

$$\begin{aligned}[T, at^m]\mu|0\rangle &= (D(at^m\mu) - at^m D(\mu)) \otimes C \\ &= (-mat^{m-1}\mu + at^m D(\mu) - at^m D(\mu)) \otimes C \\ &= (-mat^{m-1})\mu|0\rangle.\end{aligned} \tag{3.66}$$

Therefore $[T, at^m] = -mat^{m-1} \Rightarrow [T, a(z)] = \partial_z a(z)$ which proves the translation covariance axiom. From the PBW-theorem applied for Lie superalgebras (3.1.4), we have that $F(A)$ is spanned by

$$a_1 t^{-m_1} \cdots a_k t^{-m_k} \otimes C \quad a_i \in A, m_i > 0 \tag{3.67}$$

which proves the completeness axiom. Moreover, for each $a_1 t^{-m_1} \cdots a_k t^{-m_k} \otimes C \in F(A)$, the commutator formula for the Clifford affinization (3.62) yields

$$\begin{aligned}at^n(a_1 t^{-m_1} \cdots a_k t^{-m_k}|0\rangle) &= a_1 t^{-m_1} \cdots a_k t^{-m_k} at^n \otimes C \\ &= a_1 t^{-m_1} \cdots a_k t^{-m_k} \otimes at^n C \\ &= 0\end{aligned} \tag{3.68}$$

for $n \gg 0$ (or more precisely, for $n > \text{Max}\{m_i\}$). Therefore $\mathcal{F}$ is a collection of quantum fields. It remains to show that the quantum fields in $\mathcal{F}$ are mutually local, but

$$\begin{aligned}[a(z), b(w)] &= \sum_{m,n\in\mathbb{Z}} [at^m, bt^n] z^{-m-1} w^{-n-1} \\ &= \sum_{m,n\in\mathbb{Z}} \langle a, b\rangle \delta_{m,-n-1} K z^{-m-1} w^{-n-1} \\ &= \langle a, b\rangle K \sum_{m\in\mathbb{Z}} z^{-m-1} w^m \\ &= \langle a, b\rangle K \delta(z, w)\end{aligned} \tag{3.69}$$

which is local.

The equation equation $[a(z), b(w)] = \langle a, b\rangle K\delta(z, w)$ together with the Borcherds identity (3.37) implies that $a_{(0)}b = \langle a, b\rangle$ (since K acts on $F(A)$ as the identity map, we will omit it here). Therefore, identifying a with $at^{-1} \otimes |0\rangle$, we've

$$[a_\lambda b] = \langle a, b\rangle|0\rangle. \tag{3.70}$$

In expressions of this type, it will be more convenient to just write $[a_\lambda b] = \langle a, b\rangle$, i.e. 1 for the vacuum.

Notation 3.4.2. It's important to emphasize that normally we will continue writing $a(z)$ for the fields $Y(at^{-1}|0\rangle, z)$. Also, in lambda-bracket relations, the vacuum $|0\rangle$ will be omitted for convenience. Maybe also note that $a_{(n)} = at^n\big|_{F(A)}$.

3.4.1 The oscillator algebra

Definition 3.4.3. [10] The *oscillator algebra* is the infinite dimensional Lie algebra $\mathcal{A}$ with basis $\{a_m\}_{m\in\mathbb{Z}} \cup \{\mathbb{1}\}$ and relations $[a_m, a_n] = m\delta_{m,-n}\mathbb{1}$, and $\mathbb{1}$ central.

There is a natural class of representations of the oscillator algebra, called the *bosonic Fock space*.

Definition 3.4.4. [10] let m be a complex number and $\{\epsilon_j \mid j > 0\}$ be a set of nonzero complex numbers. The Fock space is a representation of the oscillator algebra $\mathcal{A}$, consisting of a vector space $B^m = \mathbb{C}[x_1, x_2, \dots]$ of polynomials in infinitely many variables and action:

$$\begin{aligned} &\text{For } \ j > 0 \\ &a_j \mapsto \epsilon_j \frac{\partial}{\partial x_j} \\ &a_{-j} \mapsto \epsilon_j^{-1} j x_j \ \ \text{(multiplication by } \epsilon_j^{-1} j x_j) \\ &a_0 \mapsto m\mathbb{1}_B \\ &\mathbb{1} \mapsto \mathbb{1}_B. \end{aligned} \tag{3.71}$$

Remark 3.4.5. It's important to note that the ϵ_j can be absorbed into redefinition of the variables x_j. However, we will be studying multiple actions of $\mathcal{A}$. So it's useful to keep the ϵ_j in the notation.

Theorem 3.4.6. [10] The representation B^m of the oscillator algebra $\mathcal{A}$ is irreducible.

Proof: For any partition $\lambda = (1^{n_1} 2^{n_2} \dots)$, we have a monomial $x_\lambda = x_1^{n_1} \cdots x_k^{n_k} \in B^m$ and an operator $a_1^{n_1} \cdots a_k^{n_k}$ which acts on B^m as the operator $x_\lambda^* = \prod_{j=1}^k \epsilon_j^{n_j} \frac{\partial^{n_j}}{\partial x_j^{n_j}}$ on B^m. Define a partial order on $\mathcal{P}$ by putting $\lambda <_m \mu$ whenever $m_j(\lambda) \le m_j(\mu)\ \forall j$.

It's clear that $x_\mu^*(x_\lambda) = \delta_{\mu\lambda} \prod_{j\ge1} \epsilon_j^{m_j(\mu)} m_j(\mu)!$ for any two partitions μ, λ with a different set of parts or satisfying $\lambda <_m \mu$. Now, suppose that $W \subset B^m$ is an invariant subspace and $v \in W$. Then we can write $v = \sum_{\lambda\in S} \alpha_\lambda x_\lambda$ where $\alpha_\lambda \in \mathbb{C}$ and $S \subset \mathcal{P}$ is finite. Let $\mu \in S$. Since S is finite, we can admit that μ is maximal with relation to the partial order $<_m$ on S. Therefore

$$x_\mu^*(v) = \alpha_\mu \prod_{j\ge1} \epsilon_j^{m_j(\mu)} m_j(\mu)! \in W \Rightarrow 1 \in W \tag{3.72}$$

and by applying $a_{-j},\ j > 0$ to 1, we can reconstruct any monomial of B^m. This shows that

$W = B^m$ and that B^m is irreducible.

Theorem 3.4.7. [10] Let V be a representation of the oscillator algebra $\mathcal{A}$. Suppose that there exists $v \in V$ such that $a_0(v) = mv,\ m \in \mathbb{C}, \mathbb{1}(v) = v$ and $a_n(v) = 0,\ n > 0$ (in this case v is said to be a *singular* vector). Then the map $B^m \to V : \prod_{j\geq 1} \left(\frac{j}{\epsilon_j}\right)^{n_j} x_1^{n_1} \cdots x_k^{n_k} \mapsto a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v)$ is injective. In particular, monomials of the form $a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v)$ are linearly independent.

Proof: Let $W = \text{Span}\{a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v) \ | n_j \geq 0, k \geq 0\}$. For $n > 0$, let's see how a_j acts on a power $a_{-j}^{n_j}(v)$. Applying the commutation relation $[a_j, a_{-j}] = j\mathbb{1}$ $(j > 0)$ repeatedly, we get

$$\begin{aligned} a_j a_{-j}^{n_j}(v) &= a_{-j} a_j a_{-j}^{n_j-1}(v) + j a_{-j}^{n_j-1}(v) \\ &= a_{-j}^2 a_j a_{-j}^{n_j-2}(v) + j2a_{-j}^{n_j-1}(v) \\ &= \cdots \\ &= a_{-j}^{n_j} a_j(v) + j n_j a_{-j}^{n_j-1}(v) \\ &= j n_j a_{-j}^{n_j-1}(v). \end{aligned} \tag{3.73}$$

Now, since $[a_j, a_i] = 0$ for $i + j \neq 0$, we get the action $a_j = j\frac{\partial}{\partial a_{-j}}$ on W for $j > 0$. By the same reason, for $j < 0$, a_{-j} acts on W just by multiplication. Therefore, we've a homomorphism of $\mathcal{A}-$modules

$$f : B^\mu \to V : \prod_{j\geq 1} \left(\frac{j}{\epsilon_j}\right)^{n_j} x_1^{n_1} \cdots x_k^{n_k} \mapsto a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v) \tag{3.74}$$

with nonzero image. Therefore, by theorem (3.4.6) and Schur's lemma, $\ker(f) = \{0\}$ which implies the claim of the theorem.

An important observation about the previous proof is that if $W = V$, then V is isomorphic to B^μ.

Now define the anti-linear map $\omega : a_n \mapsto a_{-n},\ \mathbb{1} \mapsto \mathbb{1}$. By anti-linear we mean $\omega(\alpha a_n) = \overline{\alpha} a_{-n}$. This map is an anti-involution, i.e. it's an involution $\omega^2 = \mathbb{1}_{\mathcal{A}}$ and an anti-homomorphism:

$$\omega([a_n, a_m]) = [\omega(a_m), \omega(a_n)]\ \forall m, n. \tag{3.75}$$

Therefore ω can be extended to an anti-homomorphism of $\mathcal{U}(\mathcal{A})$.

Notation 3.4.8. Let V be a representation of the oscillator algebra with singular vector v. For a partition $\lambda = (\lambda_1, \ldots, \lambda_r)$, it will be useful to write $a_{\pm\lambda}(v) = a_{\pm\lambda_1} \cdots a_{\pm\lambda_r}(v)$. In particular, if $\lambda = (1^{n_1}, \ldots, k^{n_k})$, we have

$$a_{-\lambda}(v) = a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v). \tag{3.76}$$

Theorem 3.4.9. [10] Let V be a representation of the oscillator algebra $\mathcal{A}$ generated by a singular vector v. Then, there is a unique hermitian form $\langle \cdot | \cdot \rangle$ on V satisfying the properties:

(1) $\langle \cdot | \cdot \rangle$ is contravariant with respect to ω, i.e. $\langle a\psi | \psi' \rangle = \langle \psi | \omega(a)\psi' \rangle$.

(2) $\langle v | v \rangle = 1$.

Furthermore the monomials $a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v)$ are orthogonal under $\langle \cdot | \cdot \rangle$ and

$$\langle a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v) | a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v) \rangle = \prod_{j=1}^{k} j^{n_j} j!. \tag{3.77}$$

Proof: For uniqueness, let $a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v)$, $a_{-1}^{m_1} \cdots a_{-k}^{m_k}(v)$ $(n_i, m_i \geq 0)$ be two monomials in V. If they aren't equal, we can assume that there exists an i, $(1 \leq i \leq k)$ such that $n_i > m_i$ without lost of generality. In this case, denote $\tilde{n}_j = n_j$ for $j \neq i$ and $\tilde{n}_i = 0$. Then, using (1), we get

$$\begin{aligned} \langle a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v) | a_{-1}^{m_1} \cdots a_{-k}^{m_k}(v) \rangle &= \langle a_{-i}^{n_i} a_{-1}^{\tilde{n}_1} \cdots a_{-k}^{\tilde{n}_k}(v) | a_{-1}^{m_1} \cdots a_{-k}^{m_k}(v) \rangle \\ &= \langle a_{-1}^{\tilde{n}_1} \cdots a_{-k}^{\tilde{n}_k}(v) | a_i^{n_i} \cdot a_{-1}^{m_1} \cdots a_{-k}^{m_k}(v) \rangle \\ &= \langle a_{-1}^{\tilde{n}_1} \cdots a_{-k}^{\tilde{n}_k}(v) | i^{n_i} \frac{\partial^{n_i}}{\partial a_{-i}^{n_i}} \cdot a_{-1}^{m_1} \cdots a_{-k}^{m_k}(v) \rangle \\ &= 0 \end{aligned} \tag{3.78}$$

because $n_i > m_i$. Moreover

$$\begin{aligned} a_1^{n_1} \cdots a_k^{n_k} a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v) &= \prod_{j \geq 1} j^{n_j} \frac{\partial^{n_j}}{\partial a_{-j}^{n_j}} a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v) \\ &= \prod_{j \geq 1} j^{n_j} n_j! v. \end{aligned} \tag{3.79}$$

By utilizing equations (1) and (2), we obtain equation (3.77). To establish existence, define the inner product $\langle \cdot, \cdot \rangle$ such that the monomials $a_{-1}^{n_1} \cdots a_{-k}^{n_k}(v)$ are orthogonal, while simultaneously ensuring that equation (3.77) holds. So immediately we have $\langle v | v \rangle = 1$. Now let

$\lambda = (1^{n_1}, \dots, k^{n_k}), \mu = (1^{m_1}, \dots, k^{m_k})$ and $\psi = a_{-\lambda}(v),\ \psi' = a_{-\mu}(v)$. If $j \geq 1$, we want to show that

$$\langle a_{-j}\psi|\psi'\rangle = \langle \psi|a_j\psi'\rangle. \tag{3.80}$$

There are two cases: If $j > k$, then $\langle a_{-j}\psi|\psi'\rangle = 0$ because $a_{-j}\psi \neq \psi'$. Also $a_j\psi' = a_{-\mu}a_j(v) = 0$, so we get equation (3.80). Now, if $1 \leq j \leq k$, equation (3.80) is equivalent to the identity

$$\delta_{\tilde{\lambda}\mu} z_\mu = j m_j \delta_{\lambda\tilde{\mu}} z_\lambda \tag{3.81}$$

where $\tilde{\lambda} = (1^{n_j}, \dots, j^{n_j-1}, \dots),\ \tilde{\mu} = (1^{m_1}, \dots, j^{m_j+1}, \dots)$ and $z_\nu = \prod_{j\geq 1} j^{m_j(\nu)} m_j(\nu)!$ for $\nu \in \mathcal{P}$. But it's clear that $\tilde{\lambda} = \mu \iff \lambda = \tilde{\mu}$ and in case of the equality, that $z_\mu = j m_j z_\lambda$.

3.4.2 Charged free fermions

Let $A = \mathbb{C}\psi^+ \oplus \mathbb{C}\psi^-$ be a purely odd vector superspace of dimension 2. We put on $\mathcal{A}$ a skew-supersymmetric bilinear form $\langle \cdot, \cdot \rangle$ with matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ relative to the basis $\{\psi^+, \psi^-\}$. In the last section an $\hat{A}-$module $F = F(A)$ was constructed. It's a vertex algebra with fields satisfying the relations

$$[\psi^\pm_\lambda \psi^\pm] = 0, \quad [\psi^\pm_\lambda \psi^\mp] = 1 \ \text{ and } \ [K_\lambda -] = 0. \tag{3.82}$$

Definition 3.4.10. [10] The *bozonization* of charged free fermions is the normal ordered product $\alpha =: \psi^+\psi^- :$.

It's important to note that $p(\alpha) = 0$.

Lemma 3.4.11. The bozonization of charged free fermions satisfies the identities:

$$(1)\ [\psi^\pm_\lambda \alpha] = \mp\psi^\pm \ \text{ and } \ (2)\ [\alpha_\lambda \alpha] = \lambda. \tag{3.83}$$

Proof: The first identity is immediate by the noncommutative formula (3.3.2) and relations (3.82). Now, using the skew symmetry of the lambda bracket we get $[\alpha_\lambda \psi^\pm] = \pm\psi^\pm$. By the noncommutative Wick formula

$$\begin{aligned} [\alpha_\lambda \alpha] &=: [\alpha_\lambda \psi^+]\psi^- : + p(\alpha, \psi^+) : \psi^+ [\alpha_\lambda \psi^-] : + \int_0^\lambda [[\alpha_\lambda \psi^+]_\mu \psi^-] d\mu \\ &=: \psi^+\psi^- : + : \psi^+(-\psi^-) : + \int_0^\lambda [\psi^+_\mu \psi^-] d\mu \\ &= \int_0^\lambda 1 d\mu = \lambda. \end{aligned} \tag{3.84}$$

It's important to note that $[\alpha_\lambda \alpha] = \lambda$ implies $\alpha_{(1)}\alpha = |0\rangle$ and $\alpha_{(j)}\alpha = 0,\ j \geq 0, j \neq 1$. Then $[\alpha(z), \alpha(w)] = \mathbb{1}_F \partial_w \delta(z, w) \Rightarrow [\alpha_{(m)}, \alpha_{(n)}] = m\delta_{m,-n}\mathbb{1}_F$, i.e. $\mathcal{A} = \{\alpha_{(m)}\}_{m\in\mathbb{Z}} \cup \{\mathbb{1}_F\}$ is the oscillator algebra.

Lemma 3.4.12. Define $L = \frac{1}{2} : \alpha\alpha :$, then the following identities are satisfied

$$(1)\ [L_\lambda \alpha] = (T+\lambda)\alpha \quad \text{and} \quad (2)\ [L_\lambda L] = (T+2\lambda)L + \frac{\lambda^3}{12}. \tag{3.85}$$

Remark 3.4.13. We remark that equation (2) means that $L(z)$ is a Virasoro formal distribution with central charge 1.

Proof: First we use the noncommutative Wick formula to compute

$$\begin{aligned}
[\alpha_\lambda : \alpha\alpha :] &=: [\alpha_\lambda \alpha]\alpha : + p(\alpha,\alpha) : \alpha[\alpha_\lambda \alpha] : + \int_0^\lambda [[\alpha_\lambda \alpha]_\mu \alpha] d\mu \\
&= 2\lambda\alpha + \int_0^\lambda [\lambda_\mu \alpha] d\mu \\
&= 2\lambda\alpha \quad (\text{because } [1_\mu \alpha] = 0) \\
&\Rightarrow [\alpha_\lambda L] = \lambda\alpha.
\end{aligned} \tag{3.86}$$

Now, using the skew symmetry of the lambda bracket we deduce $[L_\lambda \alpha] = (T+\lambda)\alpha$. Applying again the noncommutative Wick formula and corollary (3.3.4) we get

$$\begin{aligned}
[L_\lambda : \alpha\alpha :] &=: [L_\lambda \alpha]\alpha : + p(L,\alpha) : \alpha[L_\lambda \alpha] : + \int_0^\lambda [[L_\lambda \alpha]_\mu \alpha] d\mu \\
&=: (T+\lambda)\alpha\alpha : + : \alpha(T+\lambda)\alpha : + \int_0^\lambda [(T+\lambda)\alpha_\mu \alpha] d\mu \\
&= 2\lambda : \alpha\alpha : + : (T\alpha)\alpha : + : \alpha(T\alpha) : + \int_0^\lambda (-\mu^2 + \lambda\mu) d\mu \\
&= 4\lambda L + 2TL + \frac{\lambda^3}{6}
\end{aligned} \tag{3.87}$$

which proves the claim.

Lemma 3.4.14. The state field correspondence yields

$$L = s(L(z)) = \frac{1}{2}(: (T\psi^+)\psi^- : + : (T\psi^-)\psi^+ :). \tag{3.88}$$

Proof: Using equation (3.46) and the commutation relations between $\psi^+(z), \psi^-(z)$ we get

$$\begin{aligned}
\alpha_{(-1)} &= \sum_{m<0} \psi^+_{(m)}\psi^-_{(-m-2)} + \psi^-_{(-m-1)}\psi^+_{(-m-1)} \\
&= \sum_{m\in\mathbb{Z}} \psi^+_{(m)}\psi^-_{(-m-2)}.
\end{aligned} \tag{3.89}$$

But, $s(L(z)) = \frac{1}{2}\alpha_{(-1)}\alpha = \frac{1}{2}\alpha^2_{(-1)}|0\rangle$. Now, since $\psi^{\pm}_m|0\rangle = 0$ for $m \geq 0$ we have $\alpha_{(-1)}|0\rangle = \psi^-_{(-1)}\psi^+_{(-1)}|0\rangle$ and

$$\begin{aligned}\alpha^2_{(-1)}|0\rangle &= \alpha_{(-1)}\psi^-_{(-1)}\psi^+_{(-1)}|0\rangle \\ &= \psi^-_{(-2)}\psi^+_{(-1)}|0\rangle + \psi^+_{(-2)}\psi^-_{(-1)}|0\rangle \\ &= (T\psi^-)_{(-1)}\psi^+ + (T\psi^+)_{(-1)}\psi^- \ \text{ (by Lemma (3.3.3))} \\ &=: (T\psi^-)\psi^+ : + : (T\psi^+)\psi^- :\end{aligned} \tag{3.90}$$

which concludes the proof.

Lemma 3.4.15. The following identity is satisfied: $[L_\lambda \psi^\pm] = (T + \lambda/2)\psi^\pm$.

Proof: Using the relations (3.82) and Theorem (3.3.2) we deduce

$$[\psi^\pm_\lambda T\psi^\pm] = 0 \quad \text{and} \quad [\psi^\pm_\lambda T\psi^\mp] = \lambda. \tag{3.91}$$

Now, using the noncommutative formula we get

$$\begin{aligned}&[\psi^+_\lambda : (T\psi^+)\psi^- :] = -T\psi^+, \ \ [\psi^+_\lambda : (T\psi^-)\psi^+ :] = \lambda\psi^+, \\ &[\psi^-_\lambda : (T\psi^+)\psi^- :] = \lambda\psi^+ \quad \text{and} \quad [\psi^-_\lambda : (T\psi^-)\psi^+ :] = -T\psi^-.\end{aligned} \tag{3.92}$$

To get the correct equations, it's important to note that $p(\psi^\pm, \psi^\pm) = p(T\psi^\pm, \psi^\pm) = -1$. Therefore $[\psi^\pm_\lambda L] = \frac{1}{2}(\lambda - T)\psi^\pm \Rightarrow [L_\lambda \psi^\pm] = (T + \frac{1}{2}\lambda)\psi^\pm$ by the skew symmetry identity.

Definition 3.4.16. [10] Let $(V, |0\rangle, T, \mathcal{F})$ be a vertex algebra and $L(z) \in \mathcal{F}'$ a Virasoro formal distribution, i.e.,

$$[L_\lambda L] = (T + 2\lambda)L + \frac{\lambda^3}{12}C. \tag{3.93}$$

A field $a(z)$ is said to have of *conformal weight* $\Delta \in \mathbb{C}$ if $[L_\lambda a] = (T + \Delta\lambda)a + o(\lambda)$.

Lemmas (3.4.12) and (3.4.15) say that the the quantum fields $\psi^\pm(z), \alpha(z)$ and $L(z)$ are eigendistributions of conformal weights $1/2, 1$ and 2 relative to $L(z)$, respectively. We will not delve into the theory of conformal weights, but just say that when a field $a(z)$ has conformal weight Δ, then it's convenient to write

$$a(z) = \sum_{n \in -\Delta + \mathbb{Z}} a_n z^{-n-\Delta}, \tag{3.94}$$

i.e. $a_{(n)} = a_{n-\Delta+1}, \ n \in \mathbb{Z}$ and $a_n = a_{(n+\Delta-1)}, \ n \in -\Delta + \mathbb{Z}$. Because then $[L_0, a_n] = -na_n$.

Following this convention we will write

$$\begin{aligned}\psi^{\pm}(z) &= \sum_{n\in-\frac{1}{2}} \psi_n^{\pm} z^{-n-1/2} \\ \alpha(z) &= \sum_{n\in\mathbb{Z}} \alpha_n z^{-n-1} \\ L(z) &= \sum_{n\in\mathbb{Z}} L_n z^{-n-2}.\end{aligned} \tag{3.95}$$

Now let's summarize what we've obtained concerning these fields.

Theorem 3.4.17. Consider the Fermionic vertex algebra F of charged free fermions. The following identities are satisfied:

$$\begin{aligned}&[\psi_\lambda^{\pm}\psi^{\mp}] = 0, \quad [\psi_\lambda^{\pm}\psi^{\pm}] = 1 \\ &[\alpha_\lambda\alpha] = \lambda, \quad [\alpha_\lambda\psi^{\pm}] = \pm\psi^{\pm} \\ &[L_\lambda\alpha] = (T+\lambda)\alpha, \quad [L_\lambda\psi^{\pm}] = (T+\lambda/2)\psi^{\pm}.\end{aligned} \tag{3.96}$$

Theorem (3.4.17) can be translated to the following commutation relations for $j > 0$

$$\begin{aligned}&[\psi_m^{\pm}, \psi_n^{\pm}] = 0, \quad [\psi_m^{\pm}, \psi_n^{\mp}] = \delta_{m+n,0} \\ &[\alpha_m, \alpha_n] = m\delta_{m+n,0}, \quad [\alpha_m, \psi_n^{\pm}] = \pm\psi_{m+n}^{\pm} \\ &[L_m, \alpha_n] = -n\alpha_{m+n}, \quad [L_m, \psi_n^{\pm}] = -\left(n+\frac{m}{2}\right)\psi_{m+n}^{\pm}.\end{aligned} \tag{3.97}$$

In particular, we have

$$\begin{aligned}&(1)\ [\alpha_0, \psi_n^{\pm}] = \pm\psi_n^{\pm} \quad n \in -\frac{1}{2}+\mathbb{Z}, \\ &(2)\ [L_0, \alpha_n] = -n\alpha_n \quad n \in \mathbb{Z}, \\ &(3)\ [L_0, \psi_n^{\pm}] = -n\psi_n^{\pm} \quad n \in -\frac{1}{2}+\mathbb{Z}.\end{aligned} \tag{3.98}$$

3.5 The Boson-Fermion correspondence

3.5.1 The first part of the Boson-Fermion correspondence

Theorem 3.5.1. [10] We have a double grading $F = \bigoplus_{m\in\mathbb{Z}} \bigoplus_{k\geq 0} F_k^m$ relative to the actions of α_0 and L_0 on F respectively, i.e., $\alpha_0 v = mv$ and $L_o v = kv$ for $v \in F_k^m$. A state in the eigenspace F_k^m is said to have *charge* m and *energy* k.

Proof: By the PBW-theorem (3.1.4) we have that F is spanned by vectors of the form

$$\psi^-_{-j_t}\cdots\psi^-_{-j_1}\psi^+_{-i_s}\cdots\psi^+_{-i_1}|0\rangle \tag{3.99}$$

for $j_t>\cdots>j_1>0,\ i_s>\cdots>i_1>0$ and $i_k,j_l\in-\frac{1}{2}+\mathbb{Z}$. By the first and third equations of (3.98), we immediately get

$$\begin{aligned}\alpha_0\psi^-_{-j_t}\cdots\psi^-_{-j_1}\psi^+_{-i_s}\cdots\psi^+_{-i_1}|0\rangle &= \\ &= (-t+s)\psi^-_{-j_t}\cdots\psi^-_{-j_1}\psi^+_{-i_s}\cdots\psi^+_{-i_1}|0\rangle \\ L_0\psi^-_{-j_t}\cdots\psi^-_{-j_1}\psi^+_{-i_s}\cdots\psi^+_{-i_1}|0\rangle &\\ &= \Big(\sum i_k+\sum j_l\Big)\,\psi^-_{-j_t}\cdots\psi^-_{-j_1}\psi^+_{-i_s}\cdots\psi^+_{-i_1}|0\rangle.\end{aligned} \tag{3.100}$$

We write $m=s-t$ for the charge and $k=\sum i_k+\sum j_l$ for the energy.

Lemma 3.5.2. [10] The eigenspace $F^m=\bigoplus_{k\geq 0}F^m_k$ has a *unique* state of minimum energy $|m\rangle$, obtained from $|0\rangle$ by applying the smallest possible number of $\psi^\pm$. Explicitly, one has

$$\begin{cases} |m\rangle=\psi^+_{-\frac{2m-1}{2}}\cdots\psi^+_{-\frac{1}{2}}|0\rangle & \text{if } m\geq 0\\ |m\rangle=\psi^-_{-\frac{2|m|-1}{2}}\cdots\psi^-_{-\frac{1}{2}}|0\rangle & \text{if } m<0.\end{cases} \tag{3.101}$$

Proof: Suppose that $m\geq 0$, then each monomial (3.99) has energy $\sum_{k=1}^{m+t}i_k+\sum_{l=1}^{t}j_l$. But, both $(i_k),(j_l)$ are increasing sequences in $(-\frac{1}{2}+\mathbb{Z})\cap(0,\infty)$ and

$$\begin{aligned}\sum_{k=1}^{m+t}i_k+\sum_{l=1}^{t}j_l &\geq \sum_{k=1}^{m+t}\frac{2k-1}{2}+\sum_{l=1}^{t}\frac{2l-1}{2}\\ &= \frac{(m+t)^2+t^2}{2}.\end{aligned} \tag{3.102}$$

Therefore, to make energy as small as possible, it's necessary to take $t=0$ and $i_k=(2k-1)/2,\ (1\leq k\leq m)$, i.e. $|m\rangle=\psi^+_{-\frac{2m-1}{2}}\cdots\psi^+_{-\frac{1}{2}}|0\rangle$. For $m<0$, the proof is similar.

Lemma 3.5.3. [10] F^m is a representation of the oscillator algebra $\mathcal{A}=\{\alpha_n\}_{n\in\mathbb{Z}}\cup\{\mathbb{1}_F\}$ and the state of minimum energy $|m\rangle$ is a singular vector of F^m.

Proof: Since $[\alpha_0,\alpha_m]=0\quad\forall m\in\mathbb{Z}$, we get that F^m is invariant by the action of $\alpha_n\ \forall n\in\mathbb{Z}$, i.e F^m is an $\mathcal{A}-$module. Moreover, by the second equation in (3.98), $\alpha_j,\ j>0$ acts on F^m by reducing the energy of its states by j. In particular, this proves that $\alpha_j|m\rangle=0$ for $j>0$, while $\alpha_0|m\rangle=m|m\rangle$. So $|m\rangle$ is a singular vector of the representation F^m of the oscillator algebra.

Theorem 3.5.4. [10] For each $m \in \mathbb{Z}$, F^m is an irreducible representation of $\mathcal{A}$.

Proof: By the lemma (3.5.3) and theorem (3.4.7), there is a injective map $\eta_m : B^m \to F^m : \prod_{j\geq 1} j^{n_j} x_1^{n_1} \cdots x_k^{n_k} \mapsto \alpha_{-1}^{n_1} \cdots \alpha_{-k}^{n_k} |m\rangle$ (we're taking $\epsilon_j = 1$). So, because of theorem (3.4.6), in order to show that F^m is irreducible, it's sufficient to show that η_m is surjective. Suppose that this is not the case, i.e. $W = F^m/\mathcal{U}(\mathcal{A})|m\rangle \neq \{0\}$. Then W has a nonzero vector of minimal energy v and $\alpha_j(v) = 0 \; \forall j > 0$.

By the equation (3.46) we get

$$\begin{aligned} L_0 = L_{(1)} &= \frac{1}{2} \sum_{m<0} (\alpha_m \alpha_{-m} + \alpha_{m+1} \alpha_{-m-1}) \\ &= \frac{1}{2}\alpha_0^2 + \sum_{m>0} \alpha_{-m}\alpha_m \\ &\Rightarrow L_0(v) = \frac{1}{2} m^2 v. \end{aligned} \tag{3.103}$$

So $v \in W$ is a state of energy $m^2/2$ and by lemma (3.5.2) v must have $|m\rangle$ as its representative, which is impossible since $|m\rangle \in \mathcal{U}(\mathcal{A})|m\rangle$ and $v \neq 0$.

Notation 3.5.5. We saw in the proof of the Theorem (3.5.4) that $\sigma_m = \eta_m^{-1} : F^m \to B^m : \alpha_{-1}^{n_1} \cdots \alpha_{-k}^{n_k} |m\rangle \mapsto \prod_{j\geq 1} j^{n_j} x_1^{n_1} \cdots x_k^{n_k}$ is an isomorphism. Therefore, we have an isomorphism

$$\sigma = \bigoplus_{m\in\mathbb{Z}} \sigma_m : F \to B : \alpha_{-1}^{n_1} \cdots \alpha_{-k}^{n_k} |m\rangle \mapsto \prod_{j\geq 1} j^{n_j} x_1^{n_1} \cdots x_k^{n_k} \mathbb{1}_m \tag{3.104}$$

where $B = \bigoplus_{m\in\mathbb{Z}} B^m$ and $\mathbb{1}_m$ is the generator of B^m. To keep track of charge in B, it is useful to insert a formal variable z^m in B^m to each $m \in \mathbb{Z}$. So we let $B^m = z^m \mathbb{C}[x_1, x_2, \dots]$. In particular, this implies $\sigma_m(|m\rangle) = z^m$.

Definition 3.5.6. Let V be a vector space, the *formal completion* $\hat{V}$ is defined to be the vector space of $V-$valued formal distributions $V[[u^{\pm}]]$.

Lemma 3.5.7. Define $X^+(u) = u\psi^+(u), X^-(u) = \psi^-(u)$, so $X^+(u), X^-(u) : F \to \hat{F}$. Then we've $[\alpha_j, X^{\pm}(u)] = \pm u^j X^{\pm}(u)$.

Proof: Using the commutation relations (3.97) we get

$$\begin{aligned}[\alpha_j, X^+(u)] &= \sum_{n\in -\frac{1}{2}+\mathbb{Z}} [\alpha_j, \psi_n^+] u^{-n+1/2} \\ &= \sum_{n\in -\frac{1}{2}+\mathbb{Z}} \psi_{n+j}^+ u^{-n+1/2} \\ &= u^j X^+(u).\end{aligned} \tag{3.105}$$

Similarly for $X^-(u)$, $[\alpha_j, X^-(u)] = -u^{-j}X^-(u)$.

Now, define $\Gamma^{\pm}(u) = \sigma \circ X^{\pm}(u) \circ \sigma^{-1}$. Lemma (3.5.7) implies the commutation relations

$$\begin{aligned}\left[\frac{\partial}{\partial x_j}, \Gamma^+(u)\right] &= u^j \Gamma^+(u) \\ [x_j, \Gamma^+(u)] &= \frac{1}{j} u^{-j} \Gamma^+(u).\end{aligned} \tag{3.106}$$

Lemma 3.5.8. [10] We have the identities

$$\begin{aligned}\Gamma^+(u)\big|_{\hat{B}^m} &= z u^{m+1} \exp\left(\sum_{j\geq 1} x_j u^j\right) \exp\left(-\sum_{j\geq 1} \frac{1}{j}\frac{\partial}{\partial x_j} u^{-j}\right) \\ \Gamma^-(u)\big|_{\hat{B}^m} &= z^{-1} u^{-m} \exp\left(-\sum_{j\geq 1} x_j u^j\right) \exp\left(\sum_{j\geq 1} \frac{1}{j}\frac{\partial}{\partial x_j} u^{-j}\right).\end{aligned} \tag{3.107}$$

Proof: The reason that z appears on the right side of equation (3.107) is because $\psi^+(u) : \hat{F}^m \to \hat{F}^{m+1}$. Now, define $T_u = \exp\left(\sum_{j\geq 1} \frac{u^{-j}}{j}\frac{\partial}{\partial x_j}\right) = \prod_{j\geq 1} \exp(\frac{u^{-j}}{j}\frac{\partial}{\partial x_j})$. By Lemma (3.3.6) $T_u f(x_1, x_2, \dots) = f(x_1 + u^{-1}/1, x_2 + u^{-2}/2, \dots)$, so in particular $[x_j, T_u] = -\frac{u^{-j}}{j} T_u$. Using the identity $[a, bc] = [a, b]c + b[a, c]$ we get

$$\begin{aligned}[x_j, \Gamma^+(u)T_u] &= [x_j, \Gamma^+(u)]T_u + \Gamma^+(u)[x_j, T_u] \\ &= \frac{u^{-j}}{j}\Gamma^+(u)T_u + \Gamma^+(u)\left(-\frac{u^{-j}}{j}T_u\right) = 0.\end{aligned} \tag{3.108}$$

Therefore, $\Gamma^+(u)T_u$ has no differential part and we can write

$$\Gamma^+(u) = z f(u, x) \exp\left(-\sum_{j\geq 1} \frac{u^{-j}}{j}\frac{\partial}{\partial x_j}\right) \tag{3.109}$$

for some unknown polynomial $f(u, x) = f(u, x_1, x_2, \dots)$. But since

$$\frac{\partial}{\partial x_i} \exp\left(-\sum_{j\geq 1} u^j x_j\right) = -u^i \exp\left(-\sum_{j\geq 1} u^j x_j\right)$$

we get $\left[\frac{\partial}{\partial x_i}, \exp\left(-\sum_{j\geq 1} u^j x_j\right)\right] = -u^i \exp\left(-\sum_{j\geq 1} u^j x_j\right)$ and

$$\begin{aligned}
&\left[\frac{\partial}{\partial x_i}, \exp\left(-\sum_{j\geq 1} u^j x_j\right)\Gamma^+(u)\right] \\
&= \left[\frac{\partial}{\partial x_i}, \exp\left(-\sum_{j\geq 1} u^j x_j\right)\right]\Gamma^+(u) + \exp\left(-\sum_{j\geq 1} u^j x_j\right)\left[\frac{\partial}{\partial x_i}, \Gamma^+(u)\right] \\
&= -u^i \exp\left(-\sum_{j\geq 1} u^j x_j\right)\Gamma^+(u) + \exp\left(-\sum_{j\geq 1} u^j x_j\right) u^i \Gamma^+(u) \\
&= 0.
\end{aligned} \tag{3.110}$$

Therefore

$$\Gamma^+(u) = z \exp\left(\sum_{j\geq 1} u^j x_j\right) g\left(u, \frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \dots\right). \tag{3.111}$$

From equations (3.109) and (3.111) we get

$$\Gamma^+(u) = z c_m(u) \exp\left(\sum_{j\geq 1} u^j x_j\right) \exp\left(-\sum_{j\geq 1} \frac{u^{-j}}{j} \frac{\partial}{\partial x_j}\right) \tag{3.112}$$

where $c_m(u)$ still to be determined. To do this, apply

$$\Gamma^+(u) z^m = z^{m+1} c_m(u) \exp\left(\sum_{j\geq 1} u^j x_j\right) \cdot 1. \tag{3.113}$$

But

$$\begin{aligned}
\Gamma^+(u) z^m &= \sigma(X^+(u)|m\rangle) \\
&= \sum_{j \in -\frac{1}{2}+\mathbb{Z}} \sigma(\psi_j^+|m\rangle) u^{-j+1/2} \\
&= \sigma(\psi_{-\frac{2(m+1)-1}{2}}|m\rangle) u^{m+1} + \cdots \\
&= z^{m+1} u^{m+1} + \cdots
\end{aligned} \tag{3.114}$$

and so $c_m(u) = u^{m+1}$. The proof for $\Gamma^-(u)$ is totally similar.

Now define the maps $R(u) : \hat{B} \to \hat{B} : f(x,z) \mapsto uzf(x,uz)$. Then $R(u)z^m f(x) = uz(uz)^m f(x) = zu^{m+1} \cdot z^m f(x)$. Therefore

$$\Gamma^+(u) = R(u) \exp\left(\sum_{j\geq 1} x_j u^j\right) \exp\left(-\sum_{j\geq 1} \frac{1}{j} \frac{\partial}{\partial x_j} u^{-j}\right). \tag{3.115}$$

Similarly, $R(u)^{-1}f(x,z) = z^{-1}f(x,zu^{-1}) \Rightarrow R(u)^{-1}z^m f(x) = z^{m-1}u^{-m}f(x) = z^{-1}u^{-m} \cdot z^m f(x)$ and

$$\Gamma^-(u) = R(u)^{-1}\exp\left(-\sum_{j\geq 1} x_j u^j\right)\exp\left(\sum_{j\geq 1}\frac{1}{j}\frac{\partial}{\partial x_j}u^{-j}\right). \tag{3.116}$$

In conclusion, we have obtained the transposed action of the operators $X^{\pm}(u)$ to $\hat{B}$ through σ.

Theorem 3.5.9. [10]

$$\begin{aligned}\Gamma^+(u) &= R(u)\exp\left(\sum_{j\geq 1} x_j u^j\right)\exp\left(-\sum_{j\geq 1}\frac{1}{j}\frac{\partial}{\partial x_j}u^{-j}\right)\\ \Gamma^-(u) &= R(u)^{-1}\exp\left(-\sum_{j\geq 1} x_j u^j\right)\exp\left(\sum_{j\geq 1}\frac{1}{j}\frac{\partial}{\partial x_j}u^{-j}\right).\end{aligned} \tag{3.117}$$

3.5.2 The ring of symmetric functions

This subsection is based on ([12], Chapter I). The main objective of this subsection is to record some definitions that will be needed to work with symmetric functions, which play a major role in the Boson-Fermion-correspondence.

In the next subsection we will work with the ring of symmetric functions Λ. Let $\Lambda_n = \mathbb{Z}[x_1,\ldots,x_n]^{S_n}$ be the sub-ring of $\mathbb{Z}[x_1,\ldots,x_n]$ which is invariant under the action of the symmetric group S_n, i.e. the set of polynomials f with integer coefficients satisfying $\sigma f(x_1,\ldots,x_n) = f(x_{\sigma(1)},\ldots,x_{\sigma(n)}) = f(x_1,\ldots,x_n)$ for all $\sigma \in S_n$.

The ring Λ_n is graded by degree of the polynomials. So $\Lambda_n = \bigoplus_{k\geq 0}\Lambda_n^k$ where Λ_n^k is the sub-group of kth degree symmetric polynomials. Let $\alpha = (\alpha_1,\ldots,\alpha_n) \in \mathbb{N}^n$, we write $x^\alpha = x_1^{\alpha_1}\cdots x_n^{\alpha_n}$. If λ is a partition with length $l(\lambda) \leq n$, we may complete its coordinates with zero to get an n-tuple. So we define

$$m_\lambda(x_1,\ldots,x_n) = \frac{1}{\prod_{j\geq 1} m_j(\lambda)!}\sum_{\sigma\in S_n}\sigma(x^\lambda) \tag{3.118}$$

for $l(\lambda) \leq n$, where $\sigma(x^\lambda) = x_{\sigma(1)}^{\lambda_1}\cdots x_{\sigma(n)}^{\lambda_n}$. We claim that the set $\{m_\lambda(x_1,\ldots,x_n) \mid l(\lambda) \leq n\}$ is a basis of Λ_n. In fact, let $f = \sum_\alpha a_\alpha x^\alpha \in \Lambda_n$. Then

$$\begin{aligned}f(x_1,\ldots,x_n) &= \frac{1}{n!}\sum_{\sigma\in S_n}\sigma f(x_1,\ldots,x_n)\\ &= \frac{1}{n!}\sum_\alpha a_\alpha\left(\sum_{\sigma\in S_n}\sigma(x^\alpha)\right).\end{aligned} \tag{3.119}$$

However, the polynomial $\sum_{\sigma\in S_n}\sigma(x^\alpha)$ is $m_\lambda(x_1,\dots,x_n)$ where λ is obtained from $\alpha=(\alpha_1,\dots,\alpha_n)$ by listing its elements in decreasing order. Moreover, it's clear by construction that $\mathrm{Supp}(m_\lambda(x))\cap\mathrm{Supp}(m_\mu(x))=\varnothing$ if $\lambda\neq\mu$. Therefore, the set $\{m_\lambda(x)\ |l(\lambda)\leq n\}$ is linearly independent, and we conclude the claim.

From the preceding discussion, we see that $\{m_\lambda(x_1,\dots,x_n)|\ |\lambda|=k\}$ is a basis of Λ_n^k. For $m\geq n$, there exists a homomorphism of rings

$$\rho_{mn}:\Lambda_m\to\Lambda_n:f(x_1,\dots,x_m)\mapsto f(x_1,\dots,x_n,0,\dots,0). \tag{3.120}$$

If $l(\lambda)\leq n$, it's clear that

$$\rho_{mn}(m_\lambda(x_1,\dots,x_m))=m_\lambda(x_1,\dots,x_n,0,\dots,0)=m_\lambda(x_1,\dots,x_n) \tag{3.121}$$

and if $l(\lambda)>n$, $\rho_{mn}(m_\lambda(x_1,\dots,x_m))=0$ because each monomial $\sigma(x^\lambda)$ has a factor $x_i,(n<i\leq m)$ with nonzero power.

Now, let λ be a partition of k, then $l(\lambda)\leq k$ and the restriction $\rho_{mn}^k:\Lambda_m^k\to\Lambda_n^k$ will send basis elements to basis elements if $m\geq n\geq k$, i.e. it'll be an isomorphism. Then we define the inverse limit

$$\begin{aligned}\Lambda^k&=\varprojlim\Lambda_n^k\\&=\{(f_0,f_1,\dots,f_n,\dots)\mid f_n\in\Lambda_n^k\text{ and }\rho_{mn}(f_m)=f_n\text{ for }m\geq n\}.\end{aligned} \tag{3.122}$$

Example 3.5.10. We have

$$m_{(1^2)}=(0,x_1x_2,x_1x_2+x_1x_3+x_2x_3,\dots)\in\Lambda^2 \tag{3.123}$$

and

$$m_{(r)}=(x_1^r,x_1^r+x_2^r,x_1^r+x_2^r+x_3^r,\dots)\in\Lambda^r \tag{3.124}$$

for $r\geq 1$.

Since ρ_{mn} is an isomorphism whenever $m\geq n\geq k$, the projection $\rho_n:\Lambda^k\to\Lambda_n^k$ will be an isomorphism whenever $n\geq k$. Therefore, Λ^k is a free $\mathbb{Z}-$module of rank $p(k)$. Now we define the ring of symmetric functions as the direct sum $\Lambda=\bigoplus_{k\geq 0}\Lambda^k$.

When working explicitly in the ring of symmetric functions, will take in to consideration all the partitions involved, and take a number of variables n, large enough to carry out all the computations in Λ_n^k.

Example 3.5.11. To multiply $m_{(1)}\cdot m_{(1)}$ in Λ, it's sufficient to work in Λ_2^2. Since $(x_1+x_2)^2=x_1^2+2x_1x_2+x_2^2$, we get

$$m_{(1)}\cdot m_{(1)}=m_{(2)}+2m_{(1^2)}\in\Lambda. \tag{3.125}$$

The ring of symmetric functions has many interesting bases. Some of the functions that we will use in this work are the following:

$$\text{(1) } e_k(x_1,\ldots,x_n) = \sum_{i_1<\cdots<i_k} x_{i_1}\cdots x_{i_k} \quad \text{(elementary symmetric functions)},$$

$$\text{(2) } h_k(x_1,\ldots,x_n) = \sum_{i_1\leq\cdots\leq i_k} x_{i_1}\cdots x_{i_k} \quad \text{(complete symmetric functions)},$$

$$\text{(3) } p_k(x_1,\ldots,x_n) = \sum_{i=1}^{n} x_i^k \quad \text{(power sum symmetric functions)}.$$

Elementary and symmetric functions have generating functions given by

$$E(u) = \sum_{i\geq 0} e_i u^i = \prod_{i\geq 1}(1+x_i u) \tag{3.126}$$

and

$$H(u) = \sum_{i\geq 0} h_i u^i = \prod_{i\geq 1} \frac{1}{1-x_i u}. \tag{3.127}$$

For power sum symmetric functions, we have a generating function $P(u) = \sum_{i\geq 1} p_i u^{i-1}$. There is a important connection between generating functions of complete and power sum symmetric functions:

$$\begin{aligned} P(u) &= \sum_{i\geq 1} p_i u^{i-1} \\ &= \sum_{i\geq 1}\sum_{j\geq 1} x_j^i u^{i-1} \\ &= \sum_{j\geq 1} x_j \frac{1}{1-x_j u} \\ &= \sum_{j\geq 1} \frac{d}{dt}\log\left(\frac{1}{1-x_j u}\right) \\ &= \frac{d}{dt}\log\left(\prod_{j\geq 1}\frac{1}{1-x_j u}\right) \\ &= \frac{d}{dt}\log H(u) \end{aligned} \tag{3.128}$$

or equivalently,

$$H(u) = \exp\left(\sum_{j\geq 1} p_j u^j/j\right). \tag{3.129}$$

It's important to note that if we want to work with a finite number of variables, say n, then it's sufficient to put $x_i = 0\ \forall i \geq n$. So, for example we have

$$\sum_{i\geq 0} \rho_n(e_i) u^i = \prod_{i=1}^{n}(1+x_i u). \tag{3.130}$$

The Schur symmetric function $s_\lambda(x_1, \dots, x_n)$ is defined by

$$s_\lambda(x_1, \dots, x_n) = \frac{\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\sigma(x^{\lambda+\delta})}{\prod_{i<j}(x_i - x_j)} \tag{3.131}$$

where $l(\lambda) \leq n$. In this work we will use

$$s_\lambda(x_1, \dots, x_n) = \det(h_{\lambda_i+j-i})_{1\leq i,j\leq n} \tag{3.132}$$

to define the Schur symmetric functions. The equality between these two formulas is know as the Jacobi-Trudy formula.

The next theorem will play a central role in the final version of the Boson-Fermion correspondence (see (3.5.29)).

Theorem 3.5.12. The ring of symmetric functions $\Lambda_{\mathbb{C}} = \Lambda \otimes_{\mathbb{Z}} \mathbb{C}$ is freely generated by the set of power sum and elementary symmetric functions, i.e.

$$\Lambda \cong \mathbb{C}[p_1, p_2, \dots] \cong \mathbb{C}[e_1, e_2, \dots]. \tag{3.133}$$

Proof: See ([12], Chapter I).

Remark 3.5.13. In the next chapter we will deal with the ring of polynomials in infinitely many variables $B = \mathbb{C}[x_1, x_2, \dots]$. By the previous theorem, a formal change of variables $x_j \to p_j/j$ yields a identification $B \cong \Lambda_{\mathbb{C}}$. However, to avoid confusion we will use the symbol $S(u) = \exp(\sum_{j\geq 1} x_j u^j) = \sum_{i\geq 0} S_i(x)u^i$. The polynomials $S_i(x)$ are called *elementary Schur polynomials* and they are not symmetric functions. However, under the formal change $x_j \to p_j/j = (\epsilon_1^j + \epsilon_2^j + \cdots)$ we get $S_i(x_1, x_2, \dots) = h_i(\epsilon_1, \epsilon_2, \dots)$.

3.5.3 The second part of the Boson-Fermion correspondence

Definition 3.5.14. [10] Let $V = \bigoplus_{j\in\mathbb{Z}} \mathbb{C}v_j$ be a graded vector space. The *Dirac sea* $\bigwedge^{\infty/2}$ is defined to be the vector space of semi-infinite monomials, i.e., formal expressions

$$\psi = v_{i_0} \wedge v_{i_{-1}} \wedge v_{i_{-2}} \wedge \cdots \tag{3.134}$$

such that the following conditions are satisfied:

(1) $i_0 > i_{-1} > i_{-2} > \cdots$.

(2) The limit $m = \lim_{k\to-\infty}(i_k - k)$ exists.

The value m is called the *charge* of the semi-infinite monomial ψ. Also, the standard rule for wedge manipulation is satisfied, i.e.

$$\cdots v_m \wedge v_n \cdots = -(\cdots v_n \wedge v_m \cdots). \tag{3.135}$$

If a semi-infinite monomial ψ has charge m it will be more convenient to write

$$\psi = v_{i_m} \wedge v_{i_{m-1}} \wedge \cdots \tag{3.136}$$

so the second condition in definition (3.5.14) is equivalent to $\lim_{k\to-\infty}(i_k - k) = 0$, or $i_k = k$ for $-k \gg 0$ which we write here as condition $(2)'$.

Now, let $\psi = v_{i_m} \wedge v_{i_{m-1}} \wedge \cdots$, the degree of ψ is defined to be the sum $\deg \psi = \sum_{k=0}^{\infty}(i_{m-k} - m + k)$, which is a finite sum because of condition $(2)'$. Moreover, for each $m \in \mathbb{Z}$, writing $\lambda_k = i_{m-k+1} + k - m - 1$ implies that $\lambda = (\lambda_1, \lambda_2, \dots)$ is a partition of weight $\deg \psi$. Conversely, each partition λ of length r gives a semi-infinite monomial

$$\psi_\lambda^m = v_{\lambda_1 - m} \wedge \cdots \wedge v_{\lambda_r - m + r - 1} \wedge v_{m-r} \wedge \cdots \tag{3.137}$$

of charge m and degree $|\lambda|$. This show that partitions parametrize elements of $\bigwedge^{\infty/2} V$.

Notation 3.5.15. We will write ψ_λ^m for semi-infinite monomials of charge m and associated to the partition λ.

Now, define the *shift-operators* $\Lambda_k v_j = v_{j-k}$. For $k \neq 0$, there is a well defined action of shift operators on the Dirac sea:

$$r(\Lambda_k) v_{i_1} \wedge v_{i_2} \wedge \cdots = (\Lambda_k v_{i_1}) \wedge v_{i_2} \wedge \cdots + v_{i_1} \wedge (\Lambda_k v_{i_2}) \wedge \cdots + \text{etc} \tag{3.138}$$

which is a finite sum because because of the condition $(2)'$. It's important to note that equation (3.138) is not well defined for Λ_0, indeed we would have

$$r(\Lambda_0) v_{i_1} \wedge v_{i_2} \wedge \cdots = (1 + 1 + \cdots) v_{i_1} \wedge v_{i_2} \wedge \cdots \tag{3.139}$$

which doesn't make sense.

Example 3.5.16. We have

$$\begin{aligned}\Lambda_1\psi^0_{(3,1)} &= \psi^0_{(2,1)} + \psi^0_{(3)}\\ \Lambda_1^2\psi^0_{(3,1)} &= \psi^0_{(1^2)} + 2\psi^0_{(2)}\\ \Lambda_1^3\psi^0_{(3,1)} &= 3\psi^0_{(1)}\\ \Lambda_1^4\psi^0_{(3,1)} &= 3\psi^0_{(0)}.\end{aligned} \tag{3.140}$$

In general we can see that $\Lambda_k\psi^m_\lambda = \Lambda_1^k\psi^m_\lambda = 0$ for $k \geq |\lambda| + l(\lambda)$, i.e. Λ_k is locally nilpotent for $k > 0$. Therefore, the action of $r(\Lambda_k), (k > 0)$ on $\Lambda^{\infty/2}V$ can be exponentiated. We would like to define an action $R(\exp(y_k\Lambda_k))$ on $\Lambda^{\infty/2}V$ such that

$$R(\exp y_k\Lambda_k) = \exp(r(y_k\Lambda_k)) \tag{3.141}$$

where y_k is an indeterminate. To do this we will define

$$R(\exp y_k\Lambda_k)\psi = (\exp y_k\Lambda_k)v_{i_m} \wedge \cdots \wedge (\exp y_k\Lambda_k)v_{i_{m-r+1}} \wedge v_{m-r} \wedge \cdots \tag{3.142}$$

for $\psi = v_{i_m} \wedge \cdots \wedge v_{i_{m-r+1}} \wedge v_{m-r} \wedge \cdots$ for $k > 0$. It's important to note that $(\exp y_k\Lambda_k)v_j$ belongs not to V, but to the vector space $V[[y_k]]$. But this is not a problem since $\Lambda_k v_j = v_{j-k}$ and $\cdots v_j \wedge \cdots \wedge v_{m-r}\cdots = 0$ whenever $j \leq m-r$, so each exponential $(\cdots \wedge (\exp y_k\Lambda_k)v_j \wedge \cdots)$ inside the wedge is actually a finite sum.

Example 3.5.17. We have $\psi^0_{(3,1)} = v_3 \wedge v_0 \wedge v_{-2} \wedge v_{-3} \wedge \cdots$, and

$$\begin{aligned}&R(\exp y_1\Lambda_1)\psi^0_{(3,1)}\\ &\quad= \left(v_3 + y_1v_2 + \frac{y_1^2}{2!}v_1 + \frac{y_1^3}{3!}v_0 + \frac{y_1^4}{4!}v_{-1}\right) \wedge (v_0 + y_1v_{-1}) \wedge v_{-2} \wedge \cdots\\ &\quad= \psi^0_{(3,1)} + y_1(\psi^0_{(2,1)} + \psi^0_{(3)}) + \frac{y_1^2}{2!}(\psi^0_{(1^2)} + 2\psi^0_{(2)}) + \frac{y_1^3}{3!}(3\psi^0_{(1)}) + \frac{y_1^4}{4!}(3\psi^0_{(0)}).\end{aligned} \tag{3.143}$$

Theorem 3.5.18. [10] Let $A = \exp(y_k\Lambda_k)$ and $a = y_k\Lambda_k$, $k > 0$. We have

$$R(\exp(A)) = \exp(r(a)), \tag{3.144}$$

where r is defined by (3.138) and R is defined by (3.142).

Proof: First, let $\psi = v_{i_1} \wedge v_{i_2} \wedge \cdots$ be a semi-infinite monomial with associated partition of length r. Then we claim that

$$\frac{r(a)^k}{k!} v_{i_1} \wedge v_{i_2} \wedge \cdots = \sum_{i_1+\cdots+i_r=k} \frac{1}{i_1! \dots i_r!} a^{i_1} v_{i_1} \wedge a^{i_2} v_{i_2} \wedge \cdots . \tag{3.145}$$

This follows by induction with base case $k = 1$ being just the definition (3.138). For the induction step, suppose that the result is true for some $k \geq 1$, then

$$\begin{aligned} r(a) & \sum_{i_1+\cdots+i_r=k} \frac{1}{i_1! \dots i_r!} a^{i_1} v_{i_1} \wedge a^{i_2} v_{i_2} \wedge \cdots \\ &= \sum_{i_1+\cdots+i_r=k} \frac{1}{i_1! \dots i_r!} (a^{i_1+1} v_{i_1} \wedge a^{i_2} v_{i_2} \wedge \cdots + a^{i_1} v_{i_1} \wedge a^{i_2+1} v_{i_2} \wedge \cdots + \cdots) \\ &= \sum_{i_1+\cdots+i_r=k+1} \sum_{j=1}^{r} \frac{1}{i_1! \dots (i_j-1)! \dots i_r!} a^{i_1} v_{i_1} \wedge a^{i_2} v_{i_2} \wedge \cdots \\ &= \sum_{i_1+\cdots+i_r=k+1} \frac{k+1}{i_1! \dots i_r!} a^{i_1} v_{i_1} \wedge a^{i_2} v_{i_2} \wedge \cdots \end{aligned} \tag{3.146}$$

which proves that $\frac{r(a)^{k+1}}{(k+1)!}\psi = \sum_{i_1+\cdots+i_r=k+1} \frac{1}{i_1! \dots i_r!} a^{i_1} v_{i_1} \wedge a^{i_2} v_{i_2} \wedge \cdots$. Now

$$\begin{aligned} \exp(r(a))\psi &= \sum_{k\geq 0} \frac{r(a)^k}{k!} \psi \\ &= \sum_{k\geq 0} \sum_{i_1+\cdots+i_r=k} \frac{1}{i_1! \dots i_r!} a^{i_1} v_{i_1} \wedge a^{i_2} v_{i_2} \wedge \cdots \\ &= \sum_{i_1,\dots,i_r} \frac{1}{i_1! \dots i_r!} a^{i_1} v_{i_1} \wedge a^{i_2} v_{i_2} \wedge \cdots \\ &= \left(\sum_{i_1 \geq 0} \frac{a^{i_1}}{i_1} v_{i_1} \right) \wedge \left(\sum_{i_2 \geq 0} \frac{a^{i_2}}{i_2} v_{i_2} \right) \wedge \cdots \wedge \left(\sum_{i_r \geq 0} \frac{a^{i_r}}{i_r} v_{i_r} \right) \wedge v_{i_{r+1}} \cdots \\ &= R(A)\psi \end{aligned} \tag{3.147}$$

which proves the claim.

Define GL_∞ to be the set of infinite invertible matrices $(A_{ij})_{i,j\in\mathbb{Z}}$ with the property that $A_{ij} = \delta_{ij}$ whenever $|i|, |j| \geq N$ for some $N \geq 1$. Multiplication is well defined in GL_∞ because $(AB)_{ij} = \sum_{k=-\infty}^{\infty} A_{ik} B_{kj}$ is a finite sum. we also define a large group $\overline{GL}_\infty$ of infinite invertible matrices A with the property that for $i \geq j$, there exists an integer $N \geq 1$ such that $A_{ij} - \delta_{ij} = 0$ whenever $|i|, |j| \geq N$. The operator $\exp y_k \Lambda_k$, $k > 0$ acts on $\overline{V}$ with

matrix representation in $\overline{GL}_\infty$. In fact, its matrix looks like

$$A = \begin{pmatrix} \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \iddots \\ \vdots & 1 & y_k & \frac{y_k^2}{2!} & \frac{y_k^3}{3!} & \frac{y_k^4}{3!} & \vdots \\ \vdots & 0 & 1 & y_k & \frac{y_k^2}{2!} & \frac{y_k^3}{3!} & \vdots \\ \vdots & 0 & 0 & 1 & y_k & \frac{y_k^2}{2!} & \vdots \\ \vdots & 0 & 0 & 0 & 1 & y_k & \vdots \\ \vdots & 0 & 0 & 0 & 0 & 1 & \vdots \\ \iddots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{3.148}$$

We can define the action of GL_∞ on $\bigwedge^{\infty/2} V$ in an analogous way as for equation (3.142). But here, since $Av_j = v_j$ for $j \ll 0$, we just put

$$R(A)v_{i_1} \wedge v_{i_2} \wedge \cdots \wedge v_{i_r} \cdots = Av_{i_1} \wedge Av_{i_2} \wedge \cdots \wedge Av_{i_r} \cdots . \tag{3.149}$$

Theorem 3.5.19. Let $A \in GL_\infty$, $\psi = v_{i_m} \wedge v_{i_{m-1}} \wedge \cdots$ a semi-infinite monomial with associated partition of length r and $N \geq r$ be such that $Av_{m-j} = v_{m-j}$ for $j \geq N$. Then

$$\begin{aligned} &R(A)v_{i_m} \wedge v_{i_{m-1}} \wedge \cdots \\ &= \sum_{j_m > \cdots > j_{m-N+1}} \det\left(A^{i_m,\ldots,i_{m-N+1}}_{j_m,\ldots,j_{m-N+1}}\right) v_{j_m} \wedge \cdots \wedge v_{j_{m-N+1}} \wedge v_{m-N} \cdots \end{aligned} \tag{3.150}$$

where $A^{i_m,\ldots,i_{m-N+1}}_{j_m,\ldots,j_{m-N+1}} = (A_{j_{m-k+1},i_{m-l+1}})_{1\leq k,l\leq N}$.

Proof: Let $\psi = v_{i_m} \wedge \cdots \wedge v_{i_{m-r+1}} \wedge v_{m-r} \wedge \cdots$, $A \in GL_\infty$ and $N \geq r$ be such that $Av_{m-j} = v_{m-j}\ \forall j \geq N$. Then

$$R(A)v_{i_m} \wedge \cdots \wedge v_{i_{m-N+1}} \wedge v_{m-N} \wedge \cdots = Av_{i_m} \wedge \cdots \wedge Av_{i_{m-N+1}} \wedge v_{m-N} \wedge \cdots \tag{3.151}$$

and

$$
\begin{aligned}
&R(A)v_{i_m} \wedge \cdots \wedge v_{i_{m-N+1}} \wedge v_{m-N} \wedge \cdots \\
&= \left(\sum_{j_m=-\infty}^{\infty} A_{j_m i_m} v_{j_m} \right) \wedge \cdots \wedge \left(\sum_{j_{m-N+1}=-\infty}^{\infty} A_{j_{m-N+1} i_{m-N+1}} v_{i_{m-N+1}} \right) \wedge v_{m-N} \wedge \cdots \\
&= \sum_{j_m=-\infty}^{\infty} \cdots \sum_{j_{m-N+1}=-\infty}^{\infty} A_{j_m i_m} \cdots A_{j_{m-N+1} i_{m-N+1}} v_{j_m} \wedge \cdots \wedge v_{j_{m-N+1}} \wedge v_{m-N} \cdots \\
&= \sum_{j_m > \cdots > j_{m-N+1}} \left(\sum_{\sigma \in S_N} \mathrm{sgn}\sigma A_{\sigma(j_m) i_m} \cdots A_{\sigma(j_{m-N+1}) i_{m-N+1}} \right) v_{j_m} \wedge \cdots \\
&= \sum_{j_m > \cdots > j_{m-N+1}} \left(\det A^{i_m, \ldots, i_{m-N+1}}_{j_m, \ldots, j_{m-N+1}} \right) v_{j_m} \wedge \cdots \wedge v_{j_{m-N+1}} \wedge v_{m-N} \cdots .
\end{aligned}
\tag{3.152}
$$

Although $\exp y_k \Lambda_k$ acts on $\overline{V}$ as a matrix in $\overline{GL}_\infty$, its action on each individual element of $\bigwedge^{\infty/2} V$ is the same as the action of some matrix in GL_∞ (see example (3.5.17)).

Corollary 3.5.20. Let $A = \exp(y_k \Lambda_k), k > 0$ and $\psi = v_{i_m} \wedge v_{i_{m-1}} \wedge \cdots$ be a semi-infinite monomial with associated partition of length r. Then A acts on ψ as an element of GL_∞. Moreover, theorem (3.5.19) applies to A with $N = r$.

Proof: This is a direct consequence of Definition (3.142) and the fact that an exponential inside $\cdots \wedge \exp(y_k \Lambda_k) v_{i_{m-j+1}} \wedge \cdots$ is expanded at most to $y_k^r \Lambda_k^r / r!$.

Now, we will construct an isomorphism $F \to \bigwedge^{\infty/2} V$. To do this, for $v \in V$ and $f \in V^*$ define the *wedge operator* $\hat{v}$ and the *contraction operator* $\check{f}$:

$$
\begin{aligned}
&\hat{v}(v_{i_1} \wedge v_{i_2} \wedge \cdots) = v \wedge v_{i_1} \wedge v_{i_2} \wedge \cdots \quad \text{and} \\
&\check{f}(v_{i_1} \wedge v_{i_2} \wedge v_{i_3} \cdots) = f(v_{i_1}) v_{i_2} \wedge v_{i_3} \cdots - f(v_{i_2}) v_{i_1} \wedge v_{i_3} \cdots + \cdots .
\end{aligned}
\tag{3.153}
$$

There is a bijection between the set of partitions $\mathcal{P}$ and the set of pair $(\alpha|\beta)$ of sequences $\alpha_1 > \cdots > \alpha_r \geq 0, \beta_1 > \cdots > \beta_r \geq 0$. The correspondence is given as follows: Given a partition λ, let $r = \mathrm{Max}\{i \mid \lambda_i - i \geq 0\}$ be the number of squares in the main diagonal of T_λ, the diagram of λ. Now, define $\alpha_i = \lambda_i - i, \beta_i = \lambda'_i - i$ $(1 \leq i \leq r)$ so we write $\lambda = (\alpha|\beta)$. Conversely if $(\alpha|\beta)$ is given, then $\mu = (\beta_r - r + 1, \ldots, \beta_1)$ is a partition and we define $\lambda = (\lambda_1, \ldots, \lambda_r)$ by the equations

$$
\begin{aligned}
&\lambda_i = \alpha_i + i \quad \text{for } 1 \leq i \leq r \\
&\lambda_i = \mu'_{i-r} \quad \text{for } r+1 \leq i \leq r + \mu_1.
\end{aligned}
\tag{3.154}
$$

With this notation, we can parametrize semi-infinite monomials directly by $\lambda = (\alpha|\beta)$ using the formula

$$
\psi^m_\lambda = \check{v}^*_{-\beta_r + m} \cdots \check{v}^*_{-\beta_1 + m} \hat{v}_{\alpha_r + m + 1} \cdots \hat{v}_{\alpha_1 + m + 1} \psi^m_0 . \tag{3.155}
$$

Using this equality, it's possible to define a vector space isomorphism $\bigwedge^{\infty/2} V \to F$

$$\psi_\lambda^m \longmapsto \psi^-_{-\beta_r+m-\frac{1}{2}} \cdots \psi^-_{-\beta_1+m-\frac{1}{2}} \psi^+_{-\alpha_r-m-\frac{1}{2}} \cdots \psi^+_{-\alpha_1-m-\frac{1}{2}} |m\rangle. \tag{3.156}$$

This isomorphism is defined in such a way that the wedge operator $\hat{v}_n$ is identified with the action of $\psi^+_{-n+\frac{1}{2}}$ and the contraction operator $\check{v}_n^*$ is identified with the action of $\psi^-_{n-\frac{1}{2}}$. The reason we have introduced the Dirac sea $\bigwedge^{\infty/2} V$ is so that we can use Theorems (3.5.18) and (3.5.19) to do linear algebra on $\bigwedge^{\infty/2} V$.

Example 3.5.21. For charge $m = -2$ and partitions $\lambda = (1^3), (3)$ and $(2,1)$ we have the corresponding elements in F^{-2} respectively: $-\psi^-_{-\frac{9}{2}}\psi^-_{-\frac{1}{2}}|0\rangle$, $\psi^-_{-\frac{5}{2}}\psi^-_{-\frac{3}{2}}\psi^-_{-\frac{1}{2}}\psi^+_{-\frac{1}{2}}|0\rangle$ and $-\psi^-_{-\frac{7}{2}}\psi^-_{-\frac{3}{2}}|0\rangle$.

Lemma 3.5.22. Through the isomorphism $\bigwedge^{\infty/2} V \to F$, the action of the shift operator $r(\Lambda_k)$, $k \neq 0$ on $\bigwedge^{\infty/2} V$ is identified with the action of α_k on F.

Proof: Using the wedge and contraction operators, it can be seen that

$$r(\Lambda_k) = \sum_{j\in\mathbb{Z}} \hat{v}_i \check{v}^*_{i+k} \tag{3.157}$$

which is identified with the sum $\sum_{j\in\mathbb{Z}} \psi^+_{-j+\frac{1}{2}}\psi^-_{j+k-\frac{1}{2}}$ acting on F. But using the equation (3.46), the relations (3.97) and $\psi^\pm_{(n)} = \psi^\pm_{n-\frac{1}{2}}$, we get that $\alpha_k = \sum_{j\in\mathbb{Z}} \psi^+_{-j+\frac{1}{2}}\psi^-_{j+k-\frac{1}{2}}$.

As a corollary, the commutation relation $[r(\Lambda_k), r(\Lambda_l)] = k\delta_{k+l,0}$ is satisfied for $k, l \neq 0$. So, although $r(\Lambda_0)$ is not defined from the definition (3.138), it's convenient to put $r(\Lambda_0)\psi^m_\lambda = m\psi^m_\lambda$, so we get a copy $\{r(\Lambda_k)\}_{k\in\mathbb{Z}} \cup \{\mathbb{1}_F\}$ of the oscillator algebra.

Notation 3.5.23. Because of the identifications between the Dirac sea and the space F, we will just write $F = \bigwedge^{\infty/2} V$ with the identifications:

$$\begin{aligned} |0\rangle &\longleftrightarrow \psi_0^0 \\ \psi^+_{-n+\frac{1}{2}} &\longleftrightarrow \hat{v}_n \\ \psi^-_{n-\frac{1}{2}} &\longleftrightarrow \check{v}_n^* \\ \alpha_k &\longleftrightarrow r(\Lambda_k). \end{aligned}$$

We write $r^B(\Lambda_k) = \sigma \circ r(\Lambda_k) \circ \sigma^{-1}$ for the transposed action of shift operators to the bosonic Fock space B. Therefore (see (3.5.5))

$$\begin{aligned} r^B(\Lambda_k) &= \frac{\partial}{\partial x_k} \text{ for } k > 0 \\ r^B(\Lambda_k) &= k x_k \text{ for } k < 0. \end{aligned} \tag{3.158}$$

Also, we write $R^B(A) = \sigma \circ R(A) \circ \sigma^{-1}$ for $A \in \overline{GL}_\infty$. Consequently, by theorem (3.5.18), $R^B(\exp(y_k \Lambda_k)) = \exp(y_k r^B(\Lambda_k)) = \exp\left(y_k \frac{\partial}{\partial x_k}\right)$.

Now we can state the second part of the Boson-Fermion correspondence.

Theorem 3.5.24. [10] Through the isomorphism

$$\sigma : F \to B : r(\Lambda_{-1})^{n_1} \cdots r(\Lambda_{-k})^{n_k} \psi_0^m \mapsto \prod_{j=1}^{k} j^{n_j} x_1^{n_1} \cdots x_k^{n_k} z^m \tag{3.159}$$

We have

$$\sigma(v_{i_m} \wedge v_{i_{m-1}} \wedge \cdots) = s_\lambda(x) z^m, \tag{3.160}$$

where λ is defined by $\lambda_k = i_{m-k+1} - m + k - 1$ for $k \geq 1$ and $s_\lambda(x)$ is the Schur polynomial associated to the partition λ.

Proof: Since $r(\Lambda_k)$ is locally nilpotent, the operator $\prod_{k \geq 1} \exp(y_k r(\Lambda_k))$ is well defined.

But since $[r(\Lambda_k), r(\Lambda_l)] = 0$ for $k, l \geq 1$, $\prod_{k\geq 1} \exp(y_k r(\Lambda_k)) = \exp(\sum_k y_k r(\Lambda_k))$. Now

$$\begin{aligned}
&\sigma(R(\exp(y_1\Lambda_1 + y_2\Lambda_2 + \cdots)v_{i_m} \wedge v_{i_{m-1}} \wedge \cdots)) \\
&\quad = R^B(\exp(y_1\Lambda_1 + y_2\Lambda_2 + \cdots))\sigma(v_{i_m} \wedge v_{i_{m-1}} \wedge \cdots) \\
&\quad = R^B(\exp(y_1\Lambda_1 + y_2\Lambda_2 + \cdots))P(x) \\
&\quad = \exp\left(\sum_{k\geq 1} y_k r^B(\Lambda_k)\right) P(x) \\
&\quad = \exp\left(\sum_{k\geq 1} y_k \frac{\partial}{\partial x_k}\right) P(x) \\
&\quad = \prod_{k\geq 1} \exp\left(y_k \frac{\partial}{\partial x_k}\right) P(x) \\
&\quad = P(x+y) \quad \text{(because of lemma (3.3.6))}.
\end{aligned} \tag{3.161}$$

Therefore $\sigma(R(\exp(y_1\Lambda_1 + y_2\Lambda_2 + \cdots)v_{i_m} \wedge v_{i_{m-1}} \wedge \cdots))$ is a polynomial with constant part $P(x+y)\big|_{x=0} = P(y)$. From the other side, this constant part must be the coefficient of $\sigma(\psi_0^m)$ in the expansion of $\sigma(R(\exp(y_1\Lambda_1 + y_2\Lambda_2 + \cdots)v_{i_m} \wedge v_{i_{m-1}} \wedge \cdots))$. But, by theorem (3.5.19), this is $\det(A^{i_m, i_{m-1}, \cdots}_{m, m-1, \ldots})$ where $A = \exp(y_1\Lambda_1 + y_2\Lambda_2 + \cdots)$. Now since $\Lambda_k = \Lambda_1^k$, we have $A = \exp(y_1\Lambda_1 + \cdots) = \sum_{k\geq 0} S_k(y)\Lambda_k$ and $A = (A_{mn}) = (h_{n-m}(y))$. So, the coefficient of $\sigma(\psi_0^m)$ is

$$\begin{aligned}
\det(A^{i_m, i_{m-1}, \cdots}_{m, m-1, \ldots}) &= \det(A_{m-k+1, i_{m-l+1}})_{1\leq k,l\leq r} \\
&= \det(S_{i_{m-l+1}-m+k-1}(y))_{1\leq k,l\leq r} \\
&= \det(S_{\lambda_l+k-l}(y))_{1\leq k,l\leq r} \\
&= s_\lambda(y) \quad \text{(see Remark (3.5.13) and equation (3.132))}
\end{aligned} \tag{3.162}$$

and the theorem is proved.

Corollary 3.5.25. Let

$$S(u) = \exp\left(\sum_{j\geq 1} x_j u^j\right) \exp\left(-\sum_{j\geq 1} \frac{1}{j}\frac{\partial}{\partial x_j} u^{-j}\right) = \sum_{j\in\mathbb{Z}} S_j u^j \tag{3.163}$$

and $\lambda = (\lambda_1, \ldots, \lambda_r)$ be a partition. Then

$$s_\lambda(x) = S_{\lambda_1} \cdots S_{\lambda_r} \cdot 1. \tag{3.164}$$

Proof: It can be seen directly that $\hat{v}_k \psi_\lambda^m = \psi^{m+1}_{(m-k-1,\lambda)}$ for $k > \lambda_1 + m$. Therefore

$$\hat{v}_{m+r+\lambda_1} \cdots \hat{v}_{m+1+\lambda_r} \psi_0^m = \psi_\lambda^{m+r} \tag{3.165}$$

and by the second part of the Boson-Fermion correspondence (3.5.24), we get

$$\Gamma_{m+r+\lambda_1}\cdots\Gamma_{m+1+\lambda_r}\cdot z^m = s_\lambda(x)z^{m+r}. \tag{3.166}$$

where $\Gamma(u)=\sum_{j\in\mathbb{Z}}\Gamma_j u^j$. But, by the first part of the boson-fermion correspondence, we have

$$\begin{aligned}\Gamma(u_r)\cdots\Gamma(u_1)\cdot z^m &= R(u_r)\cdots R(u_1)S(u_r)\cdots S(u_1)z^m\\ &= u_r^{m+r}\cdots u_1^{m+1}S(u_r)\cdots S(u_1)z^{m+r}\end{aligned} \tag{3.167}$$

and comparing the coefficients of $u_r^{m+r+\lambda_1}\cdots u_1^{m+1+\lambda_r}$ on both sides we get

$$s_\lambda(x)z^{m+r} = \Gamma_{m+r+\lambda_1}\cdots\Gamma_{m+1+\lambda_r}\cdot z^m = S_{\lambda_1}\cdots S_{\lambda_r}\cdot z^{m+r} \tag{3.168}$$

and $s_\lambda(x)=S_{\lambda_1}\cdots S_{\lambda_r}\cdot 1$.

Making a formal change of variables $x_j\to p_j/j$, where $p_j=\epsilon_1^j+\epsilon_2^j+\cdots$ denote the jth power sum symmetric function, we get the vertex realization of Schur symmetric functions:

$$S(u)=\exp\left(\sum_{j\geq 1}\frac{p_j}{j}u^j\right)\exp\left(-\sum_{j\geq 1}\frac{\partial}{\partial p_j}u^{-j}\right). \tag{3.169}$$

This vertex operator acts on the bosonic Fock space $B=\mathbb{C}[p_1,p_2,\dots]$. Observe that we can identify B with the ring of symmetric functions $\Lambda_{\mathbb{C}}$ (see Remark (3.5.13))

Theorem 3.5.26. [9] Let $S(u)$ be as in the equation (3.169), then

$$S(u_r)\cdots S(u_1)\cdot 1=\prod_{1\leq i<j\leq r}\left(1-\frac{u_i}{u_j}\right)H(u_r)\cdots H(u_1). \tag{3.170}$$

Proof: First, we note that if $[x,y]$ commutes with x and y, then $e^xe^y=e^ye^xe^{[x,y]}$. This follows from the identity $[x,y^k]=y^k[x,y]$ and

$$\begin{aligned}xe^y &= e^yx+[x,e^y]\\ &= e^yx+\sum_{k\geq 0}\frac{1}{k!}[x,y^k]\\ &= e^yx+\sum_{k\geq 0}\frac{y^k}{k!}[x,y]\\ &= e^y(x+[x,y]).\end{aligned} \tag{3.171}$$

Iterating this equation, we get $x^ke^y=e^y(x+[x,y])^k\ \forall k\geq 0$ and $e^xe^y=e^ye^{x+[x,y]}=$

$e^y e^x e^{x+[x,y]}$. Now, for each i, j $(1 \le i < j \le r)$, we compute

$$\begin{aligned} &\exp\left(-\sum_{k\ge1}\frac{\partial}{\partial p_k}u_j^{-k}\right)\exp\left(\sum_{l\ge1}\frac{1}{l}p_l u_i^l\right) \\ &\quad=\exp\left(\sum_{l\ge1}\frac{1}{l}p_l u_i^l\right)\exp\left(-\sum_{k\ge1}\frac{\partial}{\partial p_k}u_j^{-k}\right)\exp\left(-\sum_{k,l\ge1}\left[\frac{\partial}{\partial p_k},\frac{p_l}{l}\right]u_j^{-k}u_i^l\right). \end{aligned} \tag{3.172}$$

But

$$\begin{aligned} \exp\left(-\sum_{k,l\ge1}\left[\frac{\partial}{\partial p_k},\frac{p_l}{l}\right]u_j^{-k}u_i^l\right) &= \exp\left(-\sum_{k,l\ge1}\frac{1}{k}\delta_{k,l}u_j^{-k}u_i^l\right) \\ &= \exp\left(-\sum_{k\ge1}\frac{1}{k}\left(\frac{u_i}{u_j}\right)^k\right) \\ &= \left(1-\frac{u_i}{u_j}\right) \end{aligned} \tag{3.173}$$

while

$$\exp\left(-\sum_{k\ge1}\frac{\partial}{\partial p_k}u_j^{-k}\right)\cdot 1 = 1 \tag{3.174}$$

for all j. Furthermore, by the equation (3.129)

$$\prod_{i=1}^{r}\exp\left(\sum_{k\ge1}\frac{1}{k}p_k u_i^k\right)\cdot 1 = H(u_r)\cdots H(u_1). \tag{3.175}$$

So, moving all the differential operators to the right and doing the necessary commutations we get the result.

Corollary 3.5.27. By extracting coefficients u^λ, $\lambda = (\lambda_1, \ldots, \lambda_r)$ on both sides of equation (3.170) we get the identity

$$S_\lambda \cdot 1 = \prod_{i<j}(1 - R_{ij})h_\lambda \tag{3.176}$$

where $R_{ij}h_\lambda = h_{(\lambda_1,\ldots,\lambda_i+1,\ldots,\lambda_j-1,\ldots,\lambda_r)}$.

3.5.4 Orthogonality

There is an Hermitian form on F, relative to which the semi-infinite monomials are orthonormal.

Lemma 3.5.28. [10] Let $\langle\cdot|\cdot\rangle$ be the hermitian form on F where the set of semi-infinite monomials forms an orthonormal set. For two semi-infinite monomials $\psi, \psi' \in F$ we have

$$\langle r(E_{ij})\psi|\psi'\rangle = \langle\psi|r(E_{ji})\psi'\rangle \tag{3.177}$$

where $E_{ij}v_k = \delta_{jk}v_i$.

Proof: Let $\psi = v_{i_1}\wedge v_{i_2}\wedge\cdots$, there are two cases: If $j \notin \{i_1, i_2, \dots\}$, then $r(E_{ij})\psi = 0 \Rightarrow \langle r(E_{ij})\psi|\psi'\rangle = 0$. But in this case we necessarily have $\psi \neq r(E_{ji})\psi'$, so $\langle\psi|r(E_{ji})\psi'\rangle = 0$. Now, if $j \in \{i_1, i_2, \dots\}$ then $r(E_{ij})\psi = v_{i_1}\wedge\cdots\wedge v_j\cdots \neq 0$ and $r(E_{ji})r(E_{ij})\psi = \psi$. Therefore $\psi' = r(E_{ij})\psi \iff r(E_{ji})\psi' = \psi$, and equation (3.177) is satisfied.

Since $\Lambda_k = \sum_{j\in\mathbb{Z}} E_{j,j+k}$, lemma (3.5.28) says that

$$\langle r(\Lambda_k)\psi|\psi'\rangle = \langle\psi|r(\Lambda_k)^\dagger\psi'\rangle \tag{3.178}$$

where $A^\dagger$ means the transpose conjugate of $A \in \overline{GL}_\infty$. But since $r(\Lambda_k)^\dagger = r(\Lambda_{-k}) = \omega(r(\Lambda_k))$, we get by Theorem 3.4.9 that $\langle\cdot|\cdot\rangle$ is the unique hermitian form on F^m which is contravariant with respect to ω satisfying $\langle\psi_0^m|\psi_0^m\rangle = 1$ (remember that $r(\Lambda_k)\psi_0^m = 0$ for $k > 0$, i.e., ψ_0^m is a singular vector of F).

Using the Boson-Fermion correspondence, we transport the Hermitian form $\langle\cdot|\cdot\rangle$ on F^m to a Hermitian form $\langle\cdot|\cdot\rangle_{B^m}$ on B^m through the isomorphism $\sigma_m : F^m \to B^m$. By equation (3.178), the transported hermitian form satisfies (for $k > 0$)

$$\begin{aligned} \langle r^{B^m}(\Lambda_{-k})\cdot P|Q\rangle_{B^m} = \langle P|r^B(\Lambda_k)\cdot Q\rangle_{B^m} \iff \\ \langle kx_k\cdot P|Q\rangle_{B^m} = \left\langle P\middle|\frac{\partial}{\partial x_k}Q\right\rangle_{B^m} \end{aligned} \tag{3.179}$$

and $\langle 1|1\rangle_{B^m} = 1$. Using equation (3.179), we get that $\langle 1|P\rangle_{B^m} = P(x)|_{x=0}\langle 1|1\rangle_{B^m} = P(x)|_{x=0}$ and

$$\begin{aligned} \langle P|Q\rangle_{B^m} &= \left\langle 1\middle|P\left(\frac{\partial}{\partial x_1}, \frac{1}{2}\frac{\partial}{\partial x_2}, \dots\right)Q\right\rangle_{B^m} \\ &= P\left(\frac{\partial}{\partial x_1}, \frac{1}{2}\frac{\partial}{\partial x_2}, \dots\right)Q(x)\Big|_{x=0}. \end{aligned} \tag{3.180}$$

Under the change of variables $x_j \to p_j/j$, equation (3.180) implies

$$\langle p_\lambda|p_\mu\rangle_{B^m} = \delta_{\lambda\mu}z_\lambda \tag{3.181}$$

where $p_\lambda = p_{\lambda_1}p_{\lambda_2}\cdots$ and $z_\lambda = \prod_{j\geq 1} j^{m_j(\lambda)}m_j(\lambda)!$. Equation (3.181) defines a Hermitian form on the Fock space $B^m = \mathbb{C}[p_1, p_2, \dots]$. It's important to note that in this case, the

action of the oscillator algebra $\mathcal{A} = \{a_n\}_{n\in\mathbb{Z}} \cup \{\mathbb{1}\}$ is (for $n > 0$):

$$\begin{aligned} a_{-n} &\mapsto p_n \\ a_n &\mapsto n\frac{\partial}{\partial p_n} \\ a_0 &\mapsto m\mathbb{1}_{B^m} \\ \mathbb{1} &\mapsto \mathbb{1}_{B^m}. \end{aligned} \tag{3.182}$$

Therefore, the anti-linear anti-involution ω of $\mathcal{A}$ defines an adjoint operator $p_n^\dagger = n\frac{\partial}{\partial p_n}$. By the second part of the Boson-Fermion correspondence (3.5.24), equation (3.181) defines the unique hermitian form with the property

$$\langle s_\lambda | s_\mu \rangle_{B^m} = \delta_{\lambda\mu}. \tag{3.183}$$

The ring of symmetric functions $\Lambda_{\mathbb{C}}$ can be identified with the Fock space $\mathbb{C}[p_1, p_2, \dots]$. So, equation (3.181) defines a hermitian form $\langle \cdot, \cdot \rangle$ on $\Lambda_{\mathbb{C}}$ and by equation (3.183), the set of Schur symmetric functions forms an orthonormal basis of $(\Lambda_{\mathbb{C}}, \langle \cdot, \cdot \rangle)$.

Using Corollary (3.5.25) and equation (3.169), get the a version of the Boson-Fermion correspondence that will be of interest in the next chapter.

Theorem 3.5.29. Let $p_j = \sum_{i\geq 1} \epsilon_i^j$ be power sum functions in ϵ and $\Lambda_{\mathbb{C}} = \mathbb{C}[p_1, p_2, \dots]$ be the ring of symmetric functions. The vertex operator

$$S(u) = \exp\left(\sum_{j\geq 1} \frac{p_j}{j} u^j\right) \exp\left(-\sum_{j\geq 1} \frac{\partial}{\partial p_j} u^{-j}\right) \tag{3.184}$$

gives a realization of Schur symmetric functions through the formula:

$$s_\lambda(\epsilon) = S_{\lambda_1} \cdots S_{\lambda_r} \cdot 1. \tag{3.185}$$

Also, the set of Schur symmetric functions form an orthonormal basis of $\Lambda_{\mathbb{C}}$ with respect to the hermitian form defined by

$$\langle p_\lambda, p_\mu \rangle = \delta_{\lambda\mu} z_\lambda \tag{3.186}$$

where $z_\lambda = \prod_{j\geq 1} j^{m_j(\lambda)} m_j(\lambda)!$. Moreover, this forms is the unique Hermitian form which is contravariant with respect to the adjoint operator $p_n^\dagger = n\frac{\partial}{\partial p_n}$.

Chapter 4

Characters of $GL_n(\mathbb{F}_q)$

4.1 Construction of Heisenberg systems

From Theorem (2.1.14) and (2.1.20) we have two sets of orbits Φ, Φ^* induced by the action of the Frobenius map $F : x \mapsto x^q$ on M and L.

For each $\phi \in \Phi$ we associate the ring $\Lambda_{\mathbb{C}}(\phi)$ of symmetric functions. We will identity

$$\Lambda_{\mathbb{C}}(\phi) = \mathbb{C}[p_1(\phi), p_2(\phi), \dots] \tag{4.1}$$

where $p_j(\phi) = \sum_{i \geq 1} \epsilon_{i,\phi}^j$. As in Theorem (2.2.17) we define the tensor product

$$\mathcal{F} = \bigotimes_{\phi \in \Phi^*} \Lambda_{\mathbb{C}}(\phi). \tag{4.2}$$

It has a basis consisting of elements $p_{\tilde{\lambda}} = \prod_{\phi \in \Phi^*} p_{\tilde{\lambda}(\phi)}(\phi)$ (we will use $\prod$ notation instead of $\otimes$) where $\tilde{\lambda} : \Phi^* \to \mathcal{P}$ denotes a weighted partition valued function, i.e.,

$$||\tilde{\lambda}|| := \sum_{\phi \in \Phi^*} |\tilde{\lambda}(\phi)| = \sum_{\phi \in \Phi^*} \sum_{i \geq 1} \tilde{\lambda}_i(\phi) < \infty. \tag{4.3}$$

There is an Hermitian form on $\mathcal{F}$ defined by

$$\langle p_{\tilde{\lambda}}, p_{\tilde{\mu}} \rangle = z_{\tilde{\lambda}} \delta_{\tilde{\lambda}\tilde{\mu}} \tag{4.4}$$

where $z_{\tilde{\lambda}} = \prod_{\phi \in \Phi^*} z_{\tilde{\lambda}(\phi)}$. Now, for a fixed partition $\lambda \in \mathcal{P}$ and orbit $\phi \in \Phi^*$, let

$$\tilde{\lambda} = \begin{cases} \phi \mapsto \lambda \\ \phi' \mapsto 0, \ \phi' \neq \phi. \end{cases}$$

Then we have a monomorphism $\Lambda_{\mathbb{C}}(\phi) \to \mathcal{F} : p_\lambda(\phi) \mapsto p_{\tilde{\lambda}}$. Identifying $p_\lambda(\phi)$ with $p_{\tilde{\lambda}}$, we get

$$\langle p_\lambda(\phi), p_\mu(\phi') \rangle = z_\lambda \delta_{\lambda\mu} \delta_{\phi\phi'} \tag{4.5}$$

so we recover the Hermitian form on $\Lambda_{\mathbb{C}}(\phi)$ obtained by the Boson-Fermion correspondence (see Theorem (3.5.29)). In particular, the vertex operator $S(\phi;u):\mathcal{F}\to\hat{\mathcal{F}}$ defined by

$$S(\phi;u)=\exp\left(\sum_{j\geq 1}\frac{p_j(\phi)}{j}u^j\right)\exp\left(-\sum_{j\geq 1}\frac{\partial}{\partial p_j(\phi)}u^{-j}\right) \tag{4.6}$$

gives the realization of Schur symmetric functions in the power sums $p_r(\phi)$, i.e. for a partition $\lambda=(\lambda_1\cdots,\lambda_r)$,

$$s_\lambda(\phi):=s_\lambda(p_1(\phi),p_2(\phi),\dots)=S_{\lambda_1}(\phi)\cdots S_{\lambda_r}(\phi)\cdot 1 \tag{4.7}$$

where $S(\phi;u)=\sum_{j\in\mathbb{Z}}S_n(\phi)u^j$.

For a weighted partition valued function $\tilde{\lambda}:\Phi^*\to\mathcal{P}$ we define $S_{\tilde{\lambda}}=\prod_{\phi\in\Phi^*}s_{\tilde{\lambda}(\phi)}(\phi)$. From the Boson-Fermion correspondence (3.5.29) and equations (4.4) and (4.5) we have

$$\langle s_\lambda(\phi),s_\mu(\phi')\rangle=\delta_{\lambda\mu}\delta_{\phi\phi'} \tag{4.8}$$

and

$$\langle S_{\tilde{\lambda}},S_{\tilde{\mu}}\rangle=\delta_{\tilde{\lambda}\tilde{\mu}}. \tag{4.9}$$

Definition 4.1.1. [9] For an orbit $\phi\in\Phi^*$ and $\xi\in\phi$, define

$$\hat{p}_n(\xi)=\begin{cases}p_{n/d(\phi)}(\phi) & \text{if } d(\phi)|n,\\ 0 & \text{otherwise.}\end{cases} \tag{4.10}$$

In particular, if $\xi,\xi'\in\phi$, then $\hat{p}_n(\xi)=\hat{p}_n(\xi')$.

Let $\xi\in L_n,\xi'\in L_m$ and $[\xi]=\phi,[\xi']=\phi'$. Then by equation (4.10) we get

$$\begin{aligned}\langle\hat{p}_n(\xi),\hat{p}_m(\xi')\rangle&=\langle p_{n/d(\phi)}(\phi),p_{m/d(\phi')}(\phi')\rangle\\&=\frac{n}{d(\phi)}\delta_{\frac{n}{d(\phi)},\frac{m}{d(\phi')}}\delta_{\phi\phi'}\\&=\frac{n}{d(\phi)}\delta_{nm}\delta_{\phi\phi'}.\end{aligned} \tag{4.11}$$

Definition 4.1.2. [9] Let $x\in M$, then we write

$$\hat{p}_n(x)=\begin{cases}(-1)^{n-1}\dfrac{1}{q^n-1}\displaystyle\sum_{\xi\in M_n^*}\overline{\langle\xi,x\rangle}\hat{p}_n(\xi) & \text{if } x\in M_n\\ 0 & \text{otherwise.}\end{cases} \tag{4.12}$$

We can see that if x, y are in the same orbit in M_n, then $\hat{p}_n(x) = \hat{p}_n(y)$. In fact, let $y = F^k(x)$, then

$$\begin{aligned}\hat{p}_n(y) &= \sum_{\xi \in M_n^*} \overline{\langle \xi, F^k x \rangle}_n \hat{p}_n(\xi) \\ &= \sum_{\xi \in M_n^*} \overline{\langle F^k \xi, x \rangle}_n \hat{p}_n(\xi) \\ &= \sum_{\xi \in M_n^*} \overline{\langle \xi, x \rangle}_n \hat{p}_n(\xi) \\ &= \hat{p}_n(x).\end{aligned} \tag{4.13}$$

The last line follows because F is an automorphism of M_n.

Lemma 4.1.3. We have

$$\hat{p}_n(\xi) = (-1)^{n-1} \sum_{x \in M_n} \langle \xi, x \rangle_n \hat{p}_n(x). \tag{4.14}$$

Proof: This follows from

$$\begin{aligned}(-1)^{n-1} \sum_{x \in M_n} \langle \xi, x \rangle_n \hat{p}_n(x) &= \frac{1}{q^n - 1} \sum_{x \in M_n} \sum_{\xi' \in M_n^*} \langle \xi, x \rangle_n \overline{\langle \xi', x \rangle}_n \hat{p}_n(\xi') \\ &= \sum_{\xi' \in M_n^*} \left(\frac{1}{q^n - 1} \sum_{x \in M_n} \langle \xi, x \rangle_n \overline{\langle \xi', x \rangle}_n \right) \hat{p}_n(\xi') \\ &= \sum_{\xi' \in M_n^*} \langle \xi, \xi' \rangle_{M_n} \hat{p}_n(\xi') \\ &= \hat{p}_n(\xi)\end{aligned} \tag{4.15}$$

because $\langle \xi, \xi' \rangle_{M_n} = \delta_{\xi\xi'}$.

Definition 4.1.4. [9] Let $n \geq 1, f \in \Phi$, we put $p_n(f) = \hat{p}_{nd(f)}(x)$ for $x \in f$.

Since $x \in M_{d(f)} \subset M_{nd(f)}$, this definition makes sense, i.e. $\hat{p}_{nd(f)}(x) \neq 0$.

Lemma 4.1.5. [9] For any two orbits $f_1, f_2 \in \Phi$, we have

$$\langle p_n(f_1), p_m(f_2) \rangle = \frac{n}{q_{f_1}^n - 1} \delta_{mn} \delta_{f_1 f_2}. \tag{4.16}$$

Proof: Let $d_i = d(f_i)$ $(i = 1, 2)$ and $x \in f_1, y \in f_2$. Then

$$\begin{aligned}\langle p_n(f_1), p_m(f_2)\rangle &= \langle \hat{p}_{nd_1}(x), \hat{p}_{md_2}(y)\rangle \\ &= (-1)^{nd_1+md_2}\frac{1}{q^{nd_1}-1}\frac{1}{q^{md_2}-1}\sum_{\substack{\xi\in M_{nd_1}\\ \xi'\in M_{md_2}}} \overline{\langle \xi, x\rangle}_{nd_1}\langle \xi', y\rangle_{md_2}\langle \hat{p}_{nd_1}(\xi), \hat{p}_{md_2}(\xi')\rangle.\end{aligned} \tag{4.17}$$

But $\langle \hat{p}_{nd_1}(\xi), \hat{p}_{md_2}(\xi')\rangle = \frac{nd_1}{d([\xi])}\delta_{nd_1,md_2}\delta_{[\xi],[\xi']}$ and

$$\begin{aligned}&\sum_{\substack{\xi\in M_{nd_1}\\ \xi'\in M_{md_2}}} \overline{\langle \xi, x\rangle}_{nd_1}\langle \xi', y\rangle_{md_2}\langle \hat{p}_{nd_1}(\xi), \hat{p}_{md_2}(\xi')\rangle \\ &\quad= \sum_{\substack{\xi\in M_{nd_1}\\ \xi'\in M_{md_2}}} \overline{\langle \xi, x\rangle}_{nd_1}\langle \xi', y\rangle_{md_2}\frac{nd_1}{d([\xi])}\delta_{nd_1,md_2}\delta_{[\xi],[\xi']} \\ &\quad= \delta_{nd_1,md_2}\sum_{\substack{\xi,\xi'\in M_{nd_1}\\ [\xi]=[\xi']}} \overline{\langle \xi, x\rangle}_{nd_1}\langle \xi', y\rangle_{nd_1}\frac{nd_1}{d([\xi])} \\ &\quad= \delta_{nd_1,md_2}\sum_{\xi\in M_{nd_1}} \overline{\langle \xi, x\rangle}_{nd_1}\left(\frac{nd_1}{d([\xi])}\sum_{j=0}^{d([\xi])-1}\langle F^j\xi, y\rangle_{nd_1}\right) \\ &\quad= \delta_{nd_1,md_2}\sum_{\xi\in M_{nd_1}} \overline{\langle \xi, x\rangle}_{nd_1}\sum_{j=0}^{nd_1-1}\langle F^j\xi, y\rangle_{nd_1} \text{ (see equation (2.12))} \\ &\quad= \delta_{nd_1,md_2}\sum_{j=0}^{nd_1-1}\sum_{\xi\in M_{nd_1}} \overline{\langle \xi, x\rangle}_{nd_1}\langle \xi, F^j y\rangle_{nd_1} \\ &\quad= \delta_{nd_1,md_2}\sum_{j=0}^{nd_1-1}(q^{nd_1}-1)\delta_{x,F^j y} \\ &\quad= \delta_{nd_1,md_2}\delta_{f_1,f_2}(q^{nd_1}-1)\frac{nd_1}{d_1} \\ &\quad= \frac{n}{q_{f_1}^n-1}\delta_{nd_1,md_2}\delta_{f_1,f_2}.\end{aligned} \tag{4.18}$$

Using equations (4.17) and (4.18) we get the result.

Definition 4.1.6. Define a gradation on $\mathcal{F}$ by putting $\deg p_n(\phi) = nd(\phi)$. The mth homogeneous component of $\mathcal{F}$ will be denoted by $\mathcal{F}_m$.

It is a direct consequence of the previous definitions that $\deg \hat{p}_n(\xi) = n,\ (\xi \in M_n^*)$, $\hat{p}_n(x) = n,\ (x \in M_n)$ and $\deg p_n(f) = nd(f)$. Moreover, from equation (4.5) we get

$$p_n(\phi)^\dagger = n\frac{\partial}{\partial p_n(\phi)} : \mathcal{F}_m \to \mathcal{F}_{m-nd(\phi)}. \tag{4.19}$$

Moreover, the equation

$$[p_m(\phi)^\dagger, p_n(\phi')] = \langle p_m(\phi), p_n(\phi')\rangle \mathbb{1}_{\mathcal{F}} \tag{4.20}$$

is satisfied. By bilinearity and Lemma (4.1.5), we get

$$[p_n(f_1)^\dagger, p_m(f_2)] = \frac{n}{q_{f_1}^n - 1}\delta_{f_1 f_2}\mathbb{1}_{\mathcal{F}}. \tag{4.21}$$

From equations (4.4) and (4.5) we see that $\langle p_{\tilde{\lambda}}, p_{t\mu}\rangle = \prod_{\phi\in\Phi^*}\langle p_{\tilde{\lambda}(\phi)}(\phi), p_{\tilde{\mu}(\phi)}(\phi)\rangle$, so we have a general rule:

$$\left\langle \prod_{m\geq 0}\mathcal{F}_m, \prod_{m\geq 0}\mathcal{F}_m \right\rangle = \prod_{m\geq 0}\langle \mathcal{F}_m, \mathcal{F}_m\rangle. \tag{4.22}$$

Theorem 4.1.7. [9] For a partition $\lambda = (\lambda_1, \ldots, \lambda_r)$, let $p_\lambda(f) = p_{\lambda_1}(f)\cdots p_{\lambda_r}(f)$. Then

$$\langle p_\lambda(f_1), p_\mu(f_2)\rangle = z_\lambda(q_{f_1}^{-1})q_{f_1}^{-|\lambda|}\delta_{\lambda\mu}\delta_{f_1 f_2} \tag{4.23}$$

where $z_\lambda(t) = \prod_{j\geq 1} j^{m_j(\lambda)}m_j(\lambda)!/(1-t^{\lambda_j})$.

Proof: The proof of this theorem closely follows the proof of Theorem (3.4.9), but using the commutator formula (4.21) and (4.22).

Lemma 4.1.8. There is a one-to-one correspondence $\Phi^* \longleftrightarrow \Phi$.

Proof: Let $n \geq 1$, we claim that there is a one-to-one correspondence between orbits $\Phi^* \to \Phi$. Let ω_n be an nth primitive root of unity, so $M_n \cong \langle\omega_n\rangle$. The character given by $\xi_n(\omega_n) = \omega_n$ generates the whole group M_n^*, so $M_n^* = \langle\xi_n\rangle$. Then we have an isomorphism $M_n^* \to M_n : \xi_n^k \mapsto \xi_n^k(\omega_n)$ which commutes the with action of the Frobenius map on M_n^* and M_n. Therefore, this map induces a bijection $\Phi^* \to \Phi$.

Lemma 4.1.9. The dimension $\dim \mathcal{F}_m$ of the mth homogeneous component $\mathcal{F}_m$ is the same as the number of weighted partition valued functions $\tilde{\lambda} : \Phi^* \to \mathcal{P}$ such that $||\tilde{\lambda}|| = m$.

Proof: Since $p_{\tilde{\lambda}} = \prod_{\phi\in\Phi^*} p_{\tilde{\lambda}(\phi)}(\phi)$, then

$$\begin{aligned}\deg p_{\tilde{\lambda}} &= \sum_{\phi\in\Phi^*} \deg p_{\tilde{\lambda}(\phi)}(\phi) \\ &= \sum_{\phi\in\Phi^*}\sum_{i\geq 1} d(\phi)|\tilde{\lambda}_i(\phi)| \\ &= ||\tilde{\lambda}||.\end{aligned} \tag{4.24}$$

But $\mathcal{F}$ is freely generated by the set $\{p_{\tilde{\lambda}} \mid \tilde{\lambda} : \Phi^* \to \mathcal{P}\}$ and the lemma follows.

Theorem 4.1.10. We have an isomorphism

$$\mathcal{F} \cong \bigotimes_{f\in\Phi} \mathbb{C}[p_1(f), p_2(f), \dots]. \tag{4.25}$$

Proof: Let $\boldsymbol{\mu} : \Phi \to \mathcal{P}$ be a weighted partition valued function of weight $||\boldsymbol{\mu}|| = m$ for a fixed $m \geq 1$. By Lemma (4.1.8), the number of such functions is the same as that of weighted partition valued functions $\tilde{\lambda} : \Phi^* \to \mathcal{P}$ of weight $||\tilde{\lambda}|| = m$. Therefore by Lemma (4.1.9), the number of weighted partition valued functions $\boldsymbol{\mu} : \Phi \to \mathcal{P}$ of weight $||\boldsymbol{\mu}|| = m$ is dim $\mathcal{F}_m$. Moreover by Theorem (4.1.7), the set of polynomials $\prod_{f\in\Phi} p_{\boldsymbol{\mu}(f)}(f),\ ||\boldsymbol{\mu}|| = m$ is a linearly independent set. But since deg $\prod_{f\in\Phi} p_{\boldsymbol{\mu}(f)}(f) = ||\boldsymbol{\mu}||$, we get that this set is a basis of $\mathcal{F}_m$ and the result follows because $\mathcal{F} = \bigoplus_{m\geq 1} \mathcal{F}_m$.

4.2 The characteristic map

4.2.1 Vertex operator for Hall-Littlewood symmetric polynomials

A consequence of Lemma (4.1.5) is that $\Lambda_{\mathbb{C}}(f) = \mathbb{C}[p_1(f), p_2(f), \dots]$ can be regarded as a representation of a "deformed" version of the oscillator algebra $\mathcal{A} = \{a_n\}_{n\neq 0} \cup \{\mathbb{1}\}$ with commutator relation given by

$$[a_n, a_m] = \frac{n}{q_f^n - 1}\delta_{nm}\mathbb{1}. \tag{4.26}$$

In [8], a version of the oscillator algebra $\mathcal{A} = \{a_n\}_{n\neq 0} \cup \{\mathbb{1}\}$ is presented, with commutation relation given by

$$[a_n, a_m] = \frac{n}{1 - t^{|n|}}\delta_{nm}\mathbb{1}. \tag{4.27}$$

In this case, the bosonic Fock space B is realized as Sym$(a_{-n} : n \geq 1)$, and the action of $\mathcal{A}$ on B is given by

$$\begin{cases} \text{For } n > 0 \\ a_{-n} \mapsto a_{-n}\cdot \quad \text{(multiplication by } a_{-n}) \\ a_n \mapsto \dfrac{n}{1-t^n}\dfrac{\partial}{\partial a_{-n}}. \end{cases} \tag{4.28}$$

Moreover, a vertex operator $H(u) : B \to B[u^{\pm}]$ is defined

$$H(u) = \exp\left(\sum_{j\geq 1} \frac{1-t^j}{j} a_{-j} u^j\right) \exp\left(-\sum_{j\geq 1} \frac{1-t^j}{j} a_j u^{-j}\right) = \sum_{j\in\mathbb{Z}} H_j z^j \tag{4.29}$$

which is a generalization of the vertex operator given in (3.5.29). As in theorem (3.4.9), there is an unique hermitian form $\langle \cdot, \cdot \rangle$ which is contravariant with respect to the anti-linear anti-involution $\omega : a_n \mapsto a_{-n}$. This hermitian form satisfies

$$\langle a_{-\lambda}, a_{-\mu} \rangle = z_\lambda(t) \delta_{\lambda\mu} \tag{4.30}$$

where $z_\lambda(t) = \prod_{j\geq 1} j^{m_j(\lambda)} m_j(\lambda)!/(1-t^{\lambda_j})$.

Theorem 4.2.1. ([8] Proposition 2.20) For any partition $\lambda = (\lambda_1, \ldots, \lambda_r)$, let $H_\lambda = H_{\lambda_1} \cdots H_{\lambda_r} \cdot 1$. Then, with respect to the Hermitian form (4.30), we have

$$\langle H_\lambda \cdot 1, H_\mu \cdot 1 \rangle = b_\lambda(t) \delta_{\lambda\mu}. \tag{4.31}$$

In particular, $\{H_\lambda \cdot 1 \mid \lambda \in \mathcal{P}\}$ is a linearly independent set.

Definition 4.2.2. Define a gradation on $B = \mathrm{Sym}(a_{-n} : n \geq 1)$ by

$$\deg(a_\lambda) = |\lambda|. \tag{4.32}$$

So the mth homogeneous component B_m of B has dimension $p(m)$, the number of partitions of m.

Theorem 4.2.3. The set $\{H_\lambda \cdot 1 \mid \lambda \in \mathcal{P}\}$ is a basis of $\mathcal{F}$.

Proof: It's a direct consequence of (4.29) that $H_n : B_m \to B_{m+n}$. Therefore $\deg(H_\lambda \cdot 1) = |\lambda|$ and the set $\{H_\lambda \cdot 1 \mid |\lambda| = m\} \subset B_m$ is a linearly independent set with $p(m)$ elements. It follows that $V_m = \mathrm{Span}\{H_\lambda \cdot 1 \mid |\lambda| = m\}$ and that $\{H_\lambda \cdot 1 \,|\lambda \in \mathcal{P}\}$ is a basis of V.

From Theorems (4.1.7) and (4.1.10), we see that the vertex operator (4.29) acts on $\mathcal{F}$, provided we make a formal change in the parameters:

$$\begin{cases} t \mapsto q_f^{-1} \\ a_{-n} \mapsto q_f^{n/2} p_n(f) \\ u \mapsto q_f^{1/2} u \end{cases} \tag{4.33}$$

where $f \in \Phi$ is chosen as before. With these changes we write the vertex operator (4.29) as

$$Q(u;f) = \exp\left(\sum_{j\geq 1} \frac{q_f^j - 1}{j} p_j(f) u^j\right) \exp\left(-\sum_{j\geq 1} q_f^{-j} \frac{\partial}{\partial p_j(f)} u^{-j}\right) = \sum_{j\in\mathbb{Z}} Q_j u^j. \tag{4.34}$$

Theorem 4.2.4. ([9], Proposition 4.1) For a partition $\lambda = (\lambda_1, \ldots, \lambda_r)$, let $Q_\lambda \cdot 1 = Q_{\lambda_1} \cdots Q_{\lambda_r} \cdot 1$. Then $\{Q_\lambda \cdot 1 \mid \lambda \in \mathcal{P}\}$ is a orthogonal basis of $\mathcal{F}$. In particular

$$\langle Q_\lambda \cdot 1, Q_\mu \cdot 1\rangle = b_\lambda(q_f^{-1}) q_f^{|\lambda|} \delta_{\lambda\mu}, \tag{4.35}$$

where the Hermitian form $\langle \cdot, \cdot \rangle$ is defined by $\langle p_\lambda, p_\mu\rangle = z_\lambda \delta_{\lambda\mu}$ and $b_\lambda(t) = \prod_{i\geq 1} \varphi_{m_i(\lambda)}(t)$.

Proof: Under the formal change of variables (4.33), we get the identification $H_n = q_f^{n/2} Q_n$, and using Theorem (4.2.1) we get the result. Moreover, the fact that $\mathcal{F} = \mathrm{Span}\{Q_\lambda \cdot 1 \mid \lambda \in \mathcal{P}\}$ is a consequence of Theorem (4.2.3).

Theorem 4.2.5. [8] Let $H(u)$ be the vertex operator (4.29). Then we have the identity

$$H_n H_{m+1} - t H_{n+1} H_m = t H_{m+1} H_n - H_{m+1} H_n. \tag{4.36}$$

Proof: Let

$$H(u,v) = \exp\left(\sum_{j\geq 1} \frac{1-t^j}{j} a_{-j}(u^j + v^j)\right) \exp\left(-\sum_{j\geq 1} \frac{1-t^j}{j} a_j (u^{-j} + v^{-j})\right), \tag{4.37}$$

so $H(u,v) = H(u,v)$. Moreover, using the rule $e^x e^y = e^y e^x e^{[x,y]}$ for the case $[[x,y],x] = [[x,y],y] = 0$, we get

$$\begin{aligned} H(u)H(v) &= H(u,v) \exp\left(-\sum_{j,k\geq 1} \frac{1-t^j}{j} \frac{1-t^k}{k} [a_j, a_{-k}]\, u^{-j} v^k\right) \\ &= H(u,v) \exp\left(-\sum_{j\geq 1} \frac{1-t^j}{j} u^{-j} v^j\right) \\ &= H(u,v) \exp \log\left(\frac{1-\frac{v}{u}}{1-\frac{tv}{u}}\right) \\ &= H(u,v)\left(\frac{1-\frac{v}{u}}{1-\frac{tv}{u}}\right) \\ &= H(u,v) i_{u,v} \frac{u-v}{u-tv}. \end{aligned} \tag{4.38}$$

Therefore

$$\begin{aligned}&(u-tv)H(u)H(v)+(v-tu)H(v)H(u)=0\Rightarrow\\&\sum_{n,m}(H_{n-1}H_m-tH_nH_{m-1}+H_{m-1}H_n-tH_mH_{n-1})z^nw^m=0\end{aligned}\tag{4.39}$$

and making a change in the parameters $n\to n+1, m\to m+1$ we get the result.

Corollary 4.2.6. ([9], Proposition 4.2) Let $f\in\Phi$ and $m\geq 1$, then

$$Q_n(f)Q_{m+1}(f)-q_f^{-1}Q_{n+1}(f)Q_m(f)=q_f^{-1}Q_{m+1}(f)Q_n(f)-Q_{m+1}(f)Q_n(f).\tag{4.40}$$

Proof: This follows from Theorem (4.2.5) and equation (4.33).

There exists a generating function for the elementary symmetric functions in the power sum functions. Recall from (3.126) of elementary symmetric functions $e_j(\epsilon)$ is

$$E(t)=\sum_{j\geq 0}e_j(\epsilon)t^j=\prod_{j\geq 1}(1+t\epsilon_j).\tag{4.41}$$

Moreover

$$\begin{aligned}P(-u)&=\sum_{i\geq 1}p_i(\epsilon)(-u)^{i-1}\\&=\sum_{i\geq 1}\sum_{j\geq 1}\epsilon_j^i(-u)^{i-1}\\&=\sum_{j\geq 1}\frac{\epsilon_j}{1+\epsilon_j u}\\&=\sum_{j\geq 1}\frac{d}{du}\log(1+\epsilon_j u)\\&=\frac{d}{du}\log\prod_{j\geq 1}(1+\epsilon_j u)\\&=\frac{d}{du}\log E(u)\end{aligned}\tag{4.42}$$

and we conclude that

$$E(u)=\exp\left(\sum_{j\geq 1}\frac{(-1)^{j-1}}{j}p_ju^j\right).\tag{4.43}$$

For $f\in\Phi$ we define the half vertex operator

$$E_r(u;f)=\exp\left(\sum_{j\geq 1}\frac{(-1)^{j-1}}{j}p_j(f)u^j\right)=\sum_{j\geq 1}E_j(f)u^j\tag{4.44}$$

which is a generating function for the elementary symmetric functions in $\Lambda_{\mathbb{C}}(f)$, the ring of symmetric functions in the variables $\{x_{f,i} \,|i \geq 1\}$. Under the adjoint operator $p_n(f)^\dagger = \frac{n}{q_f^n - 1}\frac{\partial}{\partial p_n(f)}$, we define

$$E_r^\dagger(u;f) = \exp\left(-\sum_{j\geq 1}\frac{(-1)^j}{q_f^j - 1}\frac{\partial}{\partial p_j(f)}u^j\right) = \sum_{j\geq 1} E_j^\dagger(f)u^j. \tag{4.45}$$

Theorem 4.2.7. ([9], Theorem 4.3) Let $\nu = (\nu_1, \ldots, \nu_l)$ be a partition, $r, (0 < r \leq l)$ a positive integer and $f \in \Phi$. Then

$$E_r^\dagger(f)Q_\nu(f)\cdot 1 = \sum_{1\leq i_1<\cdots<i_r\leq l} Q_{\nu-(e_{i_1}+\cdots+e_{i_r})}(f)\cdot 1 \tag{4.46}$$

where $e_i = (\delta_{ij})_{j=1}^l$ is the ith element vector. Moreover, $E_r^\dagger(f)Q_\nu(f)\cdot 1 = 0$ for $r > l$.

Proof: We use the relation $e^x e^y = e^y e^x e^{[x,y]}$, which is satisfied whenever $[[x,y],x] = [[x,y],y] = 0$.

$$\begin{aligned} &E^\dagger(u;f)\exp\left(\sum_{j\geq 1}\frac{q_f^j-1}{j}p_j(f)v^j\right) = \\ &\exp\left(\sum_{j\geq 1}\frac{q_f^j-1}{j}p_j(f)v^j\right)E^\dagger(u;f)\times \\ &\qquad\times\exp\left(-\sum_{j,k\geq 1}\frac{(-1)^j}{q_f^j-1}\frac{(-1)^k}{q_f^k-1}\left[\frac{\partial}{\partial p_j(f)},p_k(f)\right]u^jv^k\right). \end{aligned} \tag{4.47}$$

But

$$\begin{aligned} &\exp\left(-\sum_{j,k\geq 1}\frac{(-1)^j}{q_f^j-1}\frac{(-1)^k}{q_f^k-1}\left[\frac{\partial}{\partial p_j(f)},p_k(f)\right]u^jv^k\right) \\ &\quad= \exp\left(-\sum_{j,k\geq 1}\frac{(-1)^j}{q_f^j-1}\frac{q_f^k-1}{k}\delta_{jk}u^jv^k\right) \\ &\quad= \exp\left(-\sum_{j,k\geq 1}\frac{(-1)^j}{j}u^jv^j\right) = (1+uv) \end{aligned} \tag{4.48}$$

and

$$E^\dagger(u;f)\exp\left(\sum_{j\geq 1}\frac{q_f^j-1}{j}p_j(f)v^j\right) = \exp\left(\sum_{j\geq 1}\frac{q_f^j-1}{j}p_j(f)v^j\right)E^\dagger(u;f)(1+uv). \tag{4.49}$$

Using this commutation relation we get

$$E^\dagger(u;f)Q(v_1;f)\cdots Q(v_l;f)\cdot 1 = \prod_{1\le i\le l}(1+uv_i)Q(v_1;f)\cdots Q(v_l;f)\cdot 1. \tag{4.50}$$

Collecting the coefficients of $u^r v_1^{\nu_1}\cdots v_l^{\nu_l}$ on both sides of the previous equation we get that $E_r(f)^\dagger Q_\nu(f)\cdot 1 = 0$ for $r > l$, and for $r \le l$ we get equation (4.46).

In the next Lemma and Theorem, we fix $f \in \Phi$ and write $Q_n, E_n^\dagger$ and q, instead of $Q_n(f), E_n^\dagger(f)$ and q_f.

Lemma 4.2.8. ([9], Lemma 4.4) For $r \le m$ we have

$$E_r^\dagger Q_{(n^m)}\cdot 1 = \begin{bmatrix} m \\ r \end{bmatrix}_{q^{-1}} Q_{(n^{m-r},(n-1)^r)}\cdot 1. \tag{4.51}$$

Proof: The proof goes by induction in $m+r$. We will prove the base case first, in this case it's sufficient to verify (4.51) for $r = 1$ and $m \ge 1$. By equation (4.46), we have

$$\begin{aligned} E_1^\dagger Q_{(n^m)}\cdot 1 &= \sum_{i=1}^{m} Q_n^{i-1}Q_{n-1}Q_n^{m-i}\cdot 1 \\ &= \sum_{i=1}^{m} q^{-m+i}Q_{(n^{m-1},n-1)}\cdot 1 \quad \text{(by equation (4.40) for } n=m) \\ &= \begin{bmatrix} m \\ 1 \end{bmatrix}_{q^{-1}} Q_{(n^{m-1},n-1)}\cdot 1. \end{aligned} \tag{4.52}$$

Now, let r $(1 \le r \le m)$ be fixed, and suppose that the statement holds for $k < m+r$. In the case $r = m$, equation (4.46) has only one summand $Q_{(n-1)^m}\cdot 1$ on the right side, which coincides with (4.51). Now, if $r < m$ we divide (4.46) into two sums:

$$\begin{aligned} &\sum_{2\le i_2<\cdots<i_r\le m} Q_{(n^m)-(e_1+e_{i_2}+\cdots+e_{i_r})}\cdot 1 = \\ &= \sum_{2\le i_2<\cdots<i_r\le m} Q_{(n-1)}Q_{(n^{m-1})-(e_{i_2}+\cdots+e_{i_r})}\cdot 1 \\ &= Q_{(n-1)}\begin{bmatrix} m-1 \\ r-1 \end{bmatrix}_{q^{-1}} Q_{(n^{m-r},(n-1)^{r-1})}\cdot 1 \quad \text{(by the induction hypothesis)} \\ &= \begin{bmatrix} m-1 \\ r-1 \end{bmatrix}_{q^{-1}} q^{-m+r}Q_{(n^{m-r},(n-1)^r)}\cdot 1 \end{aligned} \tag{4.53}$$

because of equation (4.40) for $n = m$. The second summand is

$$\begin{aligned}
&\sum_{1\le i_1<\cdots<i_r\le m} Q_n Q_{(n^{m-1})-(e_{i_1}+\cdots+e_{i_r})}\cdot 1 = \\
&Q_n \begin{bmatrix} m-1 \\ r \end{bmatrix}_{q^{-1}} Q_{(n^{m-1-r},(n-1)^r)}\cdot 1 \quad \text{(by the induction hypothesis)} \\
&\begin{bmatrix} m-1 \\ r \end{bmatrix}_{q^{-1}} Q_{(n^{m-r},(n-1)^r)}\cdot 1.
\end{aligned} \tag{4.54}$$

Summing everything we get

$$\begin{aligned}
E_r^\dagger Q_{n^m}\cdot 1 &= \left(\begin{bmatrix} m-1 \\ r-1 \end{bmatrix}_{q^{-1}} q^{-m+r} + \begin{bmatrix} m-1 \\ r \end{bmatrix}_{q^{-1}} \right) Q_{(n^{m-r},(n-1)^r)}\cdot 1 \\
&= \begin{bmatrix} m \\ r \end{bmatrix}_{q^{-1}} Q_{(n^{m-r},(n-1)^r)}\cdot 1.
\end{aligned} \tag{4.55}$$

Theorem 4.2.9. ([9], Theorem 4.5) Let $\nu = (1^{m_1}\dots k^{m_k})$ be a partition of length l and $r \le l$ be a positive integer. Then

$$E_r^\dagger Q_\nu \cdot 1 = \sum_\mu \prod_{i\ge 1} \begin{bmatrix} \nu_i' - \nu_{i+1}' \\ \nu_i' - \mu_i' \end{bmatrix}_{q^{-1}} Q_\mu \cdot 1 \tag{4.56}$$

where the sum runs over all partitions $\mu \subset \nu$ such that the skew diagram $\theta = \nu - \mu$ is a vertical r-strip, i.e. $\theta_i \le 1$ while $|\theta| = r$.

Proof: By equation (4.46), we have

$$E_r^\dagger Q_\nu \cdot 1 = \sum_{\substack{r_1+\cdots+r_k=r \\ r_1\le m_1,\dots,r_k\le m_k}} \sum_{v_1,\dots v_k} Q_{((k^{m_k})-v_k,\dots,(1^{m_1})-v_1)}\cdot 1 \tag{4.57}$$

where each $v_j, (1 \le j \le k)$ ranges over all the vectors $e_{i_1}+\cdots+e_{i_{r_j}}$ for $1 \le i_1 < \cdots < i_{r_j} \le m_j$. By Lemma (4.2.8), we get

$$E_r^\dagger Q_\nu \cdot 1 = \sum_{\substack{r_1+\cdots+r_k=r \\ r_1\le m_1,\dots,r_k\le m_k}} \prod_{i\ge 1} \begin{bmatrix} m_i \\ r_i \end{bmatrix}_{q^{-1}} Q_{(k^{m_k-r_k},\dots,j^{m_j+r_{j+1}-r_j},\dots,1^{m_1+r_2-r_1})}\cdot 1. \tag{4.58}$$

Now, fix $r = r_1 + \cdots + r_k$ and let $\mu = (k^{m_k-r_k},\dots,j^{m_j+r_{j+1}-r_j},\dots,1^{m_1+r_2-r_1})$. Then,

setting $r_{k+1} = 0$, we get $m_j(\mu) = m_j + r_{j+1} - r_j, (1 \leq j \leq k)$. In particular

$$\begin{aligned} r_i &= r_i - r_{k+1} \\ &= \sum_{j=i}^{k} (r_j - r_{j+1}) \\ &= \sum_{j=i}^{k} (m_j - m_j(\mu)) \\ &= \sum_{j=i}^{k} (\nu'_j - \nu'_{j+1} - \mu'_j + \mu'_{j+1}) \\ &= \nu'_i - \mu'_i \end{aligned} \tag{4.59}$$

because $\nu'_{k+1} = \mu'_{k+1} = 0$. Therefore $\mu'_i = \nu'_i - r_i, (1 \leq i \leq l)$ while $r = \sum r_i, (r_i \leq m_i)$. This characterizes $\theta = \nu - \mu$ as being a vertical $r-$strip. So we conclude from equation (4.58) that

$$E_r^{\dagger} Q_\nu \cdot 1 = \sum_{\mu} \prod_{i \geq 1} \begin{bmatrix} \nu'_i - \nu'_{i+1} \\ \nu'_i - \mu'_i \end{bmatrix}_{q^{-1}} Q_\mu \cdot 1 \tag{4.60}$$

summed over all μ such that $\nu - \mu$ is a vertical strip.

4.2.2 Hall-Littlewood polynomials

In [8], it's shown that under the characteristic map $\iota(a_{-1}^{m_1} \cdots a_{-k}^{m_k}) = p_1^{m_1} \cdots p_k^{m_k}$, we have $\iota(H_\lambda \cdot 1) = L_\lambda$, where

$$L_\lambda(x_1, \ldots, x_n; t) = (1-t)^{l(\lambda)} \sum_{w \in S_n} w \left(x_1^{\lambda_1} \ldots x_n^{\lambda_n} \prod_{i<j} \frac{x_i - t x_j}{x_i - x_j} \right) \tag{4.61}$$

is a symmetric polynomial for $n \geq l(\lambda)$. It's important to note that

$$L(x_1, \ldots, x_n, 0; t) = L(x_1, \ldots, x_n; t), \tag{4.62}$$

so it make sense to talk about the L_λ in the limit $n \to \infty$. In [8], the polynomials L_λ are called Hall-Littlewood polynomials. However, in [12], the Hall-Littlewood polynomials are defined by

$$P_\lambda(x_1, \ldots, x_n; t) = \frac{1}{v_\lambda(t)} \sum_{w \in S_n} w \left(x_1^{\lambda_1} \ldots x_n^{\lambda_n} \prod_{i<j} \frac{x_i - t x_j}{x_i - x_j} \right) \tag{4.63}$$

where $v_\lambda = (1-t)^{-l(\lambda)} b_\lambda(t)$. They serve to interpolate Schur symmetric s_λ functions and monomial symmetric functions $m_\lambda(x)$, i.e.

$$\begin{aligned} P_\lambda(x_1, \ldots, x_n; 0) &= s_\lambda(x_1, \ldots, x_n) \\ P_\lambda(x_1, \ldots, x_n; 1) &= m_\lambda(x_1, \ldots, x_n). \end{aligned} \tag{4.64}$$

In this work we will choose equation (4.63) to define Hall-Littlewood polynomials. Therefore $\iota(H_\lambda \cdot 1) = b_\lambda(t)P_\lambda$, and under the formal change of variables (4.33) we get

$$\begin{aligned} q_f^{-|\lambda|}Q_\lambda(f)\cdot 1 = b_\lambda(q_f^{-1})P_\lambda(f) \Rightarrow \\ P_\lambda(f) = q_f^{-|\lambda|}b_\lambda(q_f^{-1})^{-1}Q_\lambda(f)\cdot 1 \end{aligned} \tag{4.65}$$

for $f \in \Phi$.

Notation 4.2.10. We will write $P_\lambda(f) = q_f^{-|\lambda|}b_\lambda(q_f^{-1})^{-1}Q_\lambda(f)\cdot 1$ for Hall-Littlewood polynomials in $\Lambda_{\mathbb{C}}(f)$. Moreover by Theorem (4.2.4) we have

$$\langle P_\lambda(f), Q_\mu(f)\rangle = \delta_{\lambda\mu}. \tag{4.66}$$

Theorem 4.2.11. [8] Let λ be a partition and μ be the partition obtained by removing the last column of λ. Then, if the last column of λ has r boxes, the following relation is satisfied:

$$P_\mu(f)P_{(1^r)}(f) = P_\lambda(f) + \sum_\nu g^\nu_{\mu(1^r)}(q_f)P_\nu(f) \tag{4.67}$$

in $\Lambda_{\mathbb{C}}(f)$, where $g^\nu_{\mu(1^r)}(q_f) = \prod_{i\geq 1}\begin{bmatrix}\nu'_i - \nu'_{i+1}\\ \nu'_i - \mu'_i\end{bmatrix}_{q_f^{-1}}$.

Proof: We will use the fact that the Hall-Littlewood polynomial $P_{(1^r)}$ coincides with the elementary symmetric function e_r ([12], p.209). So we write

$$P_\mu(f)P_{(1^r)}(f) = E_r(f)\cdot P_\mu(f). \tag{4.68}$$

Now, by Theorem (4.2.9) and equation (4.66),

$$\langle E_r(f)\cdot P_\mu(f), Q_\nu(f)\rangle = \langle P_\mu, E_r(f)^\dagger Q_\nu(f)\rangle \tag{4.69}$$

is nonzero only if $\nu - \mu$ is a vertical strip, and if that is the case we have

$$\langle P_\mu(f), E_r(f)^\dagger Q_\nu(f)\rangle = \prod_{i\geq 1}\begin{bmatrix}\nu'_i - \nu'_{i+1}\\ \nu'_i - \mu'_i\end{bmatrix}_{q_f^{-1}} = g^\nu_{\mu(1^r)}(q_f). \tag{4.70}$$

But, by Theorem (4.2.4) and the definition of P_λ, we see that the set of Hall-Littlewood polynomials $\{P_\lambda(f) \mid \lambda \in \mathcal{P}\}$ is a basis of $\Lambda_{\mathbb{C}}(f)$. So from equation (4.70), we see that

$$P_\mu(f)P_{(1^r)}(f) = \sum_\nu g^\nu_{\mu(1^r)}(q_f)P_\nu(f). \tag{4.71}$$

Moreover

$$g^{\lambda}_{\mu(1^r)}(q_f) = \prod_{i=1}^{\lambda_1-1} \begin{bmatrix} \lambda'_i - \lambda'_{i+1} \\ 0 \end{bmatrix}_{q_f^{-1}} \begin{bmatrix} \lambda'_{\lambda_1} - \lambda'_{\lambda_1+1} \\ \lambda'_{\lambda_1} - \mu'_{\lambda_1} \end{bmatrix}_{q_f^{-1}} = 1 \tag{4.72}$$

because $\lambda'_{\lambda_1+1} = \mu'_{\lambda_1} = 0$.

An important thing to notice in the sum (4.67) is that $\nu - \mu$ is a vertical strip implies $\nu \leq \mu$, so the sum is taken only over partitions dominated by λ.

Theorem 4.2.12. ([9], Theorem 4.7) For a fixed $f \in \Phi$, there exists an isomorphism

$$\theta : H(\mathbb{F}_q[t]_{(f)}) \otimes_{\mathbb{Z}} \mathbb{C} \to \Lambda_{\mathbb{C}}(f) : u_\lambda \mapsto q_f^{-|\lambda|} P_\lambda(f). \tag{4.73}$$

Proof: By Theorems (1.4.13) and (3.5.12) we can define the ring homomorphism

$$\psi : H(\mathbb{F}_q[t]_{(f)}) \otimes_{\mathbb{Z}} \mathbb{C} \to \Lambda_{\mathbb{C}}(f) : u_{(1^r)} \mapsto q_f^{-n(1^r)} P_{(1^r)}(f). \tag{4.74}$$

We will prove by induction in the weight of λ that $\theta = \psi$. If $\lambda = (0)$, we have $\theta(1) = 1 = \psi(1)$. Now, let $|\lambda| > 0$ and μ be the partition obtained by λ by deleting the last column of r boxes. Then by Theorem (1.4.19), we have

$$u_\mu u_{(1^r)} = u_\lambda + \sum_{\nu<\lambda} G^{\lambda}_{\mu(1^r)}(q_f) u_\lambda. \tag{4.75}$$

where $G^{\nu}_{\mu(1^r)}(q_f) = q_f^{n(\nu)-n(\mu)-n(1^r)} \prod_{i\geq 1} \begin{bmatrix} \nu'_i - \nu'_{i+1} \\ \nu'_i - \mu'_i \end{bmatrix}_{q_f^{-1}}$ if $\nu - \mu$ is a vertical strip and zero otherwise. Now, by applying ψ on both sides of this equation and using the induction hypothesis we get

$$\begin{aligned} & q_f^{-(n(\mu)-n(1^r))} P_\mu(f) P_{(1^r)}(f) \\ &= \psi(u_\lambda) + \sum_{\nu<\mu} G^{\nu}_{\mu(1^r)} q_f^{-n(\nu)} P_\nu(f) \\ &= \psi(u_\lambda) + \sum_{\nu<\mu} q_f^{-(n(\mu)-n(1^r))} \prod_{i\geq 1} \begin{bmatrix} \nu'_i - \nu'_{i+1} \\ \nu'_i - \mu'_i \end{bmatrix}_{q_f^{-1}} P_\nu(f) \end{aligned} \tag{4.76}$$

by Theorem (1.4.19). Therefore

$$\begin{aligned} \psi(u_\lambda) &= q_f^{-(n(\mu)+n(1^r))} (P_\mu(f) P_{(1^r)}(f) - \sum_{\nu<\mu} \prod_{i\geq 1} \begin{bmatrix} \nu'_i - \nu'_{i+1} \\ \nu'_i - \mu'_i \end{bmatrix}_{q_f^{-1}} P_\nu(f)) \\ &= q_f^{-(n(\mu)+n(1^r))} P_\lambda(f) \quad \text{(by Theorem (4.2.11))} \\ &= q_f^{-n(\lambda)} P_\lambda(f) \end{aligned} \tag{4.77}$$

which concludes the induction step.

Now come to the main theorem of this section.

Theorem 4.2.13. ([8], Theorem 4.8) The characteristic map $ch : \mathcal{R}(q) \to \mathcal{F}$,

$$\pi_{\boldsymbol{\lambda}} \mapsto \prod_{f\in\Phi} q_f^{-n(\boldsymbol{\lambda}(f))} P_{\boldsymbol{\lambda}(f)}(f) \tag{4.78}$$

is an isometric ring isomorphism.

Proof: The fact that ch is an isomorphism follows from Theorems (2.2.17) and (4.2.12). Now, from equation (2.33) we get

$$\langle \pi_{\boldsymbol{\lambda}}, \pi_{\boldsymbol{\mu}} \rangle = 0 \tag{4.79}$$

if $||\boldsymbol{\lambda}|| \neq ||\boldsymbol{\mu}||$, and if $||\boldsymbol{\lambda}|| = ||\boldsymbol{\mu}|| = n$ we get

$$\begin{aligned} \langle \pi_{\boldsymbol{\lambda}}, \pi_{\boldsymbol{\mu}} \rangle &= \frac{1}{|G_n|} \sum_{g\in G_n} \pi_{\boldsymbol{\lambda}}(g)\overline{\pi_{\boldsymbol{\mu}}(g)} \\ &= \frac{1}{|G_n|} |c_{\boldsymbol{\lambda}}| \delta_{\boldsymbol{\lambda}\boldsymbol{\mu}} \\ &= a_{\boldsymbol{\lambda}}^{-1} \delta_{\boldsymbol{\lambda}\boldsymbol{\mu}} \end{aligned} \tag{4.80}$$

because of Theorem (2.2.9). For the other side

$$\begin{aligned} &\langle \prod_{f\in\Phi} q_f^{-n(\boldsymbol{\lambda}(f))} P_{\boldsymbol{\lambda}(f)}(f), \prod_{f\in\Phi} q_f^{-n(\boldsymbol{\mu}(f))} P_{\boldsymbol{\mu}(f)}(f) \rangle \\ &\quad = \prod_{f\in\Phi} q_f^{-n(\boldsymbol{\lambda}(f)-n(\boldsymbol{\mu}(f)))} \langle P_{\boldsymbol{\lambda}(f)}(f), P_{\boldsymbol{\mu}(f)}(f) \rangle \\ &\quad = \prod_{f\in\Phi} q_f^{-n(\boldsymbol{\lambda}(f)-n(\boldsymbol{\mu}(f)))} q_f^{-|\boldsymbol{\lambda}(f)|} b_{\boldsymbol{\lambda}(f)}(q_f^{-1})^{-1} \delta_{\boldsymbol{\lambda}(f)\boldsymbol{\mu}(f)} \quad \text{(by equation (4.66))} \\ &\quad = \delta_{\boldsymbol{\lambda}\boldsymbol{\mu}} \prod_{f\in\Phi} q_f^{-|\boldsymbol{\lambda}(f)|-2n(\boldsymbol{\lambda}(f)} b_{\boldsymbol{\lambda}(f)}(q_f^{-1})^{-1} \delta_{\boldsymbol{\lambda}(f)\boldsymbol{\mu}(f)} \\ &\quad = a_{\boldsymbol{\lambda}}^{-1} \delta_{\boldsymbol{\lambda}\boldsymbol{\mu}}. \end{aligned} \tag{4.81}$$

Then we see that *ch* is also an isometry.

Theorem 4.2.14. ([12], chapter IV) Let ϕ be an orbit of L and $n \geq 0$, then $e_n(\phi)$ is the characteristic of a character of $G_{nd(\phi)}$, i.e. $ch^{-1}(e_n(\phi))$ is a character of $G_{nd(\phi)}$.

Proof: See ([12] p.283).
There exists another version of the Jacobi-Trudy formula

$$s_\lambda(\phi) = \det(e_{\lambda'_i - i+j})_{1\leq i,j\leq m} \tag{4.82}$$

where $m \geq l(\lambda)$ (see [12], p.41). Therefore

$$s_\lambda(\phi) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) e_{\lambda'_1 - 1 + \sigma(1)}(\phi) \cdots e_{\lambda'_n - n + \sigma(m)}(\phi). \tag{4.83}$$

But $e_{\lambda'_j - j + \sigma(j)}$ is the characteristic of a character of $G_{d(\phi)(\lambda'_j - j + \sigma(j))}$ because of the previous theorem. Therefore, the polynomial $e_{\lambda'_1 - 1 + \sigma(1)}(\phi) \cdots e_{\lambda'_n - n + \sigma(n)}(\phi)$ is the characteristic of a character of $G_{d(\phi)|\lambda|}$. In fact

$$\sum_{j=1}^{n} (\lambda'_j - j + \sigma(j)) = |\lambda| \tag{4.84}$$

for any $\sigma \in S_n$. So, we conclude that if $\tilde{\lambda} : \Phi^* \to \mathcal{P}$ is a weighted partition valued function, then $S_{\tilde{\lambda}} = \prod_{\phi \in \Phi^*} s_{\tilde{\lambda}(\phi)}(\phi)$ is the characteristic of a virtual character $\chi^{\tilde{\lambda}}$ of $G_{||\tilde{\lambda}||}$.

But, from equation (4.9) and Theorem (4.2.13), we see that $\langle \chi^{\tilde{\lambda}}, \chi^{\tilde{\mu}} \rangle = \delta_{\tilde{\lambda}\tilde{\mu}}$ and $\pm\chi^{\tilde{\lambda}}$ must be an irreducible character of $G_{||\tilde{\lambda}||}$. In fact $\chi^{\tilde{\lambda}}$ is the irreducible character (see [9]).

4.3 Formula for the character table of $GL_n(\mathbb{F}_q)$

Let $n \geq 1$, then by Theorem (4.2.14), if $\tilde{\lambda} : \Phi^* \to \mathcal{P}$ is a weighted partition valued function of weight n, $S_{\tilde{\lambda}}$ is the characteristic of a character $\chi^{\tilde{\lambda}}$ of G_n. Therefore

$$\chi^{\tilde{\lambda}} = \sum_{\boldsymbol{\mu}} \chi^{\tilde{\lambda}}_{\boldsymbol{\mu}} \pi_{\boldsymbol{\mu}} \tag{4.85}$$

where the sum ranges over all weighted partition valued functions $\boldsymbol{\mu} : \Phi \to \mathcal{P}$ such that $||\boldsymbol{\mu}|| = n$. So $\chi^{\tilde{\lambda}}_{\boldsymbol{\mu}}$ is the value of the character $\chi^{\tilde{\lambda}}_{\boldsymbol{\mu}}$ in the conjugacy class of G_n represented by $\boldsymbol{\mu}$.

Moreover, the number of weighted partition valued functions $\tilde{\lambda} : \Phi^* \to \mathcal{P}$ of weight n is the same as that of the weighted partition valued functions $\boldsymbol{\mu} : \Phi \to \mathcal{P}$ of weight n, i.e., the number of conjugacy classes of G_n. By orthogonality of the irreducible characters, the set $\{\chi^{\tilde{\lambda}} \mid ||\tilde{\lambda}|| = n\}$ is the set of irreducible characters of G_n. Therefore, the matrix $(\chi^{\tilde{\lambda}}_{\boldsymbol{\mu}})$ for $||\tilde{\lambda}|| = ||\boldsymbol{\mu}|| = n$ is the character table of G_n.

Applying ch on both sides of equation (4.85), we get

$$S_{\tilde{\lambda}} = \sum_{\boldsymbol{\mu}} \chi^{\tilde{\lambda}}_{\boldsymbol{\mu}} \prod_{f \in \Phi} q_f^{-n(\boldsymbol{\mu}(f))} P_{\boldsymbol{\mu}(f)}(f). \tag{4.86}$$

Therefore, by equation (4.66), we get

$$\chi^{\tilde{\lambda}}_{\boldsymbol{\mu}} = \left\langle S_{\tilde{\lambda}}, \prod_{f \in \Phi} q^{n(\boldsymbol{\mu}(f))} Q_{\boldsymbol{\mu}(f)} \cdot 1 \right\rangle. \tag{4.87}$$

Theorem 4.3.1. [9] Let $n \geq 1$, the character table of G_n is given by the matrix $(\chi^{\tilde{\lambda}}_{\boldsymbol{\mu}})$, where

$$\chi^{\tilde{\lambda}}_{\boldsymbol{\mu}} = \left\langle S_{\tilde{\lambda}}, \prod_{f \in \Phi} q^{n(\boldsymbol{\mu}(f))} Q_{\boldsymbol{\mu}(f)} \cdot 1 \right\rangle \tag{4.88}$$

and $\tilde{\lambda}$ (resp. $\boldsymbol{\mu}$) ranges over the set of weighted partition valued functions $\Phi^* \to \mathcal{P}$ (resp. $\Phi \to \mathcal{P}$) of weight n.

Let's do a low degree computation for $q = n = 2$. Since we are interested in weighted partition valued functions $\tilde{\lambda} : \Phi^* \to \mathcal{P}$ of weight 2, the support of $\tilde{\lambda}$ must be contained in Φ_2^*. In this case $L_2 \cong M_2^* \cong \mathbb{Z}_3$, so we can identify Φ_2^* with the set of orbits $\{\{0\}, \{1,2\}\}$ induced by the Frobenius map $x \mapsto 2x$ in $\mathbb{Z}_3$. So we have three possibilities for $\tilde{\lambda}$:

$$\tilde{\lambda} = \begin{cases} \{0\} \mapsto (1^2) \\ \text{oth} \mapsto 0 \end{cases}, \quad \tilde{\lambda} = \begin{cases} \{0\} \mapsto (2) \\ \text{oth} \mapsto 0 \end{cases}, \text{ and } \tilde{\lambda} = \begin{cases} \{1,2\} \mapsto (1) \\ \text{oth} \mapsto 0 \end{cases}. \tag{4.89}$$

From equations (4.89), (4.88) and example (2.2.6), we see that we need to compute the polynomials: $s_{(1^2)}(\{0\}), s_{(2)}(\{0\}), s_{(1)}(\{1,2\}), 2Q_{(1^2)}(t-1), Q_2(t-1)$ and $Q_1(t^2+t+1)$. For the Schur polynomials we can use the vertex operator (4.6) to get

$$\begin{aligned} s_{(1^2)}(\{0\}) &= \frac{1}{2}(p_1(\{0\})^2 - p_2(\{0\})) \\ s_{(1)}(\{1,2\}) &= p_1(\{1,2\}) \\ s_{(2)}(\{0\}) &= \frac{1}{2}(p_1(\{0\})^2 + p_2(\{0\})). \end{aligned} \tag{4.90}$$

Furthermore, using the vertex operator (4.34), we get

$$\begin{aligned} 2Q_{(1^2)}(t-1) \cdot 1 &= \frac{3}{2}(p_1(t-1)^2 - p_2(t-1)) \\ Q_{(1)}(t^2+t+1) \cdot 1 &= 3p_1(t^2+t+1) \\ Q_{(2)}(t-1) \cdot 1 &= \frac{1}{2}p_1(t-1)^2 + \frac{3}{2}p_2(t-1). \end{aligned} \tag{4.91}$$

Also we have

$$\begin{aligned} p_1(t-1) &= p_1(\{0\}) \\ p_2(t-1) &= -\frac{1}{3}(2p_1(\{1,2\}) + p_2(\{0\})) \\ p_1(t^2+t+1) &= \frac{1}{3}(p_1(\{1,2\}) - p_2(\{0\})) \end{aligned} \tag{4.92}$$

because of definitions (4.1.4), (4.1.2) and (4.1.1). Combining the last two set of equations

we get

$$\begin{aligned} 2Q_{(1^2)}(t-1)\cdot 1 &= \frac{3}{2}p_1(\{0\})^2 + \frac{1}{2}p_2(\{0\}) + p_1(\{1,2\}) \\ Q_{(1)}(t^2+t+1)\cdot 1 &= p_1(\{1,2\}) - p_2(\{0\}) \\ Q_{(2)}\cdot 1 &= \frac{1}{2}p_1(\{0\})^2 - p_1(\{1,2\}) - \frac{1}{2}p_2(\{0\}). \end{aligned} \tag{4.93}$$

Using the orthogonality relation $\langle p_\lambda(\phi), p_\mu(\phi')\rangle = z_\lambda \delta_{\phi\phi'}$, we get

$$\begin{aligned} &\langle s_{(1^2)}(\{0\}), 2Q_{(1^2)}(t-1)\cdot 1\rangle = 1 \\ &\langle s_{(1^2)}(\{0\}), Q_{(1)}(t^2+t+1)\cdot 1\rangle = 1 \\ &\langle s_{(1^2)}(\{0\}), Q_{(2)}(t-1)\cdot 1\rangle = 1 \end{aligned} \tag{4.94}$$

so the weighted partition valued function $\tilde{\lambda}$ with support in $\{0\}$ and value (1^2) is associated to the trivial character. Moreover, $GL_2(\mathbb{F}_2) \cong S_3$ so we see that $\tilde{\lambda} : \{1,2\} \mapsto (1)$ is associated to the sign representation and $\tilde{\lambda} : \{0\} \mapsto (2)$ is associated to the unique $2-$dimensional irreducible representation of S_3, which we will call "standard" representation. The character table of S_3 is the following:

	(1)	(12)	(123)
Trivial	1	1	1
Sign	1	-1	1
Standard	2	0	-1

Bibliography

[1] Charles W Curtis and Irving Reiner. *Representation theory of finite groups and associative algebras*. Vol. 356. American Mathematical Soc., 1966.

[2] Alberto De Sole and Victor G Kac. "Finite vs affine W-algebras". In: *Japanese Journal of Mathematics* 1 (2006), pp. 137–261.

[3] Karin Erdmann, Thorsten Holm, et al. *Algebras and representation theory*. Springer, 2018.

[4] Thomas W Hungerford. *Algebra*. Vol. 73. Springer Science & Business Media, 2012.

[5] I Martin Isaacs. *Algebra: a graduate course*. Vol. 100. American Mathematical Soc., 2009.

[6] Nathan Jacobson. *Basic algebra I*. Courier Corporation, 2012.

[7] Nathan Jacobson. *Basic Algebra II, 1989*.

[8] Naihuan Jing. "Vertex operators and Hall-Littlewood symmetric functions". In: *Advances in Mathematics* 87.2 (1991), pp. 226–248.

[9] Naihuan Jing and Yu Wu. "Characters of $GL_n(\mathbf{F}_n)$ and vertex operators". In: *Journal of Algebra* 653 (2024), pp. 109–132.

[10] Victor G Kac, Ashok K Raina, and Natasha Rozhkovskaya. *Bombay lectures on highest weight representations of infinite dimensional Lie algebras*. Vol. 29. World scientific, 2013.

[11] Rudolf Lidl and Harald Niederreiter. *Introduction to finite fields and their applications*. Cambridge university press, 1994.

[12] Ian Grant Macdonald. *Symmetric functions and Hall polynomials*. Oxford university press, 1998.

[13] Louis Halle Rowen. *Graduate Algebra: Noncommutative View*. Vol. 2. American Mathematical Soc., 2008.